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WESTERN INTERNATIONAL SCHOOL

Physics

Grade 10

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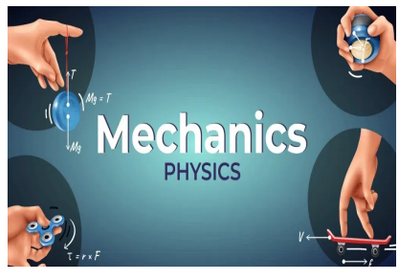
Chapter 1 Mechanics



Lesson 1: Motion in Straight Line

Contents

Section	Learning competencies
1.1 Uniform motion	• Describe the characteristics of uniform motion. • Define the terms distance, displacement, speed and velocity. • Explain the difference between distance and displacement. • Distinguish between average and instantaneous speeds and velocities.
1.2 Uniformly accelerated motion	• Define the term acceleration. • Describe the meaning of the term uniformly accelerated motion. • Explain the meaning of the unit m/s^2. • Use velocity–time graphs to determine the acceleration of an object.
1.3 Graphical description of uniformly accelerated motion	• Describe the key features of distance–time and displacement–time graphs. • Use displacement–time graphs to determine the velocity of an object. • Describe the key features of velocity–time graphs. • Use velocity–time graphs to determine the acceleration of an object and the displacement.
1.4 Equations of uniformly accelerated motion	• Describe the equations of uniformly accelerated motion. • Use these equations to solve problems. • Explain the importance of using the correct sign convention (+ or –) when dealing with velocities and accelerations. • Define the meaning of the term free fall. • Apply the equations to solve problems relating to free fall.

It is almost impossible to imagine yourself living in a world without motion. Stand still, perfectly still; are you in motion? Yes you are… the Earth is spinning at over 450 m/s and even more mind boggling, it is travelling around the Sun with a speed of 30 000 m/s! You are moving very fast.

Every physicist needs a detailed understanding of motion. From catching a ball to driving a car, motion affects our daily lives. How things move is an important aspect of physics.

This unit looks at how things move. You will learn techniques to correctly describe the motion of objects, how to calculate how a certain object will move and the fact that all motion is in fact relative.

KEY WORDS

uniform motion *the motion of an object moving at a steady speed in a straight line*

displacement *distance moved in a particular direction*

Think about this...

If an object is travelling in a circle at a steady speed why is this not considered to be uniform motion?

Activity 2.1: Distance and displacement for a journey

Using a map, design a journey from one town to another. By carefully considering the route determine the distance and the displacement for the journey.

Repeat, but this time make the journey much larger! Perhaps starting at Addis Ababa and ending up in a different continent.

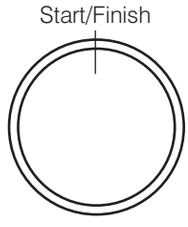


Figure 2.2 *Displacement when travelling in a circle*

Discussion activity

What would the distance and displacement be after half a lap?

What about three and a half laps?

1.1 Uniform motion

By the end of this section you should be able to:

- Describe the characteristics of uniform motion.
- Define the terms distance, displacement, speed and velocity.
- Explain the difference between distance and displacement.
- Distinguish between average and instantaneous speeds and velocities.

What is uniform motion?

In order to understand motion there are several key terms we need to understand. **Uniform motion** refers to an object moving at a steady speed in a straight line. If it is speeding up, slowing down or changing its direction then its motion is not uniform.

An example could be a bus driving at a steady 100 km/h along a straight road. The bus's motion is said to be uniform.

Distance and displacement

We have used the term displacement in the previous unit. **Displacement** is a vector quantity and so it is very different from distance.

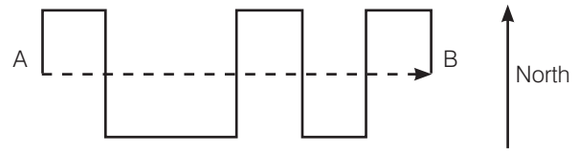


Figure 2.1 *The difference between distance and displacement for a journey*

Imagine a person travels from A to B following the black line (Figure 2.1). They would travel a distance of 32 km. This is how far they have actually travelled.

However, their displacement (the dotted line) would only be 12 km East. This is how far they have travelled in a particular direction (in this case East).

A more extreme example could be athletes running around a circular track. If they complete six laps, with each lap being 1.0 km, then they would have travelled a distance of 6.0 km. However, as they are back where they started, their displacement would be zero!

Each lap covers a distance of 1.0 km but the displacement after each lap is zero.

Speed, ,Velocity

The differences between distance and displacement are even more important when calculating average speed and average velocity.

Where d is distance travelled in meter (m), t is time taken in second(s)

$$v = \frac{\text{distance travelled}}{\text{time taken}} = \frac{d}{t}$$

Activity 2.2

How long does it take you to move around a foot ball field in your school? Five minutes, ten minutes or more? Record the time required to move around the field at least for three different trials.

Trials	Distance (d)	Time (t)	d/t
1			
2			
3			

The distance around the football field remains constant while for many practical reasons the time may not be constant. Sometimes you walk quickly and another time you may run. Therefore, your speed may not be constant. Similarly the motion of bodies, for example a car or a truck will not be uniform. To describe such kinds of motions we use the concept of **average speed**.

$$V_{av} = \frac{\text{Total Distance}}{\text{time taken}} = \frac{d}{t}$$

The SI unit of average speed is meter per second (m/s). Other non SI units such as kilometer per hour (km/hr), cm/s, etc. can also be used as units of speed.

The concept used to describe a speed that has direction is called a **velocity.** Velocity tells us how fast a body is moving in a given direction.

Velocity is a vector quantity.It depends on the displacementmade betweenfinalandinitialpositions. For examples,ifyouwalkfrom yourhometoschool andbacktohomeyourdisplacement is zero. But your distance traveled is not zero.

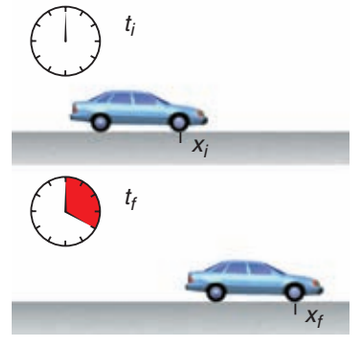


Figure 2,3
The average velocity of this car tells you how fast and in which direction it is moving.

Average velocity is displacement divided by the time interval

Consider the car in **Figure 4.** The car is moving along a highway in a straight line (the *x*-axis). Suppose that the positions of the car are x_i at time t_i and x_f at time t_f. In the time interval $\Delta t = t_f - t_i$, the displacement of the car is $\Delta x = x_f - x_i$. The **average velocity,** v_{avg}, is defined as the displacement divided by the time interval during which the displacement occurred. In SI, the unit of velocity is meters per second, abbreviated as m/s.

average velocity

the total displacement divided by the time interval during which the displacement occurred

AVERAGE VELOCITY

$$v_{av} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$$

$$\text{average velocity} = \frac{\text{change in position}}{\text{change in time}} = \frac{\text{displacement}}{\text{time interval}}$$

The average velocity of an object can be positive or negative, depending on the sign of the displacement. (The time interval is always positive.) As an example, consider a car trip to a friend's house 370 km to the west (the negative direc-tion) along a straight highway. If you left your house at 10 A.M. and arrived at your friend's house at 3 P.M., your average velocity would be as follows:

$$v_{av} = \frac{\Delta x}{\Delta t} = \frac{-370 \text{ km}}{5.0 \text{ h}} = -74 \text{ km/h} = 74 \text{ km/h west}$$

This value is an average. You probably did not travel exactly 74 km/h at every moment. You may have stopped to buy gas or have lunch. At other times, you may have traveled more slowly as a result of heavy traffic. To make up for such delays, when you were traveling slower than 74 km/h, there must also have been other times when you traveled faster than 74 km/h.

The average velocity is equal to the constant velocity needed to cover the given displacement in a given time interval. In the example above, if you left your house and maintained a velocity of 74 km/h to the west at every moment, it would take you 5.0 h to travel 370 km.

TIP

Average velocity is not always equal to the average of the initial and final velocities. For instance, if you drive first at 40 km/h west and later at 60 km/h west, your average velocity is not necessarily 50 km/h west.

Instantaneous speed and velocity

KEY WORDS

instantaneous speed *speed at a given instant in time*

instantaneous velocity *velocity at a given instant in time*

As an alternative, the terms **instantaneous speed** and **instantaneous velocity** are used. In the case of instantaneous velocity, this refers to the velocity at any given instant in time (the same is true for speed).

Instantaneous velocity is often changing. This might be due to the object getting faster, getting slower or even changing direction. This is because velocity is a vector quantity, so if the direction changes so does the velocity.

An extreme example of this is an object going around a circle at a steady speed. Here the speed of the object is constant but its velocity is always changing.

If an object is travelling with uniform motion then the instantaneous velocity (and speed) remains the same.

Summary

In this section you have learnt that:

- Uniform motion is when an object travels at constant speed in a straight line.
- Distance is a scalar quantity, whereas displacement is a vector quantity.
- Average speed = distance travelled / time taken.
- Average velocity = displacement / time taken.
- Instantaneous velocity is the velocity at any given instant in time.

SAMPLE PROBLEM Average Velocity and Displacement

PROBLEM

During a race on level ground, Andra runs with an average velocity of 6.02 m/s to the east. What is Andra's displacement after 137 s?

SOLUTION

Given: $v_{av} = 6.02$ m/s
$\Delta t = 137$ s

Unknown: $\Delta x = ?$

Rearrange the average velocity equation to solve for displacement.

$$v_{av} = \frac{\Delta x}{\Delta t}$$

$$\Delta x = v_{av}\Delta t$$

$$\Delta x = v_{av}\Delta t = (6.02 \text{ m/s})(137 \text{ s}) = \boxed{825 \text{ m to the east}}$$

CALCULATOR SOLUTION

The calculator answer is 824.74 m, but both the values for velocity and time have three significant figures, so the displacement must be reported as 825 m.

PRACTICE A

Average Velocity and Displacement

1. Heather and Matthew walk with an average velocity of 0.98 m/s eastward. If it takes them 34 min to walk to the store, what is their displacement?
2. If Joe rides his bicycle in a straight line for 15 min with an average veloc-ity of 12.5 km/h south, how far has he ridden?
3. It takes you 9.5 min to walk with an average velocity of 1.2 m/s to the north from the bus stop to the museum entrance. What is your displacement?
4. Simpson drives his car with an average velocity of 48.0 km/h to the east. How long will it take him to drive 144 km on a straight highway?
5. Look back at item 4. How much time would Simpson save by increasing his average velocity to 56.0 km/h to the east?
6. A bus travels 280 km south along a straight path with an average velocity of 88 km/h to the south. The bus stops for 24 min. Then, it travels 210 km south with an average velocity of 75 km/h to the south.
   a. How long does the total trip last?
   b. What is the average velocity for the total trip?

1.2 Uniformly accelerated motion

By the end of this section you should be able to:

- Define the term acceleration.
- Explain the meaning of the unit m/s^2.
- Use velocity–time graphs to determine the acceleration of an object

What is acceleration?

KEY WORD

acceleration *the rate of change of velocity*

The term **acceleration** has a very specific definition.

- **Acceleration is the rate of change of velocity.**

This means that whenever an object's velocity is changing it is accelerating. The faster the velocity changes, the greater the acceleration. Acceleration is the change in velocity per unit time.

It is important to note that it is a change in velocity not a change in speed. A change in velocity might be:

- getting faster
- getting slower
- changing direction.

It is possible to travel at a constant speed but with a changing velocity. For example, any object moving at a steady speed in a circle must be accelerating even though its speed is not changing. This is because when an object moves in a circle:

- its direction changes.
- This means its velocity must be changing
- and if its velocity is changing it is accelerating.

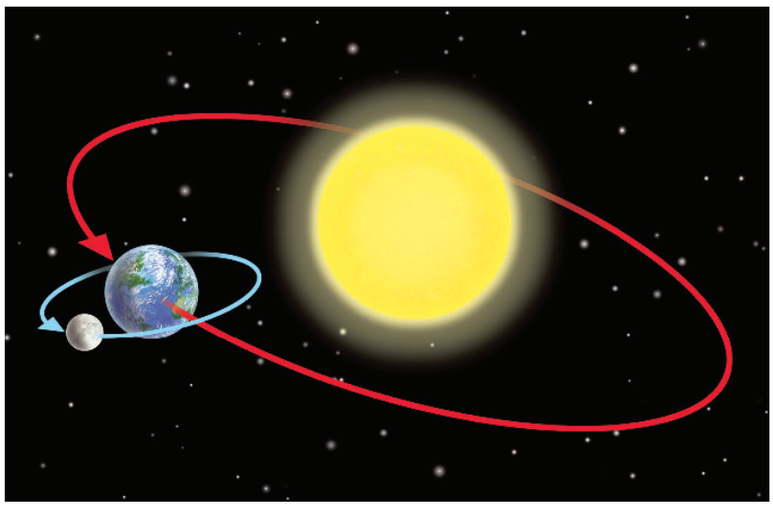


Figure 2.4 *The Earth follows a near perfect circular orbit. It travels at a fairly steady speed of around 30 000 m/s but its velocity is always changing.*

Acceleration calculations

To calculate acceleration we use:

$$\text{average acceleration} = \frac{\text{change in velocity}}{\text{time taken}} \qquad a_{av} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}$$

Acceleration has dimensions of length divided by time squared. The units of acceleration in SI are meters per second per second, which is written as meters per second squared, as shown below. When measured in these units, acceleration describes how much the velocity changes in each second.

$$\frac{(\text{m/s})}{\text{s}} = \frac{\text{m}}{\text{s}} \times \frac{1}{\text{s}} = \frac{\text{m}}{\text{s}^2}$$

Summary

In this section you have learnt that:

- Acceleration is defined as the rate of change of velocity.
- Acceleration is measure in m/s^2.
- When an object is uniformly accelerated, its acceleration remains constant.

SAMPLE PROBLEM

Average Acceleration

PROBLEM

A shuttle bus slows down with an average acceleration of $-1.8\ m/s^2$. How long does it take the bus to slow from 9.0 m/s to a complete stop?

SOLUTION

Given: $v_i = 9.0$ m/s
$v_f = 0$ m/s
$a_{avg} = -1.8\ m/s^2$

Unknown: $\Delta t = ?$

> **TIP** *Watch for implied data in problem statements, such as "starts at rest" ($v_i = 0$ m/s) or "comes to rest" ($v_f = 0$ m/s).*

Rearrange the average acceleration equation to solve for the time interval.

$$a_{avg} = \frac{\Delta v}{\Delta t}$$

$$\Delta t = \frac{\Delta v}{a_{avg}} = \frac{v_f - v_i}{a_{avg}} = \frac{0\ \text{m/s} - 9.0\ \text{m/s}}{-1.8\ \text{m/s}^2}$$

$$\boxed{\Delta t = 5.0\ \text{s}}$$

PRACTICE

Average Acceleration

1. As the shuttle bus comes to a sudden stop to avoid hitting a dog, it accelerates uniformly at $-4.1\ m/s^2$ as it slows from 9.0 m/s to 0.0 m/s. Find the time interval of acceleration for the bus.
2. A car traveling at 7.0 m/s accelerates uniformly at $2.5\ m/s^2$ to reach a speed of 12.0 m/s. How long does it take for this acceleration to occur?
3. With an average acceleration of $-1.2\ m/s^2$, how long will it take a cyclist to bring a bicycle with an initial speed of 6.5 m/s to a complete stop?
4. Turner's treadmill runs with a velocity of −1.2 m/s and speeds up at regular intervals during a half-hour workout. After 25 min, the treadmill has a velocity of −6.5 m/s. What is the average acceleration of the treadmill during this period?
5. Suppose a treadmill has an average acceleration of $4.7 \times 10^{-3}\ m/s^2$.
   a. How much does its speed change after 5.0 min?
   b. If the treadmill's initial speed is 1.7 m/s, what will its final speed be?

1.3 Graphical description of uniformly accelerated motion

By the end of this section you should be able to:

- Describe the key features of distance–time and displacement–time graphs.
- Use displacement–time graphs to determine the velocity of an object.
- Describe the key features of velocity–time graphs.
- Use velocity–time graphs to determine the acceleration of an object and the displacement.

Motion graphs

Average velocities can only tell us a certain amount of information. If we need more detail then motion graphs are used. In order to determine instantaneous velocities we can plot displacement–time graphs.

A graph is a useful way of showing how something has moved. To draw a graph, we need information about an object's displacement at different times. Table 2.1 shows the displacement of a cyclist on the way to school.

Table 2.1 *Displacement of a cyclist*

Displacement (m)	0	80	160	240	240	280
Time (s)	0	20	40	60	80	100

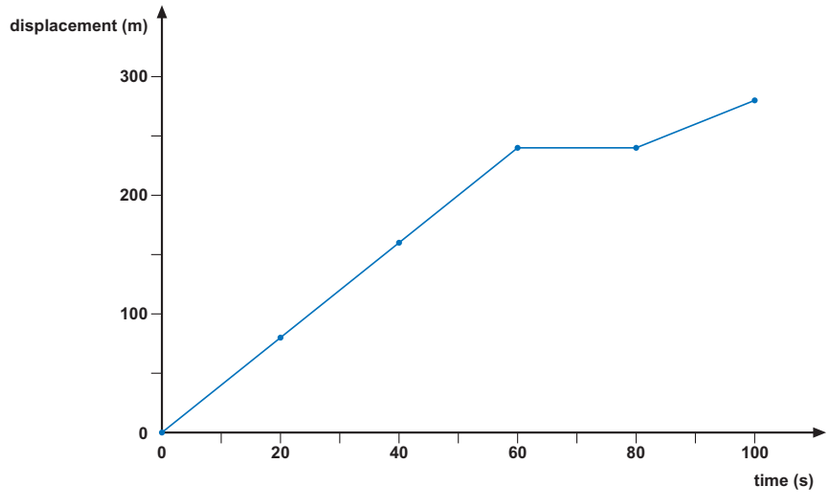


Figure 2.6 *Displacement–time graph for a cyclist*

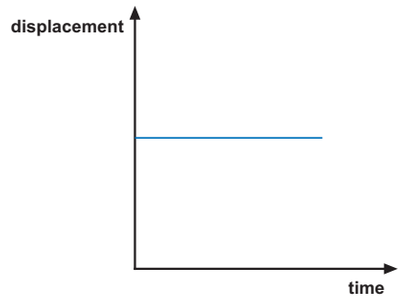
The information in the table has been used to draw the graph (Figure 2.6). Note the axes of the graph have been carefully labelled to show the quantity and unit:

- time in seconds on the *x*-axis
- displacement in metres on the *y*-axis.

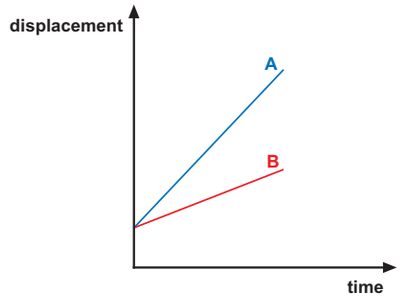
We can tell quite a lot from this graph.

- At first, the graph is a straight line sloping upwards. The cyclist went at a steady speed for the first 60 s.
- Then the graph is horizontal. The cyclist stopped for 20 s.
- Then the graph slopes upwards again, but less steeply. During the last 20 s, the cyclist moved more slowly than before.

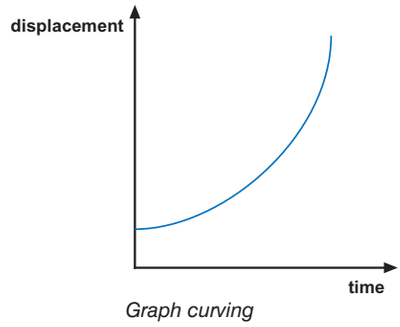
Figure 2.7 summarises how to interpret the shape of a displacement–time graph. You can see that the steeper the gradient (slope) of the graph, the greater the velocity of the moving object. A curved graph indicates that the object's velocity is changing.



Horizontal line
Constant displacement
Velocity = 0



Straight line, sloping upwards
Constant velocity
A has a greater velocity than B



Graph curving
Velocity is changing

Figure 2.7 *Different displacement–time graphs*

Calculating velocity

From the displacement–time graph, we can work out an object's velocity (as explained in the worked example):

- Velocity = gradient of displacement–time graph.

Worked example

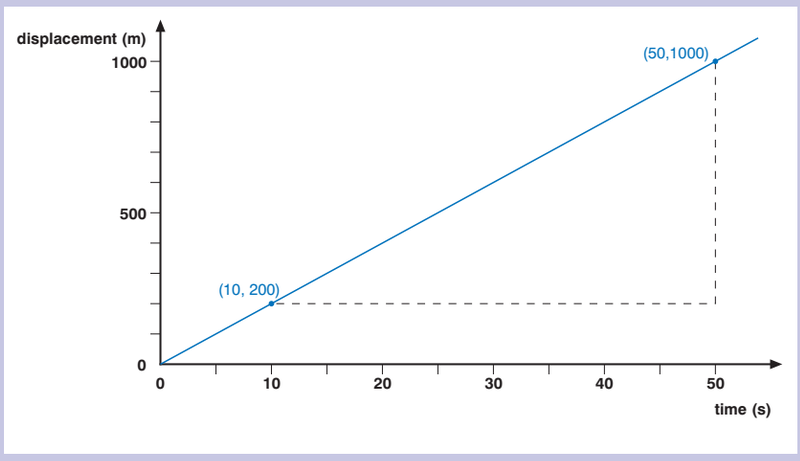


Figure 2.8 *Displacment–time graph of a taxi*

Figure 2.8 is a displacement–time graph for a moving taxi. Find its velocity.

Choose two points on the graph (they should not be too close together).

Draw horizontal and vertical lines to complete a right-angled triangle.

Calculate the displacement and time represented by these two sides of the triangle:

displacement = 1000 m – 200 m = 800 m
time = 50 s – 10 s = 40 s

Activity 2.3: Distance–time graph on way to school

Carefully sketch out a distance–time graph for your journey into school. Describe each section of your graph with a partner.

Calculate velocity in the usual way:

$$\text{velocity} = \frac{\text{displacement}}{\text{time}} = \frac{800\ \text{m}}{40\ \text{s}} = 20\ \text{m/s}$$

So the taxi is travelling at 20 m/s.

Distance–time and displacement–time graphs

Although the key features are the same, there is one big difference between distance–time and displacement–time graphs.

As distance is a scalar quantity it only goes up and up. The distance never goes down.

However, as displacement is a vector quantity it can also go down. For example, if you walk 10 m away from your friend heading North and then stop you have travelled a distance of 10 m and your displacement is 10 m North. However, if you then turn around and walk 6 m back towards your friend you will have travelled 16 m but your displacement would then be only 4 m North. This can be seen in the two examples below.

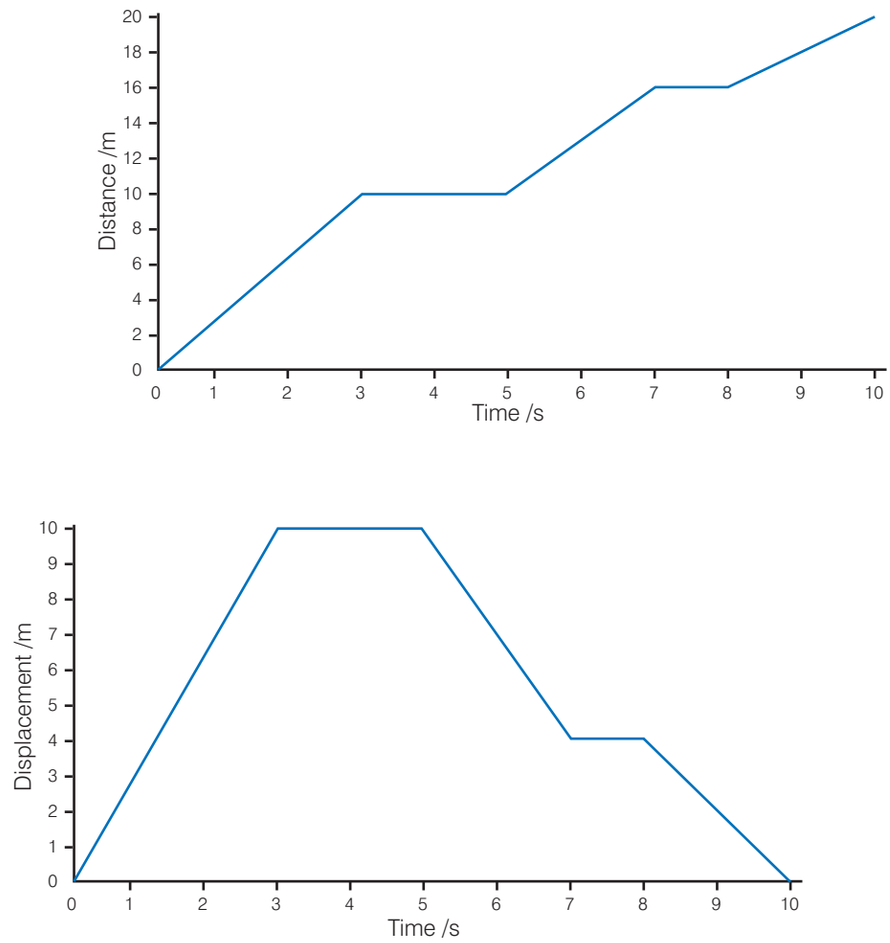


Figure 2.9 *Distance–time and displacement–time graphs for the same journey.*

You can clearly see the displacement begin to fall as you head back in the direction you came from. Eventually if you end up back by your friend your displacement will be 0 m but you will have travelled a distance of 20 m.

	Gradient	Negative gradients
Distance–time graph	Speed	No
Displacement–time graph	Velocity	Yes

Table 2.2 *A comparison between distance–time and displacement–time graphs*

This means you can get negative values from the gradient of a displacement–time graph but not from a distance–time graph. This makes sense if you think about it. You might get a negative velocity of –4 m/s but negative speeds do not make any sense.

Velocity–time graphs

Just as a displacement–time graph shows how far an object has moved, a velocity–time graph shows how its velocity changes as it travels along. Figure 2.10 shows an example; in this case, the motion of a car at the start of its journey. We can deduce several points from the graph.

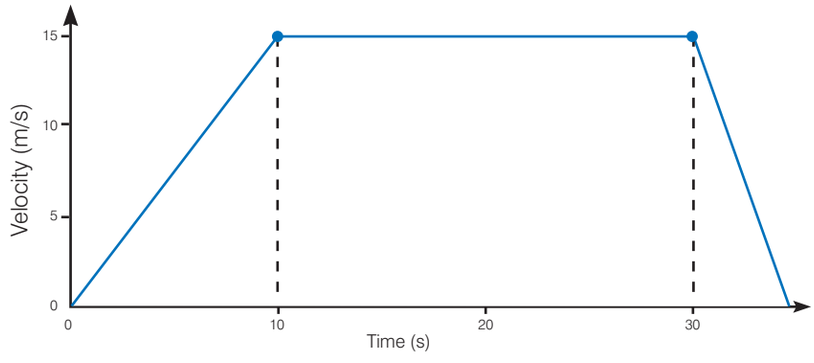


Figure 2.10 *A velocity–time graph for a car.*

At the start, the car was not moving.

- velocity = 0 when time = 0

The car accelerated at a steady rate during the first 10 s until it reached a velocity of 15 m/s.

- the graph is a straight line, sloping upwards The car travelled at 15

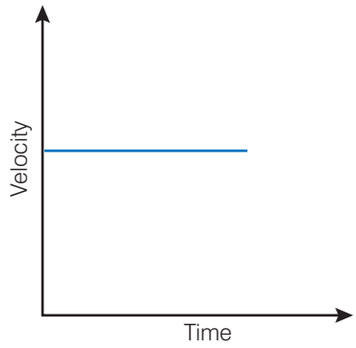
m/s for 20 s.

- the graph is horizontal, so acceleration = 0 After 30 s, the car

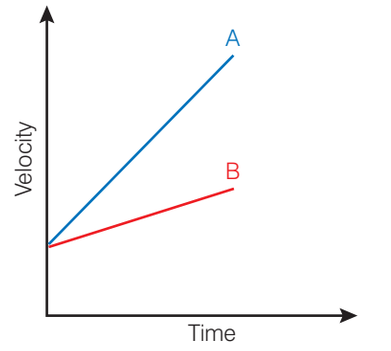
decelerated rapidly to a halt.

- graph slopes steeply down to velocity = 0

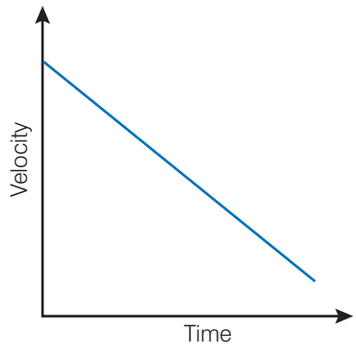
You can learn a lot from the shape of a velocity–time graph, as shown in Figure 2.11. Take care! Do not confuse these with displacement–time graphs. Always check the labels on the axes before interpreting a graph.



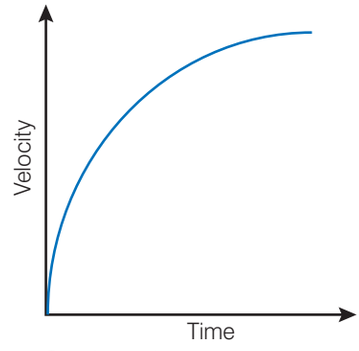
Horizontal line
Constant velocity
Acceleration = 0



Straight line, sloping upwards
Constant acceleration
A has a greater acceleration than B



Straight line, sloping downwards
Decelerating
Acceleration is negative



Graph curving
Acceleration is changing

Figure 2.11 *Four velocity–time graphs.*

If the velocity–time graph is a straight line, the object's acceleration is constant, and we say that it is moving with **uniform acceleration**.

Calculating acceleration and displacement

We can calculate two quantities from a velocity–time graph. The worked examples show how to do this.

- Acceleration is the gradient of a velocity–time graph.
- Displacement is the area under a velocity–time graph.

Worked example

Figure 2.12 shows how the velocity of a train changed as it set off from a station. Calculate its initial acceleration.

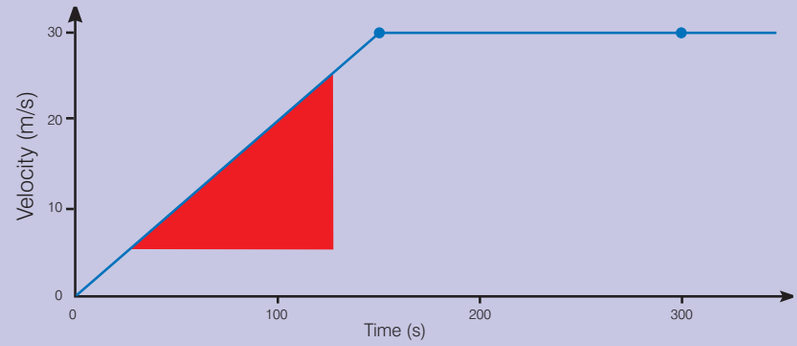


Figure 2.12 *Velocity–time graph for a train*

- Choose two points on the graph. As before, select points that are far apart.
- Complete a right-angled triangle.
- Calculate the change in velocity and the time taken:

  change in velocity = 25 m/s – 5 m/s = 20 m/s

  time taken = 125 s – 25 s = 100 s
- Calculate the acceleration:

  acceleration = gradient of graph =

  $$\frac{20 \text{ m/s}}{100 \text{ s}} = 0.2 \text{ m/s}^2$$

Calculate the distance travelled by the train during the first 300 s of its journey.

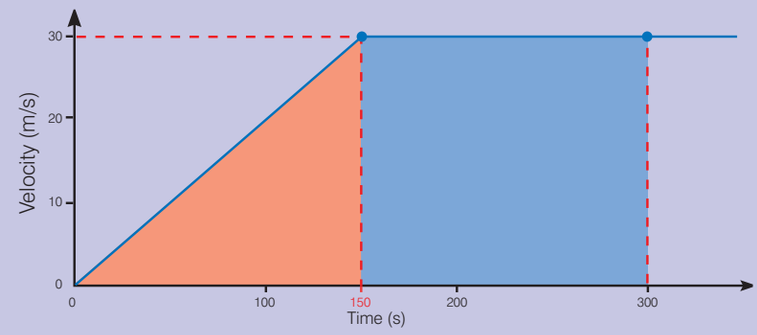


Figure 2.13 *Finding the displacement of the train from its velocity–time graph.*

Figure 2.13 shows the same graph as Figure 2.12; this time, though, we have to calculate displacement, which is equal to the area under the graph. The area is divided into two parts: a triangle and a rectangle. (Area of triangle = ½ × base × height; area of rectangle = base × height.)

displacement = area of triangle + area of rectangle

= (½ × 30 m/s × 150 s) + (30 m/s × 150 s)

= 2250 m + 4500 m

= 6750 m

KEY WORDS

uniform acceleration *a constant value of acceleration*

Think about this...

It is important to note that the area under the line may also be negative; this would indicate a negative displacement. In this case the line would dip under the *x*-axis.

Activity 2.4: Graphs that tell stories

A velocity–time graph can tell the story of a journey. Here is one driver's description of a recent trip.

'We crawled along through the city traffic at 6 m/s for five minutes. Then we left the city, and we gradually accelerated to 24 m/s in 20 s. We kept going at this speed for two minutes, but then I noticed an accident on the road ahead and I braked, so that we came to a halt in 8 s.

1 Draw a graph to represent this journey. (Remember, all the times must be in seconds.)

2 From your graph, calculate the car's acceleration and deceleration.

3 Calculate the total distance travelled by the car. Now, make up your own story and challenge a partner to draw the graph and make the calculations.

Summary

In this section you have learnt that:

- Distance–time, displacement–time and velocity–time graphs may be used to represent an object's motion.
- The gradient of a displacement–time graph is equal to the velocity of the object.
- The gradient of the line of a velocity–time graph is equal to the acceleration.
- The area under the line of a velocity–time graph is equal to the displacement.
- Acceleration is defined as the rate of change of velocity.
- Acceleration is measured in m/s^2.

Review questions

1. Draw a displacement–time graph for the following:

Displacement (m)	0	20	40	40	80	80	60	40	0
Time (s)	0	10	20	30	40	50	60	70	80

   a) Explain the different sections of the graph in as much detail as you can.

   b) Use the graph to determine the maximum velocity.

   c) Find the average velocity after 45 s.

   d) Find the instantaneous velocity at 45 s.

2. The following data were collected during a short race between two friends.

Velocity (m/s)	0	0.5	1	1.5	2	2	4	6	2	0
Time (s)	0	2	4	6	8	10	12	14	16	18

   a) Describe the different sections of the graph.

   b) Determine the acceleration over the first eight seconds.

   c) Determine the maximum acceleration.

   d) Using the graph calculate the displacement:

      i) over the first eight seconds

      ii) the total race.

   e) Find the maximum velocity reached by the runner.

1.4 Equations of uniformly accelerated motion

By the end of this section you should be able to:

- Describe the equations of uniformly accelerated motion.
- Use these equations to solve problems.
- Explain the importance of using the correct sign convention (+ or –) when dealing with velocities and accelerations.
- Define the meaning of the term free fall.
- Apply the equations to solve problems relating to free fall.

As discussed in Section 2.2, acceleration has a very specific definition.

- Acceleration is the rate of change of velocity.

This can be written as:

average acceleration = change in velocity / time taken

If the acceleration is uniform (i.e. does not change) then the average acceleration is the same as the acceleration during any given time.

So we could rewrite that equation as:

acceleration = change in velocity / time

But, only if the acceleration is constant.

To calculate change in velocity we could use the equation below:

change in velocity = final velocity – initial velocity

Or in symbols:

change in velocity = $v - u$

where

v = final velocity

u = initial velocity

This means our first equation could be written as:

$a = (v - u)/t$

where

v = final velocity

u = initial velocity

a = acceleration

t = time

This first equation is usually written as

$v = u + at$ (1)

For example, if a car is travelling at 8 m/s and accelerates with uniform acceleration at 2 m/s^2 for 6 s its final velocity will be:

$v = u + at$ *State principle or equation to be used*

$v = 8$ m/s + (2 m/s^2 × 6 s *Substitute in known values and complete calculation*

$v = 20$ m/s *Clearly state the answer with unit*

This equation is often referred to as the first equation of the equations of uniformly accelerated motion; there are four more. Remember, this only applies if the acceleration is constant.

The second equation comes from the definition of velocity: Velocity is the rate of change of displacement

This can be written as:

average velocity = displacement / time

If the acceleration is uniform then the average velocity can be found by:

average velocity = (final velocity + initial velocity / 2

So the equation becomes:

(initial velocity + final velocity) / 2 = displacement / time

Or in symbols $\boldsymbol{(u + v)/2 = x/t}$

Where x= displacement, v = final velocity, u= initial velocity, t = time

DID YOU KNOW?

The Greek symbol delta Δ is often used to represent 'change in'. So the formula for acceleration could be written as $a = \Delta v / t$.

Think about this...

To help confusing v and u, remember that u comes before v in the alphabet and so u is the initial velocity, the velocity before v!

Activity 2.5: Deriving equations

Using algebra, derive the three remaining equations from the two equations given opposite.

Symbols used in the equations

x = displacement

v = final velocity

u = initial velocity

a = acceleration

t = time

DID YOU KNOW?

These equations are often referred to as the SUVAT equations. But don't forget, they only apply if the acceleration of the object is uniform (constant).

Rather confusingly, s if often used for displacement. Be careful not to confuse this for speed!

This second equation is usually written as:

$$x = \tfrac{1}{2}(u + v)t \quad (2)$$

This gives us two of the five equations:

$$v = u + at \quad (1)$$

$$x = \tfrac{1}{2}(u + v)t \quad (2)$$

Notice that these equations only use five quantities: s, u, v, a and t.

The first one is missing s, the second one is missing a. The three remaining equations are each missing one of the remaining quantities. They are derived from the two above.

The complete set of equations in their usual form can be seen below:

$v = u + at$ (1) (no s)

$x = \tfrac{1}{2}(u + v)t$ (2) (no a)

$x = ut + \tfrac{1}{2}at^2$ (3) (no v)

$v^2 = u^2 + 2as$ (4) (no t)

$x = vt - \tfrac{1}{2}at^2$ (5) (no u)

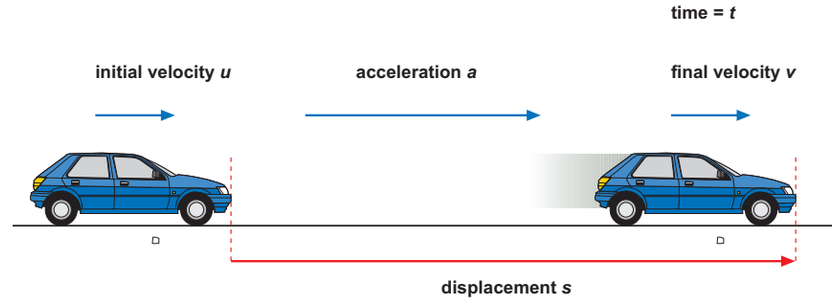


Figure 2.14 *The five quantities that appear in the equations of motion.*

Using the equations

These equations can be used to solve a range of problems regarding the motion of accelerating objects. There are lots of terms to use and so to avoid confusion it is often a good idea to draw a quick table like the one below:

Table 2.3 *A table of motion quantities*

x(m)	u (m/s)	v (m/s)	a (m/s^2)	t (s)

You can then fill in the quantities you know and this will help you select the correct equation.

For example:

A cheetah accelerates at 3 m/s^2 for 5 s. If its final velocity is 24 m/s, determine its initial velocity.

We can now fill in what we know.

Figure 2.15 *Cheetahs are the fastest land animals, reaching speeds of 120 kph!*

x(m)	*u* (m/s)	*v* (m/s)	*a* (m/s^2)	*t* (s)
	?	24	3	5

From the table you can see we don't have *s* so we have to use equation (1), the only one without *s* in it.

$v = u + at$ *State principle or equation to be used*

Rearranging to give *u* gives

$u = v - at$ *Rearrange equation to make u the subject*

$u = 24 \text{ m/s} - (3 \text{ m/s}^2 \times 5 \text{ s})$ *Substitute in known values and complete calculation*

$u = 9 \text{ m/s}$ *Clearly state the answer with unit*

Here is another example. A runner in a race decides to accelerate right up to the moment he crosses the line. He is initially travelling at 5 m/s and accelerates at 0.4 m/s^2 for 5 s. Find:

Figure 2.16 *How fast does the runner finish?*

i) The distance from the line when he decides to accelerate.

ii) His final velocity as he crosses the line.

Again we can fill in what we know.

x (m)	*u* (m/s)	*v* (m/s)	*a* (m/s^2)	*t* (s)
?	5		0.4	5

From the table you can see we don't have *v* so we have to use equation (3), the only one without *v* in it.

$s = ut + \frac{1}{2}at^2$ *State principle or equation to be used*

$s = (5 \text{ m/s} \times 5 \text{ s}) + \frac{1}{2} \times 0.4 \text{ m/s}^2 \times (5 \text{ s})^2$ *Substitute in known values and complete calculation*

$s = 30 \text{ m}$ *Clearly state the answer with unit*

Adding this to the table we get.

x (m)	*u* (m/s)	*v* (m/s)	*a* (m/s2)	*t* (s)
30	5	?	0.4	5

To find *v* we can use any equation apart from equation (5). Perhaps the best one to use is equation (1) as this does not rely on the value for *s*. You may have miscalculated this so it's better to be safe and use values you are certain of if at all possible.

$v = u + at$ *State principle or equation to be used*

$v = 5 \text{ m/s} + (0.4 \text{ m/s}^2 \times 5 \text{ s})$ *Substitute in known values and complete calculation*

$v = 7 \text{ m/s}$ *Clearly state the answer with unit*

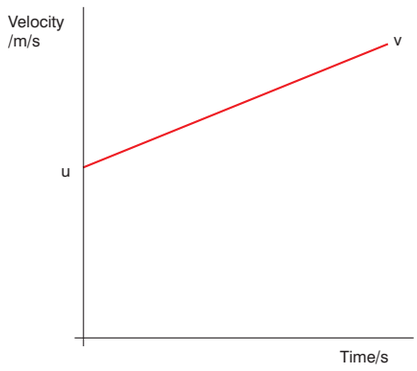


Figure 2.17 *A typical velocity–time graph*

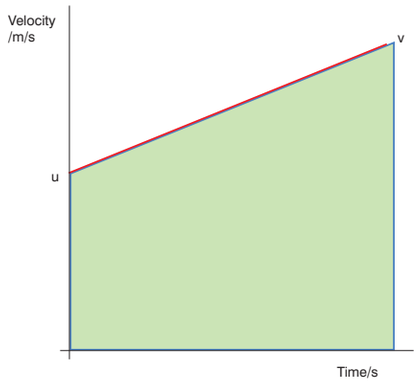


Figure 2.18 *The area under the line represents the displacement of the object.*

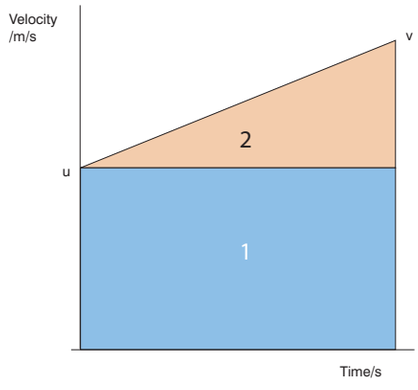


Figure 2.19 *The area can be split into two sections.*

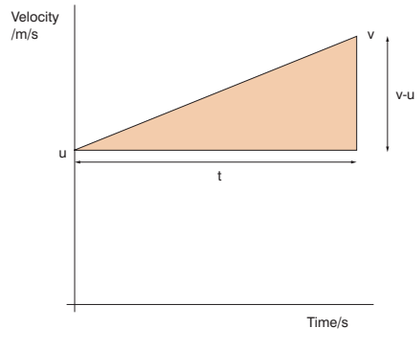


Figure 2.20 *The area of the triangle*

Velocity–time graphs for $x = ut + \frac{1}{2}at^2$

Equation (3) can be derived using ideas covered in section 2.3.

A velocity–time graph for an object with constant acceleration might look like the one in Figure 2.17. This might be a marble rolling down an inclined ramp with the velocity measured at two points along the ramp.

The gradient of the line is constant because the acceleration of the object is constant.

The total area under the graph represents the displacement of the object between these two velocities (see Figure 2.18).

This area has two sections, shown as 1 and 2 in Figure 2.19.

The area of the first section is simply $u \times t$ or ut. This added to the second area will give the displacement.

The area of the triangle (Figure 2.20) is given by:

$$\frac{1}{2}(v - u) \times t$$

From equation (1), $v = u + at$, it follows that $v - u = at$ and so the area can be expressed as:

$$\frac{1}{2}at \times t \text{ or } \frac{1}{2}at^2$$

The total area is given by the two areas added together. This gives:

$$\text{total area} = ut + \frac{1}{2}at^2$$

So, the total area is the same as the displacement:

$$x = ut + \frac{1}{2}at^2$$

If the acceleration was zero the graph would be a horizontal line; the area in this case would be just ut. In other words, $\frac{1}{2}at^2$ would be 0. Equally, if the object started from rest then u would be 0 and the graph would be just a triangle, in which case the area would be just $\frac{1}{2}at^2$ as ut would be 0.

Positive or negative?

As both velocity and acceleration are vector quantities their directions are very important. If the velocity is in the same direction as the acceleration then both could be considered to be positive. However, if they are in opposite directions then one must be negative.

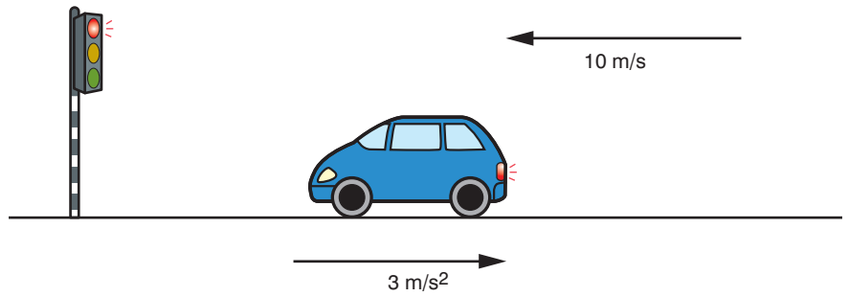


Figure 2.21 *A car braking at traffic lights*

As an example, Figure 2.21 shows a car approaching a set of traffic lights. If the car has to stop, its velocity is in one direction but the acceleration is in the opposite direction (since it is slowing down). This would give us a velocity of 10 m/s and an acceleration of –3 m/s^2.

Imagine a ball rolling up a very long slope with an initial velocity of 6 m/s. The acceleration acts down the slope and has a value of 2 m/s^2. If we wanted to find the velocity of the ball after two seconds we could use one of our equations of constant acceleration.

x (m)	u (m/s)	v (m/s)	a (m/s^2)	t (s)
	6	?	2	2

This table is wrong. We have both initial velocity and acceleration as positive. This is not right as they are in opposite directions.

If we were to use $v = u + at$ using these values we would get a final velocity of 10 m/s. The ball has got faster as it has travelled up the slope!

Instead if we decide to say the velocity up the slope is positive we get

x (m)	u (m/s)	v (m/s)	a (m/s^2)	t (s)
	6	?	–2	2

The acceleration is –2 m/s^2 as we have decided that the positive direction is up the slope.

$v = u + at$ *State principle or equation to be used*

$v = 6$ m/s + (–2 m/s^2 × 2 s) *Substitute in known values and complete calculation*

$v = 2$ m/s *Clearly state the answer with unit*

This makes much more sense! The ball has got slower.

What about if we wanted the velocity after 10 s? Filling in the table we would get:

x (m)	u (m/s)	v (m/s)	a (m/s^2)	t (s)
	6	?	–2	10

The acceleration is –2 m/s^2 as we have decided that the positive direction is up the slope.

$v = u + at$ *State principle or equation to be used*

$v = 6$ m/s + (–2 m/s^2 × 10 s) *Substitute in known values and complete calculation*

$v = -14$ m/s *Clearly state the answer with unit*

Our answer is –14 m/s. What does this mean? Because we decided to make the direction up the slope positive, –14 m/s must mean the ball has gradually slowed down, stopped and then rolled back down. After 10 s it is travelling down the slope at 14 m/s.

Think about this...

It does not really matter which one is negative as long as we think carefully about our answers. Using the car example it would be equally valid to say the velocity is –10 m/s and the acceleration is 3 m/s^2.

Figure 2.22 *Ball rolling up a slope*

Think about this...

We would have got the same answer if we had made the acceleration positive and the initial velocity in the negative direction. Except our final answer would be +14 m/s; indicating it is in the same direction as the acceleration.

Equally, using $s = ut + \frac{1}{2}at^2$ we would get a displacement of –40 m, meaning the ball is 40 m lower down the slope than when it started.

Figure 2.23 *Air resistance is very important to parachutists!*

DID YOU KNOW?

Technically, as the definition of free fall does not include any mention of velocity (either magnitude or direction), it also applies to objects initially moving upward. For example, a small marble thrown vertically up into the air is undergoing free fall on both the way up and the way down!

Figure 2.24 *The Leaning Tower of Pisa*

Free fall

Free fall is a kind of motion where the acceleration of the object is just due to the acceleration due to gravity. For this to take place we must assume that air resistance (drag) is not acting on the object. For most examples we are going to look at this as a fair assumption. Air resistance only plays an important role if the object is moving quite fast or has a very large surface area. However, there are plenty of cases when we will need to consider air resistance in the future (for example, a parachutist!).

Around 1590, there was a story about Galileo Galilei (1564–1642), an Italian scientist. It is said he climbed up the Leaning Tower of Pisa to test out his theory of free fall. He dropped two cannon balls, one large one, one small one. Everyone watching thought the larger one, that is the one with more mass, would hit the ground first. Instead they both hit the ground at the same time. Galileo had realised that all objects dropped on Earth accelerate at the same rate; it is only air resistance that slows them down.

When an object is undergoing free fall it will accelerate at 9.81 m/s^2; this is the acceleration due to gravity on the surface of the Earth. It is important to note that if we ignore air resistance then all objects, regardless of their mass, will accelerate at this rate.

This is a little counter-intuitive; our experiences work against us when thinking about free fall. If you imagine a stone and a piece of paper being dropped, it is obvious the stone will hit the ground first! However, this is due to air resistance having a greater effect on the piece of paper. Both the stone and paper initially accelerate at the same rate.

On the Moon there is no atmosphere and so no air resistance. In 1971, American astronaut David Scott simultaneously dropped a hammer and a feather from the same height to demonstrate free fall. The hammer and the feather both fell exactly at the same rate and so hit the ground at the same time!

If we ignore air resistance then the acceleration of all falling objects can be considered to be uniform. We can then use the equations of uniform acceleration to determine how long objects take to hit the ground and what their final velocity is just before impact.

For example, imagine a ball dropped from a height of 4.0 m. How long would it take to hit the ground?

x (m)	u (m/s)	v (m/s)	a (m/s^2)	t (s)
4.0	0.0 (as dropped)		9.81	?

You can see we've used the initial velocity as 0 m/s, as the ball is dropped, and the acceleration as 9.81 m/s^2.

Worked example

We don't know the final velocity of the ball so we must use equation (3) (there is no v in this equation).

$x = ut + \frac{1}{2}at^2$ *State principle or equation to be used*

$ut = 0$, as the ball was dropped, so the equation becomes:

$x = \frac{1}{2}at^2$

This can rearranged to $t = \sqrt{2x/a}$ *Rearrange equation to make t the subject*

$t = \sqrt{(2 \times 4.0 \text{ m}/9.81 \text{ m/s}^2}$ *Substitute in known values and complete calculation*

$t = 0.9$ s *Clearly state the answer with unit*

You can see from this that it does not matter what the mass of the ball is. Any object dropped from 4 m will hit the ground after 0.9 s if we ignore air resistance.

Using our equations of uniform acceleration we can also work out the final vertical velocity. Looking back at the table we now have:

x (m)	u (m/s)	v (m/s)	a (m/s^2)	t (s)
4.0	0.0 (as dropped)	?	9.81	0.9

Worked example

We could use either equation (1, (2, (4 or (5 to determine v. However, equation (4 does not rely on your calculation of time, so this is preferable.

$v^2 = u^2 + 2ax$ *State principle or equation to be used*

$v = \sqrt{(u^2 + 2ax)}$ *Rearrange equation to make v the subject*

$v = \sqrt{(0^2 + 2 \times 9.81 \text{ m/s}^2 \times 4.0 \text{ m}}$ *Substitute in known values and complete calculation*

$v = 8.9$ m/s *Clearly state the answer with unit*

The equations can also be used if the ball is thrown vertically upwards. In this case it is the same process, but u is not 0 m/s and it is very important to remember that u is in one direction and a is in the other. One will have to be negative!

For example, we can use the equations to work out how long it takes a ball thrown vertically with a velocity of 20 m/s to reach its maximum height and how high it reaches.

Looking at the table we have:

x (m)	u (m/s)	v (m/s)	a (m/s^2)	t (s)
	20	0	9.81	?

At its maximum height, the velocity of the ball will be 0 m/s. To find t we use equation (1).

Figure 2.25 *Astronaut on the Moon*

Activity 2.6: Dropping a ball

Drop a ball from several different heights and time how long it takes to hit the ground. Record your data carefully and take repeats for each height.

Using equation (3), calculate the time it actually takes to hit the ground. Compare the actual times with your readings and comment on your findings.

DID YOU KNOW?

The acceleration due to **gravity** varies all over the globe. At sea level it ranges between 9.79 m/s^2 and 9.83 m/s^2 depending on location. It also changes with altitude (although not by very much). So we often use a standard value of exactly 9.80665 m/s^2.

KEY WORDS

gravity *the force of attraction between an object in the Earth's gravitational field and the Earth itself*

Think about this...

If you drop an object, the displacement before it hits the ground is given by $x = \frac{1}{2}at^2$. If you take a as 10 m/s^2 (close enough), this becomes $x = 5t^2$ If it takes 1 s to hit the ground then it must have fallen 5 m, 2 s means 20 m, etc. This is handy to work out to approximate height of bridges or depth of wells. Just make sure nobody is standing underneath!

$v = u + at$ *State principle or equation to be used*

$t = (v - u)/a$ *Rearrange equation to make t the subject*

$t = (0 \text{ m/s} - 20 \text{ m/s}) / -9.81 \text{ m/s}^2$ *Substitute in known values and complete calculation*

$t = 2.0$ s *Clearly state the answer with unit*

A similar process gives $x = 14$ m. Try it for yourself!

An object in free fall produces very distinctive displacement–time and velocity–time graphs. The displacement–time and velocity-time graphs for an object in free fall can be seen in Figure 2.26.

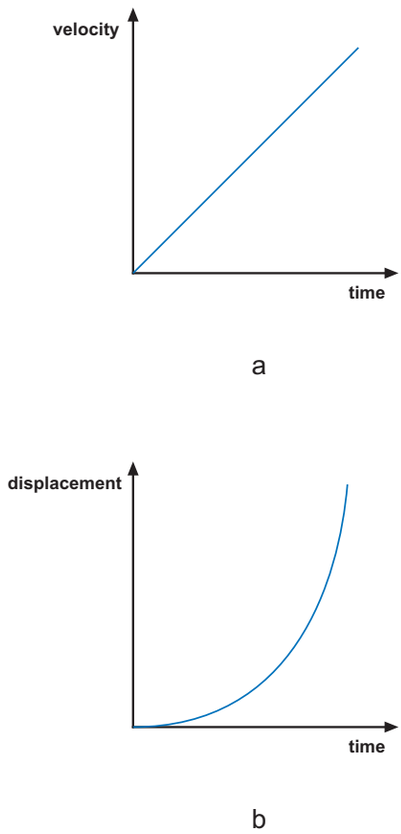


Figure 2.26 *Motion graphs for objects undergoing free fall*

Summary

- There are five equations that describe uniformly accelerated motion; these can be used to solve a range of problems.
- The directions of the velocity and the acceleration of an object are important when deciding whether they are positive or negative values.
- When an object accelerates under gravity it is said to be in free fall.
- The equations of uniform acceleration can be used to solve problems relating to free fall.

Review questions

1. What are the five equations that describe uniform accelerated motion?
2. A bus accelerates from 10 m/s to 18 m/s over 3 s. Find:
   a) The distance the bus covers whilst it is accelerating.
   b) The acceleration of the bus.
3. A runner slows down after completing a race. Her deceleration is 0.25 m/s^2. After 5 s she is travelling at 4 m/s, determine her initial velocity.
4. A stone is dropped off a tall building. It takes 5.3 s to hit the ground. Determine the height of the building.
5. Explain what is meant by free fall.

Summary

In this section you have learnt that:

- A frame of reference refers to a certain point of view depending on the position and motion of the observer.
- The laws of motion only apply if the reference frame of the observer is stationary or moving at a constant velocity.
- The velocity of an object depends on the frame of reference of the observer.
- The relative velocity between one moving object and another is given by the difference between their velocities.

End of unit questions

1. How long will a bus take to travel 150 km at an average speed of 40 km/h?
2. A cheetah can run at 30 m/s, but only for about 12 s. How far will it run in that time?
3. It takes a cheetah just 3 s to reach its top speed of 30 m/s. What is its acceleration?
4. Table 2.4 shows how the displacement of a runner changed during a sprint race. Draw a displacement–time graph to show this data, and use it to deduce the runner's speed in the middle of the race.

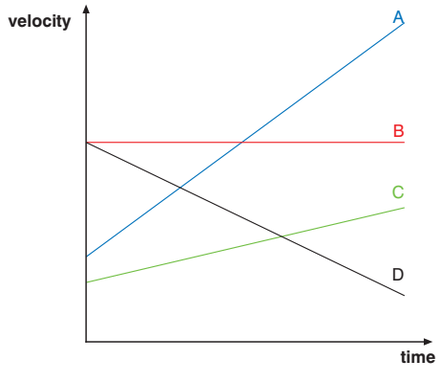


Figure 2.31 *Velocity–time graphs for four cars.*

Table 2.4 *Data for a sprinter during a race*

Displacement (m)	0	4	10	20	50	80	105
Time (s)	0	1	2	3	6	9	12

5. Figure 2.31 shows how the velocity of four cars changed as they travelled along a straight road. Give reasons for your answers to these questions:
   (a) Which car was travelling at a steady speed?
   (b) Which car was decelerating?
   (c) Which car had the greatest acceleration?
6. Table 2.5 shows how the velocity of a car changed during part of a journey along a main road.
   (a) Draw a velocity–time graph for the journey.
   (b) Write a brief description of the journey.
   (c) The car's speed changed during two parts of the journey. Calculate its acceleration at these times.

Table 2.5 *Data for part of a car journey – see Question 6*

Velocity (m/s)	16	20	24	24	24	21	18
Time (s)	0	10	20	40	60	70	80

7. A taxi is travelling at 15 m/s. Its driver accelerates with acceleration 3 m/s^2 for 4 s. What is its new velocity?
8. A car accelerates from 20 m/s to 30 m/s in 10 s.
   (a) Calculate the car's acceleration using $v = u + at$.
   (b) Draw a velocity–time graph to show the car's motion. Find the distance it travels by calculating the area under the graph.
   (c) Check your answer by using the equation $s = ut + \frac{1}{2}at^2$.

9. A truck gradually starts off from rest with a uniform acceleration of 2 m/s^2. It reaches a velocity of 16 m/s. Using the equation $v^2 = u^2 + 2as$, calculate the distance it travels while it is accelerating.

10. Table 2.6 shows values of the displacement and velocity of a falling object. Copy and complete the table, and use it to draw displacement–time and velocity–time graphs for the object. (Take g = 10 m/s^2.)

Table 2.6 *The motion of a falling object – see Question 10*

Time *t* (s)	0	1	2	3	4
Displacement *s* (m)	0	5	20		
Velocity *v* (m/s)	0	10	20		

11. A stone is dropped from the top of a 45 m high building. How fast will it be moving when it reaches the ground? And what will its velocity be?

12. Two cars A and B are moving along a straight road in the same direction with velocities of 25 km/h and 40 km/h, respectively. Find the velocity of car B relative to car A.

13. An aircraft heads North at 320 km/h relative to the wind. The wind velocity is 80 km/h from the North. Find the velocity of the aircraft relative to the ground.

14. Two aircraft P and Q are flying at the same speed, 300 m/s. The direction along which P is flying is at right angles to the direction along which Q is flying. Find the magnitude of the velocity of the aircraft P relative to aircraft Q.

15. A train travelling along a straight track starts from rest at point A and accelerates uniformly to 20 m s^{-1} in 20 s. It travels at this speed for 60 s, then slows down uniformly to rest in 40 s at point C. It stays at rest at C for 30 s, then reverses direction, accelerating uniformly to 10 m s^{-1} in 10 s. It travels at this speed for 30 s, then slows down uniformly to rest in 10 s when it reaches point B.
    a Plot a graph of the motion of the train.
    b Use your graph to calculate:

        i the train's displacement from point A when it reaches point C
        ii the train's displacement from point A when it reaches point B
        iii the train's acceleration each time its speed changes.

Lesson 2 Forces and Newton's Laws

Contents

Section	Learning competencies
2.1. Forces in nature	• List some of the forces that occur in nature and categorise them as contact or non-contact. • State Newton's first law. • Explain the relationship between mass and inertia. • State Hooke's law and distinguish between elastic and inelastic materials. • Experimentally determine and describe the force constant of a spring.
2.2.Newton's second law	• Distinguish between resultant force and equilibrant force. • Describe the effect of a force acting on a body. • Apply Newton's second law (as $F_{net} = ma$) to solve problems. • Resolve forces into rectangular components and compose forces acting on a body using component methods. • Describe the terms weight and weightlessness (including distinguishing between weight and apparent weight). • Calculate the weight and apparent weight of an object in a range of situations.
2.3.Frictional forces	• Explain the causes of frictional forces. • Describe the differences between limiting friction, static friction and kinetic friction. • Draw free body diagrams for objects on inclined planes (to include frictional forces) and use these diagrams to solve problems.
2.4. Newton's third law	• State Newton's third law. • Describe experiments to demonstrate it and give examples of where it is applicable.

Forces are all around us. From keeping us standing on the Earth, to the Earth moving around the Sun. We experience forces every day of our lives.

This unit looks at different types of forces, how they interact and what effect they have on motion. This is a large topic encompassing

some of the most important work ever carried out by Physicists. You will look into Newton's laws, Hooke's work on springs, and even learn how to calculate your mass and weight on different planets.

2.1 Forces in nature

By the end of this section you should be able to:

- List some of the forces that occur in nature and categorise them as contact or non-contact.
- State Newton's first law.
- Explain the relationship between mass and inertia.
- State Hooke's law and distinguish between elastic and inelastic materials.
- Experimentally determine and describe the force constant of a spring.

What are forces?

In simple terms, a force is a **push** or a **pull**. You might push a book across the desk or gravity might pull objects towards the centre of the Earth.

There are plenty of different examples of forces. However, if you look deeper, forces fall into just four groups:

- Electromagnetic forces, dealing with charged objects, atomic interactions and whenever objects come into contact.
- Gravity, which relates to all objects that have mass, from an apple falling to the ground to the Earth orbiting the Sun.
- Finally, two forces dealing with interactions within the nucleus of atoms. These are called the strong nuclear force and the weak nuclear force. Although very important we rarely encounter these forces in our day to day lives.

Below are some examples of common forces.

Table 3.1 *Some examples of forces*

Friction	Drag forces (including air resistance and water resistance)
Electrostatic attraction or repulsion	Thrust
Buoyant force (upthrust)	Gravitational attraction
Weight	Tension
Contact force (reaction)	Magnetic attraction or repulsion

All forces are vector quantities. This means they all have both a **magnitude** and a **direction,** and are often represented in diagrams as arrows. The size of the arrow represents the magnitude of the force and the way it is pointing shows the direction it is acting. The SI derived unit of force is the newton (N).

Figure 3.4 on the next page, is called a **free body diagram**. These kinds of diagrams are really useful when dealing with forces. It

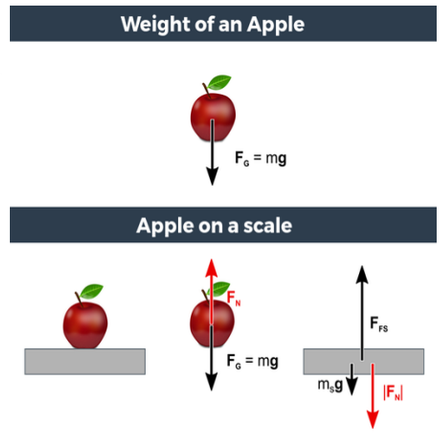


Figure 3.1 *Weight is a common force we experience every day.*

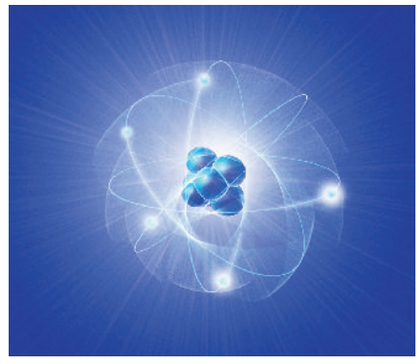


Figure 3.2 *Forces play an important role in keeping atoms together.*

Activity 3.1: Categorising forces

Categorise all the forces listed in Table 3.1 as contact or non-contact.

Figure 3.3 *Forces pull stars together to form gigantic galaxies.*

DID YOU KNOW?

All forces are measured in **newtons**, named after Sir Isaac Newton (more on him later). He was born in 1642 and in his famous book *Principia Mathematica* he made significant advances in understanding motion. He also developed key theories on gravity and optics, and invented an entire new branch of mathematics: calculus.

KEY WORDS

pull *movement towards a force*

push *movement away from a force*

contact forces *forces where objects must touch before the force has an effect*

newton *SI unit of force*

non-contact forces *forces where objects are not required to touch for the force to have an effect*

is important that you consider all the forces acting and draw the arrows approximately to scale. In this case the weight of the stone is greater than the air resistance.

Contact or non-contact

Forces can be categorised as either **contact** or **non-contact**. Some forces act over a distance and so the objects involved do not need to be touching. Other forces need objects to touch before their effects can be noticed.

If you push your hands together you can feel a contact force (this is really an electrostatic repulsion between the electrons in the atoms in your hands). The same is true when you kick a ball.

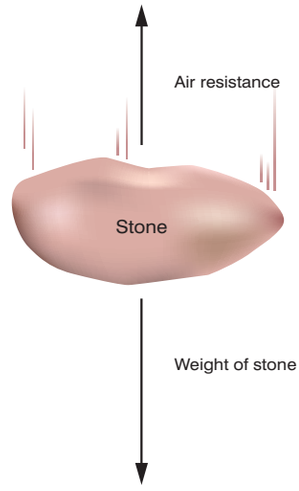


Figure 3.4 *Forces acting on a stone falling through the air*

Figure 3.5 *Kicking a ball demonstrates a contact force.*

Several forces act over a distance, the most obvious being gravitational attraction. The Earth is kept in orbit around the Sun even though they are 150 million km apart!

It is not just gravity; magnetic forces can also act over distances, for example, two magnets attracting each other.

Figure 3.7 *Magnets can attract or repel each other without being in contact.*

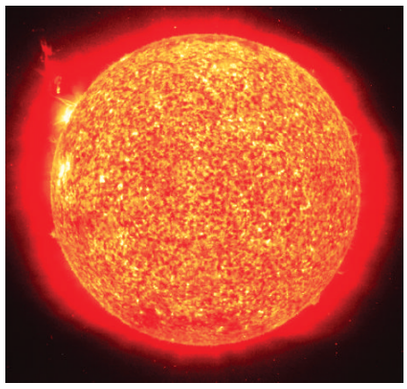


Figure 3.6 *Despite being 150 million km away the Sun's gravity still has a significant influence on the motion of the Earth.*

2.2 What effect do forces have?

The famous ancient Greek, Aristotle, did a great deal to help develop the idea of science. However, he got forces all wrong! He thought that forces were needed to make objects move, that is, there cannot be any movement unless a force is acting.

This idea makes a lot of sense in our experience. If we push a block along it will keep on moving, but if we stop pushing the block it will slow down and stop. The problem is that on Earth whenever objects are in motion there are other forces acting, namely weight, friction and/or drag. These forces have an effect on the motion of the object.

It is true that forces and motion are linked but forces *do not simply make objects move*.

It was not until the famous English physicist, Sir Isaac Newton, came along, some 2000 years after Aristotle, that we developed a more complete understanding of forces. Newton took some of the ideas developed by Galileo and constructed three laws that describe how motion and forces are related.

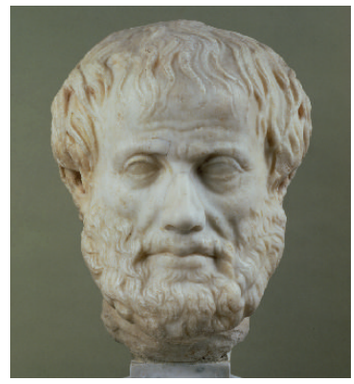


Figure 3.8 *Aristotle developed some excellent theories but his ideas about forces were wrong!*

Newton's first law of motion explains what effect forces have on objects. It states:

- **An object will remain at rest or travelling at a constant velocity unless acted upon by an external force.**

This takes a bit of reading but what it means is that forces don't make objects move but they do make objects change the way they are moving.

An object will remain at rest unless a force makes it start to move. It will then continue to move at the same velocity until another force slows it down. So using our block example from earlier, when we stop pushing it the block slows down because friction is acting on it. If there was no friction it would continue at the same velocity until another force acted on it.

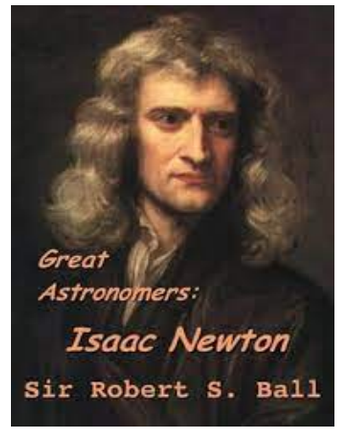


Figure 3.9 *Sir Isaac Newton – perhaps the greatest physicist of all time.*

The use of the term velocity here is also important. It means an object moving around a curve or in a circle must have a force acting on it. Whenever an object moves in a circle its velocity is changing (because velocity is a vector quantity) and so according to Newton's first law there must be a force acting on it.

Newton's first law means a force is always required to make an object:

- **speed up**
- **slow down**
- **or change direction.**

If an object is not doing any of these, then we can conclude there are no overall forces acting on it. This might mean remaining stationary but it also means travelling at a steady speed in a straight line.

Think about this...

Newton's first law means if you were to throw a tennis ball in space, far away from any stars and planets, it would continue to travel at a steady speed in a straight line forever! (Well until it got near another object and then its gravity would start to have an effect).

2.3 Mass and inertia

Newton's first law means that objects have a tendency to resist any changes to their motion. They will remain stationary or at constant velocity unless a force acts on them.

This is referred to as the **inertia** of an object. It is defined as:

- **The property of an object to remain at rest or moving at a steady speed in a straight line.**

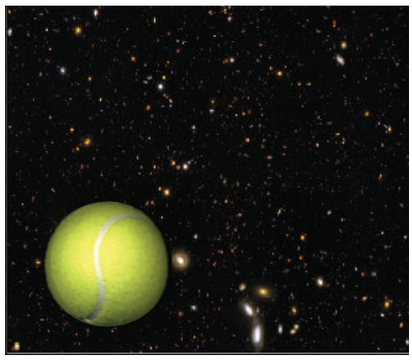


Figure 3.10 *With no friction or air resistance to slow it down, a ball thrown in space will travel at a steady speed in a straight line.*

You may have experienced this on a bus or train. If you are standing still and the vehicle moves you tend to fall backwards. This is because as it moves your feet are pulled along due to friction, but the rest of your body resists this change in motion; it wants to stay at rest.

The same is true when the bus/train stops suddenly; you tend to 'fly forward'. You're not really flying forward, you just keep moving at the same speed as the vehicle slows down.

The inertia of an object depends on its mass. **The greater the mass of the object, the greater its inertia.**

This is why it is easy to kick a small stone. Because it has a small mass and so a small inertia, only a small force is required to change the motion of the stone. However, a large boulder has a great deal more mass. If you kicked a boulder chances are it wouldn't move (and you'd have a sore toe!). It has much more mass, so it has a much greater inertia and a much larger force is required to change its motion.

KEY WORDS

inertia *the tendency of an object to resist changes to its motion*

Figure 3.11 *The greater the mass the greater the inertia. The large boulder has a much greater inertia.*

DID YOU KNOW?

Inertia comes from the Latin word, "iners", meaning idle, or lazy. Newton used this word to illustrate that objects were lazy; they did not want to move unless a force was applied to them.

Activity 3.2: Observing inertia

Try these simple observations (Figure 3.12).

- Place a book on a cloth on a smooth table. Pull the cloth quickly. The book remains at rest.
- Place a coin on a small card. Support the card on the edges of a table so that its sides stick out. Hit the card firmly with one finger. The coin stays where it is.
- Put some water in a bucket or can. Spin it around quickly, in a vertical circle. Although the can is upside down at the top of the circle, no water falls out.

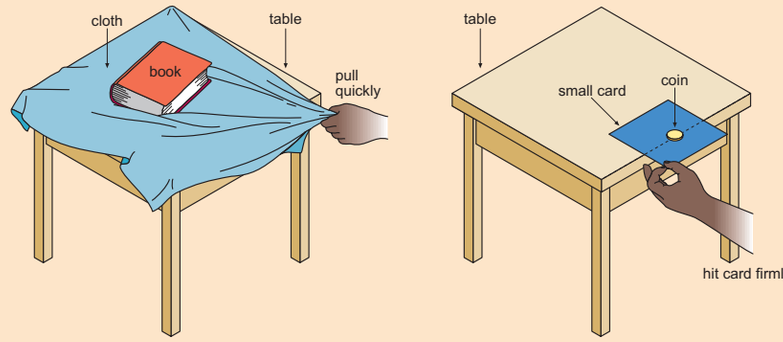


Figure 3.12 *Demonstrating inertia*

2.4 Other effects of forces

If more than one force acts on an object it can also *change the shape* of the object. Two parallel equal and opposite forces can either stretch or compress an object.

Forces can also twist or bend an object if applied in different directions.

Robert Hooke was another physicist working in London around the same time as Newton. He was investigating methods for making more precise clocks. He was interested in the effect forces had on springs.

Hooke used springs fixed at one end (this provided an upward force) and applied a force to the bottom of the spring to stretch it (this force is sometimes called the **load**).

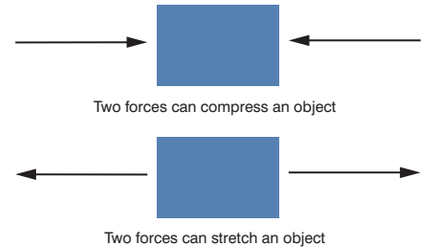


Figure 3.13 *Some possible effects of two equal and opposite forces*

Figure 3.14 *Robert Hooke was a great scientist and was a rival of Isaac Newton.*

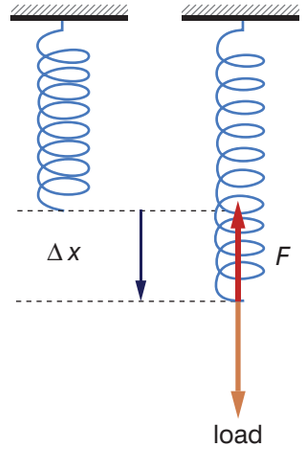


Figure 3.15 *Applying a force to a spring fixed at one end causes it to extend.*

KEY WORDS

load *a force applied to a spring*

extension *the increase in length of a spring*

Hooke found that the greater the force applied to the spring the greater the **extension**. Not only that, he found that the extension of the spring was directly proportional to the force applied. This is often referred to as **Hooke's law**.

This means when he applied twice the force the spring would extend twice as far. Three times the force, the spring would extend three times as far.

Hooke's experiments are easy to repeat in a lab. Figure 3.17 on the next page, shows a simple experimental arrangement you could use to test his findings.

Table 3.2 *Some results from an experiment on stretching a spring*

Force applied (N)	Length of spring (cm)	Extension (cm)
0	10.0	0.0
1	11.5	1.5
2	13.0	3.0
3	14.5	4.5
4	16.0	6.0
5	18.5	8.5
6	22.0	12.0

DID YOU KNOW?

Hooke is perhaps best known for Hooke's law (also called the law of elasticity), but like Newton he made several other valuable contributions. He is often described as the father of microscopy, making several important discoveries. Hooke also came up with the term cell to describe the basic unit of life.

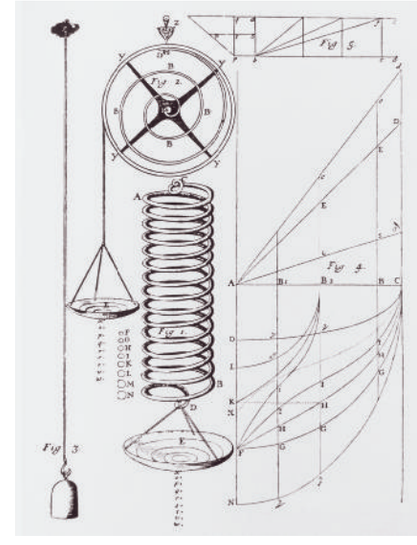


Figure 3.16 *Some of Robert Hooke's original drawings of his experiment*

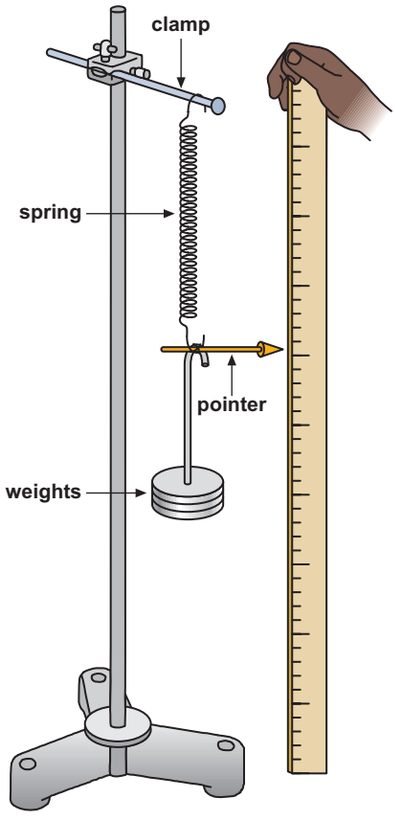


Figure 3.17 *Investigating how force affects the extension of a spring*

Plotting these results on a graph will produce one like that in Figure 3.18. With Hooke's law experiments it is not uncommon to see it the other way around, with extension plotted against force applied, so make sure you look carefully at the axis!

Any relationship that is **directly proportional** will produce a straight line graph with the line going through the **origin**. However, it is worth remembering it does not have to be at 45°. Figure 3.19 shows three directly proportional relationships.

Looking at Figure 3.19, what is different about the springs to produce different slopes? Some springs are stiffer than others. A stiffer spring will not extend as far when a force is applied to it. Looking at the graph, which is the stiffest spring?

If you answered spring A you'd be correct. Spring C is the least stiff; it is the easiest to extend. Let's look at why, but this time just using two springs instead of three.

Figure 3.20 shows the results collected for two different springs. Spring A is stiffer than spring B.

Consider the same force applied to each spring – force *F*. You can see from the second graph that this force causes spring B to extend more than spring A. Therefore you can conclude that spring A is stiffer than spring B.

Think about this...

Extending twice as far does not mean the spring is now double its length. It just means the extension is twice the size. Take a spring 15 mm long; if 2 N caused a 3 mm extension then 4 N would cause a 6 mm extension. With a load of 4 N the spring would be 21 mm long.

KEY WORDS

Hooke's Law *the force applied to a spring is directly proportional to its extension up to the elastic limit*

directly proportional *a relationship where both variables increase (or decrease) at the same time*

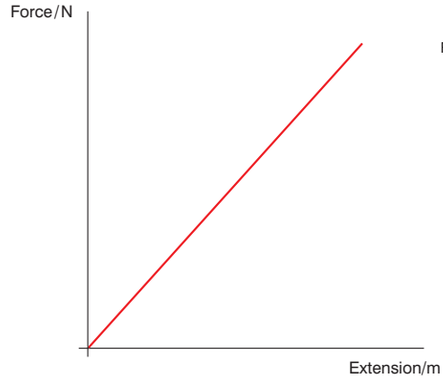


Figure 3.18 *A graph showing that force is directly proportional to extension*

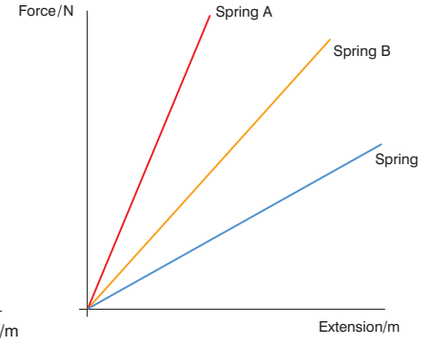


Figure 3.19 *Three directly proportional relationships for three different springs*

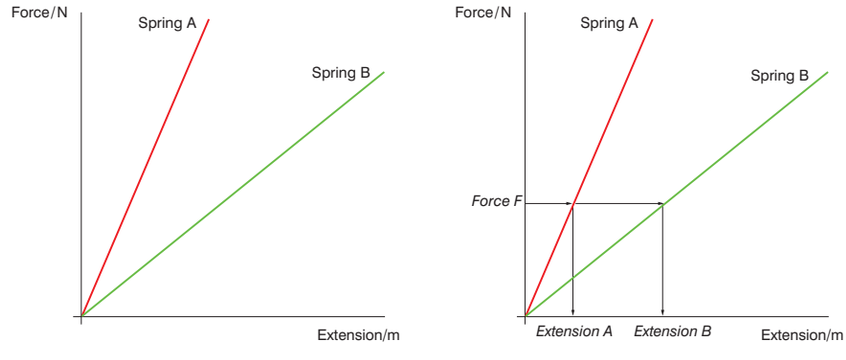


Figure 3.20 *Results collected for two different springs*

The **spring constant** is a measure of the **stiffness** of a spring. It is given the symbol *k*. A stiff spring might have a spring constant of 1000 N/m and a less stiff spring might have a spring constant of 15 N/m.

You can determine the spring constant of any given spring by using the force–extension graph.

The **gradient** of the line is equal to the spring constant. The steeper the line, the higher the gradient, the greater the spring constant and the stiffer the spring!

Using the data and graph below we can determine the spring constant for the spring.

Table 3.3 *Typical force and extension data*

Force (N)	Extension (m)
0	0.00
1	0.05
2	0.10
3	0.15
4	0.20
5	0.25
6	0.30

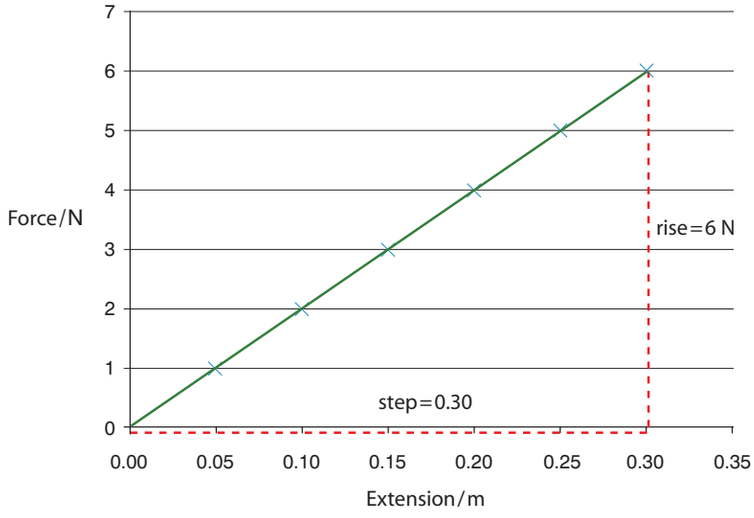


Figure 3.23 *Force–extension graph using data from Table 3.3*

The gradient of the line = rise/step *State principle or equation to be used (determine the gradient of the line)*

gradient = 6 N / 0.3 m *Substitute in known values and complete calculation*

gradient = 20 N/m *Clearly state the answer with unit*

Therefore, *k* = 20 N/m. *Make clear the gradient is also equal to k*

Spring balances

The relationship between force and extension is used to great effect in **spring balances**. These are very simple devices designed to measure forces. They are often used to determine the weight of an object.

Think about this...

The spring constant of 15 N/m means you would need to apply a force of 15 N to extend the spring by 1 m. 30 N would cause an extension of 2 m, etc. If *k* = 1000 N/m, then 1000 N would be needed to extend the spring by 1 m. 500 N would cause a 50 cm extension, etc.

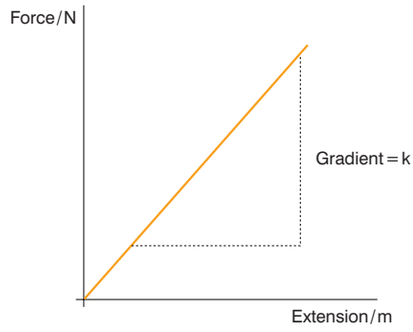


Figure 3.21 *Using a force–extension graph to determine the spring constant*

Figure 3.22 *The springs used in car suspension systems need to have a high spring constant.*

KEY WORDS

origin *the point of intersection of the axes of a graph*

Figure 3.24 *Two different examples of spring balances*

Spring balances work on the principle that the greater the force applied the greater the extension. This means it is easy to construct a simple scale and pointer next to the spring. When a force is applied (e.g. the weight of an object) the spring will extend to a pre-determined length.

Activity 3.3: Making a spring balance

You can make a spring balance of your own.

- You need a spring, and a container for the objects you are going to weigh (Figure 3.25).
- You also need a scale, next to the spring. Make a cardboard pointer, and attach it to the bottom of the spring, so that it will move past the scale.
- First, you must **calibrate** the spring balance. Hang some known loads on the meter. Mark their values on the scale. Mark the scale in equal divisions.
- Now use your meter to weigh other objects.

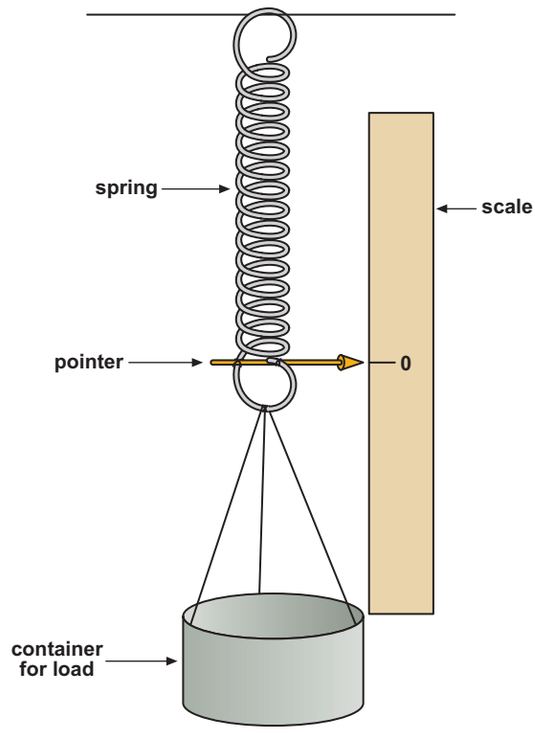


Figure 3.25 *Making a spring balance*

DID YOU KNOW?

Spring balances are often called newtonmeters (or forcemeters). That definitely would not have pleased Hooke! He and Newton were scientific rivals and did not get on at all well.

Think about this...

What would be different about the spring constant of a spring in a spring balance used to weigh heavy objects?

KEY WORDS

gradient *the slope of a line on a graph*

spring balance *device used to measure force via the extension of a spring*

spring constant *a measure of the stiffness of a spring*

stiffness *the amount of force required to stretch or compress a spring*

The elastic limit

If we keep on applying force will the spring keep extending forever? Obviously at some point the spring will break, but before it does it behaves slightly differently. It begins to stretch more easily and eventually it will stretch so far that it will not return to its original length when the force is removed.

So far we have been dealing with what is called **elastic deformation** of the spring. This happens when the force applied to the spring is directly proportional to the extension and when you remove the force the spring returns to its original length.

A spring will only stretch elastically up to a certain point. This point is called the **elastic limit**. After this limit is reached the deformation is said to be **plastic**.

Plastic deformation means that the force is no longer proportional to the extension and when you remove the force the spring no longer returns to its original length; it has been permanently stretched.

The graph below shows you how to indentify the two different types of deformation.

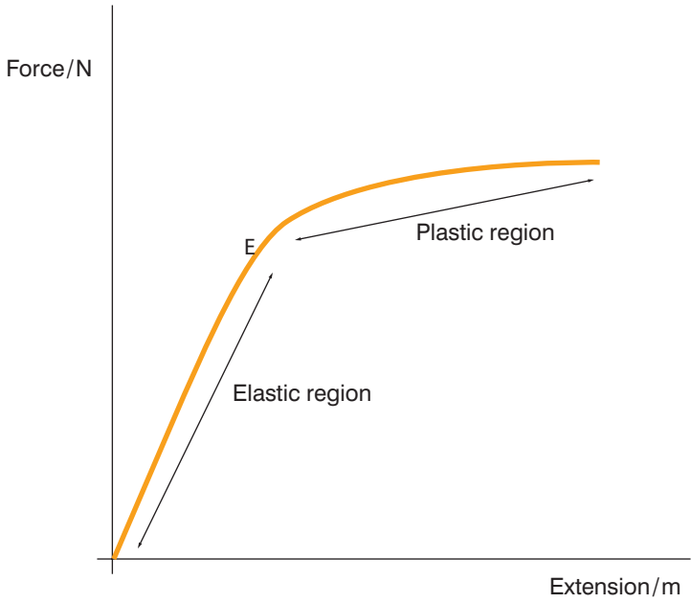


Figure 3.26 *Elastic and plastic deformation of a spring*

E on the graph is the elastic limit. Below the elastic limit the deformation is **elastic**. Above the elastic limit plastic deformation occurs.

Hooke's findings about springs led to the law of elasticity, which is more commonly called Hooke's law. This only applies if the spring is below its elastic limit and so may be written as:

- **The force applied is directly proportional to the extension of a spring up to the elastic limit.**

Different springs have different elastic limits depending on their shape, thickness, material, etc. All materials have an elastic limit; think about a wooden or plastic ruler. If you bend it a little bit it will return to its original length. If you apply too much force it will bend so far it snaps; you've gone beyond the elastic limit for the ruler.

DID YOU KNOW?

The shorthand way of writing directly proportional is to use this symbol: $\propto$. This means we could write Hooke's law as $F \propto \Delta x$ up to the elastic limit.

KEY WORDS

calibrate *to compare a measuring device with a known standard*

elastic deformation *where the force applied is directly proportional to the extension and where the object will return to its original length when the force is removed*

elastic limit *the point up to which a spring will stretch elastically*

plastic deformation *where the force applied is not directly proportional to the extension and where the object will not return to its original length when the force is removed*

Summary

In this section you have learnt that:

- Forces can either be classed as contact or non-contact. Examples of forces include friction, drag, weight, gravitational attraction and contact forces.
- Newton's first law states: "An object will remain at rest or travelling at a constant velocity unless acted upon by an external force".
- Inertia is the tendency of an object to resist changes to its motion. The greater the mass of an object the greater its inertia.
- Hooke's law states: "The force applied to a spring is directly proportional to the extension of the spring up to the elastic limit".
- The stiffer the spring the greater the spring constant (k; measured in N/m).
- Elastic deformation means when forces are removed the object will return to its original length. Plastic deformation means when the forces are removed the object does not return to its original length; it is permanently stretched.

Review questions

1. Give some examples of forces and classify them as contact or non-contact.
2. State Newton's first law and explain what it means.
3. Describe Hooke's law and define the following terms: elastic deformation, elastic limit and plastic deformation.
4. Sketch two force vs. extension graphs, one for a stiff spring the other for a much weaker spring.

2.5 Newton's second law

By the end of this section you should be able to:

- Distinguish between resultant force and equilibrant force.
- Describe the effect of a force acting on a body.
- Apply Newton's second law (as $F_{net} = ma$) to solve problems.
- Resolve forces into rectangular components and compose forces acting on a body using component methods.
- Describe the terms weight and weightlessness (including distinguishing between weight and apparent weight).
- Calculate the weight and apparent weight of an object in a range of situations.

What if more than one force is acting?

There are often several forces acting on an object. As all forces are **vector** quantities we can add them up using the techniques covered in Unit 1.

The overall force acting on any object is referred to as the **resultant force**. This is often called the net force or F_{net}. It is defined as:

- **The vector produced when two or more forces act upon a single object.**

It is calculated by vector addition of the forces acting upon the object.

For example, consider two forces A and B acting on an object. They will produce a resultant force. In the two examples below forces A and B give rise to a resultant, force C.

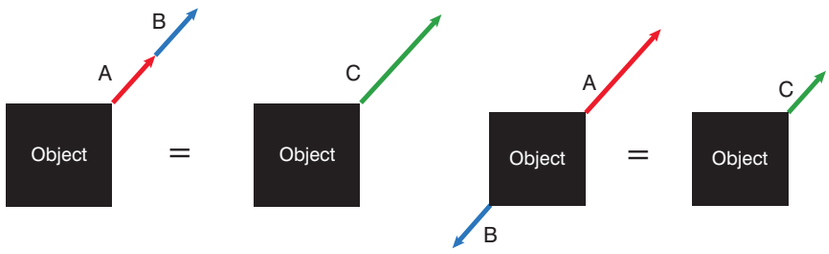


Figure 3.27 *Different resultant forces acting on an object*

If the forces are parallel it is easy to determine the resultant vector. However, if the forces are not parallel (as in Figure 3.28) we then use scale diagrams, parallelogram rules or the mathematical techniques covered in Unit 1 to determine the magnitude and direction of the resultant force.

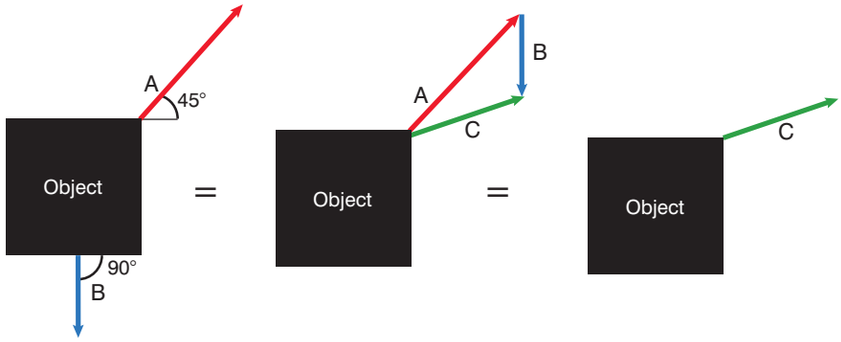
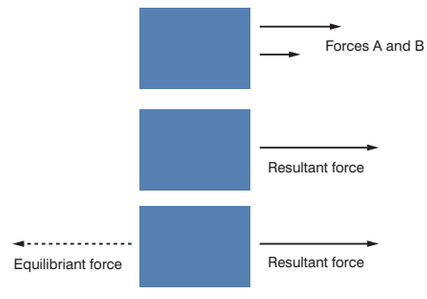


Figure 3.28 *Non-parallel forces leading to a resultant force*

Sometimes it is helpful to know the **equilibrant force**. This is the force you need to apply to a system to cancel out the resultant force. This will result in there being no net force acting on an object.

Figure 3.30 *An equilibrant force will cancel out the resultant force acting on an object.*



KEY WORDS

equilibrant force *the force required to cancel out the resultant force*

resultant force *the overall force acting on an object*

DID YOU KNOW?

The equilibrant force for any system is always equal in magnitude to the resultant force but it acts in the opposite direction. This just cancels out the effect of the resultant force. This can be written as: $F_{net} = -F_{equilibrant}$

Figure 3.29 *There are several forces acting on an aircraft in flight.*

Worked example

Two forces are acting on a boat. One force of 400 N is due to current in the river, acting downstream. The other force due to the propeller has a magnitude of 500 N and acts at an angle of 50° to the river bank. Determine the resultant force acting on the boat.

Figure 3.31 *Boat crossing a river*

Figure 3.32 *Two force vectors acting on the boat*

We could determine the resultant force using a **scale diagram**. However, on this occasion we are going to find the resultant force mathematically.

In order to determine the resultant force we must first **resolve** the 500 N into horizontal and vertical components using trigonometry.

- Vertical component:
  sin θ = opp / hyp *State principle or equation to be used (trigonometry)*
  hyp x sin θ = opp *Rearrange to make the opp side the subject*
  500 N x sin 50° = 383 N ↑. *Substitute in known values and complete calculation, then clearly state the answer with unit*
- Horizontal component:
  cos θ = adj / hyp *State principle or equation to be used (trigonometry)*
  hyp x sin θ = adj *Rearrange to make the adj side the subject*
  500 N x cos 50° = 321 N →. *Substitute in known values and complete calculation, then clearly state the answer with unit*

We can then add the horizontal forces to give the resultant horizontal force.

- Resultant horizontal force:
  $F_{net\ horizontal}$ = 321 N → + 400 N → *Determine the net horizontal force (note the directions)*
  $F_{net\ horizontal}$ = 721 N →. *Clearly state the answer with unit*

We can then use **Pythagoras's theorem** to determine the magnitude of the resultant force and **trigonometry** to determine the direction.

- Magnitude of resultant force:
  $F_{net\ horizontal}$ = 721 N → *Clearly state known values*
  $F_{net\ vertical}$ = 383 N ↑ *Clearly state known values*
  F_{net} = 383 N ↑ + 721 N → *Determine the net force (note the directions)*
  $F_{net}^2 = 383^2 + 721^2$ *Apply Pythagoras's theorem*
  F_{net}^2 = 666 530 *Solve for* F_{net}^2
  $F_{net} = \sqrt{666\ 530}$ *Rearrange for resultant (take square root) and solve*
  F_{net} = 816 N *Clearly state the answer with unit*
- Direction of resultant force:
  $F_{net\ horizontal}$ = 721 N → *Clearly state known values*
  $F_{net\ vertical}$ = 383 N ↑ *Clearly state known values*
  tan θ = opp / adj *State principle or equation to be used (trigonometry)*
  $\theta = \tan^{-1}$ (opp / adj) *Rearrange equation to make θ the subject*
  $\theta = \tan^{-1}$ (721 / 383) *Substitute in known values and complete calculation*
  θ = 62° *Clearly state the answer with unit*

This is the angle between the resultant and the vertical component. The angle between the resultant force and the river bank is 90°– 62° = 28°.

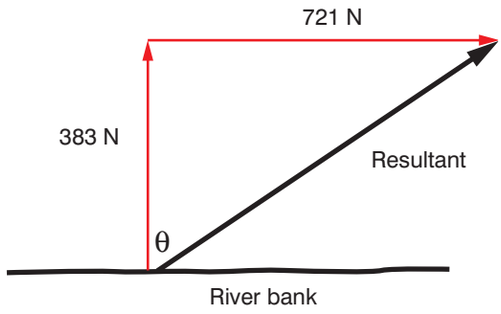


Figure 3.33 *Determining the resultant force*

In this situation the equilibrant force would be 816 N acting in the opposite direction to the resultant force.

Forces and acceleration

A force is required to make an object:

- **speed up**
- **slow down**
- **change direction.**

If an object does any of these things we can say it is **accelerating**. In other words, forces cause objects to accelerate, or more precisely if there is a resultant force acting on an object, then that object will accelerate. The forces are said to **unbalanced**.

If there are **balanced forces** acting on an object then there is no resultant force and so the object will not accelerate.

Figure 3.34 *Any object going around a bend is accelerating; the forces are unbalanced and so there must be a resultant force acting on it.*

KEY WORDS

trigonometry *a type of mathematics that deals with the relationships between the sides and angles of triangles*

accelerating *where an object is speeding up, slowing down or changing direction*

balanced forces *where the forces acting on a body cancel each other out and there is no resultant force*

inversely proportional *a relationship where one variable increases as the other decreases and vice versa*

unbalanced forces *where the forces acting on a body do not cancel out and there is a resultant force*

Newton's second law: whenever a resultant force acts on an object it will accelerate and this acceleration takes place in the same direction as the force. If you push an object to the left it will accelerate towards the left.

Through careful experiment and investigation he also worked out that if you double the resultant force then the acceleration of the object will also double. In other words the force applied is directly proportional to the acceleration (as long as everything else remains constant).

He also determined that the acceleration of the object also depends on the object's mass. The greater the mass the greater the inertia, and so the lower the acceleration. In fact if you double the mass the acceleration will halve and vice versa. An object with a quarter of the mass will accelerate at four times the rate if the same force is

applied. This relationship is called **inversely proportional**. As one quantity doubles the other halves.

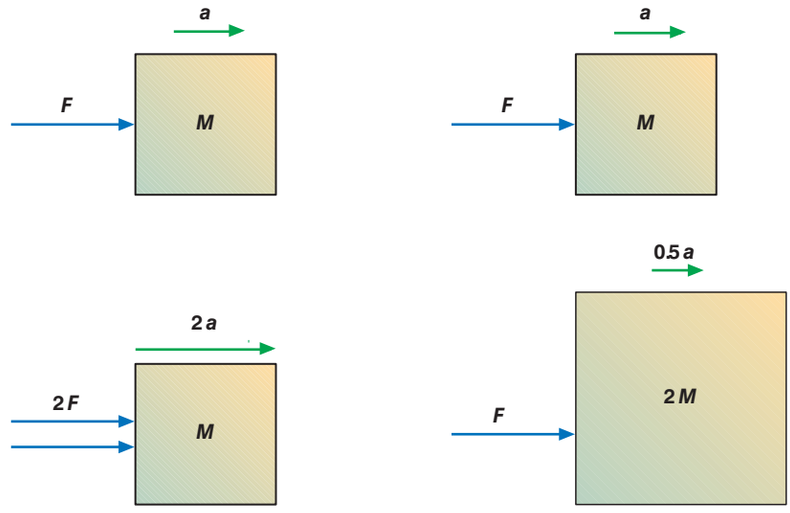


Figure 3.35 *The effects of force and mass on acceleration*

As long as the mass of the object remains constant then Newton's second law can be expressed as:

- **The acceleration of an object is directly proportional to t.. resultant force acting on the object.**

and

- **This acceleration occurs in the direction of the resultant forc. .**

(Remember, this only applies if the mass of the object is constant.)

This gives us:

Resultant force = mass of objec. × acceleration of object

$F_{net} = ma$

We can use this equation to determine the resultant force required to make a car of mass 1200 kg accelerate at 2 m/s^2.

Resultant force = mass of objec. × acceleration of object *State principle or equation to be used (Newton's second law)*

$F_{net} = ma$ *Simplify statement to symbols*

F_{net} = 1200 k. × 2 m/s^2 *Substitute in known values and complete calculation*

F_{net} = 2400 N *Clearly state the answer with unit*

We can use the equation to determine the acceleration of a soccer ball if we know the applied resultant force. A footballer may strike a ball of mass 400 g with a force of 200 N.

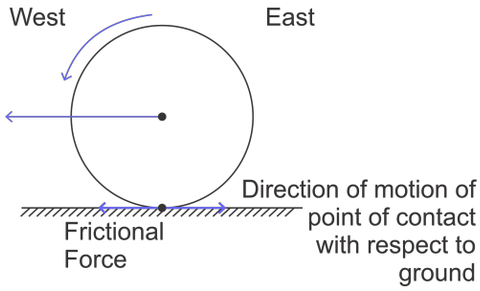


Figure 3.36 *When you apply the brakes, a force is generated in the opposite direction to motion. You accelerate in this direction and so slow down.*

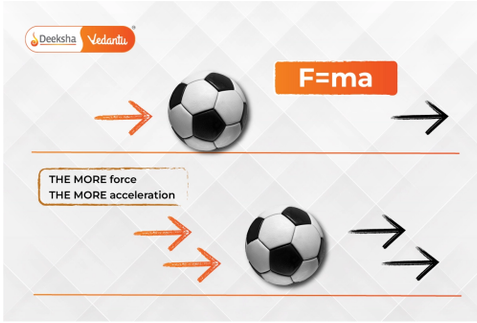


Figure 3.37 *The greater the force applied to the ball the greater its acceleration.*

$F_{net} = ma$ *State principle or equation to be used (Newton's second law)*

$a = F_{net} / m$ *Rearrange equation to make a the subject*

m = 400 g, which is 0.4 kg *Ensure all values are in SI units*

a = 200 N / 0.4 kg *Substitute in known values and complete calculation*

a = 500 m/s^2 *Clearly state the answer with unit*

This acceleration will be in the same direction as the resultant force.

$F_{net} = ma$ with several forces

If several forces are acting on an object then in order to determine its acceleration we must first determine the resultant force.

To determine the acceleration we would use $F_{net} = ma$.

$F_{net} = ma$ *State principle or equation to be used (Newton's second law)*

$a = F_{net} / m$ *Rearrange equation to make a the subject*

The resultant force in Figure 3.38 is 30 N → *Determine resultant by simple calculation of net force*

a = 30 N / 4.0 kg *Substitute in known values and complete calculation*

a = 7.5 m/s^2 to the right *Clearly state the answer with unit*

To determine the acceleration we would again use $F_{net} = ma$. Except in this case we must subtract the forces to determine the resultant force.

$F_{net} = ma$ *State principle or equation to be used (Newton's second law)*

$a = F_{net} / m$ *Rearrange equation to make a the subject*

The resultant force in Figure 3.39 is 20 N → *Determine resultant by simple calculation of net force*

- a = 20 N / 2.0 kg *Substitute in known values and complete calculation*
- a = 10 m/s^2 in the direction of the 50 N force *Clearly state the answer with unit*

This process can be repeated for forces at an angle and for problems involving more than two forces.

If you know the acceleration of the object you can also determine the magnitude and direction of the resultant forces. For example, two people are pushing a 60 kg trolley along. One applies a force of 40 N and the trolley accelerates at 2.0 m/s^2. Determine the size of the force applied by the other person.

$F_{net} = ma$ *State principle or equation to be used (Newton's second law)*

F_{net} = 60 kg × 2 m/s^2 *Substitute in known values and complete calculation*

F_{net} = 120 N *Clearly state the answer with unit* The resultant force is 120 N.

$F_{net} = F_1 + F_2$ *Express net force in terms of F_1 and F_2*

DID YOU KNOW?

Newton's second law may be used to define the newton as the unit of force. Using $F_{net} = ma$ and making sure the units are all correct (force in N, mass in kg and acceleration in m/s^2), we can say that a force 1 N is the force required to give a mass of 1 kg an acceleration of 1 m/s^2. Or 1 N is equivalent to 1 kg m/s^2.

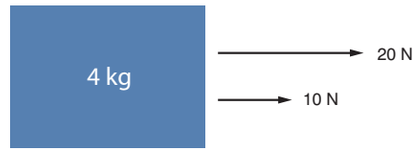


Figure 3.38 *Two forces acting on an object*

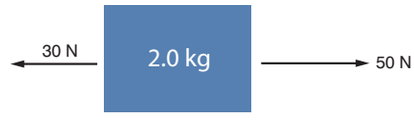


Figure 3.39 *Two forces acting on an object in different directions*

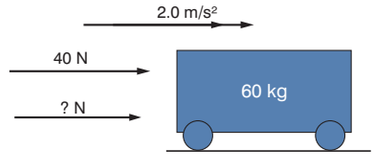


Figure 3.40 *Trolley being pushed by two people*

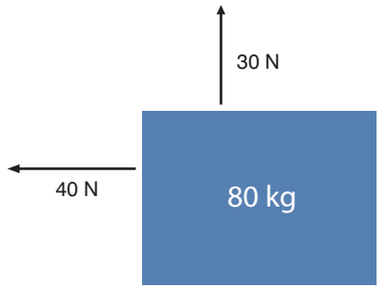


Figure 3.41 *Two forces acting on an object at right angles*

$120\ \text{N} = 40\ \text{N} + F_2$ *Substitute in known values and complete calculation*

$F_2 = 80\ \text{N} \rightarrow$ *Clearly state the answer with unit*

The same technique may be used to determine the acceleration of an object with two forces acting on it at right angles. For example:

First we must determine the resultant force using Pythagoras's theorem.

$a^2 = b^2 + c^2$ *State principle or equation to be used (Pythagoras's theorem)*

$F_{net}^{\ 2} = (40\ \text{N})^2 + (30\ \text{N})^2$ *Substitute in known values*

$F_{net}^{\ 2} = 2500$ *Solve for $F_{net}^{\ 2}$ then solve for F_{net} by taking square root*

$F_{net} = 50\ \text{N}$ *Clearly state the answer with unit*

Then using $F = ma$ we get:

$a = F_{net} / m$ *Rearrange F = ma to make a the subject*

$a = 50\ \text{N} / 80\ \text{kg}$ *Substitute in known values and complete calculation*

$a = 0.63\ \text{m/s}^2$ *Clearly state the answer with unit*

Trigonometry should then be used to determine the direction of this acceleration; this is in the same direction as the result force (37° to the horizontal – check it for yourself).

KEY WORDS

mass *a measure of the quantity of matter*

weight *the force experienced by an object due to the gravitational pull of the Earth*

DID YOU KNOW?

The kilogram is defined as the same amount of mass as the international prototype kilogram. This is a platinum–iridium block held in Paris, France.

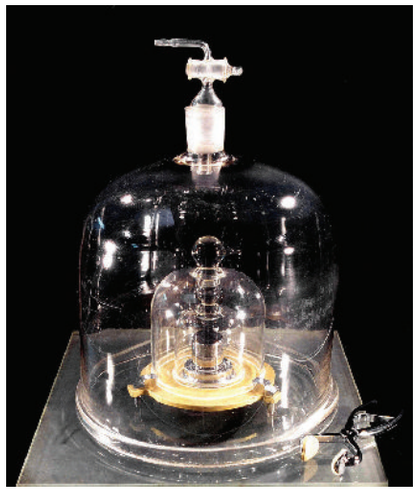


Figure 3.42 *The international prototype kilogram has a mass of exactly* ***one*** *kilogram.*

Mass and weight

Mass and **weight** are two terms that are frequently confused. We often say we are going to weigh something and then record its mass in kg!

We must make sure we don't muddle the two; they are very different.

Mass is a scalar quantity and it is a measure of the quantity of matter. The more mass the more stuff (the more matter). Remember the ineria of an object depends on its mass, you can think of mass as a measure of an object's inertia. Mass is measured in kilograms (kg).

Weight is a force and so it's a vector quantity, measured in newtons (N). It is the force we experience due to the gravitational pull of the Earth pulling on our mass. Weight is directed towards the centre of the Earth.

We can calculate the weight of an object using:

- weight = mass × gravitational field strength
- $w = mg$

On the surface of the Earth the gravitational field strength is around 9.81 N/kg. We will use 10 N/kg in the following examples to make the mathematics a little easier.

A person with a mass of 70 kg will have a weight of:

$w = mg$ *State principle or equation to be used*

$w = 70\ \text{kg} \times 10\ \text{N/kg}$ *Substitute in known values and complete calculation*

w = 700 N (actually more like 687 N if we use g = 9.81 N/kg).
Clearly state the answer with unit

If the gravitational field strength changes then the weight of the object will change but its mass will stay the same. The gravitational field strength varies a little around the Earth. This is for two reasons.

Firstly the amount of mass between you and the centre of the Earth changes depending on where you are. If there is a particularly dense pocket of material underneath you this will increase the gravitational field strength slightly. The reverse is also true, if there is large pocket of gas or lower density material underneath you the gravitational field strength will go down.

The distance from the centre of the Earth also affects g; it gets smaller the further away from the centre of Earth you get. This change is quite small, you need to move really far away before is becomes noticeable. Even at the top of the tallest mountain g is still around 9.8 N/kg.

Remember, only the weight of the object will change; its mass will stay the same. This is also true if we consider different planets. Taking our astronaut as an example, if he stands on the Moon his mass is still 70 kg (there is still the same amount of matter). However, on the Moon the gravitational field strength is much less than that on Earth. This is because the Moon has much less mass and so a weaker gravitational field. The value for g on the moon is just 1.6 N/kg.

His weight on the Moon would be:

$w = mg$ *State principle or equation to be used*

w = 70 kg × 1.6 N/kg *Substitute in known values and complete calculation*

w = 112 N *Clearly state the answer with unit*

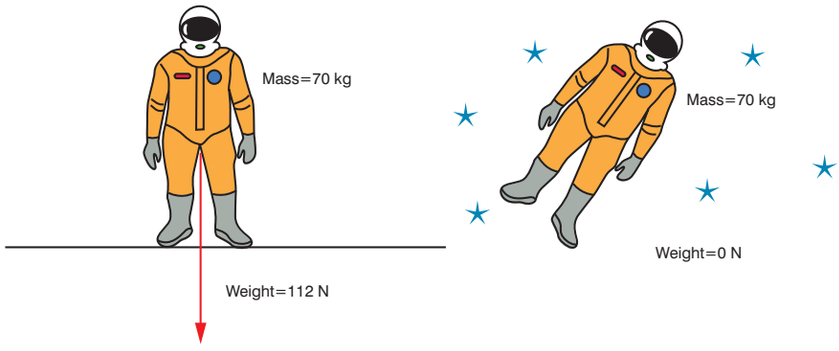


Figure 3.45 *Astronaut on the Moon and in deep space*

In deep space, far away from any planets and stars, the gravitational field strength is pretty much zero. In this case his mass would still be 70 kg. However, his weight would be 0 N; he is weightless.

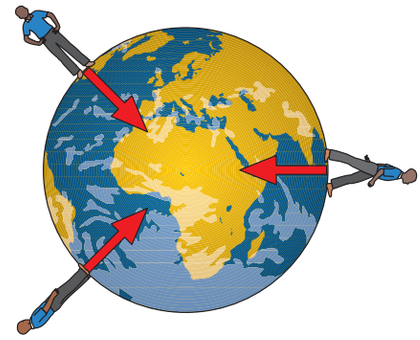


Figure 3.43 *Weight pulls all objects towards the centre of the Earth.*

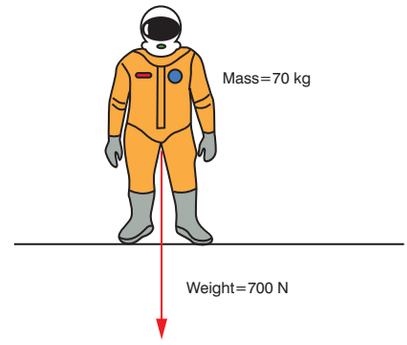


Figure 3.44 *Weight and mass*

Think about this...

The value for gravitational field strength is the same value as the acceleration due to gravity (9.81). This can shown by considering an object of mass m dropped from a height above the ground. From Newton's second law we know the acceleration will be equal to $a = F_{net} / m$. We also know the force accelerating the object is the weight of the object so we could write $F_{net} = mg$. Combining these two equations gives us: $a = mg / m$, the m's cancel giving $a = g$!

DID YOU KNOW?

Slight variations in the gravitational field strength are used to look for oil and gas deposits. Because the oil and gas is less dense than the surrounding rock this causes a small dip the gravitational field strength above the deposit. This dip may be detected with sensitive equipment.

Figure 3.46 *Astronauts on the Moon can carry very large packs due to the Moon's weak gravity.*

Figure 3.47 *Astronauts in the International Space Station are not truly weightless.*

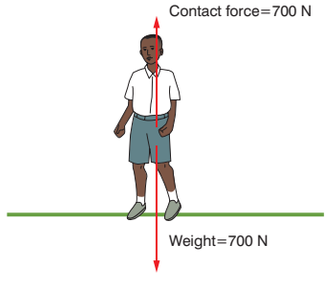


Figure 3.48 *Weight and contact force cancel out*

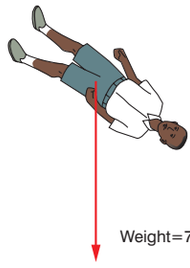


Figure 3.49 *This diver would experience apparent weightlessness for a brief period of time.*

True weightlessness and apparent weightlessness

You are only truly weightless if the gravitational field strength is zero. Even astronauts in orbit around the Earth are not truly weightless. There is still a gravitational pull due to the Earth; they still have a weight. So why do they float around?

When we are standing on the ground our weight pulls us vertically downwards towards the centre of the Earth. We push down on the Earth and the Earth pushes back up with a contact force. These two forces cancel out so there is no resultant force (this is why we don't accelerate towards the centre of the Earth; if the ground was not there then we would!).

It is this contact force we feel. We don't notice the pull of gravity. If you take this contact force away by jumping off a tall diving board, our weight accelerates us downwards but we don't feel it. It feels like we are weightless, but we are not!

- **Apparent weightlessness is when the only force acting is your weight.**
- **Real weightlessness is when your weight is zero.**

You get a similar feeling when a car goes over a humpback bridge or when an aircraft climbs or descends. We notice the change in the contact force and this makes us feel like our weight has changed.

Another common example is when you are in a lift. If the lift is not accelerating the two forces are equal, as shown in Figure 3.51.

If the lift accelerates upwards then there must be a net force acting on it. A net force also needs to act on you as you are inside the lift! Imagine the net force acting on you is 200 N (assuming your mass is 70 kg this would give an acceleration of 2.86 m/s^2).

The floor would push you up harder; the contact force would have to increase to 900 N. This provides the extra 200 N. You feel heavier, even though your weight has not changed. It would feel like your weight is 900 N. This is referred to as your **apparent weight**; your **real weight** is still 700 N.

The same is true if the lift were to accelerate downwards. Again imagine the net force on you is 200 N. In this case the contact force would drop 200 N to 500 N. You would feel like your weight has dropped! Your apparent weight would be 500 N.

You can use Newton's second law to determine your apparent weight in an accelerating lift. Taking a person of mass 55 kg then their weight would be:

$w = mg$ *State principle or equation to be used*

$w = 55\text{ kg} \times 10\text{ N/kg}$ *Substitute in known values and complete calculation*

$w = 550\text{ N}$ *Clearly state the answer with unit*

If this person is in a lift accelerating vertically upwards at 2 m/s^2 then the net force acting on the person would be:

$F_{net} = ma$ *State principle or equation to be used (Newton's second law)*

$F_{net} = 55\text{ kg} \times 2\text{ m/s}^2$ *Substitute in known values and complete calculation*

$F_{net} = 110\text{ N}$ *Clearly state the answer with unit*

This force would come from an increase in the contact force. The contact force would have to go up to 660 N (550 N + 110 N). This would be your apparent weight.

Figure 3.50 *The contact force we experience changes dramatically on an exciting roller coaster ride.*

Think about this...

You only notice this effect when the lift accelerates. When the lift is travelling at a steady speed the forces are balanced again (from Newton's first law).

KEY WORDS

apparent weight *the resultant of an object's real weight and any contact forces acting on the object*

real weight *the force experienced by an object solely due to the gravitational pull of the Earth*

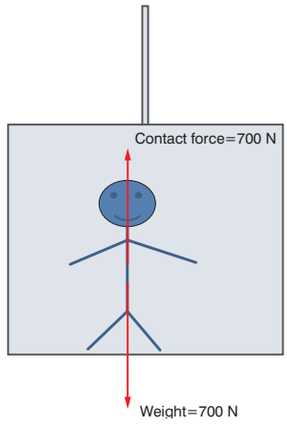


Figure 3.51 *Contact force and weight in a stationary lift.*

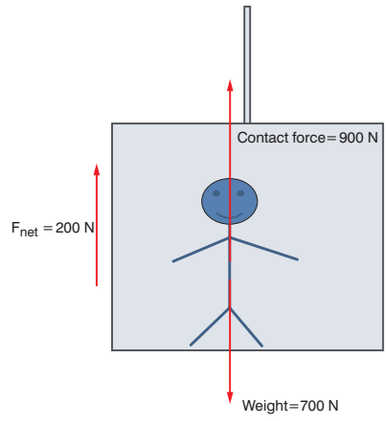


Figure 3.52 *Accelerating upwards requires a net vertical force.*

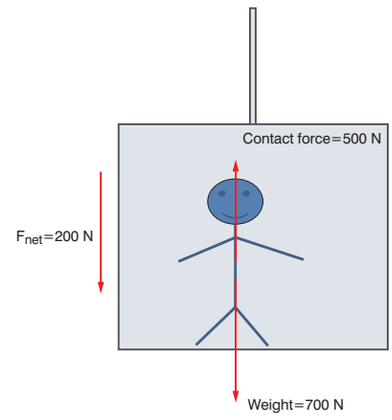


Figure 3.53 *Accelerating downwards requires a net vertical force. This time you would feel like your weight has dropped.*

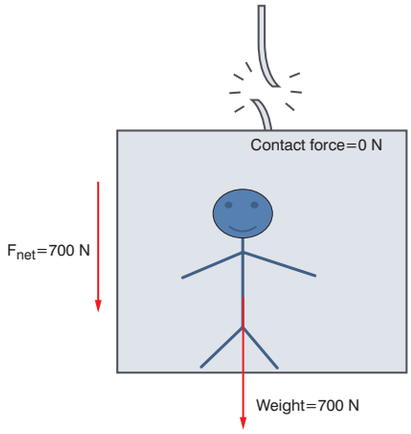


Figure 3.54 *If the contact force is zero you would be apparently weightless.*

If the lift was accelerating downwards at 2 m/s^2 then your apparent weight would be 440 N. This would give a net force vertically downwards equal to 110 N.

If the lift cable were to snap then as the lift accelerates towards the ground the contact force would fall to zero! The floor would stop pushing you up. You would feel like you are weightless. Your apparent weight would be 0 N; you would be apparently weightless.

Figure 3.55 *A photo of the infamous 'Vomit Comet'*

DID YOU KNOW?

As part of astronaut training trainees take a flight in an aircraft commonly called the Vomit Comet! This aircraft accelerates downwards at 9.81 m/s^2; this means the contact force inside the aircraft falls to zero. All the occupants become apparently weightless for around 30 s (until the aircraft needs to pull up again).

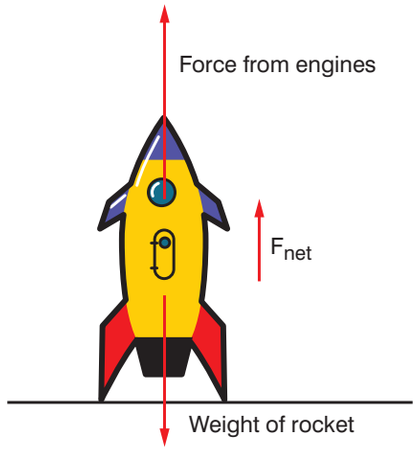


Figure 3.56 *A free body diagram for the forces acting on a rocket at take off.*

$F_{net} = ma$ considering the weight of the object

We must always think carefully when solving $F_{net} = ma$ problems. Take for example a rocket of mass 15 000 kg. If the engines provide a force of 200 000 N what would its acceleration be?

- $F_{net} = ma$
- $a = F_{net} / m$
- a = 200 000 N / 15 000 kg
- a = 13.3 m/s^2

This is wrong! We've not used the **resultant force**. Remember **free body diagrams** really help to identify all the forces acting on an object.

You can see the resultant force is equal to:

F_{net} = force from engines – weight of rocket *Express F_{net} in terms of all forces acting*

F_{net} = 200 000 N – (15 000 kg × 10 N/kg) *Substitute in known values*

F_{Net} = 200 000 N – 150 000 N *Solve calculation in brackets then complete calculation*

F_{Net} = 5000 N *Clearly state the answer with unit*

This would give us an acceleration equal to:

$F_{net} = ma$ *State principle or equation to be used (Newton's second law)*

$a = F_{net} / m$ *Rearrange equation to make a the subject*

a = 5000 N / 15 000 kg *Substitute in known values and complete calculation*

a = 0.3 m/s^2 *Clearly state the answer with unit*

We must always make sure we consider **all the forces** involved before determining the resultant force acting on an object.

KEY WORDS

free body diagrams *are used to gain an understanding of the forces (or sometimes the fields) acting on an object*

Summary

In this section you have learnt that:

- The overall force acting on an object is called the resultant force. The equilibrant force is the force that needs to be applied to cancel out the resultant force.
- A resultant force will cause an object to accelerate in the same direction as the resultant force.
- Newton's second law states: "Force is directly proportional to acceleration, as long as the mass remains constant, and the acceleration is in the same direction of the force". This gives us $F_{net} = ma$.
- In order to determine the resultant force, the forces acting on the object may need to be resolved then combined together again.
- Mass is a measure of the amount of matter measured in kg, whereas weight is a force measured in N caused by gravity pulling on an object's mass.

DID YOU KNOW?

When large rockets take off their acceleration usually increases for the first few minutes of their flight. The acceleration starts off quite low then increases as the rocket burns fuel. This has a significant effect on its acceleration for two reasons. Firstly the weight drops and so this increases the resultant force acting and secondly as the object has less mass its acceleration will be greater (remember acceleration and mass are inversely proportional).

Review questions

1. Explain what is meant by the terms resultant force and equilibrant force.
2. Describe Newton's second law.
3. Copy and complete Table 3.4.

Table 3.4

Force (N)	Mass (kg)	Acceleration (m/s^2)
100	40	
	60	10
1000		25
	0.2	10
30	600	

4. Figure 3.57 shows the forces acting on three different objects. For each:
   (a) calculate the resultant force acting;
   (b) say whether the forces are balanced or unbalanced;
   (c) calculate the object's acceleration.
5. Explain the differences between mass and weight.

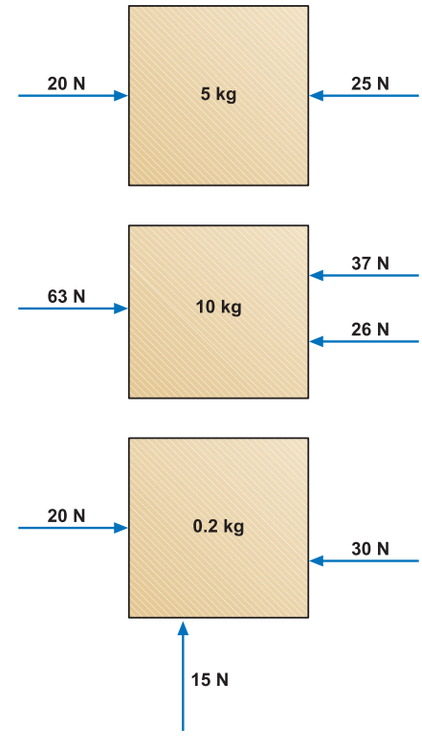


Figure 3.57 *See Question 4*

2.6 Frictional forces

By the end of this section you should be able to:

- Explain the causes of frictional forces.
- Describe the differences between limiting friction, static friction and kinetic friction.
- Draw free body diagrams for objects on inclined planes (to include frictional forces) and use these diagrams to solve problems.

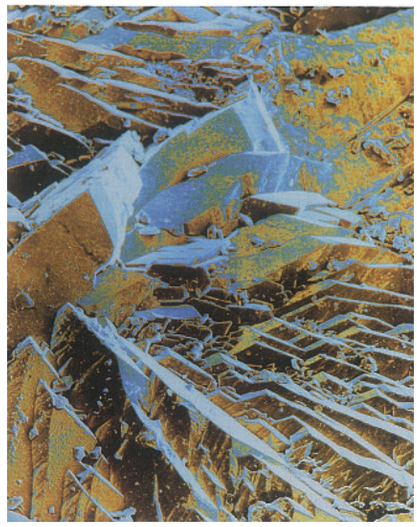


Figure 3.58 *Even very smooth surfaces have a rough texture at the microscopic level.*

What causes friction?

Friction is a force we experience every single day. Without friction even the simplest of actions, like walking, would be impossible. Friction occurs whenever two solids rub against each other. It is a contact force and it always tends to act in a direction opposing motion.

It is caused by tiny bumps in the surface of the two objects knocking and locking together. No surface is perfectly smooth. This is obvious if you look closely at sandpaper but you need to look really close at smoother objects like a metal sheet.

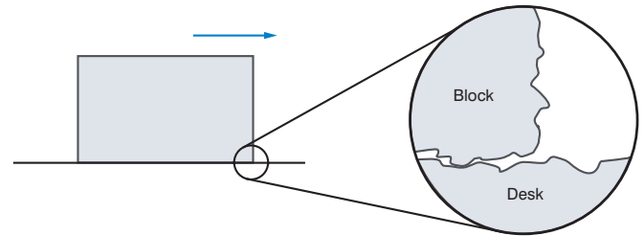


Figure 3.59 *The bumps on the surfaces of material knock together causing friction.*

When magnified, you can see all the small bumps in the surface of a material. It is these bumps that cause friction.

KEY WORDS

friction *the force generated when solids slide or attempt to slide over each other*

Figure 3.60 *Sandpaper is very rough. Sliding over sandpaper generates a great deal of friction.*

Different types of friction

There are two different types of friction. It depends on if the objects in contact are moving or if they are stationary.

Static friction

- **This is the frictional force between two objects that ar. in contact and trying to move past each other, but not yet moving.**

Imagine gently pushing a heavy book on a desk. At first it does not accelerate. This is because the force you are applying is cancelled out due to static friction. As you gradually increase the force the static friction also increases and the book remains stationary. If

you continue to push harder, eventually the book will slide. The maximum value of the static friction, i.e. the value just before sliding occurs, is called the **limiting friction**.

Kinetic friction (sometimes called dynamic friction)

- **This is the frictional force between two objects sliding over each other.**

It always acts in the opposite direction to motion.

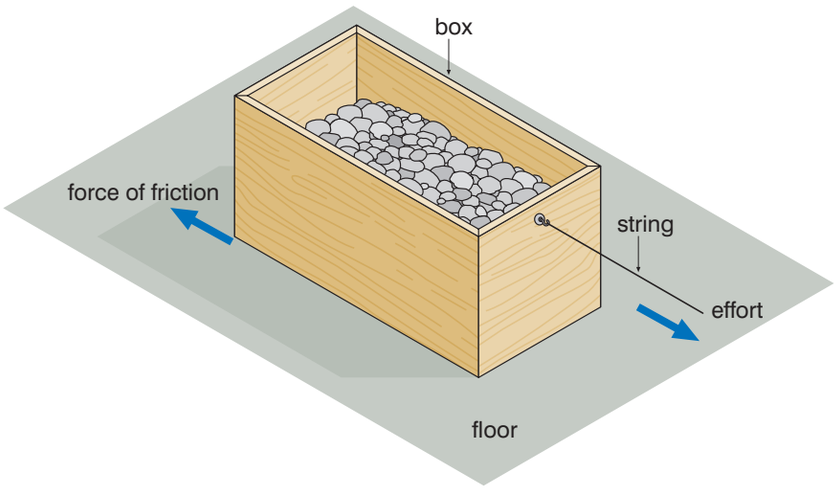


Figure 3.61 *Kinetic friction always acts in the opposite direction to motion.*

The force of friction usually drops when objects start moving and so it is often the case that kinetic friction is less than the limiting friction of a surface.

Think about this...

Friction only happens when solids rub together. This means that there is no such thing as friction with the air or friction through water; both of these examples are types of drag. This is a different type of force.

Figure 3.62 *The friction between snow and ski is very small. This allows professional skiers to reach some very high speeds.*

Factors affecting the frictional force

There are several factors affecting the force of friction between objects.

Perhaps the most obvious is the **roughness** of the surface. The rougher the surface, the greater the friction. In simple terms the bumps on the surface are bigger or more frequent. This causes them to lock together more easily or more often.

You might think the weight of the object affects the friction. A heavier object will push down harder on the surface locking the bumps together harder and so increasing the force of friction. This is generally true but actually it is the contact force that affects the friction. Think about the lift example covered in the previous section. When the lift is accelerating downwards the weight stays the same but the contact force (and so the frictional force) would drop. This is especially important when considering objects on slopes (more on this later).

KEY WORDS

kinetic friction *the frictional force between two objects sliding over each other*

limiting friction *the maximum value of static friction*

static friction *the frictional force between two objects that are trying to move against each other but are not yet moving*

roughness *a measure of the texture of a surface*

DID YOU KNOW?

Friction is really just another example of the electrostatic force. It is caused by the electrons in the atoms in the bumps repelling each other.

Think about this...

The surface area of the objects in contact with each other does not affect the frictional force between them. Although there is a greater area in contact, the weight of the object is more spread out and so there is no change in the frictional force.

KEY WORDS

coefficient of friction *a ratio representing the friction between two surfaces*

The friction force between objects can be calculated using the following equation:

- $F_f = \mu N$

where:

F_f is the frictional force.

μ is a constant called the **coefficient of friction**, which depends on the roughness of the two surfaces. A high coefficient of friction would mean that the surfaces are very rough and so this would lead to a high frictional force. Materials have a static coefficient of friction and a kinetic coefficient of friction, depending on the type of friction being calculated.

N is the normal contact force acting on the block. Normal in this case means at right angles to the surface. If the block is horizontal and there is no vertical acceleration then the normal contact force is equal to the weight.

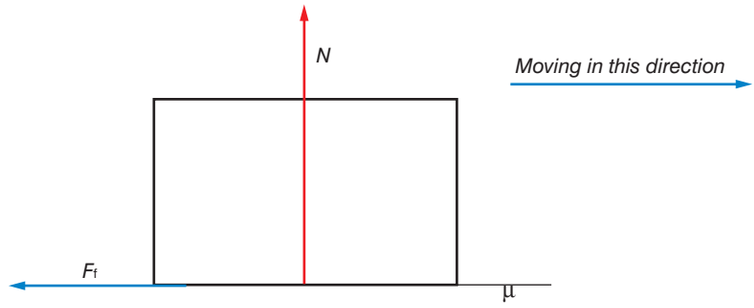


Figure 3.63 *Factors affecting friction*

Table 3.5 *Examples of the static friction coefficient between materials*

Materials rubbing together		μ_{static}
Aluminium	Steel	0.61
Concrete	Rubber	1.00
Concrete	Wood	0.62
Steel	Teflon	0.04
Wood	Wood	0.45

Worked example

The kinetic coefficient of friction between rubber and asphalt is 0.8. Calculate the force of friction acting on a rubber block of mass 2.0 kg as it is pulled along a level road at a steady speed.

$F_f = \mu_{kinetic} N$ *State principle or equation to be used*

$F_f = 0.8 \times N$ *Substitute in known value for* $\mu_{kinetic}$

As the road is level the normal contact force is equal to the weight of the rubber block. In this case the weight = 20 N (2 kg x 10 N/kg)

$F_f = 0.8 \times 20$ N *Substitute in known values and complete calculation*

$F_f = 16$ N *Clearly state the answer with unit*

A 12 kg block of wood is stationary on a horizontal concrete slab. The maximum coefficient of static friction between wood and concrete is 0.65 (this occurs at the limiting friction). What force needs to be applied in order to slide the block along.

$F_f = \mu_{static} N$ *State principle or equation to be used*

$F_f = 0.65 \times N$ *Substitute in known constant for* μ_{static}

As the block is level the normal contact force is equal to the weight of the wood. In this case the weight = 120 N

$F_f = 0.65 \times 120$ N *Substitute in known values and complete calculation*

$F_f = 78$ N *Clearly state the answer with unit*

DID YOU KNOW?

Teflon has one of the lowest coefficients of friction of any material. It was accidentally invented by an American named Roy Plunkett in 1938. The use of Teflon was important in America's development of the atomic bomb. Nowadays its low friction makes it ideal for non-stick frying pans!

Figure 3.64 *Non-stick frying pans have a very low friction coefficient.*

Activity 3.4: Measuring friction

- Tie a block of wood with string to the hook of a spring balance. Place the block on a table. Pull the balance gradually parallel to the table. Note its reading when the block just starts to move.
- Repeat and take the average of the results.

You have measured the maximum force of static friction (the limiting friction).

- Now pull the balance until the block moves steadily along. Note the reading.
- Repeat several times and take the average.

You have measured the force of dynamic friction.

- Which is greater?

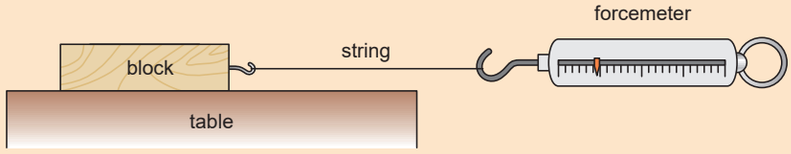


Figure 3.65 *Measuring friction using a spring balance*

You must ensure you pull the block along at a steady speed. This tells us the forces are balanced and the reading on the spring balance is the same as the frictional force.

KEY WORDS

inclined plane *a sloping surface or ramp*

Friction and inclined planes

If an object is resting on an **inclined plane** the normal contact force is reduced (the weight stays the same). This means the frictional force is also reduced.

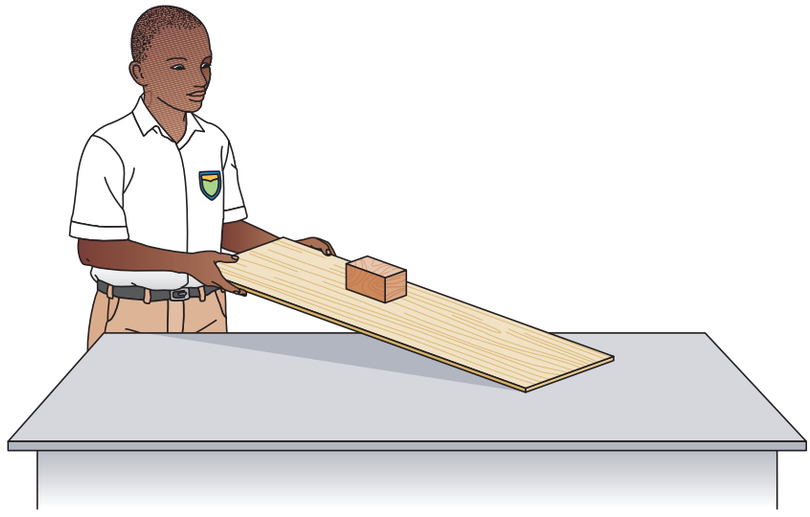


Figure 3.66 *A wooden block on a slope*

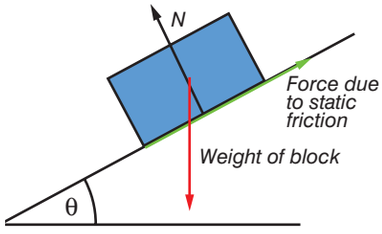


Figure 3.67 *Forces acting on a block on a slope*

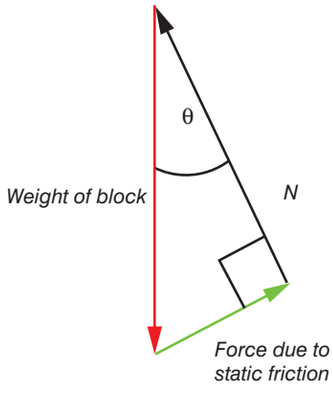


Figure 3.68 *The three forces form a triangle with no resultant force.*

Let's assume the block is not sliding down the ramp. If we consider the forces acting on the object we can see that there are three different forces.

As the object is not accelerating (in this case it is stationary) we can conclude from Newton's first law that there is no resultant force acting.

These three forces must form a triangle, as shown in Figure 3.68.

The normal contact force is given by:

- $N = w \cos \theta$

This is always true regardless of if the object is in equilibrium or not. As a result, as the angle of the slope increases the normal contact force falls and so does the force due to friction. If the slope was vertical then the force due to friction would be 0 N.

In order for the block to remain stationary (i.e. the forces remain balanced) then the force due to static friction must equal:

- $F = w \sin \theta$

where w is the weight of the block, F is the force due to static friction and θ is the angle of the slope.

However, the force of friction is also equal to:

- $F_f = \mu_{static} N$

In this case $N = w \cos \theta$, so:

- $F_f = \mu_{static}\, w \cos \theta$

As the angle of the slope increases, cos θ gets smaller. This means the frictional force that can be provided also falls (as all the other variables are constant). At the same time the force required to keep the object stationary (w sin θ) increases.

This means as the slope gets steeper eventually the block will accelerate down the slope as the forces can no longer be balanced; the limiting friction has been reached and exceeded.

If the object is accelerating down the slope then there must be a resultant force acting on the object.

This resultant force is equal to the difference between w sin θ and the force due to kinetic friction.

Think about this...

Putting the two equations for the friction force equal to each other: $w \sin\theta = \mu_{static}\, w \cos\theta$. This can be rewritten as $\tan\theta = \mu_{static}$ and so the maximum angle of the slope before the block will slide is given by $\theta = \tan^{-1} \mu_{static}$. The higher the maximum value for μ_{static} the steeper the slope can before the object slides down the slope.

- $F_{net} = w \sin\theta - \mu_{kinetic} N$

Take, for example, a block of wood of mass 30 kg accelerating down a concrete slope inclined at 45°. We could use the formula above to calculate the acceleration of the block. The $\mu_{kinetic}$ between the wood and the slope is = 0.45.

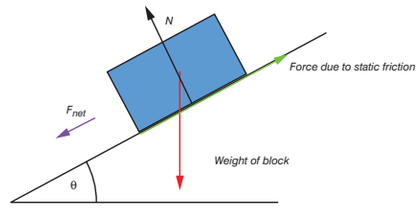


Figure 3.69 *The object slides due to a resultant force.*

First we need to find the resultant force:

- $F_{net} = w \sin\theta - \mu_{kinetic} N$ *Express F_{net} in terms of other forces*

In this case the weight of the block is 300 N (from $w = mg$) and the normal contact force is 212 N (from $N = w \cos\theta$).

$F_{net} = 300\ \text{N} \times \sin 45° - (0.45 \times 212\ \text{N})$ *Substitute in known values and complete calculation*

$F_{net} = 117$ N *Clearly state the answer with unit*

The acceleration of the block can then be calculated using Newton's second law.

$F_{net} = ma$ *State principle or equation to be used (Newton's second law)* $a = F_{net} / m$ *Rearrange equation to make a the subject*

$a = 117\ \text{N} / 30\ \text{kg}$ *Substitute in known values and complete calculation*

$a = 3.3\ \text{m/s}^2$ *Clearly state the answer with unit*

Reducing friction

In order to reduce the friction between objects there are two techniques that can be used.

Polishing

Polishing or sanding down an object reduces the size of the bumps on the surface. This makes it smoother and so the coefficient of friction drops.

Figure 3.70 *Polishing reduces the roughness of a surface.*

Figure 3.72 *Without oil vehicle engines would heat up too much and possibly seize up entirely.*

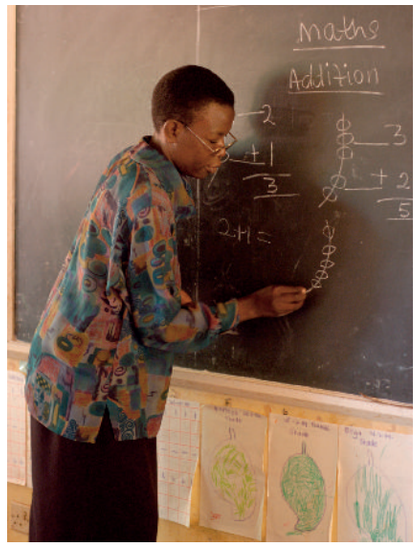


Figure 3.73 *Friction between board and chalk causes the chalk to gradually wear away*

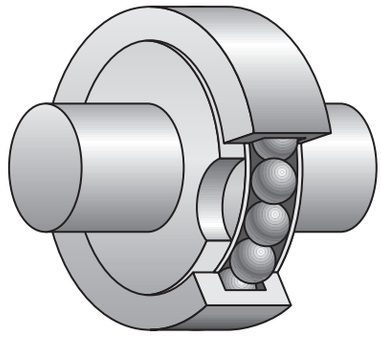


Figure 3.74 *Ball bearings ensure that a wheel turns smoothly on its axle.*

Lubrication

Lubricating between the surfaces rubbing together also reduces friction. Commonly used lubricants include oil, water and even graphite.

The lubricant effectively fills the gaps between the materials, preventing them from bumping into each other and so allowing them to slide over each other easily.

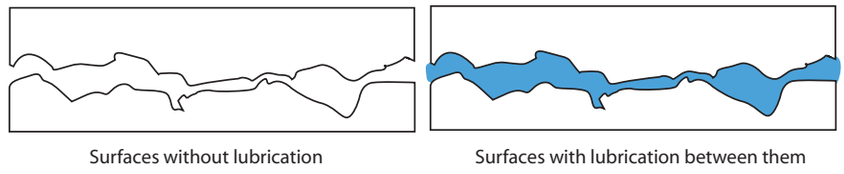


Figure 3.71 *Lubrication keeps the surfaces apart.*

The effects of friction

Friction causes a heating effect. When you rub your hands together friction between them causes them to warm up. This has many applications but also causes several problems.

Advantages of friction

Is friction always a problem? No. We could not walk if there was no friction. Our feet would slip, just as they do on ice, banana skin or very smooth surfaces. Rubber-soled shoes and car tyres have 'tread' on them to increase friction. Smooth tyres tend to skid, especially on wet, greasy or icy roads.

The brakes on a bicycle, car or other vehicle make use of friction. The brake pads press on the wheels, slowing them down.

Figure 3.73 shows one situation where friction is useful. Without friction, the teacher's chalk would not mark the board.

Disadvantages of friction

When two parts of any machine rub against each other, the friction between them causes heat, noise and wear. The heat produced in fast-moving machines may be so great that the parts become red-hot.

Friction is reduced by lubrication with grease, oil or graphite. Bicycles and sewing machines need oil regularly. The engine of a motor car has a case at the bottom, called a sump, which is full of oil. This covers all the moving parts in the engine. If the engine has too little oil, the pistons and cylinders become so hot that they join together.

A bicycle wheel must turn freely. If there is friction between the wheel and its axle, the bicycle will be harder to ride. Ball bearings between the wheel and axle allow the wheel to turn freely – see Figure 3.74.

Summary

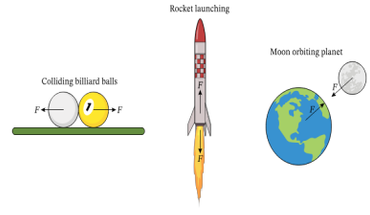
In this section you have learnt that:

- Friction is a force generated when solids either attempt to slide or slide over each other.
- Friction is caused by bumps in the surface of the materials.
- Static friction occurs when objects try to move past each other. Kinetic friction occurs when objects slide over each other; it acts in the opposite direction to motion.
- Frictional forces can be calculated using $F = \mu N$ (where N is the normal contact force – this reduces if the object is on an inclined plane).

Review questions

1. Describe the causes of friction and the factors that affect it.
2. Explain the difference between static friction and kinetic friction.
3. If the static friction between wood and concrete is 0.62, determine the force required to make a wooden block of mass 2 kg start to slide.
4. Give two examples in which friction is useful and two where it is a disadvantage.

2.7 Newton's third law



By the end of this section you should be able to:

- State Newton's third law.
- Describe experiments to demonstrate it and give examples of where it is applicable.

The third law of motion

Newton's third law deals with what happens when you apply a force. It is perhaps the most counter-intuitive of Newton's three laws.
It states:

- **If body A exerts a force on body B then body B will exert an equal and opposite force on body A.**

In simple terms this means whenever you push an object it pushes back with an equal and opposite force; essentially forces come in pairs. You can't apply a force to an object without that object applying the same force back onto you.

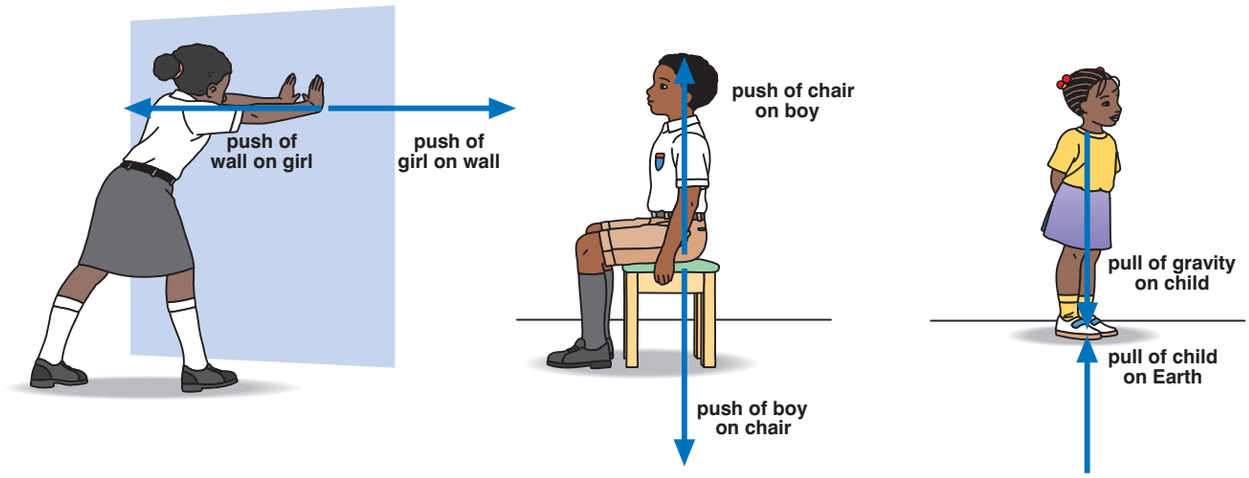


Figure 3.75 *Examples of Newton's third law in action.*

The pairs of forces are often called a **Newton's pair** or an action and reaction pair. It is important to notice that they are equal and opposite.

- **Equal: same magnitude**
- **Opposite: opposite direction**

If you push **down** on the desk with a force of **10 N** the desk pushes back **up** with a force of **10 N**. This applies to all forces!

It may seem strange but the gravitational attraction of the Earth on a satellite is exactly the same size as the pull on the Earth from the satellite. The same is true at ground level. If you hold a stone above the Earth then it pulls the Earth up with the same force that the Earth pulls to stone down. When you drop it the stone appears to fall but both the stone and the Earth experience the same force. However, the stone's acceleration is much, much greater as it has much less mass.

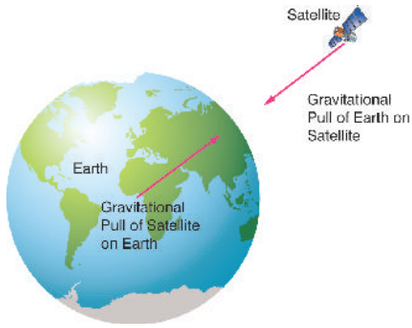


Figure 3.76 *An example of Newton's pairs*

KEY WORDS

Newton's pair *a pair of equal and opposite forces acting between two objects*

To correctly identify Newton's pairs it is worth remembering that the pairs of forces must fit the following four criteria:

- **equal in magnitude**
- **opposite direction**
- **act on different bodies**
- **same type of force**

So, for example, consider a book on a desk.

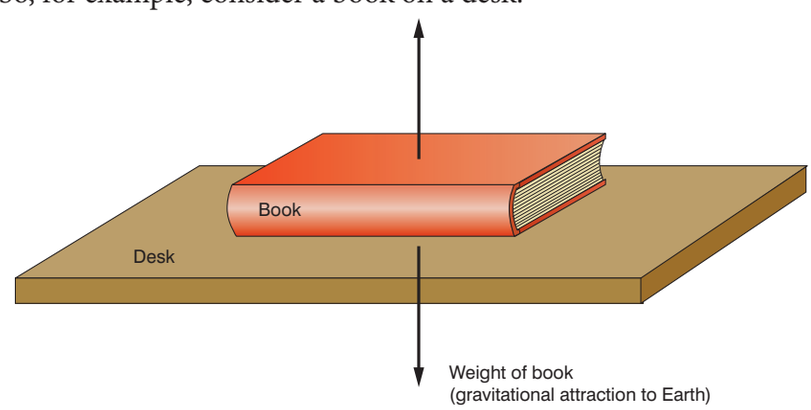


Figure 3.77 *Two forces acting on a book, but they are not a Newton pair.*

Figure 3.77 shows two forces acting on the book, but they are not an action–reaction pair. They are equal and opposite but they do not act on different bodies and they are not the same type of force.

So, where are the Newton's pairs in this example?

Table 3.6 *Newton's pairs for a book on a desk*

Force	Newton's pair
Contact force on book from desk	Contact force on desk from book
Weight of book (gravitational attraction of the Earth pulling on the book)	Gravitational attraction of the book pulling on the Earth

The book pushes down on the desk and pulls the Earth upward due to gravitational attraction. These are the pairs to the two forces in Figure 3.77. If we draw three free body diagrams (Figure 3.78) we can more easily see the pairs of forces.

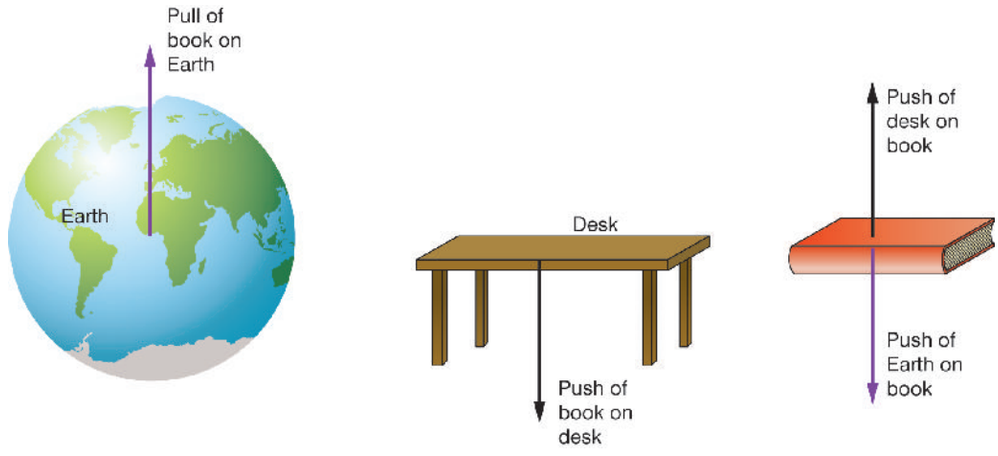


Figure 3.78 *Thetw o pairs of forces*

There are two more pairs of forces not included in Figure 3.78. Can you work out what they are? (Hint: they do not involve the book).

Applications of the third law

Newton's third law is incredibly important to motion. Applications such as rockets, jet engines, cars and even just walking around rely on this law.

When you walk you push backwards on the ground; at the same time the ground pushes forward on you and so you accelerate forwards! The same is true with car tyres.

With a rocket or jet engine hot gases are blasted out of the back; they are in essence pushed out. This results in an equal and opposite force on the engine pushing it forward.

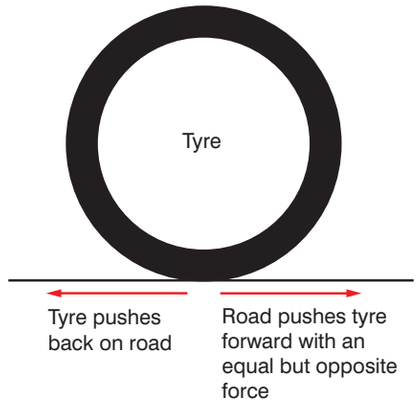


Figure 3.79 *Newton's third law in action*

Figure 3.80 *Without Newton's third law space rockets would not be able to move!*

Activity 3.5: Discovering equal and opposite forces

With a partner get a rope and two skateboards. Both stand on a board some distance apart and hold a rope between you. If one of you holds the rope and the other one pulls it towards them, who moves?

You both will! An equal and opposite force is exerted on the puller. If he pulls with twice as much force he will experience twice as much force pulling him forwards.

Summary

In this section you have learnt that:

- Newton's third law states: "If body A exerts a force on body B then body B will exert an equal and opposite force on body A".
- Newton's third law means forces always come in pairs.

SECTION REVIEW

1. A 6.0 kg object undergoes an acceleration of 2.0 m/s^2.
   a. What is the magnitude of the net force acting on the object?
   b. If this same force is applied to a 4.0 kg object, what acceleration is produced?
2. A child causes a wagon to accelerate by pulling it with a horizontal force. Newton's third law says that the wagon exerts an equal and opposite force on the child. How can the wagon accelerate? (Hint: Draw a free-body diagram for each object.)
3. Identify the action-reaction pairs in the following situations:
   a. A person takes a step.
   b. A snowball hits someone in the back.
   c. A baseball player catches a ball.
   d. A gust of wind strikes a window.
4. The forces acting on a sailboat are 390 N north and 180 N east. If the boat (including crew) has a mass of 270 kg, what are the magnitude and direction of the boat's acceleration?

Lesson 3 Work Energy and Power

Contents

Section	Learning competencies
3.1.Mechanical work	• Describe the necessary conditions for work to be done by a force (including work done by a force F acting on a body at an angle of θ). • Use $W = F s \cos\theta$ to solve problems. • Calculate the work done against gravity, the work done by a frictional force and the work done by a variable force. • Distinguish between negative and positive work.
3.2. Work–energy theorem	• Explain the relationship between work and energy. • Derive the relationship between work and kinetic energy and use this to solve problems. • Show the relationship between work and potential energy as $W = -\Delta U$ and use this to solve problems. • Describe gravitational potential energy and elastic potential energy. • Explain mechanical energy as the sum of kinetic and potential energy.
3.3. Conservation of energy	• State the law of conservation of mechanical energy. • Revise the term collision and distinguish between elastic and inelastic collisions. • Solve problems involving inelastic collisions in one dimension using the laws of conservation of mechanical energy and momentum. • Explain the energy changes that take place in an oscillating pendulum and an oscillating spring–mass system. • Describe the use of energy resources, including wind energy, solar energy and geothermal energy. • Explain the meaning of the term renewable energy.
3.4. Mechanical power	• Solve problems relating to the definition of power. • Show that the kWh is also a unit of work. • Express the formula of mechanical power in terms of average velocity.

Bouncing a ball involves some complex energy changes and transfers. No matter what surface you drop the ball on to, it will never return to its original height. Why is this? In simple terms, some of the ball's energy has been transferred into the air and ground. After the bounce it has less energy than it did before, and so it can't return to its original height.

This unit looks at work and energy, how it comes in different forms, and how you can transform it and transfer it. However, no matter how hard we try, we can't make any more energy then there is to start with, nor can we destroy any.

Figure 4.1 *These fishermen are working hard, but what does the term working hard mean?*

With dwindling global energy resources and continuously increasing demand, energy issues will play a very significant role in the next 20 years.

3.1. Mechanical work

By the end of this section you should be able to:

- Describe the necessary conditions for work to be done by a force (including work done by a force F acting on a body at an angle of θ).
- Use $W = F\,s\cos\theta$ to solve problems.
- Calculate the work done against gravity, the work done by a frictional force and the work done by a variable force.
- Distinguish between negative and positive work.

What is work?

The term **work** is used all the time in everyday language. You might go to work, a device may stop working, you might complete schoolwork, or work hard to solve a problem. However, in physics, work means something very specific.

You might describe someone performing a physically demanding task as working hard. This is closer to the truth than it first appears. In physics the term work (or often **work done**) is another way of saying **energy** is being transferred from one object to another or transformed from one type to another.

- **Work done = energy transferred**

This means, like energy, work done is measured in **joules**. (The joule is the SI derived unit of energy). The more energy transferred the more work has been done. Work is a scalar quantity, just like energy.

DID YOU KNOW?

The term work was first used in the 1830s by the French mathematician Gaspard-Gustave Coriolis. He is more famous for giving his name to the Coriolis effect. This explains the rotation of large weather systems like hurricanes and cyclones.

Figure 4.2 *The forklift is transferring energy to the box as it lifts it up. It is doing work.*

Calculating work done

Look back at the fishermen in Figure 4.1. As they pull the rope along they are transferring energy to their catch at the end of the rope. The harder they pull or the greater distance they travel, the more energy they transfer, the more work they do.

Mechanical work is defined as the amount of energy transferred by a force acting through a distance. We can calculate work done using the following equation:

- $W = F\,s$

W = work done in J.

F = average force applied (it is assumed to be constant) in N.

s = the distance moved in the direction of the force in m.

Notice we usually use *s* instead of *d* or *x*. This is because the direction of the distance moved is really important. The distance travelled has to be in the direction of the force. Either in the same direction or in the opposite direction (this is often referred to as the distance against the force). If you are not pulling or pushing against a force then you are not doing any work. We use *s* because the distance has a specific direction; therefore it can be considered to be a displacement.

DID YOU KNOW?

One joule is defined as the work done when a force of 1 N moves through a distance of 1 m. So 1 J = 1 N × 1 m.

- **Mechanical work may be defined as the product of displacement and the force in the direction of the displacement.**

In both examples in Figure 4.3 work is being done, energy is being transferred or transformed. The first example involves pulling a trolley along the ground against a frictional force of 2 N. The second involves lifting a 2 N book. In both cases the distance moved against the force is 3 m and so 6 J of work has been done.

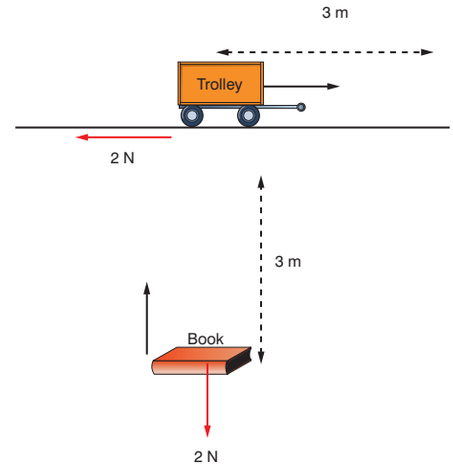


Figure 4.3 *Two examples of doing work, for example lifting a book to place it on a shelf or pushing a shopping trolley through a store.*

- $W = F\,s$
- $W = 2\ \text{N} \times 3\ \text{m}$
- $W = 6\ \text{J}$

Looking at the second example the direction of the force is vertically downwards (it is the weight of the book). Therefore it is only the vertical distance moved that is important.

Look at Figure 4.4. Assuming the book weighs 2 N and there are no other forces acting, how much work is done in each case?

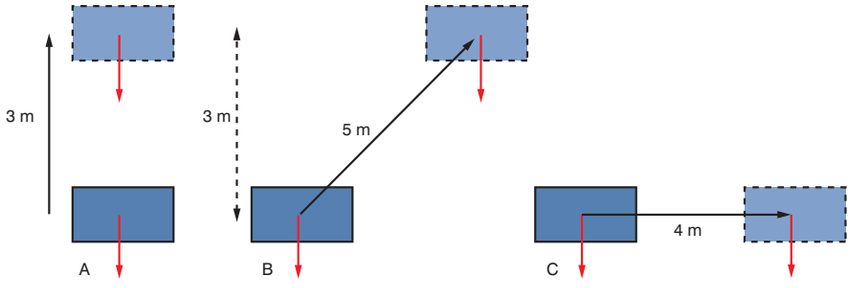


Figure 4.4 *The distance moved must be in the opposite direction to the force.*

In example A the work done is simple to calculate: $W = F\,s$, $W = 6$ J.

Example B is more complex and serves to illustrate the importance of working against the force. The book has been moved 5 m. However, it has only been moved 3 m vertically. It is this distance, the distance against the force, which we use in our calculation.

$W = F\,s$ *State principle or equation to be used (definition of mechanical work)*

$W = 2\ \text{N}. \times 3\ \text{m}$ *Substitute in known values and complete calculation*

$W = 6\ \text{J}$ *Clearly state the answer with unit*

So in both A and B the work done is 6 J. The energy transferred to the book is 6 J in each case.

DID YOU KNOW?

As well as mechanical work you can do electrical work on an object. The equation for electrical work done is $W = VIt$, where V is the potential difference in volts, I is current in amperes and t is time in seconds.

KEY WORDS

energy *the stored ability to do work*

joule *the SI unit of work and energy*

work / work done *the amount of energy transferred when an object is moved through a distance by a force*

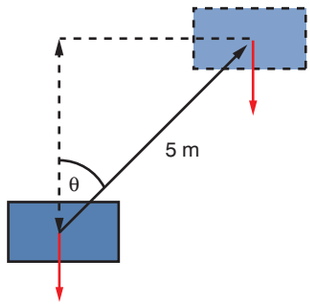


Figure 4.5 *θ is the angle between the direction of movement and the direction of the force.*

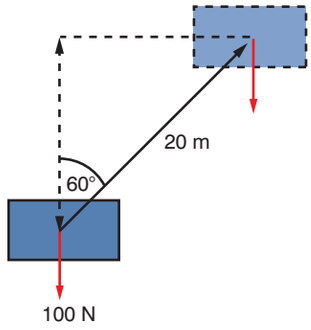


Figure 4.6 *A box lifted up at an angle*

Think about this...

If the angle between the force and distance moved is 0° (i.e. they are parallel) then cos θ = cos 0° = 1. The equation $W = F\,s \cos \theta$ becomes $W = F\,s$, as used in the earlier examples.

In example C in Figure 4.4 the book moves 4 m. However, it does not move any distance against the force (it does not move vertically). Therefore
$s = 0$ m.

- $W = F\,s$
- $W = 2\ \text{N} \times 0\ \text{m}$
- $W = 0\ \text{J}$

So in example C no work has been done. No energy has been transferred to the book.

A more complex version of the work equation can be seen below.

- $W = F\,s \cos \theta$

s is the distance travelled.

θ is the angle between the force and the direction of movement.

If you think about this equation, $s \cos \theta$ is really the distance moved in the direction of the force.

For example, Figure 4.6 shows a 100 N box lifted 20 m at an angle of 60° to the vertical.

The work done would be:

$W = F\,s \cos \theta$ *State principle or equation to be used (definition of mechanical work)*

$W = 100\ \text{N} \times 20\ \text{m} \times \cos 60°$ *Substitute in known values and complete calculation*

$W = 1000\ \text{J}$ *Clearly state the answer with unit*

Doing work against gravity, friction, and gravity and friction!

Gravity

Work is often done against gravity. Whenever you lift up an object you are doing work against the force of gravity. In this case the force you are working against is the weight of the object. We can adapt our work done equation for working against gravity:

- $W = F\,s$
- Work done against gravity = weight × vertical distance moved (or $\boldsymbol{W_{gravity} = w \times h}$)

The work done in lifting a 60 kg mass vertically 3 m can be found using the work done equation:

$W_{gravity} = w \times h$ *State principle or equation to be used*

$w = mg$, $w = 60\ \text{kg} \times 10\ \text{N/kg} = 600\ \text{N}$ *Calculate weight from known values*

$W_{gravity} = 600\ \text{N} \times 3\ \text{m}$ *Substitute in known values and complete calculation*

$W_{gravity} = 1800\ \text{J}$ *Clearly state the answer with unit*

Remember, it must be the vertical distance moved and weight acts vertically.

Friction

Whenever you push an object along the ground you are working against a force of kinetic friction.

Kinetic friction always acts in the opposite direction to motion. In Unit 3 we learnt that:

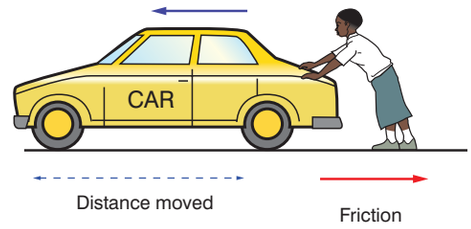


Figure 4.7 *Working against friction*

- $F_{friction} = \mu_{kinetic} N$

This is the force you are working against. We can adapt our work done equation for working against frictional forces:

- $W = F\,s$
- Work done against friction = force due to kinetic friction × distance moved
- $W_{friction} = \mu_{kinetic} N \times s$

For example, we can determine the work done in pushing a 100 kg wooden block 30 m across a horizontal concrete floor with $\mu_{kinetic}$ = 0.48

- $W_{friction} = \mu_{kinetic} N \times s$

In this case the normal contact force is equal to the weight (as the floor is horizontal) and so

$N = w = mg$ *Express N in terms of weight*

$N = 100\text{ kg} \times 10\text{ N/kg}$ *Substitute in known values and complete calculation*

$N = 1000\text{ N}$ *Clearly state the answer with unit*

$W_{friction} = \mu_{kinetic} N \times s$ *Express $W_{friction}$ in terms of frictional force and distance moved*

$W_{friction} = 0.48 \times 1000\text{ N} \times 30\text{ m}$ *Substitute in known values and complete calculation*

$W_{friction} = 14\,400\text{ J or } 14.4\text{ kJ}$ *Clearly state the answer with unit*

This energy has been transformed into **heat energy** where the block and surface rub together.

KEY WORDS

heat energy *energy that is transferred between two objects as a result of their difference in temperature*

Gravity and friction

If you were to push or pull on object up a ramp then you end up doing work against both friction and gravity!

In this case the total work done could be found using the following equation:

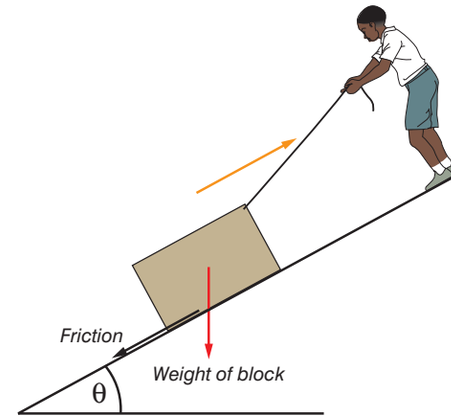


Figure 4.8 *Working against friction and gravity*

- **Total work done = work done against gravity + Total work done = work done against friction**

Work done against gravity = weight × vertical distance moved.

- $W_{gravity} = w \times h$

Work done against friction = force due to kinetic friction × distance moved up ramp.

- $W_{friction} = \mu_{kinetic} N \times s$

Think about this...

Using the equations in Unit 3 and trigonometry can you show how we might expand the final equation to:
$W_{total} = (w \times s \sin \theta) +$

$(\mu_{kinetic} \times w \cos \theta \times s)$

KEY WORDS

graph *a drawing showing how two or more sets of numbers are related to each other*

area under the line *the area between the line on a graph and the axes*

calculus *a type of mathematics that deals with rates of change*

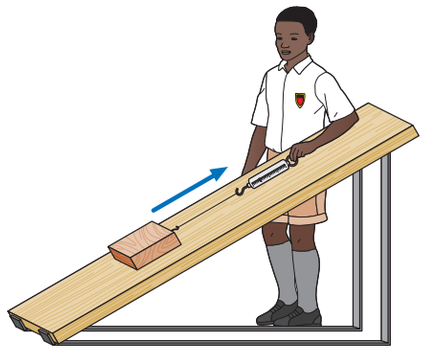


Figure 4.9 *Pulling an object up a ramp*

So:

- $W_{total} = (w \times h) + (\mu_{kinetic} N \times s)$

We have to be very careful in considering the distances we use in this equation; h has to be the vertical distance, as this is the distance moved against gravity, whereas s must be the distance moved up the slope as friction acts down the slope.

Worked example

Using the wooden block earlier we can determine the work done if the block was pulled 20 m up a ramp at an angle of 30°.

- **Total work done = work done against gravity + work done against friction.**

Work done against gravity:

$W_{gravity} = w \times h$ *Express $W_{Gravity}$ in terms of force (weight) and distance moved (height lifted)*

In this case $w = mg$ = 100 kg × 10 N/kg = 1000 N. h = vertical distance moved, which, using trigonometry, = $s \sin \theta$ = 20 m × sin 30° = 10 m.

$W_{gravity}$ = 1000 N × 10 m *Substitute in known values and complete calculation*

$W_{gravity}$ = 10 000 J *Clearly state the answer with unit*

Work done against friction:

$W_{friction} = \mu_{kinetic} N \times s$ *Express $W_{friction}$ in terms of frictional force and distance moved*

In this case $\mu_{kinetic}$ = 0.48, s = 20 m and $N = w \cos \theta$ (see Unit 3) = 1000 N x cos 30° = 866 N

$W_{friction}$ = 0.48 × 866 N × 20 m *Substitute in known values and complete calculation*

$W_{friction}$ = 8313.6 N or 8300 N *Clearly state the answer with unit*

Total work done:

$W_{total} = W_{gravity} + W_{friction}$ *Simple expression of total work done*

W_{total} = 10 000 J + 8300 J *Substitute in known values and complete calculation*

W_{total} = 18 300 J *Clearly state the answer with unit*

What if the force varies?

If the force applied *varies* we can't use the $W = F s \cos \theta$ equation to find the work done. We need a different technique to calculate the work done.

We can plot a **graph** of the force applied against the distance travelled against the force.

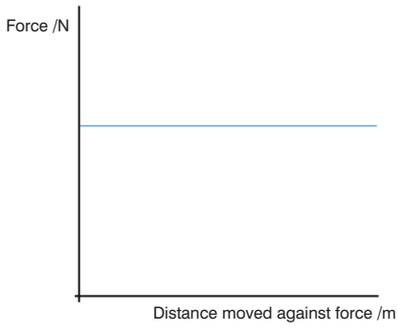


Figure 4.10 *A graph showing a constant force acting over a distance*

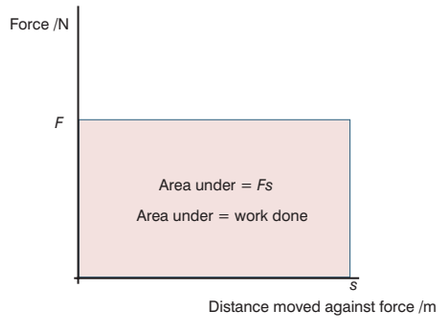


Figure 4.11 *The area under a force vs. distance moved graph is equal to the work done.*

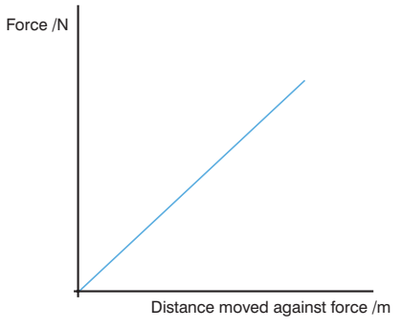


Figure 4.12 *A graph showing a force that increases as the distance moved increases*

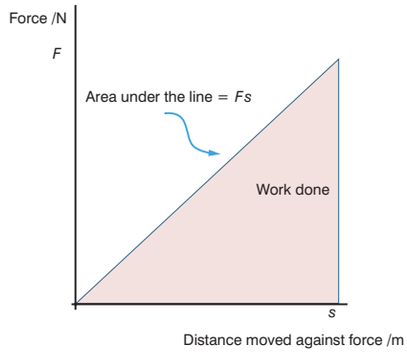


Figure 4.13 *The area under the line still represents the work done.*

The **area under the line** is equal to $F\,s$; it is equal to the work done. Increasing the distance moved or increasing the force both increases the area under the line and so more work has been done.

What if the force was not constant but gradually increasing? You might get a graph that looks like Figure 4.12.

In this case the area under the line is a triangle. This area is still equal to the work done.

What if the force varied in a more complex way? Take, for example, Figure 4.14. This might be a varying force of friction as a box is dragged over different surfaces.

Remember the area under the line is still equal to the work done. But how do we calculate it?

In order to determine the area under the line we need to count the squares under the line and then use this to calculate the work done.

Take a small square under the line and calculate the area of this square. For example, if the square is 20 N high and 0.1 m across the area is equal to:

- area of one square = 20 N × 0.1 m
- area of one square = 2 J.

DID YOU KNOW?

You could use some powerful mathematics called **calculus** to determine the area under the line. Newton invented this kind of mathematics to help him solve complex problems relating to the motion of objects.

Think about this...

Hooke's law produces a graph very similar to Figure 4.12. In fact the area under the line in this case represents the work done on the spring. That is, the energy stored by the spring. You can work out the energy stored using the equation $W = \frac{1}{2}F\Delta x$.

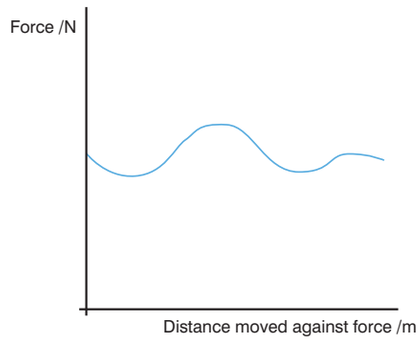


Figure 4.14 *A graph showing a force that changes in a complex way as distance increases*

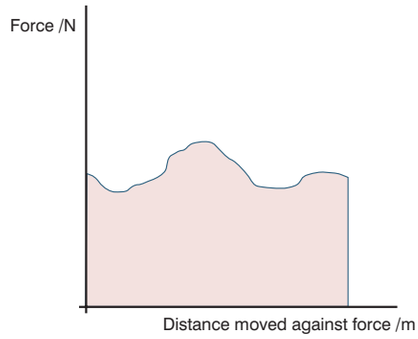


Figure 4.15 *No matter how complex the force vs. distance moved graph, the area under the line is still equal to the work done.*

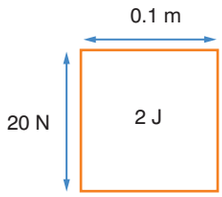


Figure 4.16 The area of one square represents a small amount of the total work done.

This small square represents 2 J of work done. We need to count up all the squares and then multiply this by 2 J to determine the total work done. For example, if there are 100 squares the total work done would be:

- **total work done = number of squares × work done for each square**
- total work done = 100 × 2 J = 200 J.

If there were 500 squares the total work done would be 1000 J, etc.

You must be careful when counting the squares. You need to make a few estimations near the line. For example:

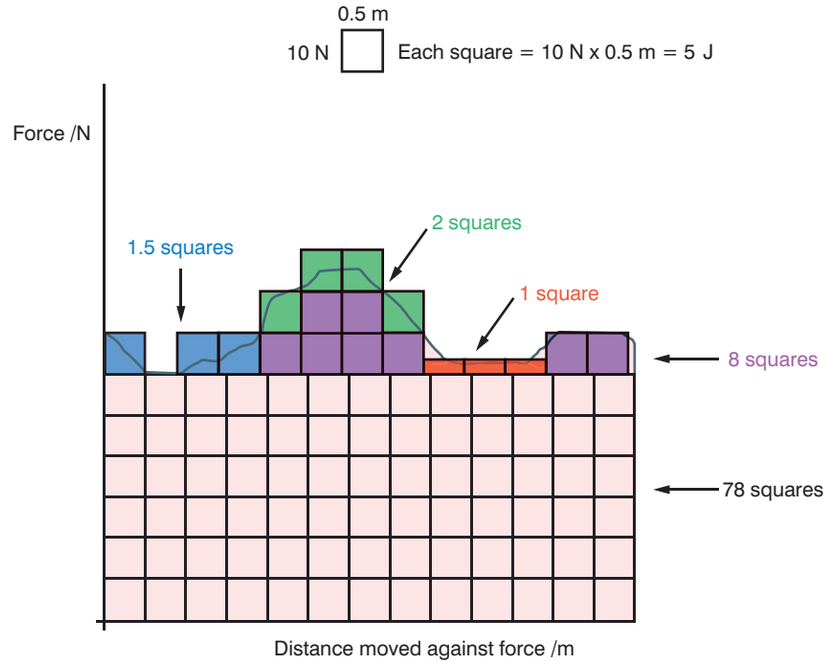


Figure 4.17 Counting the squares often involves some estimation close to the line.

In Figure 4.17 there are a total of 90.5 squares. We have had to estimate some of those near the line. The three small red areas add up to one complete square, the four green areas add up to two squares, etc.

In this case the total work done is equal to:

- total work done = number of squares × work done for each square
- total work done = 90.5 × 5 J
- total work done = 452.5 J (approximately 450 J).

Although this is only an approximate value if you are careful counting the squares you will get very close to the true value of the work done.

+W or –W?

Work may be expressed as a **positive** or **negative** value. Remember, work is a scalar quantity and the opposite sign does not mean the opposite direction.

Instead, whether the work is positive or negative depends on whether or not the object gains or loses energy.

In both cases in Figure 4.18 the work done is 500 J. In the first case we can say work is done *on* the box. It gains 500 J of energy.

In the second case the box loses 500 J of energy. We can express this as –500 J or we could say the work done *by* the box is 500 J.

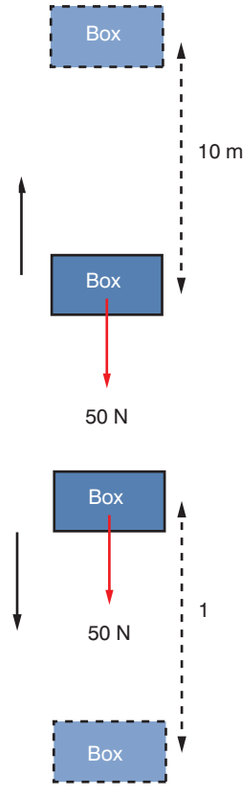


Figure 4.18 *Work being done on or by a moving box*

Summary

In this section you have learnt that:

- Work done is another way of saying energy transferred.
- Mechanical work is done whenever you move a force through a distance.
- The work done may be found using the equation: $W = F\,s\,\cos\theta$
- Work done may be positive or negative depending on whether the object in question gains or loses energy.

KEY WORDS

negative *less than zero*
positive *greater than zero*

Review questions

1. Explain the meaning of the term work done and give an example of where work is done.

2. Calculate the total work done in the following examples:
   a) A 20 kg log lifted 2 m into the air
   b) Thirty 6 kg boxes lifted onto a shelf 1.5 m high
   c) A car of mass 1400 kg pushed 50 m along a road ($\mu_{kinetic} = 0.3$)
   d) A concrete slab of mass 200 kg pulled 10 m up a slope at an angle of 30° to the horizontal ($\mu_{kinetic} = 0.6$).
3. Describe in detail how you would determine the work done by a varying force.
4. Explain the difference between positive and negative work done.

Figure 4.19 *It takes energy to play football.*

3.2 Work–energy theorem

By the end of this section you should be able to:

- Explain the relationship between work and energy.
- Derive the relationship between work and kinetic energy and use this to solve problems.
- Show the relationship between work and potential energy as $W = -\Delta U$ and use this to solve problems.
- Describe gravitational potential energy and elastic potential energy.
- Explain mechanical energy as the sum of kinetic and potential energy.

Energy vs. work?

Energy and work are really just different ways of looking at the same thing. The energy of an object is a mathematical representation of the amount of work an object can do. Whereas work is any energy transferred to or from the object, energy refers to the total amount of work the object could theoretically do. In algebraic terms:

- $\Delta E = W$

Both energy and work are scalar quantities measured in joules.

DID YOU KNOW?

A common definition for energy is capacity to do work. The more energy an object has the more work it can do! The term energy comes from the Greek word 'energeia' meaning activity or operation.

Forms of energy

There are several different **forms** of energy. These include:

Table 4.1 *Different types of energy*

Kinetic energy	Gravitational potential energy
Heat energy	Elastic potential energy (strain)
Sound energy	Chemical energy
Electromagnetic energy (light)	Nuclear energy
Electrical energy	

Activity 4.1: Energy examples

Can you give examples of where you might come across each of the forms of energy listed in Table 4.1?

The forms of energy on the left hand side of Table 4.1 are all energies associated with a kind of movement, whereas the forms of energy on the right are all to do with storing energy due to the particular arrangement of objects. **Remember, all forms of energy are scalar quantities measured in joules.**

KEY WORDS

forms *types*

motion *the act of moving or the way an object moves*

Kinetic energy

Any object in **motion** has a kinetic energy (E_k). The amount of energy depends on the mass of the moving object and how fast it is travelling. Kinetic energy is calculated using the equation below:

- **kinetic energy** = $\frac{1}{2}mv^2$

For example, a car of mass 1000 kg travelling at 12 m/s will have a kinetic energy of:

$E_k = \frac{1}{2}mv^2$ *State principle or equation to be used*

$E_k = \frac{1}{2} \times 1000\ \text{kg} \times (12\ \text{m/s})^2$ *Substitute in known values and complete calculation*

$E_k = 72\,000$ J or 72 kJ *Clearly state the answer with unit*

An object with double the mass travelling at the same speed will have twice the kinetic energy. Mass and kinetic energy are directly proportional. However, if you double the velocity of an object its kinetic energy will increase by a factor of four (2^2). This relationship is not directly proportional; instead E_k is directly proportional to v^2. If the velocity increases by a factor of five the E_k will increase by a factor of 25 (5^2).

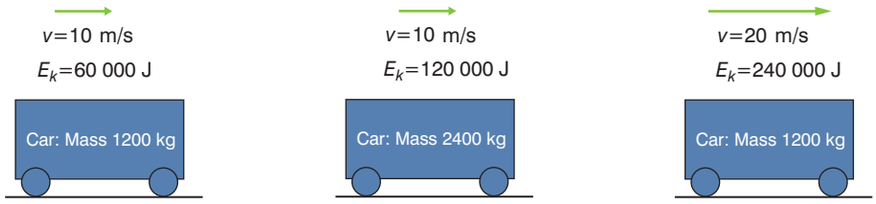


Figure 4.20 *The effect of mass and velocity on the kinetic energy of an object*

Why does $E_k = \frac{1}{2}mv^2$?

This equation comes from combining Newton's first and second laws of motion and one of the equations for constant acceleration.

Part of the work–energy theorem states:

- **If an external force acts upon an object it will cause its kinetic energy to change from E_{k1} to E_{k2}. The net work done on a body equals its change in kinetic energy.**

This statement should make sense. Work done is energy transferred. If a resultant force is applied to an object it will accelerate (Newton's first law). As a result it will change its kinetic energy and this change will be equal to the energy transferred (or work done).

In terms of equations we have:

- Work done = change in kinetic energy
- $W = \Delta E_k = E_{k2} - E_{k1}$
- $W = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$
- $W = \frac{1}{2}m(v_2^2 - v_1^2)$

This does not show where $E_k = \frac{1}{2}mv^2$ comes from; however, we can derive this equation another way to show that it is valid.

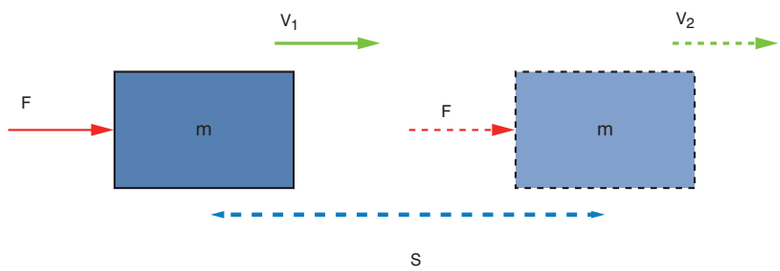


Figure 4.22 *Deriving $E_k = \frac{1}{2}mv^2$*

Activity 4.2: Kinetic energy of a car

Determine the kinetic energy of the car used in the worked example if it were travelling firstly at 16 m/s and then at 24 m/s.

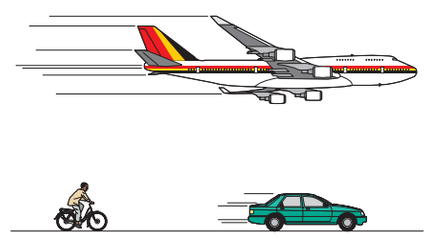


Figure 4.21 *All moving objects have kinetic energy. In this picture the aircraft has the most E_k.*

Think about this...

Because $E_k \propto v^2$ the velocity of a moving car has a significant impact on its stopping distance. Travelling at 50 km/h it may take 25 m to stop (depending on road conditions, etc). Double that, travelling at 100 km/h and it takes a massive 75 m to stop, much more than double the distance. This is because the brakes have to do more than double the work (as there is more than double the E_k and so the force has to act over a much greater distance).

Worked example

A car of mass 800 kg is travelling at 12 m/s. The car accelerates over a distance of 240 m. The net force causing this acceleration is 200 N. Determine the work done on the car and its final velocity.

W = F s *State principle or equation to be used (definition of mechanical work)*

W = 200 N x 240 m *Substitute in known values and complete calculation*

W = 48 000 J *Clearly state the answer with unit*

You can calculate the final velocity in a number of different ways (including use one of the equations of constant acceleration). In this case we will use:

$W = \frac{1}{2}m(v^2 - u^2)$

$2\ W / m = v^2 - u^2$ *Rearrange equation to give $v^2 - u^2$ on right hand side*

$v^2 = (2\ W / m) + u^2$ *Rearrange equation to make v^2 the subject*

$v^2 = (2 \times 48\ 000\ J / 800\ kg) + (12\ m/s)^2$ *Substitute in known values and complete calculation for v^2*

$v^2 = 264$ *Solve for v^2 then take the square root to complete*

v = 16 m/s *Clearly state the answer with unit*

Starting from Newton's second law:

- $F_{net} = m\ a$

Our defining equation for work done:

- $W = F\ s$

So we could substitute in for F and we get:

- $W = m\ a\ s$

From the equations of constant acceleration we have:

- $v^2 = u^2 + 2as$ This can be written as:
- $as = (v^2 - u^2) / 2$

Combining this with our previous equation we get:

- $W = m\ (v^2 - u^2) / 2$ Or
- $W = \frac{1}{2}m(v^2 - u^2)$.

Activity 4.3: Final velocity

Check the final velocity in the worked example using one of the equations of constant acceleration.

KEY WORDS

potential energy *the ability of an object to do work as a result of its relative position*

stored energy *the potential ability of an object to do work as a result of its relative position or shape change*

Potential energies

As previously mentioned the second column in our table of energies contains some different kinds of **potential energy**. They are effectively **stored energies**. They are all due the particular organisation or position of parts of the object/system of objects. The potential energy of an object is usually given the symbol U.

- **Potential energy = *U***

If an object has a potential energy it can be thought of as storing some energy. This energy has the potential to do some work, i.e. the potential energy might be transformed into another form of energy and so work would be done (remember work done is just another way of saying energy has been transferred).

Imagine an object has a potential energy of 1000 J. If this object did 300 J of work then the potential energy remaining after the work has been done will be 700 J. In other words:

- Work done by object = drop in potential energy of object

Or in symbols:

- $W = -\Delta U$

Equally, if work is done on the object then its potential energy might increase (it is also fair to say its kinetic energy may also increase). This is really just another way of saying work done is equal to the energy transferred; we just need to think carefully about where that energy has come from.

Gravitational potential energy

Perhaps the most common potential energy is **gravitational potential energy** (GPE). Any object with mass in a **gravitational field** has a GPE. How much GPE depends on three factors, its mass, the gravitational field strength (g) and its position in the field.

We usually deal with GPE with reference to the surface of the Earth. Therefore, on the ground an object has 0 J of GPE.

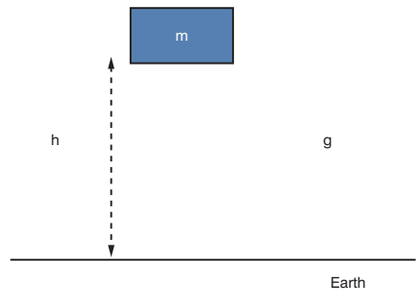


Figure 4.23 *Factors affecting the GPE of an object*

- **Gravitational potential energy = *mgh***

m = mass in kg.

g = gravitational field strength (on Earth this is 10 N/kg or more precisely 9.81 N/kg).

h = height above the ground.

For example, an object of mass 30 kg at a height of 12 m has a GPE equal to:

GPE = mgh *State principle or equation to be used*

GPE = 30 kg × 10 N/kg × 12 m *Substitute in known values and complete calculation*

GPE = 3600 J *Clearly state the answer with unit*

An object with double the mass at the same height above the ground will have twice the GPE. Equally, an object twice as high above the ground will have double the GPE. Mass and height above the ground are both directly proportional to the GPE of the object.

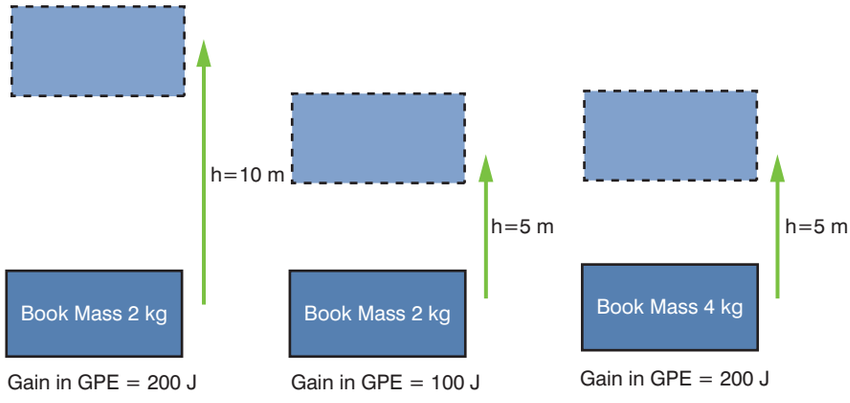


Figure 4.24 *The effect of mass and height above the ground on the GPE of an object*

If you think about when you do work by lifting up an object, you are transferring GPE to the object you are lifting. Looking back at the equations we can see they are both saying the same thing.

Activity 4.4: Calculating GPE

How high above the ground would a 10 kg object need to be to have the same GPE as the 30 kg object in the example?

Think about this...

If an object has 0 J when on the ground how much GPE will an object have at the bottom of a well? It takes energy to lift the object out of the well. Work is done on the object and it gains energy to end up with 0 J. This must mean the GPE at the bottom of the well is less than 0 J. It must be a negative number! This is often referred to as a potential well.

KEY WORDS

gravitational field *the space around an object in which the object's gravitational effect can be felt*

gravitational potential energy *the energy an object has due to its relative position above the ground*

Figure 4.25 *A child's spring toy stores EPE.*

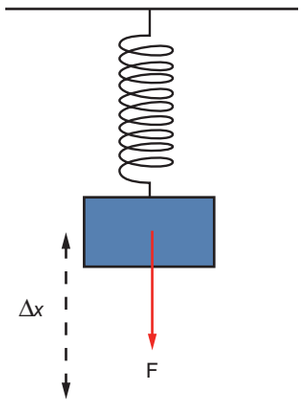


Figure 4.26 *Factors affecting the EPE of an object*

Activity 4.5: Energy stored in a spring

Determine the energy stored in a spring that has a spring contact of 15 N/m and is extended by 20 cm.

Figure 4.27 *An aircraft flying through the air has both kinetic and potential energy.*

- $W_{gravity} = w \times h$
- **GPE** = mgh

The energy gained by the mass (or the work done on the mass) is equal to weight multiplied by the vertical distance moved (the height above the ground).

Elastic potential energy

Another common potential energy is **elastic potential energy** (EPE), sometimes called strain energy. This is the energy associated with any object that has been **stretched** or **compressed**. Think about compressing a spring in a toy; it will store energy, which it converts into kinetic energy as it bounces.

The amount of EPE stored in the spring depends on the force applied and the distance moved (i.e. the extension of the spring). Think back to the Hooke's law force vs. extension graphs studied in Unit 3. The area under the line is equal to the work done on the spring. This gives us the equation for EPE:

- **Elastic potential energy** = ½ $F \Delta x$

F = force in N.

Δx = extension of spring in m.

For example, if a force of 100 N causes a spring to extend by 40 cm the energy stored in the spring will be equal to:

EPE = ½ $F \Delta x$ *State principle or equation to be used*

EPE = ½ × 100 N × 0.4 m *Substitute in known values and complete calculation*

EPE = 20 J *Clearly state the answer with unit*

There is an alternative equation for EPE that includes the spring constant of the spring rather than the force applied. From Hooke's law

- $F = k \Delta x$

We can combine this with our equation for EPE and we get:

- Elastic potential energy = ½ $k \Delta x \Delta x$
- **Elastic potential energy** = ½ $k \Delta x^2$

Total energies and energy changes

The total mechanical energy of a system is the sum of all the possible kinetic and potential energies.

- **Total mechanical energy = Σkinetic energy + Σpotential energy**
- **Total mechanical energy** = $\Sigma E_k + \Sigma U$

An aircraft cruising at 10 000 m will have a both a kinetic energy (as it is moving) and a potential energy (in this case GPE as it is above the ground). Its total mechanical energy will be its E_k + GPE.

Summary

In this section you have learn that:

- When an object does work, the work done is equal to the change in energy of the object. $W = \Delta E$. Or $W = -\Delta U$ if there is change in potential energy.
- Any moving object has a kinetic energy given by $E_k = \frac{1}{2}mv^2$.
- Potential energies are 'stored energies'. For example, GPE and EPE.
- GPE = mgh and EPE = $\frac{1}{2}F\Delta x$ (or $\frac{1}{2}k\Delta x^2$).
- The total mechanical energy of an object is given by the sum of its kinetic and potential energies.

Think about this...

Heat is another form of energy. The aircraft will also contain a certain amount of heat energy. However, this does not count towards its mechanical energy. More on heat in Unit 7.

KEY WORDS

compressed *pressed or squeezed into a smaller space*

elastic potential energy *the energy stored in a spring as a result of it being stretched or compressed*

stretched *made longer or wider by the application of force*

Review questions

1. Use the work–energy theorem ($W = \Delta E$) to show how $W = \frac{1}{2}m(v_2^2 - v_1^2)$.
2. Calculate the kinetic energy of the following objects:
   a) a 75 kg human running at 8 m/s
   b) a 3 g bullet travelling at 400 m/s
   c) a car of mass 1200 kg that travels 60 m in 3 s.
3. Explain what is meant by the term potential energy and give four different examples of potential energies.
4. Calculate:
   a) the GPE of a 15 kg wooden block 6 m above the ground
   b) the height of the wooden block if it were to have a GPE of 300 J.
5. Calculate the energy stored in a spring when it is compressed 5 mm by a 60 N force.
6. Determine the mechanical energy of a bird of mass 200 g flying at 12 m/s at a height of 50 m above the ground.

3.3 Conservation of energy

By the end of this section you should be able to:

- State the law of conservation of mechanical energy.
- Revise the term collision and distinguish between elastic and inelastic collisions.

Figure 4.28 *A burning candle transforms chemical energy into heat and light energy.*

- Solve problems involving inelastic collisions in one dimension using the laws of conservation of mechanical energy and momentum.
- Explain the energy changes that take place in an oscillating pendulum and an oscillating spring–mass system.
- Describe the use of energy resources including, wind energy, solar energy and geothermal energy.
- Explain the meaning of the term renewable energy.

The law of conservation of energy

Perhaps the most important idea in all of physics, the **law of conservation of energy**, states:

- **The total energy of a closed system must remain constant.**

In essence this means energy cannot be *created* or *destroyed* only transferred from one place to another or transformed from one type to another. The energy has been **conserved**; it has not changed in value.

For example, when a candle burns we might say it 'gives out' heat and light. What we really mean to say is that the chemical energy in the candle is transformed into heat and light. The energy has not been created just transformed. Importantly, the amount of each type of energy must balance. If 200 J of chemical energy was converted into heat and light then there must be 200 J of heat and light energy, not 198 J or 202 J, exactly 200 J! Energy cannot be created or destroyed.

We often use terms like 'wasted energy' or 'lost energy' and we might say 'it's run out of energy'. In these cases we mean transformed into a form we don't need or can't use. Most energy is eventually transformed into heat. This is often wasted as it is not used by the device but transferred to the surroundings; the energy has not been destroyed.

Let's think about what happens to the potential energy of a 5.0 kg mass when it is dropped from a height of 10 m. The total energy of a system must stay the same, but as the mass falls it 'loses' GPE. This GPE is converted into other forms. If we assume that the air resistance is negligible then the GPE will be converted into kinetic energy. The further it falls the faster is goes and the higher its kinetic energy.

Throughout the drop the total mechanical energy will be 500 J. When the mass hits the floor the kinetic energy will then be converted into 500 J of heat and sound energy.

Figure 4.29 *Filament bulbs 'waste' quite a lot of energy as heat.*

DID YOU KNOW?

The term closed system refers to a situation where objects are isolated from their wider surroundings. It is an idealised enironment as the only totally closed system in the universe itself!

Think about this...

In reality the block will hit the ground with just less than 500 J of kinetic energy. What would have happened to the rest of the energy?

Kinetic energy and momentum

Kinetic energy and **linear momentum** are two quantities that are very closely related. They both relate to moving objects with mass and both increase if the mass and/or the velocity of the objects increase, but not by the same proportion.

There are a few other important differences. Table 4.2 summarises some of the key points about kinetic energy and linear momentum.

Table 4.2 *Comparing linear momentum and kinetic energy*

	Momentum	Kinetic energy
Unit	kg m/s	J
Type of quantity	Vector	Scalar
Equation	$p = mv$	$E_k = \frac{1}{2}mv^2$
Effect if mass doubles	Doubles	Doubles
Effect if velocity doubles	Doubles	Quadruples ($2^2 = 4$)
Conserved in collisions as long as no external force acts	Yes, always	Possibly, but not always

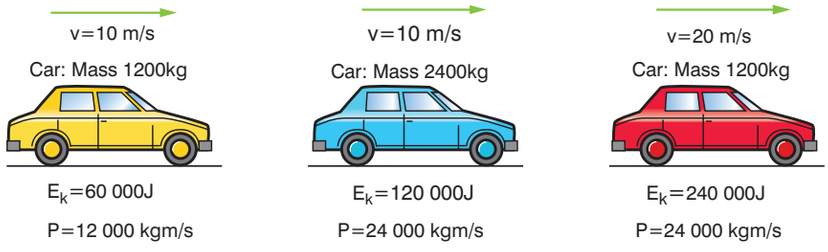


Figure 4.31 *The effect of changing mass and velocity on momentum and kinetic energy*

Look carefully at Figure 4.31. You can see that both momentum and kinetic energy are directly proportional to the mass of the moving object. Double the mass and both the momentum and the kinetic energy double. However, if the velocity doubles, the momentum doubles, but the kinetic energy goes up by four.

Elastic and inelastic collisions

Energy and momentum are two factors that are always conserved in collisions between objects. However, the energy may be transformed (for example, into heat and sound) and as a result the kinetic energy may not always be conserved.

We briefly looked at elastic and inelastic collisions in Unit 3. In an elastic collision the kinetic energy is conserved. In an inelastic collision the kinetic energy is not conserved.

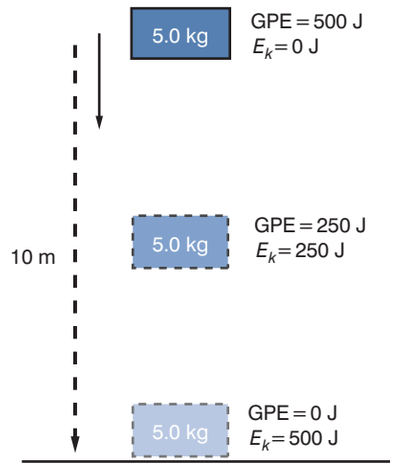


Figure 4.30 *As an object falls GPE is transformed into E_k.*

Activity 4.6: Energy changes

Describe all the energy changes when a football is dropped onto the ground. Why does the ball not return to its original height?

Activity 4.7: Kinetic energy and momentum

Calculate the kinetic energy and momentum of a mass of 10 kg travelling first at 6 m/s then at 12 m/s. Repeat for a mass of 20 kg.

KEY WORDS

conserved *neither increased nor destroyed*

closed system *a situation where objects are isolated from their environment*

law of conservation of energy *law stating that energy cannot be created or destroyed but is converted from one type to another*

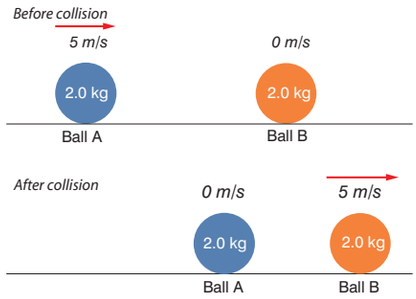


Figure 4.32 *A perfectly elastic collision*

Activity 4.8: Collisions

Show that both kinetic energy and momentum are conserved in the collision shown in Figure 4.33. (Remember, momentum is a vector quantity, whereas kinetic energy is a scalar.).

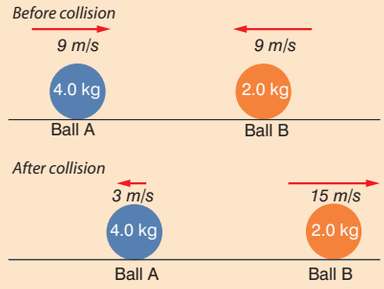


Figure 4.33 *Is this an elastic collision?*

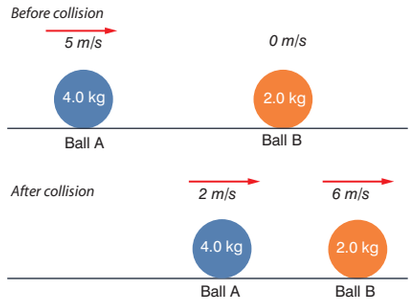


Figure 4.34 *An inelastic collision*

For example, Figure 4.32 shows a perfectly elastic collision. Both kinetic energy and momentum are conserved.

Momentum before = $m_A v_A + m_B v_B$ = (2.0 kg × 5 m/s) + (2.0 kg × 0 m/s) = 10 kg m/s → *Calculate momentum before as sum of momentum of A and momentum of B*

Momentum after = $m_A v_A + m_B v_B$ = (2.0 kg × 0 m/s) + (2.0 kg × 5 m/s) = 10 kg m/s → *Calculate momentum after as sum of momentum of A and momentum of B*

- **Momentum before = momentum after; momentum has been conserved.**

Kinetic energy before = $\frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2$ = (0.5 × 2.0 kg × (5 m/s)²) + (0.5 × 2.0 kg × (0 m/s)²) = 25 J *Calculate kinetic energy before as sum of KE of A and KE of B*

Kinetic energy after = $\frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2$ = (0.5 × 2.0 kg × (0 m/s)²) + (0.5 × 2.0 kg × (5 m/s)²) = 25 J *Calculate kinetic energy after as sum of KE of A and KE of B*

- **Kinetic energy before = kinetic energy after; kinetic energy has been conserved and therefore it is a perfectly elastic collision.**

Momentum is always conserved but kinetic energy is not. Figure 4.34 shows an example of an inelastic collision.

Momentum before = $m_A v_A + m_B v_B$ = (4.0 kg × 5 m/s) + (2.0 kg × 0 m/s) = 20 kg m/s → *Calculate momentum before as sum of momentum of A and momentum of B*

Momentum after = $m_A v_A + m_B v_B$ = (4.0 kg × 2 m/s) + (2.0 kg × 6 m/s) = 20 kg m/s → *Calculate momentum after as sum of momentum of A and momentum of B*

- **Momentum before = momentum after; momentum has been conserved.**

Kinetic energy before = $\frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2$ = (0.5 × 4.0 kg × (5 m/s)²) + (0.5 × 2.0 kg × (0 m/s)²) = 50 J *Calculate kinetic energy before as sum of KE of A and KE of B*

Kinetic energy after = $\frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2$ = (0.5 × 4.0 kg × (2 m/s)²) + (0.5 × 2.0 kg × (6 m/s)²) = 44 J *Calculate kinetic energy after as sum of KE of A and KE of B*

- **Kinetic energy before > kinetic energy after; kinetic energy has been lost and therefore it is not a perfectly elastic collision.**

In this example 6 J has been converted into heat and sound and so kinetic energy is not conserved and the collision is not perfectly elastic.

Energy in oscillating systems

We have seen that when an object falls its GPE is converted into kinetic energy. The same is true if you throw an object into the air. Here the kinetic energy is transformed into GPE as it rises.

In oscillating systems kinetic energy is continuously being transformed into potential energy and vice versa. If there are no energy losses (e.g. no losses as heat) then the total mechanical energy will stay the same and this process will go on forever!

Take, for example, a pendulum as it swings.

As it is lifted to A the pendulum gains GPE. It is then released and the gain in GPE is converted into E_k. At B it is travelling fastest, it has the most E_k but also the lowest GPE. It then rises to C, losing E_k and gaining GPE as it does so. Figure 4.36 shows how the potential energy and kinetic change over time.

From the graph you can see that the total mechanical energy stays the same. As the potential energy falls the kinetic energy increases and vice versa.

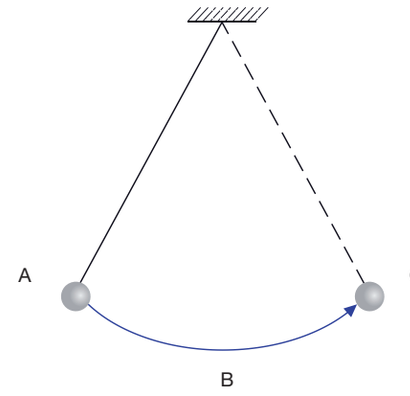


Figure 4.35 *A simple pendulum transforms GPE into* E_k *and then back again.*

- **The total mechanical energy = kinetic energy + potential energy**

Another example of an oscillating system is a mass–spring system. In simple terms this is just a mass on the end of a spring. However, the suspension in a car is a more complex example of a mass–spring system.

In this case the potential energy may not be GPE, instead it may be EPE.

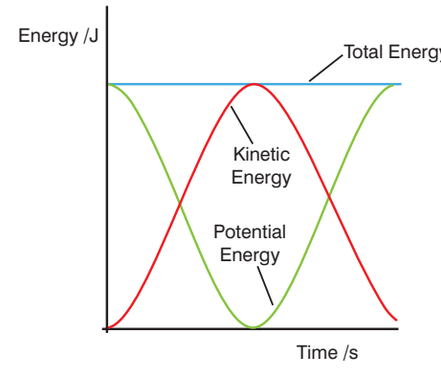


Figure 4.36 *Graph showing how the potential energy and kinetic energy of oscillating systems are related*

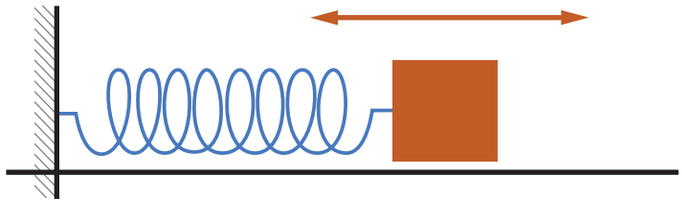


Figure 4.37 *An example of a mass–spring system*

As the spring is compressed the EPE increases and the mass slows down (its E_k decreases). Eventually the mass will stop; at this point the EPE is at its maximum and the E_k is zero. The mass then accelerates as EPE is converted into E_k. This process continues.

A more complex example might be a mass–spring system oscillating vertically like the one shown in Figure 4.38.

In this case the kinetic energy is changed into GPE and EPE. In any case the total mechanical energy of the system remains then same.

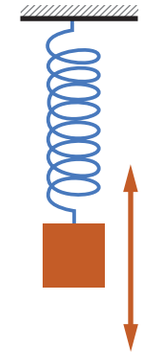


Figure 4.38 *A vertically oscillating mass–spring system*

Energy resources

Every country demands a huge amount of energy, from fuel to run cars and other vehicles, to gas for cooking and heating and, of course, electrical energy. A source of energy that may be used by a country or individuals within that country is commonly referred to as an **energy resource**. Energy resources are very precious commodities, perhaps the most obvious being oil.

Selecting which energy resources to use is often a very difficult decision. There are lots of factors to consider, chief among them

KEY WORDS

energy resource *a source of energy that can be used by a country or its population*

Figure 4.39 *A simple generator*

being the availability of the resource, the economics involved and the subsequent environmental impact (more on this later).

Energy resources are often used to generate electricity. Electricity is exceptionally useful as it is quite simple to transfer a vast amount of energy from one place to another (all you need is a suitable wire!) and it can be easily transformed into most other forms of energy. Most methods of electricity generation involve a rotating turbine. This turbine turns a generator (a magnet or series of magnets inside coils of wire). This generator converts kinetic energy into electrical energy.

Globally the most common method for generating electricity involves the burning of **fossil fuels** such as coal, oil and natural gas. The chemical energy contained within these fuels is released as heat (through burning), this heat is used to turn water into steam, this steam then turns a turbine to generate electricity. Large fossil fuel power stations can generate up to 4 billion joules per second!

However, such a global reliance on fossil fuels is problematic for two main reasons.

- Fossil fuels are a finite energy resource. Eventually we will run out of coal, oil and natural gas.
- Burning fossil fuels produces several atmospheric pollutants, including sulphur dioxide and perhaps more worryingly, carbon dioxide. Carbon dioxide (CO_2) is a powerful greenhouse gas. It is thought the increase in CO_2 output is a significant factor in man-made global warming, heating up the entire planet and leading to dramatic changes to weather and climate.

Think about this...

Remember energy cannot be created or destroyed, so when we talk about generating energy we really mean converting it from one form into electrical energy.

Figure 4.40 *An example of a coal-fired power station*

Ethiopia has few proven fossil fuel resources. However, some people estimate that there is considerable potential for oil and natural gas exploration in the future.

In a nuclear power station uranium is used as a fuel. Inside the reactor there is a complex nuclear reaction (fission – splitting the atom). This process generates heat, which is used to turn water to steam, etc. The only real difference between a nuclear power station and a coal-fired one is the method for generating the heat. In a nuclear reactor a great deal of heat can be produced per kg of uranium, and so nuclear plants can generate vast amounts of electricity. As no fuel is 'burnt' there are no greenhouse gases produced; however, this process produces radioactive waste. This waste will remain dangerous for millions of years.

KEY WORDS

fossil fuels *fuels that are produced by the action of high temperature and pressure on organic materials over millions of years*

renewable resource *an energy resource that does not involve a fuel that will run out*

Renewable energy resources

Resources that do not involve a fuel that will eventually run out are referred to as **renewable**. Table 4.3 includes a selection of some of the forms of renewable energy resources. This is not a definitive list; other forms include tidal (energy from tidal movements), wave (energy from water waves) and biomass (burning organic matter specifically grown for the task).

Table 4.3 Comparison of some renewable energy resources

Type	Description	Positives	Negatives
Wind	The Sun heats the Earth's surface. This heating is uneven and so creates convection currents. This leads to areas of higher and lower pressure and wind moves between them. The wind turns large turbines and this generates electricity.	Relatively inexpensive – just running costs. Does not produce any greenhouse gases.	Not a consistent supply. When there is no wind there is no electricity generated. A large number of turbines are needed to generate a significant amount of power.
Geothermal	Heat from processes inside the Earth is used to turn water into steam. Water is pumped down into 'hotspots' in the Earth's crust. It is turned to steam and this steam is used to turn turbines to generate electricity.	Only small amount of greenhouse gases are released (due to gases trapped inside the Earth being released in the process). Can generate a significant amount of power.	Only certain locations are suitable for geothermal power plants (see next section). Initial construction can be expensive.
Hydroelectric	Falling water turns turbines to generate electricity. In order to provide a sufficient drop in height large dams are often constructed. The water builds up behind the dam and is then released through turbines.	Only a small amount of greenhouse house gases are produced. Very large amounts of energy can be generated with relatively small running costs. Hydroelectric plants tend to have longer lives than thermal power stations.	Construction of large dams can damage the local environment. This may affect a significant number of the local inhabitants (animal and human). Initial construction can be very expensive and is limited to only certain sites. Generation may be affected by extended droughts.
Solar (photovoltaic)	The first type of solar power converts the energy in sunlight directly into electrical energy (via photovoltaic cells).	No greenhouse gases. Very low running costs.	Construction often involves the use of a large quantity of toxic materials. Photovoltaic cells remain very expensive. Only a relatively small amount of energy is generated per km^2.

Summary

In this section you have learnt that:

- The law of conservation of energy states that energy cannot be created or destroyed, just converted from one type to another.
- In elastic collisions both kinetic energy and linear momentum are conserved. In an inelastic collision only momentum is conserved.
- In oscillating systems (such as simple pendulum or mass–spring systems) potential energy is continuously transformed into kinetic energy and back again.
- A renewable energy resource is one that does not involve a fuel that will eventually run out.
- Wind, solar, geothermal and hydroelectric energy resources all offer significant benefits; however, they each have their drawbacks.

Figure 4.43 *The high top of mountain offers significant geothermal potential.*

Review questions

1. State the law of conservation of energy and explain why it is not correct to describe energy as being lost.
2. Use the principle of conservation of momentum to determine if the collision in Figure 4.44 is elastic or inelastic. If inelastic, calculate the amount of energy converted into heat and sound.
3. Describe the energy changes as a pendulum swings. If the pendulum has a mass of 50 g and is lifted so that it has a GPE of 0.1 J calculate:
   a) its increase in height
   b) the velocity of the bob as it passes through the bottom of the swing (assume no energy losses).
4. Explain what is meant by the term renewable energy resource and give three examples.
5. Describe how hydroelectric power may be used to generate electricity. Include the advantages and disadvantages of using this resource.

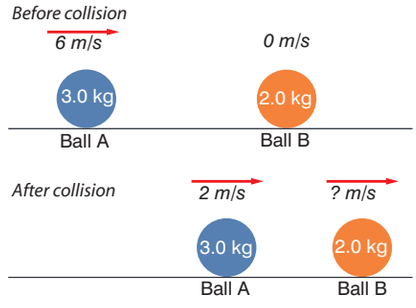


Figure 4.44 *What type of collision?*

KEY WORDS

per second *a measurement of rate*

power *the rate of doing work*

watt *the unit of power*

kilowatt-hour *a unit of energy*

3.4 Mechanical Power

By the end of this section you should be able to:

- Solve problems relating to the definition of power.
- Show that the kWh is also a unit of work.
- Express the formula of mechanical power in terms of average velocity.

What is power?

Power, like work, is another term that is frequently used in everyday language. It's a term that is often misused when maybe energy or velocity would be more appropriate.

In physics power has a very specific definition.

- **Power is the rate of doing work.**

As discussed in Unit 2, rate means **per second**. In other words, power is the work done per second. A greater power means more work is done per second or more energy is transferred per second.

Imagine two cars racing up a hill. If the cars have exactly the same mass, when they reach the top of the hill they would both have done the same amount of work. However, the more powerful car will be the winner (the one that can do the most work per second) as it will get to the top of the hill first!

An equation for average power is:

- **Power = work done / time taken**
- $P = W/t$

P = average power in W.

W = work done in J.

t = time in s.

Power is measured in **watts** (or kilowatts, etc). As energy is in joules and time in seconds, 1 watt is equal to 1 joule per second. A 4.0 kW motor can do 4000 J of work per second. The watt is the SI derived unit of power.

For example, a kettle uses 168 000 J of electrical energy in two minutes. Its average power can be found using the equation:

$P = W/t$ *State principle or equation to be used (definition of power)*

In this case the time taken is two minutes, which is 120 s.

P = 168 000 J / 120 s *Substitute in known values and complete calculation*

P = 1400 W or 1.4 kW *Clearly state the answer with unit*

Think about this...

Technically the equation is for average power. However, if the rate of doing work is constant (for example, if the force you are working against and the speed of movement both remain constant) then the average power is the same as the actual power.

DID YOU KNOW?

The watt is named after the Scotsman James Watt. He was instrumental in the engineering of the late 18th century. In particular his developments on steam engines are widely credited to have brought about the industrial revolution.

If the same kettle were to run for five minutes how much work would the kettle do?

- $P = W/t$ *State principle or equation to be used (definition of power)*
- $W = P \times t$ *Rearrange equation to make W the subject*

In this case the time taken is five minutes, which is 300 s and $P = 1400$ W

- $W = 1400$ W.$\times$ 300 s *Substitute in known values and complete calculation*
- $W = 420\,000$ J or 420 kJ *Clearly state the answer with unit*

This work would be transferred to the water and surroundings as heat energy.

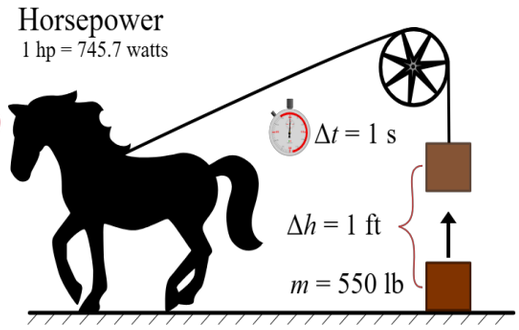


Figure 4.45 *One 'horsepower' is around 750 W.*

Activity 4.10: The power of a student

You do work when you run up stairs, because you have to move your weight upwards. The faster you run, the greater your power.

- Weigh a volunteer student.
- Use a stopwatch to measure the time the student takes to run up a flight of stairs.
- Count the number of stairs. Measure the *vertical* height of one stair, and calculate the total height of the stairs.
- Calculate the work done (= weight $\times$ height).
- Calculate the student's power $\left(= \frac{\text{work done}}{\text{time taken}}\right)$

Figure 4.46 *Lifting an apple around 1 m into the air transfers about 1 J of GPE to the apple.*

Activity 4.11: Power calculation

Use the equation for power to show that 1 kWh is equal to 3.6 million J.

The joule, the watt and other units

We have already mentioned the Joule as the standard unit of energy and the **watt** as the unit of power.

However, a joule is quite a small unit. Lifting an apple around 1 m in the air and you would do 1 J of work. It's not much. When we deal with large-scale energy usage, in particular electricity demands and generation, an alternative unit is used.

The **kilowatt-hour** is an alternative unit of energy. It is the energy transformed by a 1 kW device in 1 hour. This means 1 kWh is equivalent to 3.6 million J.

We can still use our equation for power but we must consider the units carefully.

DID YOU KNOW?

The joule was named after the English physicist James Prescott Joule. He was born on Christmas Eve in 1818 and he has been described by some as the quintessential physicist. He conducted a series of incredibly precise experiments that led to the theory of conservation of energy.

Table 4.4 *Comparing the joule and the kilowatt-hour*

Joule	Kilowatt-hour
• Work done = power × time • [J] = [W] x [s]	• Work done = power × time • [kWh] = [kW] x [h]
Work done in J	Work done in kWh
Power in W	Power in kW
Time in s	Time in h

For example, how much work is done by a 500 W motor running for 30 minutes?

In joules:

$W = P \times t$ *State principle or equation to be used (definition of power in terms of W)*

In this case the time taken is 30 minutes, which is 1800 s, and P = 500 W.

W = 500 W.× 1800 s *Substitute in known values and complete calculation*

W = 900 000 J or 900 kJ *Clearly state the answer with unit*

In kilowatt-hours:

$W = P \times t$ *State principle or equation to be used (definition of power in terms of W)*

In this case the time taken is 30 minutes, which is 0.5 hours, and P = 500 W, which is 0.5 kW.

W = 0.5 kW.× 0.5 h *Substitute in known values and complete calculation*

W = 0.25 kWh *Clearly state the answer with unit*

As well as the joule and kilowatt-hour, Table 4.5 lists some other commonly used units of energy.

Table 4.5 *Different energy units*

Unit	Application	Equivalent value (J)
Electronvolt (eV)	Sub-atomic particles and particle accelerators	1.6×10^{-19}
Erg (erg)	Using cm, grams and seconds instead of m, kg and s	1.0×10^{-7}
Kilocalorie (kcal)	Energy contained within foods	4.2×10^{3}
Kilowatt-hour (kWh)	Unit of energy used by electricity suppliers or when comparing large-scale energy demands (GWh is also used).	3.6×10^{6}
Tonne of oil equivalent (toe)	Another large-scale unit. It is the value of the chemical energy contained within one tonne of crude oil.	4.2×10^{10}
Megaton (MT)	Nuclear weaponry; 1 MT is the energy released by 1 million tonnes of TNT exploding (the largest recorded detonation was around 50 MT).	4.2×10^{15}

Power and velocity

Imagine a car travelling along at a steady speed. Its engine is still running and it is still using fuel but the kinetic energy of the car is not changing. Where is the chemical energy going? It can't be destroyed.

For objects to move at steady speed through the air a force needs to be applied. Remember, forces don't make things move they make them change the way they are moving. In the case of an object moving through the air at a steady speed there must be no net force acting on in it. The force from the engine must cancel out the resistive forces of kinetic friction and air resistance (drag).

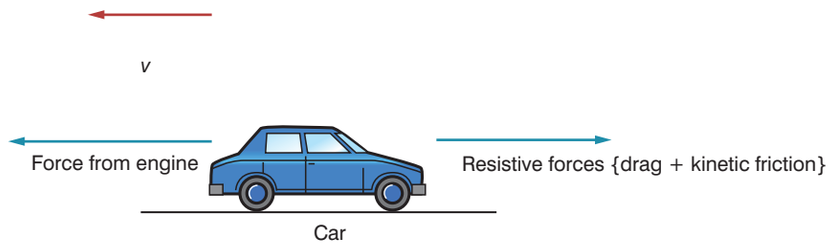


Figure 4.47 *For a car to move at a steady speed there must be a force from the engine.*

A force is being moved through a distance so work must be being done, but this energy is not transferred into the kinetic energy of the car as this is constant.

Instead the energy is transferred into two places:

- Heat energy (road – due to friction)
- Kinetic energy (including sound) of the air. A very turbulent wake is created behind the car.

If the engine is doing 4000 J of work per second then 4000 J of energy is transferred to the road and the air every second.

We can look at this process more mathematically by combining the equations for mechanical work and power and we get:

- Power = work done / time
- Power = force × distance moved against force /time
- Average velocity = distance moved against force /time So
- **Power = force × velocity**
- ***P = F v***

So, for a car to travel at 15 m/s against a force of 6000 N the power from its engine needs to be:

- $P = 6000\ \text{N} \times 15\ \text{m/s}$
- $P = 90\ 000\ \text{W}$

This means the engine is converting 90 000 J of energy per second.

Looking at it another way, in order for a train to travel at 20 m/s its engine may have a power output of 800 000 W. This can be used to determine the force from the engine and so the magnitude of the resistive forces acting on the train.

$P = F\,v$ *State principle or equation to be used*

$F = P / v$ *Rearrange equation to make F the subject*

F = 800 000 W / 20 m/s *Substitute in known values and complete calculation*

F = 40 000 N *Clearly state the answer with unit*

Summary

In this section you have learnt that:

- Power is defined as the rate of doing work (power = work done / time taken).
- Power is measured in watts (or kW) and 1 W is 1 joule per second.
- The scientific unit of work/energy is the joule. However, other units are commonly used, including the kilowatt-hour (kWh).
- For a moving object, $P = Fv$.

Figure 4.48 *A train travelling at high speed does a great deal of work against air resistance and kinetic friction.*

Review questions

1. What is the definition of power, state its units and give two different equations for calculating the power of an object.
2. Calculate the power of the following:
   a) a motor that does 24 000 J of work in two minutes
   b) a crane that lifts a 60 kg mass 100 m in 60 seconds.
3. Calculate the work done in J by the following:
   a) a 10 kW heater running for 15 minutes
   b) two 100 W light bulbs on for 24 hours.
4. Recalculate the values in question 2, but this time express the work done in kWh.
5. Derive $P = Fv$.
6. Determine the power output from an aircraft travelling at 200 m/s working against resistive forces of 1000 N.

End of unit questions

1. State the law of conservation of energy and describe a situation where $W = -\Delta U$ could be used to illustrate this law.
2. Determine the work done when a forklift truck lifts a box of mass 350 kg a height of 2 m.
3. Calculate the work done if a boulder of mass 100 kg is rolled 40 m up a slope at an angle of 20°. Assume the force of friction is negligible.

4. As a block falls through the air by 40 m it does work equal to –1800 J. Determine the mass of the block.
5. Calculate the kinetic energy of a ball of mass 50 g travelling at 30 m/s. How much work will need to be done to stop the ball?
6. A mass of 2.0 kg is hung off a spring, which extends 2 cm. Determine the energy stored in the spring.
7. A spring is used to launch a ball vertically into the air. The spring has a spring constant of 200 N/m and is compressed by 5 cm. A ball of mass 10 g is placed just above the spring. Calculate:
   a) the energy stored in the spring
   b) assuming the spring transfers all of its energy to the ball, the velocity of the ball just as it launches
   c) the height reached by the ball assuming all the E_k is converted into GPE.
8. Describe the energy changes in a mass–spring system that is oscillating horizontally. Explain how this changes if the system is vibrating vertically.
9. An 8.0 kg ball travelling at 4 m/s collides head on with a 3 kg ball travelling at 14 m/s. The balls bounce off each other and travel back the way they came. The 8.0 kg ball travels away at 2 m/s. Calculate:
   a) the velocity of the 3 kg ball after the collision
   b) the kinetic energy before and after the collision.
   c) State whether or not the collision is elastic and explain your answer.
10. Summarise the advantages and disadvantages of using the following energy resources to generate electricity:
    a) coal
    b) geothermal
    c) wind
11. A man raises 100 kg from the floor to a height of 2 m in 2.5 s. What is the work done and the power developed?
12. A petrol engine raises 200 kg of water in a well from a depth of 7 m in 6 s. Show that the engine is developing about 2.33 kW of power.
13. It is proposed to use a small waterfall to turn an electricity generator. 10 m^3 of water fall 50 m per minute. Only one-fifth of its energy can be obtained usefully. Show that the water can develop 16.7 kW.
14. 300 kg of water are lifted 10 m vertically in 5 s. Show that the work done is 30 kJ and that the power is 6 kW.
15. Calculate the resistive forces acting on a sports car if it is travelling at a steady speed of 25 m/s when the engine is providing 200 kW.

Lesson 4: Fluid Statics

Contents

Section	Learning competencies
4.1. Air pressure	• Define the term air pressure and use the definition to solve related problems. • Describe atmospheric pressure and explain its variation with altitude. • Explain how to measure atmospheric pressure and show that 760 mmHg is equal to one atmosphere.
4.2. Fluid pressure	• Define the term fluid and state the similarities and differences between liquids and gases. • Define the term density and relative density and determine each for a given body. • Explain how the pressure in a liquid at rest varies. • Apply the formula $p = h\rho g$ and use it to solve problems (including determining the pressure inside a fluid taking into account atmospheric pressure). • State Pascal's principle, and apply it to solve problems and explain applications (such as the hydraulic lift). • Explain the use of a manometer. • Demonstrate an understanding of, distinguish between and calculate atmospheric, gauge and absolute pressure. • State Archimedes's principle and the principle of flotation. • Distinguish between true weight and apparent weight of a body. • Calculate the buoyant force acting on the body in a fluid and explain why bodies float or sink. • Calculate the density of a floating body or density of a fluid using the flotation principle.

How does a massive ocean liner, made of steel, float on the water, yet a tiny penny sinks? Why is it when you go swimming you can feel the water pushing up on you, yet you can't feel the massive weight of the column of air on top of your head? This is all down to fluid statics, the study of the density and pressure in stationary liquids and gases.

From simply breathing in and out, to the blood pumping through your veins, pressure in liquids and gases plays an important role in our lives. Without atmospheric pressure our blood would simply boil and life on Earth would not even be possible.

In this unit we will investigate atmospheric pressure, look into what causes pressure in liquids and gases, explore the factors that affect it and learn how to use a range of simple pieces of equipment to measure pressure.

4.1. Air pressure

By the end of this section you should be able to:

- Define the term air pressure and use the definition to solve related problems.
- Describe atmospheric pressure and explain its variation with altitude.
- Explain how to measure atmospheric pressure and show that 760 mmHg is equal to one atmosphere.

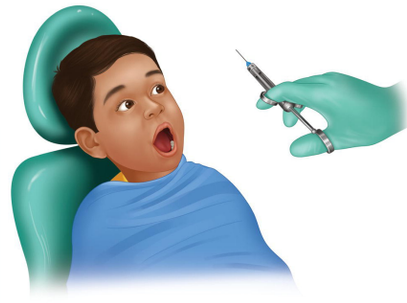


Figure 6.1 *Injections don't hurt much because the needle exerts a very high pressure on the skin.*

Under pressure

If you've ever had an injection you will have noticed how easy it is for the doctor to push the needle through your skin. This is because the needle has a very sharp point and so when the doctor exerts a relatively small force the needle creates a great deal of pressure on the skin.

Figure 6.2 *The area over which the force is acting affects the pressure it exerts.*

Pressure is defined as the amount of *force acting per unit area*.

- **Pressure is equal to force per unit area.**

If a large force acts on a small area it creates a greater pressure. For example, most animal predators have pointed teeth. When a crocodile or shark bites into its prey, the pressure is very large and so the teeth sink in!

The reverse is also true. A large vehicle like a tractor or truck may have some very large tyres. These increase the area over which the force is acting and so reduce the pressure. This means it is less likely for the tractor to sink into the mud and get stuck.

The pressure exerted by a force may be calculated using the equation below:

- pressure = force / area
- $p = F / A$

p = pressure in Pa.

F = force in N.

A = area in m^2.

Pressure is measured in pascals. One pascal is equal to a pressure of 1 N per square metre (1 N/m^2). The pascal is the SI derived unit of pressure (this includes all forms of pressure).

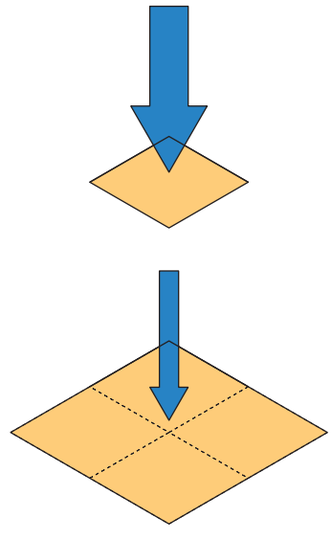


Figure 6.3 *A large force pressing on a small area creates greater pressure than a smaller force on a larger area.*

Worked example

A boy weighs 500 N and the soles of his feet have an area of 0.05 m^2. Determine the pressure he exerts when he stands a) on both feet and b) on one foot.

DID YOU KNOW?

The pascal is named after Blaise Pascal. He was a French physicist most noted for his experiments with barometers in the mid-17th century (a barometer is an instrument to measure air pressure; more on this later).

Activity 6.1: Pressure

Complete the table below:

Force	Area	Pressure
720 N	4.0 m^2	
	0.02 m^2	240 kPa
5.0 N		1.0 Pa

Activity 6.2: Your own pressure

Stand on a piece of squared paper. Carefully draw around your feet (or get a partner to do this for you).

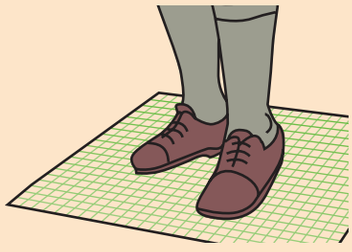


Figure 6.4 *In order to determine the pressure you exert you need to measure the area of your feet!*

Use this to work out the area of your feet (to do this count the number of squares and multiply by the area of each square).

Measure your weight in N (or your mass in kg and multiply by 10 N/kg), then use the equation $p = F / A$ to determine your pressure.

On both feet:

$p = F / A$ *State principle or equation to be used (definition of pressure)*

p = 500 N / 0.05 m^2 *Substitute in known values and complete calculation*

p = 10 000 Pa or 10 kPa *Clearly state the answer with unit*

If he stands on one foot his weight will be the same but the area will be halved to 0.025 m^2:

$p = F / A$ *State principle or equation to be used (definition of pressure)*

p = 500 N / 0.025 m^2 *Substitute in known values and complete calculation*

p = 20 000 Pa or 20 kPa *Clearly state the answer with unit*

We didn't really need to do that last calculation. We can see that if the area halves and the force stays the same then the pressure doubles (pressure is inversely proportional to area).

Worked example

A book rests on a desk. Its covers measures 20 cm by 25 cm. It exerts a pressure of 100 Pa. Determine the mass of the book.

$p = F / A$ *State principle or equation to be used (definition of pressure)*

$F = p \times A$ *Rearrange equation to make F the subject*

In order to find the mass we first need to find the weight of the book. This is the force the book exerts on the desk. The area of the book is 0.20 m × 0.25 m = 0.05 m^2.

F = 100 Pa × 0.05 m^2 *Substitute in known values and complete calculation*

F = 5 N *Clearly state the answer with unit*

This is the weight of the book so to find its mass we use $w = mg$.

$w = mg$ so $m = w / g$ *State equation and rearrange equation to make m the subject and solve*

m = 5 N / 10 N/kg *Substitute in known values and complete calculation*

m = 0.5 kg or 500 g *Clearly state the answer with unit (ideally kg)*

What causes air pressure?

Although we can't feel it in our day to day lives, air has mass. This means it also has a weight. One cubic metre of air has a mass of about 1 kg and so a weight of 10 N. The simplest way to think about air pressure is to treat it as the pressure due to the weight of the air above pushing down on a certain area. This may seem like a silly idea but actually it is pretty close to the truth.

A more complete picture involves thinking about the actual air particles. These are in constant motion, they are travelling in different directions and some travel faster than others. When the air particles are near a surface some will bounce into it and so exert a force on the surface. It is this force that gives rise to a pressure.

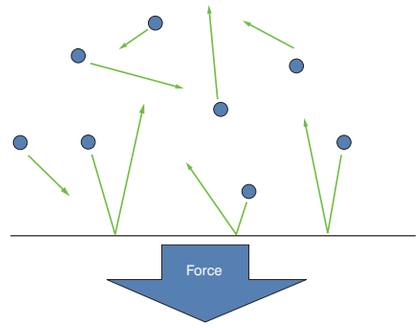


Figure 6.5 *Air particles crashing into a surface apply a force to that surface.*

Atmospheric pressure

The **atmosphere** is the layer of air that surrounds the Earth. Above your head right now there is a column of air about 40 km tall. The exact height is quite hard to determine due to the fact that as the height above the ground increases the air gets thinner and thinner until there is practically no air.

This column of air has a weight, which presses down on you and it is this that gives rise to atmospheric pressure.

We don't normally notice atmospheric pressure. If you move your hands up and down you can't really feel it, but it is definitely there! The reason we don't feel it is because not only does it push on you equally from all directions (left, right, front and back) but our bodies push back out.

There is a kind of **equilibrium** between the pressure in our bodies and the surrounding atmosphere. If you went somewhere where the pressure was much greater than atmospheric pressure our bodies would be crushed. For example, deep-sea submarines have to be very strong to withstand the crushing effect caused by the pressure of the water.

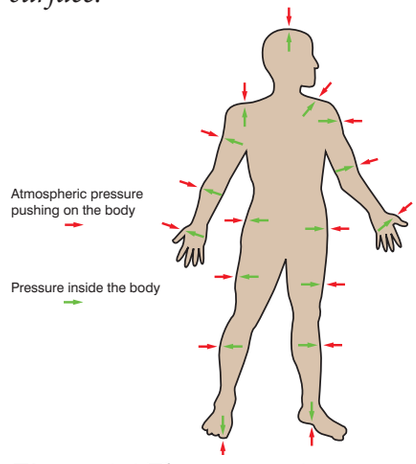


Figure 6.6 *The pressure in our bodies pushes back against atmospheric pressure.*

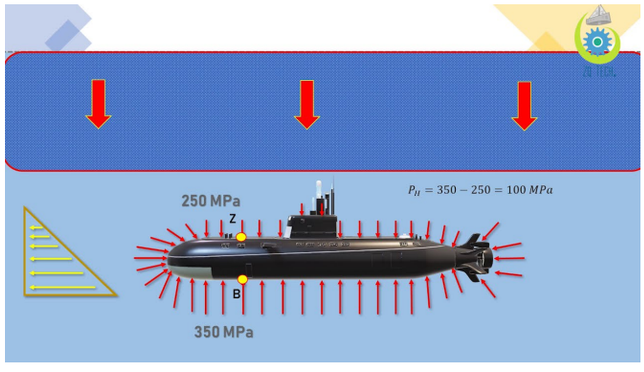


Figure 6.7 *A deep-sea submarine has to withstand very high pressures.*

The reverse is also true. If you went somewhere where the pressure was very low (e.g. into space without a pressurised space suit) the pressure inside our bodies would push outwards with some very nasty effects!

Figure 6.8 *Without a pressurised space suit this astronaut would experience severe difficulties.*

How big is atmospheric pressure?

The weight of the column of air above 1 m^2 at ground level is around 101 000 N! This means atmospheric pressure at ground level is around 101 kPa. This is often referred to as 1 atmosphere or 1 atm:

- **1 atm = 101 kPa**

Think about this...

In fact the pressure inside our bodies is generally a bit higher than atmospheric pressure. Think about how you already know this and why do you think this is important?

DID YOU KNOW?

The planet Venus has a much denser atmosphere that we do on Earth. The pressure on the surface is around 90 atm! That is 9 MPa or 9 million N per square metre. That is the same pressure you would experience if diving to a depth of nearly 1 km under water.

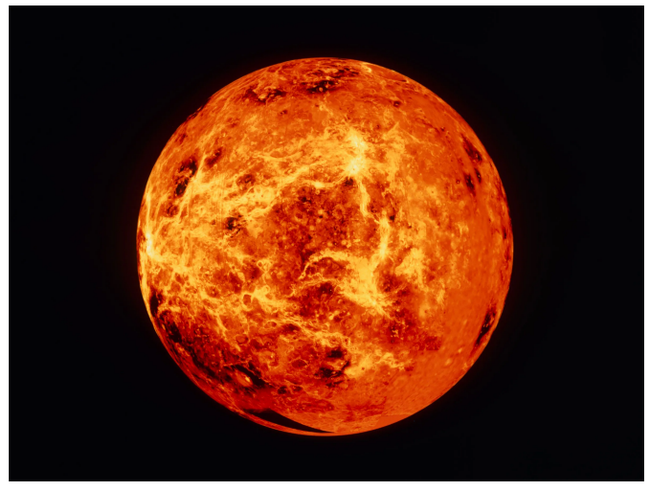


Figure 6.9 *Venus has a much greater atmospheric pressure than the Earth.*

101 000 Pa is a very large pressure, but we rarely notice it in our day to day lives. It is about the same as having a medium-sized elephant balance on the top of your head!

In the mid-17th century a German named Otto Von Guericke (who was mayor of Magdeburg) invented a **vacuum pump**. This clever machine removed the air from inside a chamber and so the force due to atmospheric pressure could really be seen.

Von Guericke used his pump to removed air from inside two brass hemispheres touching each other. With the air removed the pressure from the atmosphere squeezed the two hemispheres together. With no counter pressure from the air inside, the hemispheres were locked tightly together. In 1654, in front of Emperor Ferdinand III, he demonstrated how tightly by using thirty horses in two teams of 15 to try to separate the hemispheres. They couldn't do it!

Figure 6.10 *Magdeburg hemispheres*

Figure 6.11 *Teams of horses could not pull the hemispheres apart.*

KEY WORDS

atmosphere *the layer of air surrounding the Earth*

vacuum pump *a machine for removing the air from inside a chamber*

The atmospheric pressure pushing the two hemispheres together was too strong. the hemispheres could not be separated. It was not until air was allowed back inside the hemispheres that the difference in pressure was small enough to allow them to be pulled apart.

What effect does altitude have on atmospheric pressure?

The actual atmospheric pressure in the room today might be a bit higher or lower than 1 atm. The heating effect from the Sun causes small changes in pressure due to the uneven heating of the Earth's surface. This leads to high or low pressure weather systems. You can think of a high pressure system as meaning there is a slightly greater mass of air above your head than on an average day.

Figure 6.12 *Differences in atmospheric pressure can lead to powerful storms.*

The height above sea level, or altitude, also has a significant effect on atmospheric pressure. Imagine climbing a tall mountain; the higher you get, the smaller the column of air above you. This means there is a smaller mass of air above you and so less weight pushing down.

As altitude increases the atmospheric pressure decreases.

Table 6.1 shows how the pressure varies with altitude. You can see that it is not a simple relationship and it depends on temperature changes and position on the Earth.

Table 6.1 *Pressure at different altitudes*

Altitude (m)	Approx. pressure (Pa)
0	101 000
1000	90 000
2000	79 000
5000	54 000
10 000	26 000
15 000	12 000
20 000	6000
25 000	1300
30 000	270

Figure 6.13 *As you climb a mountain the surrounding atmospheric pressure drops.*

Activity 6.3: Atmospheric pressure

Fill a glass to the very top and then place a card on top of it. Make sure there is no air trapped between the glass the card.

While holding the card carefully turn the glass over and then let go of the card.

It should stay in place! Atmospheric pressure is pushing the card up and preventing the water from rushing out.

Activity 6.4: The effect of altitude

Plot a graph of altitude against atmospheric pressure using the information in Table 6.1.

DID YOU KNOW?

At 1 atm, water boils at 100 °C. However, if the atmospheric pressure drops so does the boiling point. At the top of tall mountains the pressure is so low water will boil at 75 °C! In the mid-19th century explorers used this fact to determine their altitude.

Figure 6.14 *A barometer*

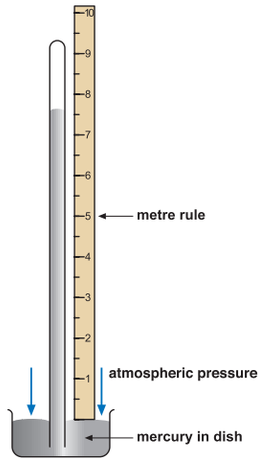


Figure 6.15 *A diagram of a simple barometer*

DID YOU KNOW?

The volume of an object is affected by the temperature and the surrounding pressure. Chemical reactions also depend on pressure and temperature. In order to ensure experiments are conducted under the same conditions across the globe the International Union of Pure and Applied Chemistry (IUPAC) define standard temperature and pressure as a temperature of 0 °C and a pressure of 100 kPa. However, several different organisations use slightly different values!

Measuring atmospheric pressure

There are several instruments used to measure atmospheric pressure. The most common is a **barometer**.

Think about the mercury in the dish. On the outside, air pressure is pressing down on the mercury. On the inside, the column of mercury in the tube is pressing down with an equal pressure. If these pressures were not equal, the level of mercury in the tube would alter until the pressures were balanced.

Until about 1650 the rise of liquid up a tube was explained by saying that the vacuum 'sucks up' the liquid. This is not so – a vacuum cannot suck, because there is nothing there to do the sucking! The rise is due to air pressure on the surface of the liquid outside.

A mercury barometer is long and inconvenient, heavy, and contains a liquid that is hazardous and easily spilt. Therefore, an **aneroid barometer** is commonly used. (*Aneroid* means *without liquid*.) It is compact and portable. A flat circular metal box, with only a little air inside, is the important part (Figure 6.16). A spring prevents its sides from being pushed in. The box is corrugated to make it strong, so that it does not collapse under air pressure. When the pressure changes, the upper face of the box moves. The movement is magnified several hundred times by a system of levers, which move a pointer over a circular scale, graduated in centimetres. It is graduated by comparing its readings with those of a mercury barometer.

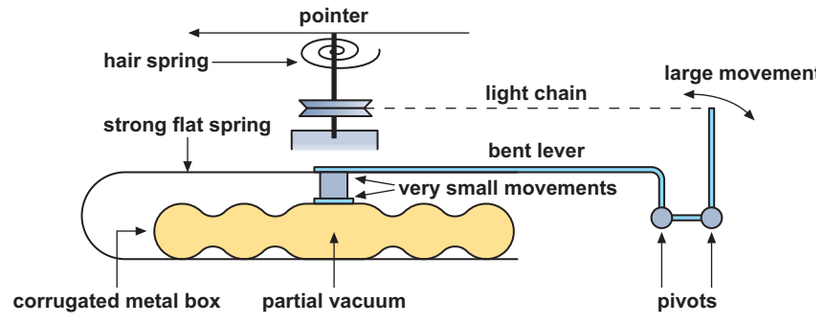


Figure 6.16 *The construction of an aneroid barometer*

Why 760 mmHg?

Pressure is often expressed in the units of mmHg. If the atmospheric pressure is equal to 1 atm then the height of the column of mercury in a barometer is 760 mm. We can prove this mathematically.

The column of mercury will have a weight and the weight must equal the force due to the atmospheric pressure pushing up on the bottom of the column. Let's imagine a column of mercury 760 mm tall with a radius of 5 mm. This exerts a force equal to its weight. The weight is given by $w = mg$ and we can determine the mass of the column from its density and volume ($\rho = m / V$ and so $m = \rho V$ and $V = \pi r^2 h$ as this is the volume of a cylinder). So:

- $V = \pi r^2 h$

- $V = \pi(0.005\ \text{m})^2 \times (0.76)\ \text{m}$
- $V = 5.97 \times 10^{-5}\ \text{m}^3$

The density of mercury is 13 570 kg/m^3 so its mass is:

- $m = \rho V$
- $m = 13\ 570\ \text{kg/m}^3 \times 5.97 \times 10^{-5}\ \text{m}^3$
- $m = 0.81\ \text{kg}$

Therefore the weight of the column can be found:

- $w = mg$
- $w = 0.81\ \text{kg} \times 9.81\ \text{N/kg}$
- $w = 7.9\ \text{N}$

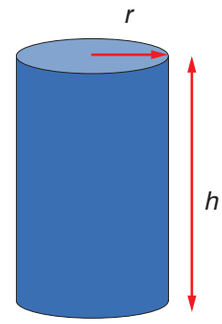


Figure 6.17 *The volume of a cylinder*

This weight must equal the force due to the pressure on the bottom of the column. So we can use the pressure equation to determine the pressure required to support of column of this height.

- $p = F/A$

As it is a cylinder the area of the base of the column is given by $A = \pi r^2$ so:

- $p = F / \pi r^2$
- $p = 7.9\ \text{N} / \pi \times (0.005\ \text{m})^2$
- $p = 101\ 000\ \text{Pa}$ or 101 kPa

You can repeat the calculation above for columns with different radii; the answers are always the same! You can combine all the steps into one big equation:

- $p = \rho\ \pi r^2 h\ g / \pi r^2$

The areas cancel, which shows that the area of the column does not matter. Any column will reach the same height. This gives us:

- $p = \rho\ h\ g$ (more on this equation later).

You might ask, why use mercury? Mercury is quite toxic and needs to be handled very carefully; why not use water instead? This is because water has a much lower density than mercury (around 1000 kg/m^3 vs. 13 600 kg/m^3). This means for that atmospheric pressure can support a column of water around 10 m tall! This would make our barometer far too large to be practical.

Some uses of air pressure

There are several uses for air pressure. Most rely on creating a **pressure difference** by pumping air into or out of a chamber. Pumping air into a chamber creates a greater pressure and pumping air out of a chamber creates a lower pressure.

If you create an area of lower pressure then the atmospheric pressure is larger in relative terms. As a result air is *pushed* in due to the greater force from the atmospheric pressure. Notice that there is no such thing as sucking to pull air into a machine.

Think about this...

As atmospheric pressure can support a column of water 10 m high this is also the maximum height to which a column of water can be drawn up by a vacuum pump (i.e. by creating a pressure difference). For any higher, water pumps must be used.

KEY WORDS

aneroid barometer *a device for measuring atmospheric pressure that uses a corrugated metal box rather than liquid*

barometer *a device for measuring atmospheric pressure*

pressure difference *the relative value of the pressure of gas in different chambers*

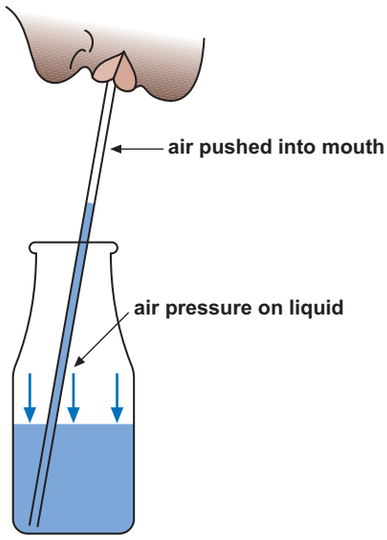


Figure 6.18 *Atmospheric pressure pushes the drink up the straw*

KEY WORDS

common pump *a pump that relies on atmospheric pressure to move water*

drinking straw a *thin tube used to suck liquids into the mouth*

force pump *a pump that relies on atmospheric pressure and compressed air to move water, often to a great height*

lift pump *a pump that relies on atmospheric pressure to move water*

suction pad *a round rubber pad that relies on atmospheric pressure to stick to smooth surfaces*

Uses of air pressure

A **suction pad** is a round rubber pad, perfectly flat on one side. Wet this side and press the pad against a window or smooth wall, pushing out all the air from under it. The pad sticks firmly. Atmospheric pressure holds it in place. The pads are used to lift and move large sheets of plate glass, metal and plastics, to put notices on windows, and on many toys, e.g. arrows, that stick to walls.

If you drink through a **drinking straw**, you are making use of atmospheric pressure. You suck on the air inside the straw. Therefore the atmospheric pressure outside is greater than the pressure inside, and liquid is pushed up (Figure 6.18).

A **lift pump (common pump)** is often used to raise water from wells. A piston moves up and down a tube (Figure 6.19). There is a valve in the piston and also one at the end of the tube. A valve is usually made of leather, and has brass on it to make it heavy. The valves are normally shut. They let water pass upwards but not downwards.

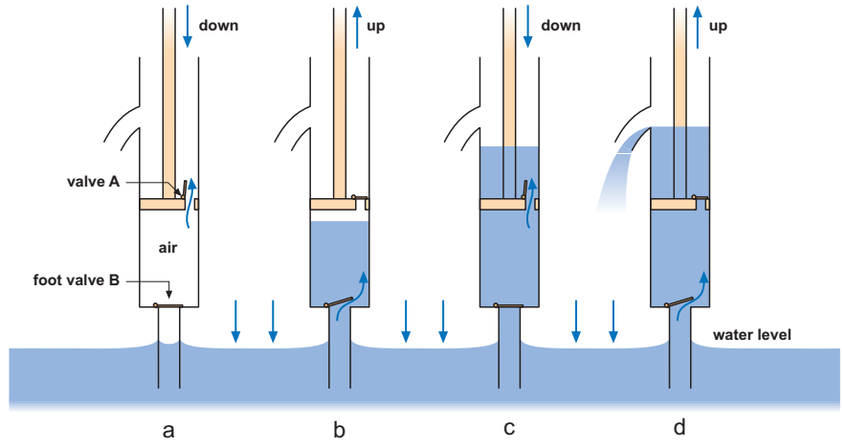


Figure 6.19 *The four stages in the operation of a lift pump*

- *Downstroke:* Foot valve B closes under its own weight. Valve A opens and lets air pass through it into the space above the piston.
- *Upstroke:* Valve A closes under its own weight. The pressure under the piston is less than atmospheric. Atmospheric pressure forces water into the tube through B.
- *Downstroke:* Valve B closes and A opens. Water passes through A into the space above the piston.
- *Upstroke:* Valve A closes and water is lifted up the tube and out of the spout. More water passes through B to keep the pump filled.

Atmospheric pressure determines the height to which water can be pumped. Even a perfect pump can raise water only 10.4 m. In practice, because of leaks at the valves and piston and of dissolved gases from the water, most pumps raise water about 7 m only. Delivery of water is not continuous.

A **force pump** can pump water to a great height. Some, used by firemen, can force water hundreds of metres high.

There is a foot valve B (see Figure 6.20), as in the lift pump, but it has a solid piston and a delivery tube at the bottom of the pump. There is a valve A in the delivery tube where it joins a chamber.

- *Upstroke:* The pressure in the tube under the piston becomes less. Valve A closes and foot valve B opens. Water is forced through B into the tube by atmospheric pressure.
- *Downstroke:* B closes. Valve A opens; water is forced through it and the delivery pipe into chamber C. The pressure on the piston (and not atmospheric pressure) determines the height to which the water is pumped.

The force pump itself delivers water only on the downstroke; the flow of water stops on the upstroke. However, the air trapped in chamber C is compressed during every downstroke. The pressure of this air continues to force out water during the upstroke, and therefore the pump delivers a steady stream of water.

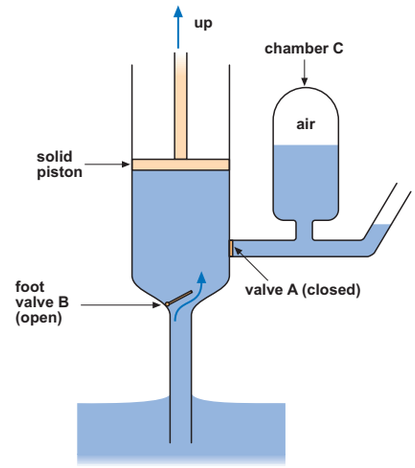


Figure 6.20 *A force pump*

Bicycle pump

The handle moves a piston in a metal cylinder (Figure 6.21). There is a cup-shaped leather or rubber washer on the end of the piston. This acts as a valve and lets air move in one direction only. The soft edge of the washer fits closely to the sides of the cylinder.

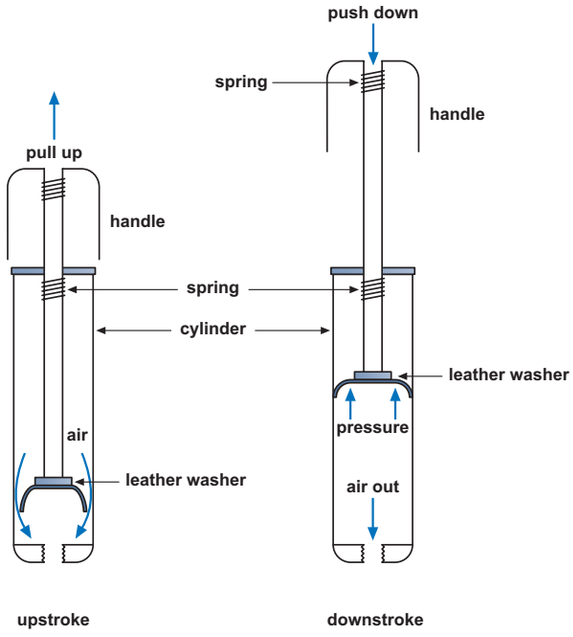


Figure 6.21 *How a bicycle pump works*

- *Upstroke:* The pressure below the piston is reduced. Atmospheric pressure forces air between the washer and the wall of the cylinder.
- *Downstroke:* The pressure below the piston is increased. The washer is pressed tightly against the walls of the cylinder, making it airtight. When the pressure rises above the pressure inside the tyre, the tyre valve opens and air is forced into the tyre.

KEY WORDS

siphon *a tube which can move liquid using the difference between the pressure of the liquid and atmospheric pressure*

A bicycle pump with the washer reversed acts as a vacuum pump or suction pump.

Siphon

A **siphon** is a convenient way of removing liquid from a container such as an aquarium or petrol tank.

Activity 6.5: To show the action of a siphon

- Fill a tall jar with water. Submerge a long rubber tube so that it fills with water.
- Leave one end in the water, close the other end with the fingers (to prevent the water running back), and lift it out of the jar. Lower this end until it is below the water level in the jar. Open it and let water flow out into a second jar (Figure 6.22).

The water flows so long as the end C is below water level A. The further C is below A, the faster is the flow of water.

- Now raise the second jar until it is higher than the first. Water flows in the other direction. (The tubing must always be full of water and its ends must be under the water.)

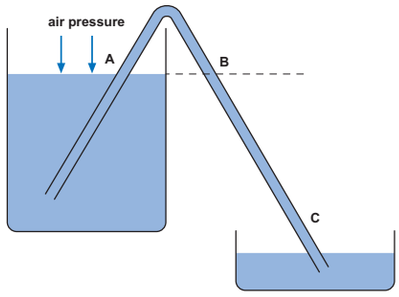


Figure 6.22 *A siphon*

Activity 6.6: Vacuum cleaner

Can you explain how a simple vacuum cleaner works? Remember that it does not suck up the dust.

How a siphon works

The pressure at A and B is atmospheric. Therefore the pressure at C is atmospheric pressure plus the pressure due to the column of water BC. Hence, the pressure at C is greater than atmospheric and the water can push its way out against the atmosphere.

Summary

In this section you have learnt that:

- Pressure is defined as the force acting per unit area. It can be calculated using the equation $p = F/A$.
- Pressure is measured in pascals (Pa), where 1 Pa equals a pressure of 1 newton per square metre.
- Atmospheric pressure is caused by the weight of the column of air above you pushing down on you. On a typical day this is equal to 101 kPa.
- As your altitude increases the atmospheric pressure decreases.
- A barometer is a simple instrument used to measure atmospheric pressure. The pressure from the atmosphere pushes the fluid up the tube.
- 1 atm is equal to 760 mmHg.

Review questions

1. Define pressure and states its units.
2. A wooden block of mass 2.0 kg is 20 cm thick, by 10 cm wide by 30 cm tall. Calculate the minimum and maximum pressure this block could exert on a surface.
3. Explain the causes of atmospheric pressure and why it changes with altitude.
4. Describe how a barometer works and show that at 1 atm the height of a column of mercury would equal 760 mm.
5. Calculate the pressure in Pa if the reading from a barometer is 820 mmHg.

4.2. Fluid pressure

By the end of this section you should be able to:

- Define the term fluid and state the similarities and differences between liquids and gases.
- Define the term density and relative density and determine each for a given body.
- Explain how the pressure in a liquid at rest varies.
- Apply the formula $P = h\rho g$ and use it to solve problems (including determining the pressure inside a fluid taking into account atmospheric pressure).
- State Pascal's principle, and apply it to solve problems and explain applications (such as the hydraulic lift).
- Demonstrate an understanding of, distinguish between and calculate atmospheric, gauge and absolute pressure.
- State Archimedes's principle and the principle of flotation.
- Distinguish between true weight and apparent weight of a body.
- Calculate the buoyant force acting on the body in a fluid and explain why bodies float or sink.
- Calculate the density of a floating body or density of a fluid using the flotation principle.

KEY WORDS

flow *smooth unbroken movement of a substance*

fluid *a substance that will flow e.g. gases, liquids*

gases *substances in a state of matter where particles can move about randomly and are widely spaced, with no bonds between them*

liquids *substances in a state of matter where there are weak bonds between the particles which are close together but can still move*

4.2.1 What are fluids?

Can you name a **fluid**? I suspect you came up with either water, an oil of some sort, petrol or maybe something like milk. However, I doubt many, if any, of you came up with air. In physics a fluid refers to a substance that will **flow** along a pipe. In common use fluids tend to mean just liquids. However, in science fluids include all **gases** as well as **liquids.**

Figure 6.23 *Natural gas is a fluid.*

Another characteristic of fluids is that they can change their shape. This means they always take the shape of the container they are put in. For example, consider a rectangular glass box. A liquid and gas will both fill the bottom of the container; however, a solid will not.

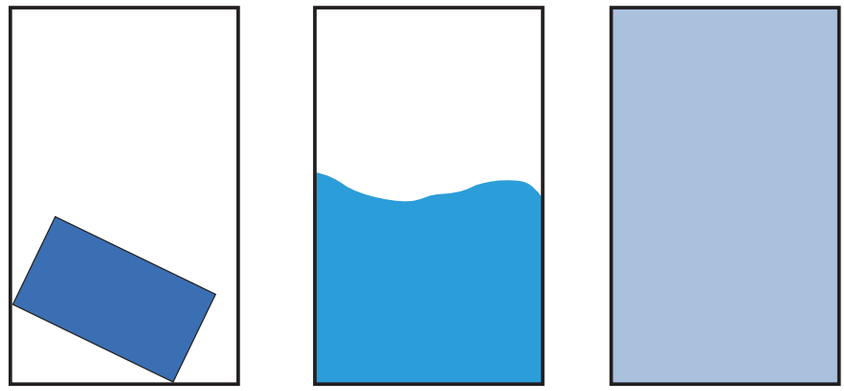


Figure 6.24 *Fluids take the shape of their container.*

KEY WORDS

incompressible *where the volume of a substance stays the same when force is applied*

density *the mass per unit volume of a substance*

There are still some very important differences between liquids and gases. Perhaps the most important is the fact that gases can be compressed by forces. You can squeeze a balloon filled with air and its volume will go down. However, liquids are **incompressible**; effectively this means the volume of a liquid stays the same when force is applied.

Table 6.2 summarises the key properties of liquids and gases.

Table 6.2 *Liquids and gases*

	Liquid	Gas
Particles	Quite close together, with no set pattern; particles can move past each other.	Far apart with no set pattern; particles can move past each other.
Bonding	Weak bonds between the particles	No bonding between the particles
Can flow / change their shape to match a container	Yes	Yes
Compressible	No; the particles are already close together.	Yes; there is lots of space between the particles.

4.2.2 Fluid density

The **density** of any fluid may be calculated using the standard equation for density:

- density = mass / volume
- $\rho = m / V$
- **Density is defined as mass per unit volume**.

As the particles are closer together in a liquid, liquids have higher densities than gases. Table 6.3 includes some typical densities of fluids.

***Table 6.3** Densities of fluids*

Fluid	Density (kg/m^3)
Mercury	13 600
Honey	1400
Water	1000
Sea water	1020
Diesel	950
Alcohol	800
Petrol	740
Air	1.20
Carbon dioxide	1.98
Nitrogen gas	1.25

Think about this...

Density is also often measured in g/cm^3. 1 g/cm^3 is equal to 1000 kg/m^3. How would convert from g/cm^3 to kg/m^3 and vice versa?

KEY WORDS

relative density *the ratio between the density of two substances*

Both temperature and pressure have an effect on the volume of a fluid. Therefore the densities in Table 6.3 are at standard atmospheric pressure (101 kPa) and a temperature of 20 °C.

For liquids it is fair to assume that the density is uniform throughout the liquid (as they are incompressible). However, for large volumes of gas the density increases as the gas gets closer to the surface of the Earth (due to gravity). This is most noticeable in the Earth's atmosphere. As the altitude increases the air gets less dense; the air is described as getting thinner.

What's relative density?

The term **relative density** is often used to compare the density between two fluids. In most cases this involves comparing the density of a fluid to that of water; however, it could be any other substance.

The relative density of a substance is the ratio between its density and the density of water. For example, if something has a relative density of two it means it is twice as dense as water. A relative density of 0.25 means it has ¼ of the density of water. You can calculate relative density using:

- relative density = density of substance / density of water.

The relative density of alcohol would be:

- relative density = density of substance / density of water
- relative density = 800 kg/m^3 / 1000 kg/m^3
- relative density = 0.8.

Notice relative density has no units since it is a ratio.

If we are comparing two identical volumes of fluids then the relative density can be calculated as the ratio of the masses of the same volume of fluid:

- relative density = mass of substance / mass of equal volume of water

Activity 6.7: Relative density

Determine the relative density of:

1. mercury
2. carbon dioxide
3. petrol
4. honey.

Activity 6.8: Measuring relative density

This method uses a density bottle (Figure 6.25) to find the relative density of a liquid. A density bottle has a ground-glass stopper, which fits exactly. There is a small hole in the stopper through which liquid and air can flow out when the stopper is put in the neck of the bottle. (This means that no air bubbles can be trapped under the stopper, which would give a false result.)

- Weigh a clean, dry density bottle with its stopper (mass = A).
- Fill with water and put in the stopper. Water should come out of the hole in the stopper. Dry the outside of the bottle and weigh it again (mass = B).
- Pour out the water, rinse with some of the liquid whose relative density is to be found. Fill with the liquid, put in the stopper, dry carefully and weigh (mass = C).

$$\text{Relative density} = \frac{\text{mass of liquid}}{\text{mass of water}} = \frac{(C - A)}{(B - A)}$$

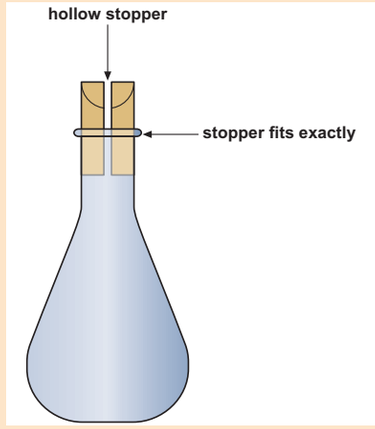


Figure 6.25 *A density bottle*

DID YOU KNOW?

If the relative density of a substance is relative to the density of water it is often called specific gravity. If the object has a specific gravity greater than 1, it will sink in water (more on this later).

4.2.3 Pressure in fluids

We've already discussed atmospheric pressure but if we investigate pressure in fluids in general we find there are two key points to consider:

- **Pressure increases with depth.**
- **At any given depth the pressure is equal in all directions.**

Pressure and depth

In any fluid the pressure increases with depth. The taller the column of the fluid above you, the greater the pressure it exerts. You can see this by conducting a very simple experiment.

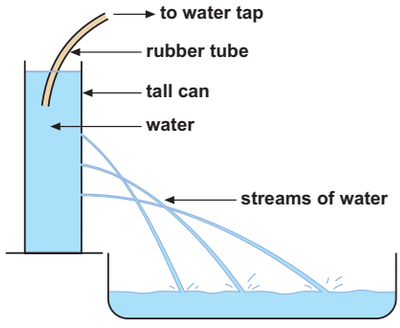


Figure 6.26 *The effect of depth on pressure*

Activity 6.9: Pressure and depth

Take a tall tin can and carefully make several holes going up one side (three or four should do it).

Quickly fill the tin with water and observe how the water squirts out of the holes.

You will notice that the stream from the bottom hole travels further. This is because the water is under more pressure at the bottom of the can.

We have already derived an equation for pressure in fluids in Section 6.1:

- $p = h\rho g$
- p = pressure in Pa
- h = depth of fluid in m
- ρ = density of fluid in kg/m^3
- g = gravitational field strength (9.81 N/kg)

Be careful not to mix up p and ρ (the Greek letter rho); make sure you look carefully before completing any calculations.

To recap:

Figure 6.28 shows a tank, filled with water of density ρ to a depth h. The base of the tank has area A. What is the pressure on the bottom of the tank?

The pressure is caused by the weight of the water in the tank, pressing down on the bottom.

- Volume of water = $h \times A$
- Mass of water = volume × density = $\rho \times h \times A$
- Weight of water = mass × g = $\rho \times h \times A \times g$
- Pressure = weight / area = $\rho \times h \times A \times g / A = \rho \times h \times g$.

This equation shows that the pressure increases with depth (h); in fact the pressure exerted by the fluid is directly proportional to the depth of fluid. Dive twice as deep and the pressure exerted by the water above you is doubled.

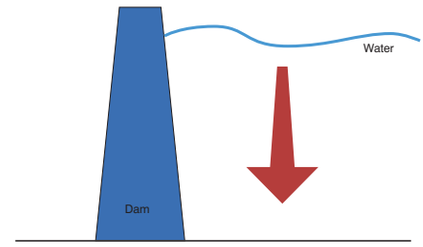


Figure 6.27 *Dams have to be thicker at the bottom in order to withstand the greater pressure.*

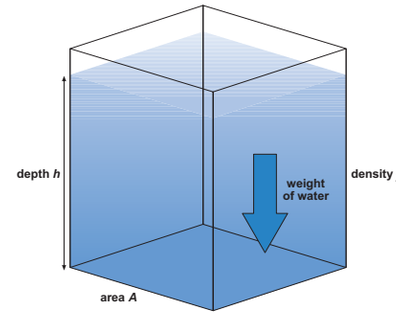


Figure 6.28 *The pressure on the bottom of the tank is caused by the weight of the water above it.*

Worked example

Calculate the pressure exerted by the water at the bottom of a swimming pool 6 m deep.

$p = h\rho g$ *State principle or equation to be used (pressure in fluids)*

p = 6 m × 1000 kg/m^3 × 10 N/kg *Substitute in known values and complete calculation*

p = 60 000 Pa *Clearly state the answer with unit*

Calculate the force this pressure would exert on a concrete block with an area of 3 m^2

$p = F / A$ *State principle or equation to be used (definition of pressure)*

$F = p \times A$ *Rearrange equation to make F the subject*

F = 60 000 Pa × 3 m^2 *Substitute in known values and complete calculation*

F = 180 kN *Clearly state the answer with unit*

Activity 6.10: Pressure calculations

Using information in the density table (Table 6.3) calculate the pressure exerted by the fluid in the following situations:

1. Diving in sea water to a depth of 15 m.
2. The base of a column of mercury 760 mm tall.

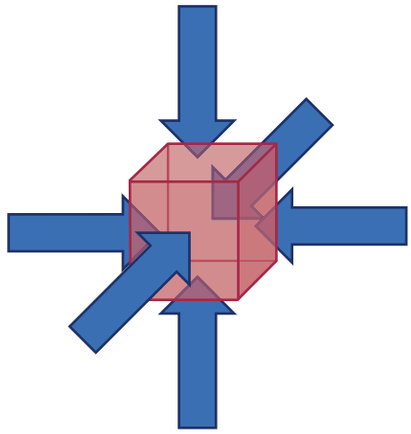


Figure 6.29 *Pressure is the same in all directions on a small cube.*

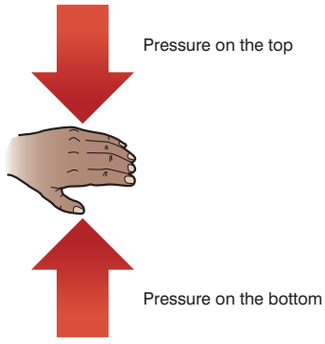


Figure 6.30 *The pressure on the top and on the bottom of your hand is essentially the same.*

DID YOU KNOW?

This topic is called fluid statics (or hydrostatics), meaning we are dealing with stationary fluids. A moving fluid exerts less pressure on its surroundings. This is studied in hydrodynamics and is very important when it comes to keeping aircraft in the air.

Pressure acts equally in all directions

In fluids, despite the pressure being caused by the column of fluid above you, the pressure acts equally in *all directions*. If you imagine a very small cube placed under water, the pressure on each cube face would be same.

If you hold your hand up horizontally in front of you the pressure on the top is the same as the pressure on the bottom.

Technically there is a very small difference in the height of the column of air on the top compared to the bottom (the thickness of your hand) but essentially the pressure is the same.

Activity 6.11: Pressure in a can

You can show this using another tin can. This time make four or five holes at the same depth around the bottom of the can.

Again quickly fill it with water and you can see all the streams of water are the same. In other words the pressure is the same in all directions inside the can.

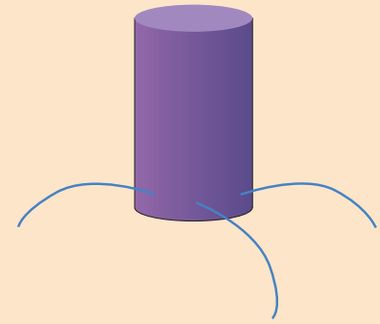
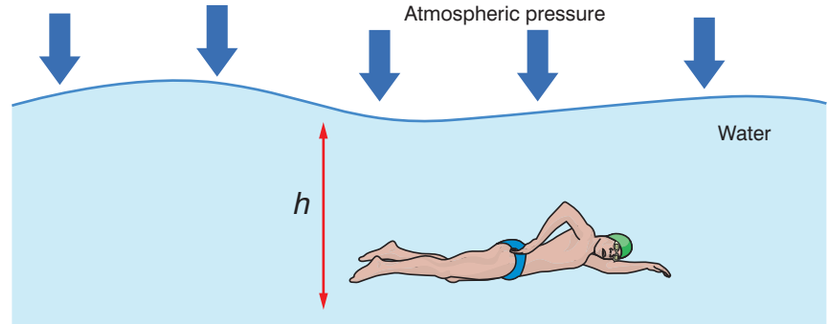


Figure 6.31 *Pressure is the same in all directions.*

What about the effect of atmospheric pressure?

If you go swimming the pressure acting on you is not just due to the water above you. You must not forget to include atmospheric pressure.



$$p = p_{atm} + p_{fluid}$$

Figure 6.32 *The total pressure on a swimmer*

The pressure on the swimmer would be the sum of the pressure due to the fluid and the atmospheric pressure. In terms of an equation this could be written as:

$$p = p_{atm} + h\rho_{fluid}g$$

4.2.4 Pascal's principle

Activity 6.12: Water transmits pressure

- Take two syringes of different sizes. Connect them with plastic or rubber tubing. Fill the syringes and the tube with water (Figure 6.33).
- Press one syringe with one hand, and the other with the other hand. Feel how their forces differ.

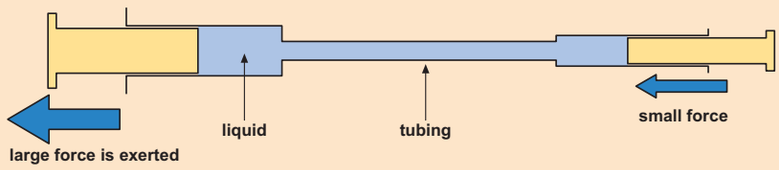


Figure 6.33 *Using water pressure to magnify a force.*

If you conduct the experiment above the difference in the forces is clear.

This difference comes down to the fact that liquids are incompressible; this means they can transfer pressure from one place to another. The force applied to the smaller syringe creates a pressure inside the liquid. This pressure is transferred throughout the liquid and is the same value everywhere. This pressure acts on the larger syringe and because the area of the syringe is larger the force exerted is also greater. Remember, from Unit 4, energy cannot be created or destroyed. Just like the simple machines studied in unit 5, if the output force gets bigger it must move through a smaller distance.

This phenomenon is referred to as **Pascal's principle** and it states:

- **The pressure applied to an enclosed fluid is transmitted to every part of the fluid, as well as to the walls of the container without reducing in value.**

Pascal's principle is used in the design and construction of simple **hydraulic machines**. Figure 6.34 shows two different sized pistons, which form part of a hydraulic system.

If a force is applied to the left hand piston it will create a pressure inside the fluid.

- $p = F_1 / A_1$

This pressure is transferred throughout the liquid. It is the same everywhere.

- p on the left = p on the right.

The piston on the right has a much larger area. The force from this piston is equal to:

- $F_2 = p \times A_2$

Worked example

Determine the pressure acting on a diver 20 m below the surface.•

$p = p_{atm} + h\rho_{fluid}g$ *Express total pressure in terms of atmospheric pressure and pressure from fluid*

- In this case, h = 20 m and ρ_{fluid} = 1000 kg/m^3

p = 101 000 Pa + (20 m $\times$ 1000 kg/m^3 $\times$ 10 N/kg) *Substitute in known values and complete calculation*

p = 301 000 Pa or 301 kPa *Clearly state the answer with unit*

Think about this...

Discuss with a partner why this effect does not happen in gases.

KEY WORDS

hydraulic machines *machines that rely on the incompressibility of liquids to do work*

Pascal's principle *principle stating that the pressure applied to an enclosed fluid is transmitted to every part of the fluid without reducing in value*

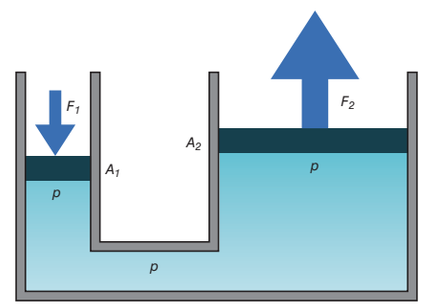


Figure 6.34 *Pascal's principle*

KEY WORDS

hydraulic lift *a hydraulic machine used to raise heavy objects*

hydraulics presses *a hydraulic machine used to shape metal or compress materials into smaller volumes*

hydraulic brakes *a mechanism which uses fluid to transfer pressure from a foot pedal to push brake pads onto brake discs*

As A_2 is much bigger than A_1, F_2 will also be bigger than F_1. In fact if the piston has double the area the force will be doubled. If the piston has ten times the area the force will be 10 times greater!

For example, let's imagine the areas are:

$A_1 = 2\ m^2 \quad A_2 = 6\ m^2$

If a force of 100 N is applied on A_1 then the force at A_2 will be 300 N (three times bigger). Let's prove it though calculation:

$p = F / A$ *State principle or equation to be used (definition of pressure)*

$p = F_1 / A_1$ *Relate to this context*

$p = 100\ N / 2\ m^2$ *Substitute in known values and complete calculation*

$p = 50\ N/m^2$ *Clearly state the answer with unit*

From Pascal's principle the pressure is the same throughout the liquid so:

$p = F_2 / A_2$ *State principle or equation to be used (definition of pressure expressed in this context)*

$F_2 = p \times A_2$ *Rearrange equation to make F_2 the subject*

$F_2 = 50\ N/m^2 \times 6\ m^2$ *Substitute in known values and complete calculation*

$F_2 = 300\ N$ *Clearly state the answer with unit*

As the pressure is same throughout the fluid we can summarise the relationship between the forces and areas in the following equation:

- $F_1 / A_1 = F_2 / A_2$

4.2.5 Hydraulic machines

Pascal's principle has many applications; one of the simplest is the hydraulic lift. This is used to lift a heavy object (such as a car)
off the ground. Just like our example, a small force is applied to a smaller area piston. This creates a pressure inside a hydraulic fluid, which is transferred to a larger area piston. This piston creates a much larger force and, if the object to be lifted sits on top of the large piston, it can be easily lifted by the smaller force at the smaller area piston.

Other examples include hydraulic presses and hydraulic brakes (in cars).

Hydraulic presses are used to shape metal (e.g. make motor-car bodies), to press waste paper or cotton wool into bales of small size, to press oil from oil seeds, and to lift cars so that work can be done easily underneath.

Activity 6.13: A hydraulic lift

- **Inner tube method:**

  Use the inner tube of a bus or lorry tyre. Take out the valve, and fit about 1.5 metres of rubber tubing to the tube over the metal valve. Put a funnel into the other end of the rubber tubing. Place a large wooden board on the flat inner tube, and stand on the board. Pour water into the funnel. The inner tube fills with water and lifts you.

- **Polythene bag method:**

  Connect some rubber tubing to a closed polythene bag. Place a brick with its largest surface on the bag. Blow into the tubing. The brick is lifted.

  Turn the brick so that a smaller surface is on the bag. A larger pressure is needed to lift the brick as much as before.

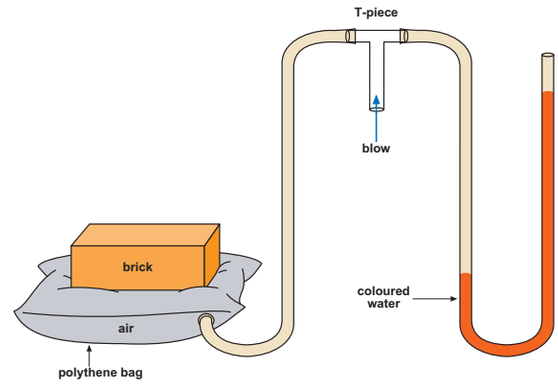


Figure 6.35 *A hydraulic lift operated by air pressure*

The hydraulic press (Figure 6.36) changes a small force into a large one. It consists of a cylinder and a piston, of large diameter, joined by a pipe to a second cylinder and a piston of small diameter. Water or oil is pumped into the small cylinder, and it lifts the large piston with an enormous force. A release valve lets the liquid run away after the piston has done its work.

Think about this...

Why is it a serious problem if air bubbles get into the hydraulic brake lines of a car?

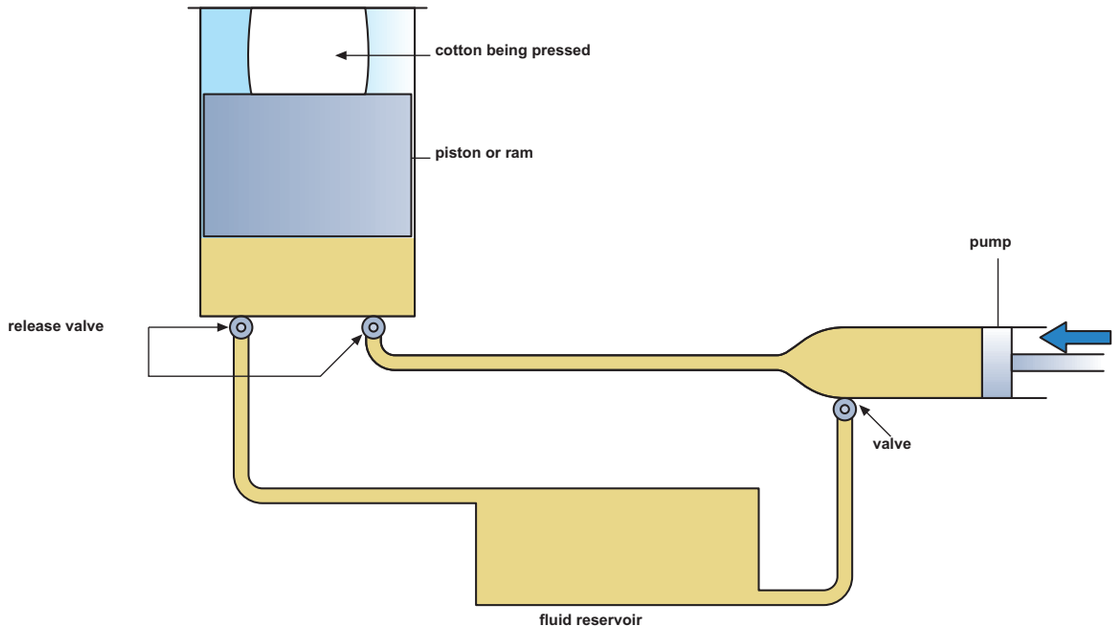


Figure 6.36 *A hydraulic press, used to compress a bale of cotton*

A car's **hydraulic brakes** work in a similar way. By pressing the foot on the brake pedal, a small force is applied to a piston with a small diameter. The pressure is transmitted through oil pipes to pistons of large diameter on the car wheels. These push the brake pads against the brake discs to stop the wheels.

Figure 6.37 *Pressure gauges may read absolute pressure or gauge pressure.*

KEY WORDS

absolute pressure *the actual pressure at a given point*

gauge pressure *the difference between absolute pressure and atmospheric pressure*

What is the difference between atmospheric, gauge and absolute pressure?

When it comes to measuring the pressure of a fluid there are several different terms you may come across. These include atmospheric pressure, gauge pressure and absolute pressure.

Absolute pressure

The **absolute pressure** is the actual pressure at a given point. It is the true pressure of a system if all of the factors are taken into account (including atmospheric pressure).

Atmospheric pressure

Atmospheric pressure has already been discussed. It is the pressure of the surrounding air when measured at the surface of the Earth. It has a value of 101 kPa. Atmospheric pressure varies depending on the temperature, the altitude above sea level and the impact of weather systems.

Gauge pressure

Pressure gauges often give readings of **gauge pressure** rather than absolute pressure. Gauge pressure is the pressure difference between a system and atmospheric pressure.

If the pressure gauge reads 25 kPa it would mean 25 kPa *above* atmospheric pressure (giving 126 kPa in total). If the gauge was disconnected it would read 0 Pa even though the absolute pressure is still 101 kPa.

Gauge pressure can be calculated using the equation below:

- $p_g = p_s - p_{atm}$
- p_g = gauge pressure
- p_s = system pressure (the absolute pressure of the system being measured)
- p_{atm} = atmospheric pressure

This is often used to determine the absolute pressure of the system. For example, if a compressed gas was measured and the gauge pressure of the system was 49 kPa then the absolute pressure would be:

- $p_g = p_s - p_{atm}$ *so* $p_s = p_g + p_{atm}$
- p_s = 49 000 Pa + 101 000 Pa
- p_s = 150 000 Pa

As gauge pressure is relative to atmospheric pressure it is possible to obtain negative readings. A reading of –10 kPa would mean 10 kPa *below* atmospheric pressure.

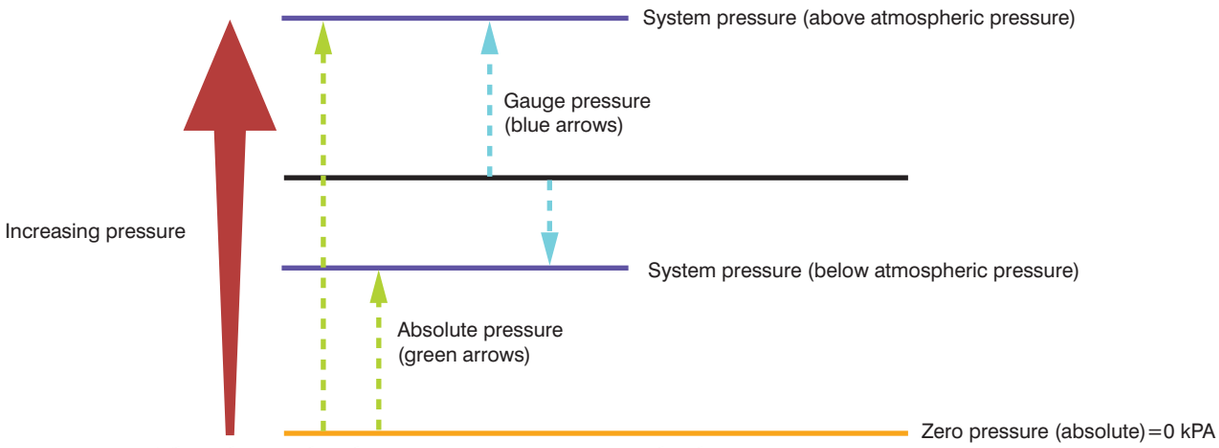


Figure 6.38 *The relationship between gauge pressure, absolute pressure and atmospheric pressure*

4.2.6. Measuring pressure

We have already looked at simple and aneroid barometers. However, there are a number of other ways to measure the pressure of a fluid. Most modern techniques use electronic pressure sensors. However, there are two other common mechanical techniques.

KEY WORDS

Bourdon gauge *an instrument for measuring the pressure of a gas*

manometer *a U-shaped tube filled with liquid which is used to measure pressure*

Bourdon gauge

A **Bourdon gauge** is a more practical instrument for measuring the pressure of a gas (Figure 6.39). Inside the gauge is a flattened tube with one end sealed. The tube is coiled round in a spiral. The open end is connected to, say, the gas supply. As the gas presses in, it causes the spiral tube to uncurl slightly. This makes the needle move round the dial, indicating the pressure.

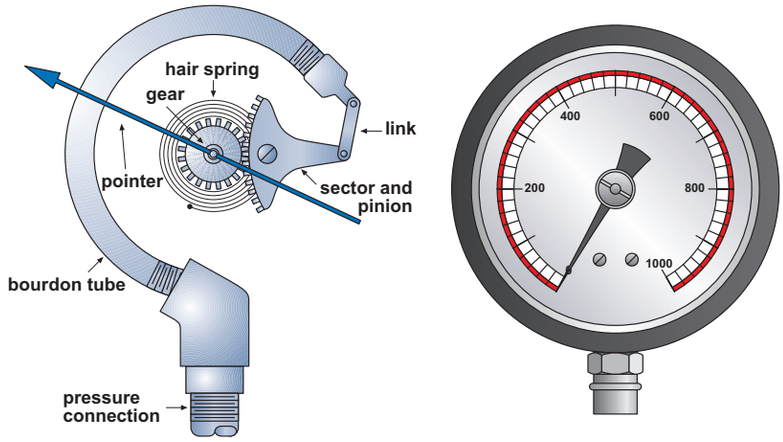


Figure 6.39 A Bourdon Gauge

A Manometer

A manometer is a simple instrument often used to measure the pressure of a gas supply. It comprises a U-shaped tube open at both ends. The tube is filled with a liquid (this is often coloured to make it easier to see).

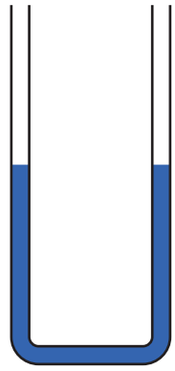


Figure 6.40 *A simple manometer*

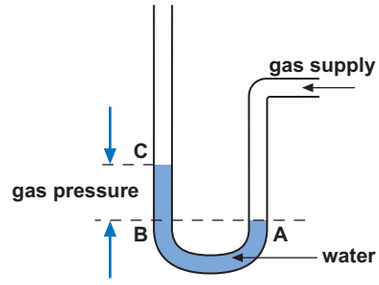


Figure 6.41 *Using a manometer to measure the pressure of the gas supply*

Figure 6.42 *A manometer being used*

Think about this...

If using a manometer to measure the pressure of higher pressure gases why is it a good idea to use mercury? Why is water usually used for gas supplies?

If one side of the manometer is connected to a system under pressure, the liquid will move. For example, if one end was connected to a gas supply the liquid would be pushed down as the supply is at a greater pressure than the surrounding atmosphere.

The height difference between B and C can then be used to determine the pressure of the gas supply.

- pressure of gas = atmospheric pressure + pressure due to the column of liquid BC
- pressure of gas = $p_{atm} + h_{BC}\rho g$

For example, if a water-filled manometer was connected to a gas supply and the height difference (BC) was 9 cm the pressure of the gas would be:

- pressure of gas = $p_{atm} + h_{BC}\rho g$
- pressure of gas = 101 000 Pa + 0.09 m × 1000 kg/m^3 × 10 N/kg
- pressure of gas = 101 900 Pa

This would most likely be expressed as a gauge pressure of 900 Pa.

4.2.7 Forces in fluids

Objects seem less heavy in water. For example, it is easy to hold up a friend horizontally in a swimming pool. Try doing this in air!

Figure 6.43 *Despite their large mass elephants appear to be lighter underwater.*

There is a force from the water that pushes you up, acting against gravity. This force is called a **buoyant force** (or sometimes **upthrust**). It arises due to the fact that as pressure increases with depth if you immerse an object in a fluid the pressure on the bottom will be greater than the pressure on the top.

This can be shown by considering the equation, $p = h\rho g$. The difference in pressure can be found by using:

- $\Delta p = \Delta h\rho g$

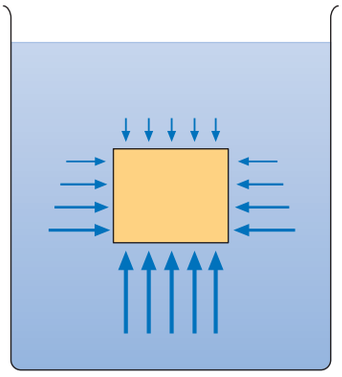


Figure 6.44 *The pressure is greater at a greater depth in water, so there is a bigger force on the lower surface of the block than on the upper surface.*

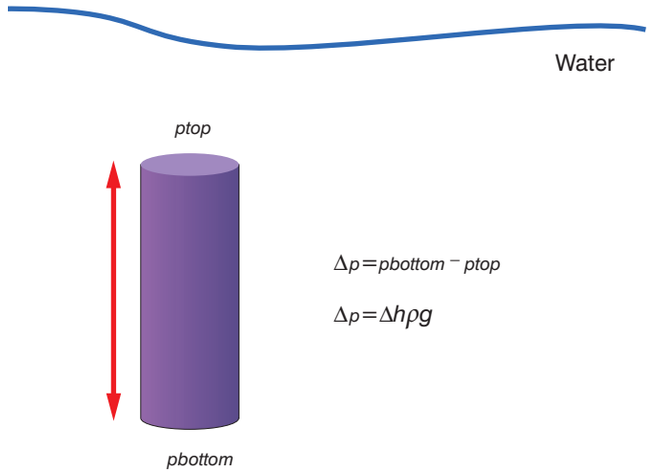


Figure 6.45 *The pressure difference = Δhρg*

This difference in pressure means there is a difference in force acting on the top and bottom of the object. The force on the bottom is greater and so there is net force upwards.

If you hold a cork underwater and then release it the buoyant force accelerates it towards the surface of the water. Equally if you drop a stone in the water it accelerates through the water much more slowly than it did through the air as the buoyant force means the net force acting on the stone is reduced.

The size of the buoyant force (F_b) depends on a number of factors including the *density of the fluid* and the *volume of the object*.

Buoyant forces are not just limited to liquids. Air also provides a buoyant force but it is very small (as the density of air is much less than that of water). In order for it to have a significant effect the volume of the object must be huge. Hot air balloons 'float' in the air due to the buoyant force of the air pushing them up, acting against their weight.

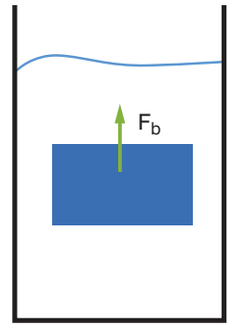


Figure 6.46 *The pressure difference leads to a force acting vertically upwards.*

Figure 6.47 *The buoyant force from the air keeps the hot balloon in the air.*

Apparent weight

As we mentioned earlier, objects immersed in water (or any liquid) appear to weigh less. Obviously their weight has not changed ($w = mg$) but they now have an apparent weight. The buoyant force pushes upwards, acting against the objects weight and so the weight appears to drop.

The apparent weight may be calculated using the equation below:

- apparent weight = weight – buoyant forc.

Gases (like air) also provide a buoyant force but it is usually too small to need thinking about.

This equation is more commonly used to determine the buoyant force acting on an object:

- buoyant force = weight – apparent weigh.

KEY WORDS

buoyant force *a force from the water which pushes a body upwards against gravity*

upthrust *a force from the water which pushes a body upwards against gravity*

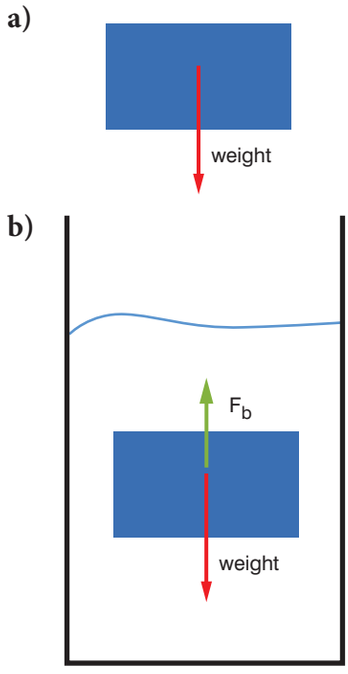


Figure 6.48 *The forces acting on an object a) in air b) in water*

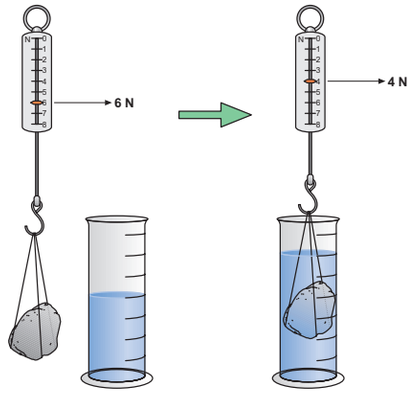


Figure 6.49 *Measuring the buoyant force acting on a stone*

KEY WORDS

Archimedes's principle
principle stating that the weight of the fluid displaced by an object is equal to the buoyant force acting on it

Using a forcemeter we can easily determine the buoyant force acting on a stone (see Figure 6.49).

Here the buoyant force is equal to:

buoyant force = weight – apparent weight *State principle or equation to be used*

buoyant force = 6.0 N – 4.0 N *Substitute in known values and complete calculation*

buoyant force = 2.0 N *Clearly state the answer with unit*

4.2.8 Archimedes's principle

You probably know the story of Archimedes in his bath. King Hiero had ordered a new gold crown, in the shape of a wreath of leaves. The crown was the correct weight, but he suspected that the jeweller had cheated him by mixing silver with the gold. Could Archimedes find a way of checking the crown without damaging it?

Archimedes was in his bath when he thought of the solution. As everyone knows, when you get in the bath, the water level rises because your body displaces some of the water. Archimedes, seeing how he could put this to use, leapt from the bath and ran down the street shouting 'Eureka!' which means 'I have it!'

Here is how Archimedes tested the crown. He put a weight of gold equal to the crown, and known to be pure, into a bowl which was filled with water to the brim. Then the gold was removed and the king's crown put in, in its place. This caused the bowl to overflow.

Archimedes was using the fact that gold is denser than silver, so it takes up less space. He found that the new crown had a greater volume than one made of pure gold. It was indeed a cheat, and the jeweller was punished.

Archimedes realised that when an object is immersed in a liquid it displaces a certain volume of the liquid.

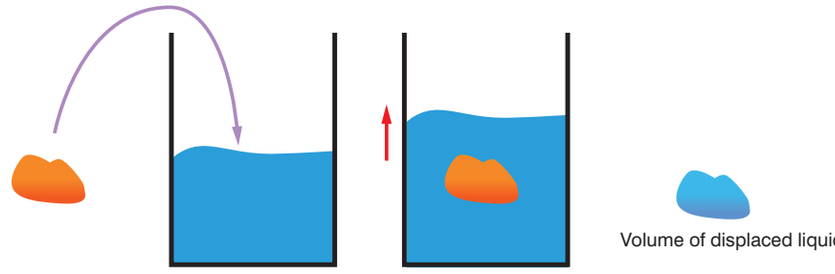


Figure 6.50 *A stone placed in a beaker of water will cause the level of water to rise as it displaces its own volume.*

He determined that the weight of the displaced fluid was equal to the buoyant force. Or in his own words:

- **Any object, wholly or partially immersed in a fluid, is buoyed up by a force equal to the weight of the fluid displaced by the object.**

In other words, the buoyant force acting on an object is equal to the weight of the displaced liquid.

- buoyant force = weight of displaced fluid

The greater the volume of liquid displaced the greater the buoyant force.

Activity 6.14: Testing Archimedes's principle

Use thin thread to tie an object (a stone, metal weight or glass stopper is suitable) to the hook of a newtonmeter (a spring balance). Note its weight.

- Weigh a beaker.
- Place an overflow can on the bench and fill it with water. When no more water drips out of the can, place the weighed beaker under its spout (see Figure 6.51).
- Lower the object carefully into the water until it is partially immersed. Note the apparent weight of the object.
- Weigh the beaker with the displaced water in it.
- Replace the beaker and water under the spout. Lower the object into the can until it is totally immersed but not touching the bottom of the can. Note the apparent weight of the object.
- Weigh the beaker and the displaced water.

You have to find the buoyant force on the object, and compare it with the weight of water displaced.

Upthrust = weight of object in air – weight of object in water

Weight of displaced water = weight of beaker with water – weight of empty beaker

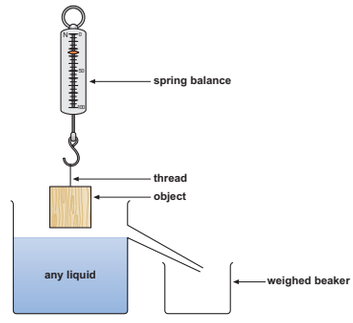


Figure 6.51 *Testing Archimedes's principle*

We can modify our equation for apparent weight in light of Archimedes' principle:

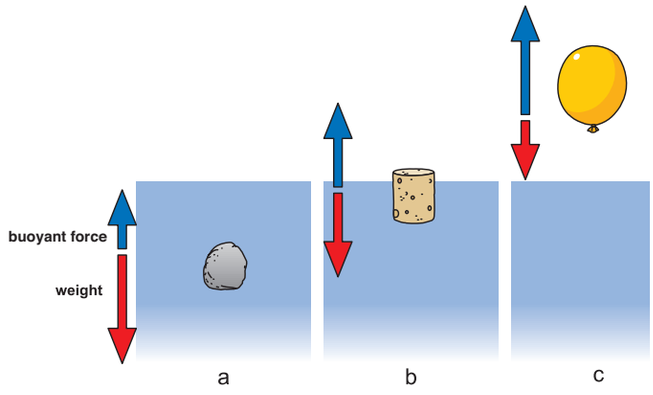
- apparent weight = weight – buoyant force
- buoyant force = weight of displaced fluid
- apparent weight = weight – weight of displaced fluid

KEY WORDS

law of flotation *law stating that if the buoyant force is equal to the weight of the object then the object will float*

4.2.9 Floating and sinking

Whether or not an object floats or sinks depends on the weight of the object and the size of the buoyant force acting on the object.



a – weight is greater than buoyant force, so the stone sinks

b – weight is equal to the buyoyant force, so the cork floats

c – weight is less than the buoyant force so the balloon rises

Figure 6.52 *The relative sizes of the buoyant force and the weight determine whether an object will float or sink.*

In order to float an object must displace a volume of fluid (liquid or gas) equal to its own weight. This is called the **law of flotation**.

If the weight of the volume of fluid displaced is equal to the weight of the object then the object will float.

A large steel ship is able to float because it displaces such a large volume of water. This volume of water has the same weight as the ship.

When you step into a small boat you might notice the boat sinks down a little in the water. This is because as the weight of the boat increases it needs to displace a greater volume of liquid in order to float, and so it sinks down lower in the water. A heavily loaded boat sits much lower in the water than a lightly loaded boat.

A solid block of steel sinks in water. A ship with the same mass floats on the surface.

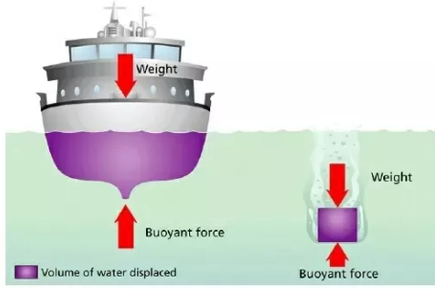


Figure 6.53 *The ship floats due to the law of flotation.*

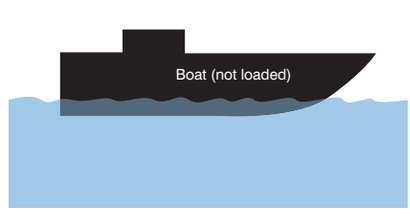


Figure 6.54 *A boat that is not heavily loaded displaces a smaller volume of liquid in order to float.*

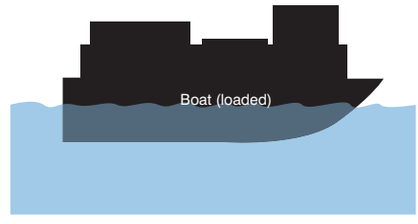


Figure 6.55 *A boat that is heavily loaded needs to displace a much larger volume of water in order to float.*

In the late 19th century greedy ship owners were overloading their ships and several ships sank as a result. The Englishman Samuel Plimsoll developed the waterline (or more commonly the Plimsoll line). This was a line that by law must be painted on all large ships. For safety reasons, when the ship is fully loaded the level of the water must not be above the Plimsoll line.

Figure 6.56 *The Plimsoll line on a ship*

Worked example

A toy submarine has a weight of 6.2 N in air. When immersed in water it has a weight of 4.6 N. Determine the buoyant force and the weight of water displaced

buoyant force = weight – apparent weight *State principle or equation to be used*

buoyant force = 6.2 N – 4.6 N *Substitute in known values and complete calculation*

buoyant force = 1.6 N *Clearly state the answer with unit*

weight of displaced fluid = buoyant force *Make it clear the two quantities are equal from Archimedes's principle*

- weight of displaced fluid = 1.6 N *Clearly state the answer with unit*

Think about this...

If you look carefully at the image of the Plimsoll line you can see that there are several different lines depending on whether the ship is in fresh water, salt water, cold water (North Atlantic) or warm water (tropical). Why is this?

What about density?

If, even when fully immersed, the weight of the volume of liquid displaced is less than the weight of the object, then the object will sink. A small cube of steel does not displace enough water to float. However, if you hammer out the steel into a bowl shape it displaces a greater volume of water and so will float.

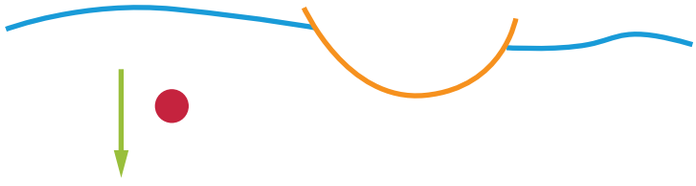


Figure 6.57 *The same mass of steel will sink or float depending on its shape and so the amount of fluid it displaces.*

In other words, if the density of the object is greater than the density of the fluid it will sink.

This means we need to consider the relative density between the object and the liquid. If the relative density is less than one the object will float (as the weight of the object will be less than the weight of the volume of liquid it displaces). If the relative density is more than one the object will sink (as the weight of the object will be more than the weight of the volume of liquid it displaces). We can modify our previous equations to include the density of the object and the density of the fluid.

Worked example

A floating wooden block has a volume of 0.4 m^3 and displaces 0.3 m^3 of water. Determine the density of the block.

$\rho_{object} V_{object} = \rho_{fluid} V_{fluid}$ *State principle or equation to be used (a version of Archimedes's principle)*

$\rho_{object} = \rho_{fluid} V_{fluid} / V_{object}$ *Rearrange equation to give* ρ_{object}

$\rho_{object} = (1000\ kg/m^3 \times 0.3\ m^3) / 0.4\ m^3$ *Substitute in known values and complete calculation*

$\rho_{object} = 750\ kg/m^3$ (or a relative density of 0.75) *Clearly state the answer with unit*

- $w = mg$ and $\rho = m / V$
- weight of object = $m_{object} g$ and so weight of object = $\rho_{object} V_{object} g$
- weight of displaced liquid = $m_{fluid} g$ and so weight of displaced fluid = $\rho_{fluid} V_{fluid} g$

If the object is floating then:

- buoyant force = weight of displaced liquid = weight of object

So:

- $\rho_{object} V_{object} g = \rho_{fluid} V_{fluid} g$

The g's cancel, giving:

- $\rho_{object} V_{object} = \rho_{fluid} V_{fluid}$

This equation only applies if the object is floating.

Summary

In this section you have learnt that:

- A fluid is any substance that can flow. This includes gases as well as liquids.
- Gases may be compressed but liquids are incompressible.
- Density is defined as the mass per unit volume and it may be calculated using the equation $\rho = m / V$. Density is measured i n kg/m^3.
- The relative density of a substance is the density of the substance compared to another (e.g. compared to water).
- In fluids the pressure increases with depth and is the same in all directions.
- In fluids the pressure due to the fluid is equal to $p = h\rho g$. The total pressure is equal to the pressure due to the fluid plus atmospheric pressure.
- Pascal's principle states that liquids transfer pressure from one place to another without any reduction in pressure.
- Gauge pressure is the difference between absolute pressure and atmospheric pressure.
- A manometer is a simple U-shaped tube filled with liquid used to measure pressure.
- The apparent weight of a body is equal to the weight of the object minus the buoyant force acting on it.
- Archimedes's principle states that the weight of the displaced fluid is equal to the buoyant force acting on the object.
- The principle of flotation states if the buoyant force (or weight of displaced fluid) is equal to the weight of the object then the object will float.
- If the object is floating then the density of the floating object can be calculated from: $\rho_{object} V_{object} = \rho_{fluid} V_{fluid}$ where
  V_{fluid} is the volume of the displaced fluid.

Review questions

1. Explain what is meant by the term fluid and give three examples.
2. Calculate the pressure caused by sea water when diving to a depth of 100 m. What is the total pressure acting on the diver?
3. State Pascal's principle and describe one of its applications.
4. Two pistons are connected together to make a hydraulic lift. The smaller piston has an area of 0.05 m^2 and the larger piston has an area of 2 m^2. Calculate the following:
   a) The pressure in the fluid and the force at the larger piston if the force on the smaller piston is 50 N.
   b) The pressure in the fluid and the force from the smaller piston required to lift a car of mass 1200 kg.

End of unit questions

1. An elephant has a mass of 3200 kg. Each of its feet covers an area equal to 0.08 m^2. Calculate the pressure from each foot.
2. Describe what causes pressure in gases in terms of the particles in the gas.
3. Describe some similarities and difference between liquids and gases.
4. How deep under water would you need to be in order to be at double atmospheric pressure?
5. Explain the meaning of the terms atmospheric pressure, absolute pressure and gauge pressure.
6. Describe the use of a manometer and calculate the pressure of a gas supply that causes a column of water 15 cm high.
7. State Archimedes's principle and explain how this leads to the law of flotation.
8. Explain why a heavily loaded boat sinks lower in the water.
9. The weight of an object is measured in air to be 7.0 N. The object is then immersed in water and its apparent weight is measured to be 4.0 N. Determine the buoyant force and state whether or not the object floats.
10. A large ocean liner floating in the sea has a volume of 375 000 m^2 and displaces 50 000 m^2 of sea water. Determine the density and mass of the ship. Explain why, despite being made of metal, the ship is able to float.

Chapter 2 Thermodynamics

Lesson 1 Temparature and Heat

KEY WORDS

atoms *the smallest parts of a chemical element that can take part in a reaction*

average kinetic energy *the mean kinetic energy of all the particles in a substance*

ions *an atom or group of atoms that has acquired an electrical charge by gaining or losing one or more electrons*

molecules *the smallest unit of a substance, consisting of a group of atoms, which retains the chemical and physical properties of the substance*

particles *small units of matter such as atoms, ions and molecules*

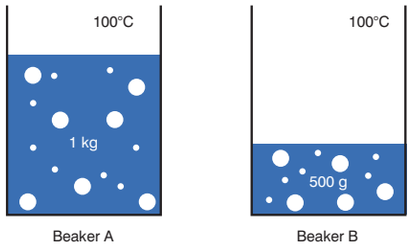


Figure 1. *Two beakers of boiling water*

Figure 2. *What is the difference between heat and temperature?*

1.Temperature and heat

By the end of this section you should be able to:

- Explain the difference between heat and temperature.
- Convert temperature from one scale to others.

What is heat?

When we cook food, we might say we are heating it up. he temperature of the food increases. It seems like heat and temperature are the same thing, but they are not!

We already know that matter is made up of moving **particles** (**molecules**, **atoms** and **ions**). In solids these particles are tightly bonded together and so they can only vibrate, whereas in fluids (liquids and gases) the particles can move around more freely.

Heat is one form of energy; it is therefore measured in joules and is a *scalar* quantity. Heat is a flow of energy from hotter regions to colder ones.

Imagine two beakers of boiling water. Beaker A contains 1 kg of boiling water and beaker B contains 0.5 kg of boiling water. They are both at the same temperature, 100 °C, but there are more particles in beaker A and so there is more energy contained within it than with beaker B.

When we heat up a substance, we are transferring energy to the substance. This means one of two things could happen.

- **The particles of he substance gain kinetic energy and so move more rapidly.**

What is temperature?

Temperature is a measure of 'hotness'. The higher the temperature, the hotter the object. The complication is that 'hotter' may not mean more heat when comparing two objects.

The temperature of a substance is a measurement of the **average kinetic energy** of the particles within the substance. If the particles in a substance have a higher average kinetic energy then the object is at a higher temperature. That is to say if the particles are, on average, moving faster then the object is at a higher temperature. The water molecules in a glass of water at 50 °C are, on average, moving faster than those in a glass at 20 °C.

DID YOU KNOW?

At room temperature all gases will have the same average kinetic energy. This means the lower mass gas particles are, on average, travelling faster. One of the reasons there is very little helium in the atmosphere is because they are so light a significant number of helium atoms are going fast enough to escape the Earth's gravitational pull and float off into space. The heavier gases, like oxygen, nitrogen and carbon dioxide are on average moving slower and so don't escape.

KEY WORDS

absolute zero *the temperature at which a substance has no thermal energy*

Celsius scale *a temperature scale where the freezing point of water is fixed at 0 degrees and the boiling point at 100 degrees C*

Kelvin scale *a temperature scale that uses absolute zero as one of its fixed points*

Figure 7. *The Swedish astronomer Anders Celsius first proposed the Celsius scale in 1742.*

2.Temperature Scales

A range of temperature scales have been used in the past, though scientists now tend to deal with the **Kelvin scale** (K) or the **Celsius scale** (°C). The Fahrenheit scale (°F) is still used by the United States but is rarely used by the scientific community.

To design a temperature scale two fixed reference points have to be used. The scale is then based on these points with a certain number of jumps in between them.

For example, in the Celsius scale, the freezing point of water is given as 0 °C , with the second fixed point being the boiling point of water – the difference between the two fixed points is divided into 100 equal divisions so the boiling point of water is 100 °C.

Figure 6.*The freezing point and boiling point of water were the two fixed points used on the Celsius scale.*

The Kelvin scale uses **absolute zero** as one of its fixed points. This is the temperature at which a substance will have no thermal energy and it is not possible to get a lower temperature (0 K or –273.15 °C). The Kelvin scale has a units symbol of K; there is no degree symbol included.

The Kelvin and the Celsius scales are often used together as they have the same scale division. This means a *change* of 20 K is the same as a change of 20 °C.

The Kelvin scale may seem unusual as it uses fixed points that we are not familiar with but, importantly, temperatures measured in the Kelvin scale are directly proportional to the average kinetic energy of the particles present. For example, the particles in a block

of iron have on average twice as much kinetic energy at 200 K than at 100 K.

The Fahrenheit scale uses one fixed point as the temperature of an ice, water and ammonium chloride mixture (0 °F) – the second fixed point is normal body temperature (98 °F).

The diagram in Figure 8. shows how these temperature scales compare. We will use the Celsius and Kelvin scale in our calculations.

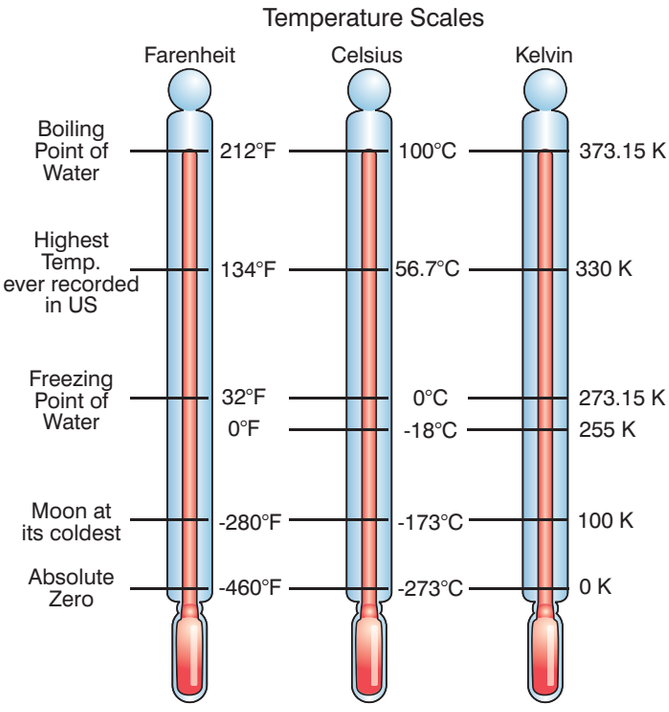


Figure 8. *Comparing different temperature scales*

CELSIUS-KELVIN TEMPERATURE CONVERSION

$$T = T_C + 273.15$$

Kelvin temperature = Celsius temperature + 273.15

Kelvin temperatures for various physical processes can range from around 1 000 000 000 K (10^9 K), which is the temperature of the interiors of the most massive stars, to less than 1 K, which is slightly cooler than the boiling point of liquid helium. The temperature 0 K is often referred to as *absolute zero*. Absolute zero has never been reached, although laboratory experiments have reached temperatures of just a half-billionth of a degree above absolute zero.

DID YOU KNOW?

The Celsius scale is named after the Swedish astronomer Anders Celsius. In 1742 he proposed the Celsius temperature scale, with one key difference. He set his lower fixed point (the freezing point of water) as 100 °C and the boiling point as 0 °C. This meant using his scale the number got smaller as the substance got hotter! The scale was reversed the year after he died.

Think about this...

At absolute zero a substance will have zero internal energy. What does this tell you about the kinetic energies and potential energies of the particles? Do you think it will be possible to reach absolute zero?

CELSIUS-FAHRENHEIT TEMPERATURE CONVERSION

$$T_F = 9/5\,T_C + 32.0$$

Fahrenheit temperature = (9/5× Celsius temperature) + 32.0

The number 32.0 in the equation indicates the difference between the ice point value in each scale. The point at which water freezes is 0.0 degrees on the Celsius scale and 32.0 degrees on the Fahrenheit scale.

Temperature values in the Celsius and Fahrenheit scales can have positive, negative, or zero values.

Summary:

- Heat is energy transferred from hotter regions to cooler ones.
- The temperature of a substance is an indication of the average kinetic energy of the particles and the Celsius and Kelvin scales are both temperature scales.
- On heating, the particles of a substance move faster and move further apart so that a substance expands on heating.
- CLSIUS-KELVIN temperature conversion:
  T = TC + 273.15
- CELSIUS-FAHRENHEIT TEMPERATURE CONVERSION
  $T_F = 9/5T_C + 32.0$

Temperature conversion between Degree Celsius and Fahrenheit Scale

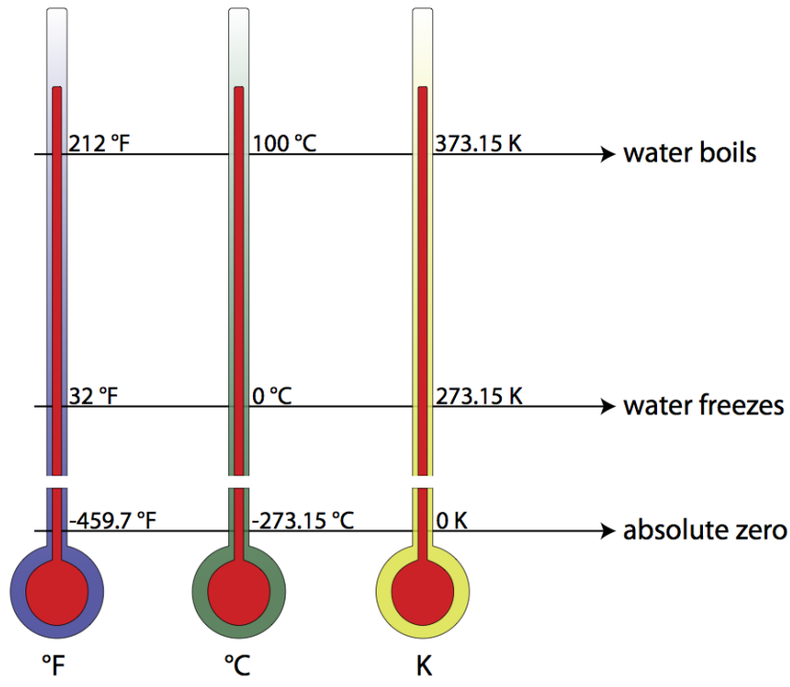


Figure 7.14 The fixed points of Fahrenheit, Celsius and Kelvin temperature scales.

Exercise 7.9: Using the fixed points from Figure 7.6

☞ Derive the formula to convert temperature in Celsius scale to Fahrenheit scale.
☞ Derive the formula to convert temperature in Fahrenheit scale to Celsius scale.

$$\frac{\text{T}_\text{C} - \text{ice point}}{\text{steam point} - \text{ice point}} = \frac{\text{T}_\text{F} - \text{ice point}}{\text{steam point} - \text{ice point}} \tag{7.1}$$

$$\frac{T_C - 0}{100 - 0} = \frac{T_F - 32}{212 - 32} \quad \Rightarrow = \quad \frac{T_C}{100} \quad \frac{T_F - 32}{180} \tag{7.2}$$

From this relation we determine the following two formulas that help to convert temperature from degree Celsius scale to Fahrenheit scale and vice versa.

$$T_C = \frac{5}{9}(T_F - 32) = \frac{(T_F - 32}{1.8} \tag{7.3}$$

$$T_F = \frac{9}{5}T_C + 32^o F = 1.8T_C + 32 \tag{7.4}$$

Temperature Conversion Between Degree Celsius and Kelvin Scale

Exercise 7.10

1. The size of one Kelvin is identical to that of one Celsius degree. Discuss and reason out.
2. The temperature difference of 5 ^oC is equal to the temperature difference of 5 K. Discuss.
3. What is the minimum possible temperature in nature?

Denoting the Celsius temperature scale by T_C and the Kelvin temperature scale by T_K and using the ice and steam points of both scales;

$$\frac{\text{T}_\text{C} - \text{ice point}}{\text{steam point} - \text{ice point}} = \frac{\text{T}_\text{K} - \text{ice point}}{\text{steam point} - \text{ice point}} \tag{7.5}$$

$$\frac{T_C - 0}{100 - 0} = \frac{T_K - 273.15}{373.15 - 273.15} \Rightarrow \frac{T_C}{100} = \frac{T_K - 273.15}{100} \Rightarrow T_C = T_K - 273.15 \tag{7.6}$$

$$T_C = T_K - 273.15, \qquad T_K = T_C + 273.15 \tag{7.7}$$

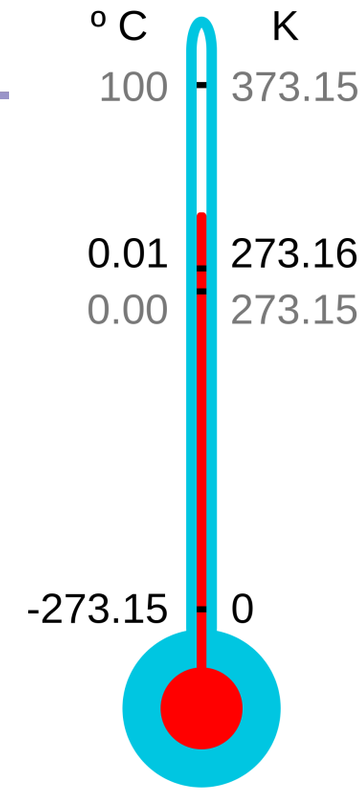


Figure 7.15 The fixed points of Celsius and Kelvin temperature scales.

Example 7.3

Water is boiled to a temperature of 72 ^OC. What does a Kelvin scale read for this value?

Given: $T_C = 72^oC$

Required: T_K =?

Solution:

Inserting the value of T_C

$$T_K = T_C + 273.15 = (72 + 273.15)K$$

$$\therefore T_K = 345.15K$$

SAMPLE PROBLEM

Temperature Conversion

PROBLEM

What are the equivalent Celsius and Kelvin temperatures of 50.0°F?

SOLUTION

Given: $T_F = 50.0°\text{F}$

Unknown: $T_C = ?$ $T = ?$

Use the Celsius-Fahrenheit equation to convert Fahrenheit into Celsius.

$$T_F = \frac{9}{5}T_C + 32.0$$

$$T_C = \frac{5}{9}(T_F - 32.0)$$

$$T_C = \frac{5}{9}(50.0 - 32.0)°\text{C} = 10.0°\text{C}$$

Use the Celsius-Kelvin equation to convert Celsius into Kelvin.

$$T = T_C + 273.15$$

$$T = (10.0 + 273.15)\text{K} = 283.2\ \text{K}$$

$T_C = 10.0°\text{C}$
$T = 283.2\ \text{K}$

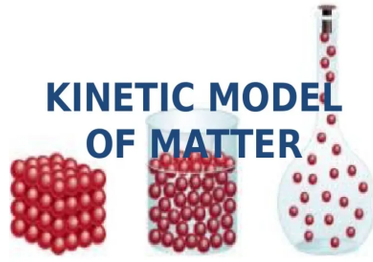
Practice: **Temperature Conversion**

1. The lowest outdoor temperature ever recorded on Earth is –128.6°F,recorded at Vostok Station, Antarctica, in 1983. What is this temperature on the Celsius and Kelvin scales?
2. The temperatures of one northeastern state range from 105°F in the sum-mer to –25°F in winter. Express this temperature range in degrees Celsius and in kelvins.
3. The normal human body temperature is 98.6°F. A person with a fever may record 102°F. Express these temperatures in degrees Celsius.
4. A pan of water is heated from 23°C to 78°C. What is the change in its temperature on the Kelvin and Fahrenheit scales?
5. Liquid nitrogen is used to cool substances to very low temperatures. Express the boiling point of liquid nitrogen (77.34 K at 1 atm of pres-sure) in degrees Celsius and in degrees Fahrenheit.

Lesson 2: The Kinetic Model of Matter

Objectives: By the end of this section you should be able to:

- **how to describe matter in three states (solid, liquid and gas)**
- **how the kinetic model can be used to explain changes of state and the behavior of gases**



Snow

Young people usually enjoy snow. You may live in a country where snow is rarely seen. Alternatively, you may be snow-bound for several months of the year. If you do experience snow, you will know the excitement of the first fall of the winter. Everyone rushes out to have snowball fights, or to go tobogganing or skiing.

Snow is remarkable stuff. It is simply frozen water. Yet people such as the Inuit who live among snow have many different words for it, depending on how it is packed down, for instance. This can be vital information if you are interested in winter sports, since it determines the avalanche risk.

We are familiar with the changes that happen when snow or ice melts. A white or glassy solid changes into a transparent, colorless, runny liquid. Heat the liquid and it 'vanishes' into thin air. Although this sounds like a magic trick, it is so familiar that it does not strike us as surprising. The Earth is distinctive among the planets of the solar system in being the only planet on which water is found to exist naturally in all three of its physical states.

In this lesson, we will look at what happens when materials change their state - from solid to liquid to gas, and back again. By thinking about the particles, the atoms and molecules of which the material
is made, we can build up a picture or model that describes changes of state and explains some of the things we observe when materials change from one state to another.

1. States of matter

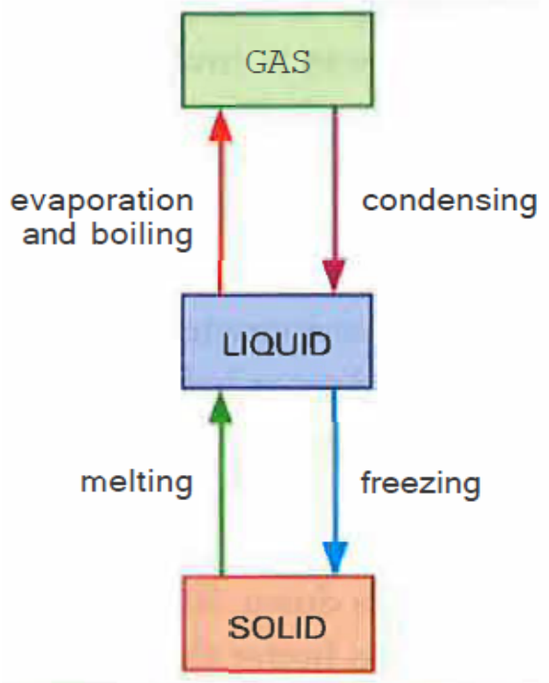
We think of matter as existing in three states, ***solid***, ***liquid*** **and** ***gas***. What are the characteristic properties of each of these three states? We need to think about shape and size. Table 1 shows how these help us to distinguish between solids, liquids and gases. It may help you to think about ice, water and steam as examples of the three states of matter.

Changes of state

Heat a solid and it melts to become a liquid. Heat the liquid and it boils to become a gas. Cool the gas and it becomes first a liquid and then a solid. These are *changes of state*. The names for these changes are shown in Figure:

melting - from solid to liquid *boiling* - from liquid to gas *condensing* - from gas to liquid

freezing - from liquid to solid. Another term for a liquid changing to a gas is **evaporation.** We will see the difference between evaporation and boiling shortly.



2. The kinetic model of matter

In this section, we will look at a **model** for matter that provides one way in which we can answer the following questions:

Why does it take time for a solid to melt? Why does it not change instantly into a liquid? Why does it take longer to boil a liquid than to melt a solid? Why do different substances melt at different temperatures?
Why do different substances have different boiling points?

The particles of which matter is made are very tiny. They may be atoms, molecules or ions, but we will simplify things by disregarding these differences and referring only to *particles*. We will also picture a material as consisting of large numbers of identical particles. Thus we are considering a pure substance whose particles are all the same, rather than a mixture that contains two or more types of particle. We will also picture the particles as simple spheres, although in reality they might have more complicated shapes. The molecules of a polymer, for example, may be like long thin strings of spaghetti, rather than like small, round peas.

The idea that matter is made up of spherical particles is a great simplification, but we can still use this idea to find answers to the questions listed above. Later, we will think about whether or not we are justified in using such a simplified model.

The model we are going to consider is called the **kinetic model of matter.** The word 'kinetic' means 'related to movement'. In this model, the things that are moving are the particles of which matter is made. The model thus has an alternative name: the *particle model of matter.*

Arrangements of particles

Figure 5 shows how we picture the particles in a solid, a liquid and a gas. For each picture, we will think about two aspects (see Table **3**)**:** how the particles are arranged, and how the particles are moving. (Because these are pictures printed on paper, it is hard to represent the motion of the particles. You may have access to software or video images that can show this more clearly)

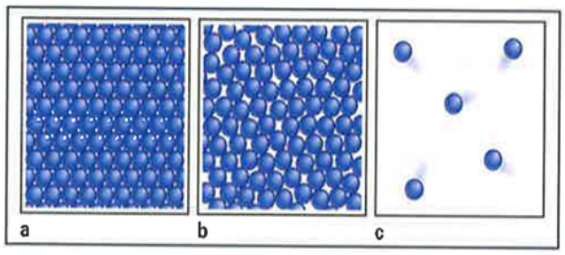
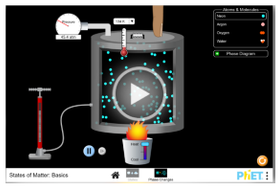


Figure 5. Representations of **a.** solid, **b.** liquid and **c.** gas. The arrangement and motion of the particles change as the solid is heated to become first a liquid and then a gas.

Lab simulation: https://phet.colorado.edu/en/simulations/filter

Click States of Matter: Basics ,Click play the video=>Click States=>Choose Solid, Liquid, and gas



3.Evidence for the kinetic model

We cannot look down a microscope and see the particles that make up matter. We certainly cannot hope to see the particles of a gas as they rush around. However, in the 1820s, the movement of the particles of a gas was investigated by a Scottish botanist, **Robert Brown.** He was using a microscope to study pollen grains when he noticed tiny particles jiggling about. At first he thought that they might be alive, but when he repeated his experiment with tiny grains of dust, suspended in water, he saw that they also moved around. This motion is now known as **Brownian motion,** and it happens because the moving particles are constantly buffeted by the fast-moving particles of the air.

State	Arrangement of particles	Movement of particles
Solid	The particles are packed closely together. Notice that each particle is in close contact with all of its neighbors. In a solid such as a metal, each atom may be in contact with 12 neighboring atoms.	Because the particles are so tightly packed, they cannot move around. However, they do move a bit. They are able to vibrate about a fixed position. The hotter the solid, the more they vibrate.
Liquid	The particles are packed slightly less closely together (compared with a solid). Each particle is still in close contact with most of its neighbors, but fewer in the case of a solid. The general arrangement is slightly more jumbled and disorderly.	Because the particles are slightly less tightly packed than in a solid, they have the opportunity to move around within the bulk of the liquid. Hence the particles are both vibrating and moving from place to place.
Gas	Now the particles are widely separated from one another. They are no longer in contact, unless they collide with each other. In air, the average separation between the particles is about ten times their diameter.	The particles are now moving freely about, bouncing off one another and off the walls of their container. In air at room temperature, their average speed is about 500 m/s.

Table l The arrangement and movement of particles in the three different states of matter. Compare these statements with the diagrams shown in Figure 5.

Lesson 3 Heat

Objectives: By the end of this section you should be able to:

- Identify different units of heat energy.
- Define the terms specific heat capacity and heat capacity.
- Calculate the amount of heat energy absorbed or liberated by a body using $Q = mc\Delta T$.
- Calculate the heat capacity of a body.
- Use the relationship heat lost = heat gained to solve problems involving heat exchange.
- Define the terms latent heat, latent heat of fusion and latent heat of vaporisation.
- Solve problems involving change of state.

1. Quantity of heat, specific heat capacity and heat capacity

What are the units of energy?

As discussed in Unit 4, the scientific unit of energy is the joule (J).
You may see another unit of energy called the calorie (cal).

One calorie is the quantity of heat energy required to increase the temperature of 1 g of water by 1 °C. The amount of energy in joules required to increase the temperature of 1 g of water by 1 °C is 4.18 J and so: **1 cal = 4.2 J**

The calorie is less frequently used now but you will see later that its definition is connected to the work we do later in this section.

What is meant by the term specific heat capacity?

If we were heating a substance to raise its temperature, the amount of heat energy required would depend on three things:

1. The **substance** being heated. A given mass of aluminium will require more energy to raise its temperature by 1 K than the same mass of wood.
2. The **mass** of the substance. The greater the mass of the substance, the more heat energy will be required to raise its temperature.
3. The temperature rise required. For a given mass of a particular substance, a large temperature increase will require a larger amount of heat energy than a small increase in temperature.

- Each substance has a specific heat capacity (c), which is defined as:
  The heat energy required to raise the temperature of 1 kg of a given substance by 1 K.

The units of specific heat capacity are J/kg K and Table 3 shows the specific heat capacities of some materials. We can see that metals tend to have lower specific heat capacities than non-metals and that water has a notably high value.

Substance	Specific heat capacity (J/kg K)
iron	470
copper	420
brass	380
aluminium	910
water	4200
rubber	1700
glass	670

Table 3. Some different specific heat capacities

Worked example

Calculate the quantity of heat energy required to heat a 1.00 kg block of iron from 290 K to 320 K. The specific heat capacity of iron is 470 J/kg K.

$Q = mc\Delta T$ *State principle or equation to be used (from definition of specific heat capacity)*

In this case, m = 1.00 kg, c = 470 J/kg K, ΔT = (320 K – 290 K) = 30 K

Q = 1.00 kg × 470 J/kg K × 30 K *Substitute in known values and complete calculation*

Q = 14 100 J = 14.1 kJ *Clearly state the answer with unit*

So, 14.1 kJ of heat energy would be required to increase the temperature of a 1.00 kg iron block by 30 K. Equally, if the 1.00 kg iron block cooled by 30 K, the iron block would have to lose 14.1 kJ of heat energy to the surroundings.

Worked example

Calculate the quantity of heat energy lost from a 580 g sample of water if it cools from 333 K to 278 K. The specific heat capacity of water is 4200 J/kg K.

$Q = mc\Delta T$ *State principle or equation to be used (from definition of specific heat capacity)*

In this case, m = 0.58 kg, c = 4200 J/kg K, ΔT = (333 K – 278 K) = 55 K

Q = 0.58 kg × 4200 J/kg K × 55 K *Substitute in known values and complete calculation*

Q = 13 398 J = 134 kJ *Clearly state the answer with unit*

So, as the 580 g sample of water cools from 333 K to 290 K, it would lose 134 kJ of heat energy to its surroundings.

From the definition of specific heat capacity, the quantity heat energy required (Q) to increase the temperature of a substance is found using the equation below:

- $c = Q / m\Delta T$

This is usually written as:

- $\boldsymbol{Q = mc\Delta T}$

m = mass of substance (kg)

c = specific capacity of substance (J/kg K)

ΔT = change in temperature (K). Remember, a change in temperature of 1 °C is the same as a change in temperature of 1 K. So, in this case °C or K may be used.

Activity 6: Specific heat calculations

Calculate the specific heat capacity of a 2.0 kg block of a solid that requires 63 700 J to raise its temperature by 35 K.

A solid has a specific heat capacity of 800 J/kg K. How much heat energy would be released a 250 g sample of this solid if its temperature falls from 310 K to 260 K?

How can we find the specific heat capacity of a substance?

There are several methods we can use to determine the specific heat capacity of a substance, but remember that to calculate this value we will always need to know the **mass** of the substance, the **amount of heat energy** supplied to it as well as its **starting temperature** and **final temperature**. All of these slightly different approaches involve heat exchange. The heat from a hot body is used to warm a colder body. In approaching these heat exchange calculations we use the principle that the heat energy lost from the hot body will equal the heat gained by the cold body.

- **Heat energy lost by hotter body= heat energy gained by colder body**

What is the heat capacity of a body?

We have seen lots of examples of specific heat capacity. The word *specific* here tells us that this is the heat energy required to increase the temperature of 1 kg of a substance by 1 K. In other words, this is specific to *1 kg* of the substance.

The **heat capacity** of a body is the defined as the energy required to raise the temperature of the given body by 1 K; the mass of the body is not considered, only the energy required to raise its temperature by 1 K.

- **heat capacit. = $Q / \Delta T$**

Let us look again at the calculation to determine the specific heat capacity of the steel bolt.

The experimental data shows us that the steel bolt lost 4410 J and its temperature fell by 75 K, or, to increase its temperature by 75 K, 4410 J of heat energy would be required.

So, if 4410 J of energy causes a 75 K rise, the heat capacity would be:

- heat capacit. = $Q / \Delta T$
- heat capacity = 4410 J / 75 .
- heat capacity = 58.8 J/K.

Note that the units do not include a mass term.

The specific heat capacity of steel is 470 J/kg K and if the heat capacity of a body is known, the specific heat capacity of the material of which it is made can be found provided the mass of the body is known.

KEY WORDS

heat capacity *the energy required to raise the temperature of a body by 1 K*

- **Specific heat capacity = heat capacity of body / mass of bod.**

  Let's try this with the steel bolt used in the last specific heat capacity experiment.

  So, for the steel bolt of mass 0.125 kg:

  - specific heat capacity = heat capacity of body / mass of body
  - specific heat capacity = 58.8 J/K / 0.125 kg
  - specific heat capacity = 470 J/kg K

Activity 7.8: Heat capacity of a ball

A plastic ball experiences a temperature rise of 10 K when 600 J of heat energy are supplied to it. Calculate the heat capacity of the ball. What else would you need to know in order to calculate the specific heat capacity of the ball?

We will now try another example:

A solid block requires 3000 J of heat energy to increase its temperature by 60 K. Calculate the block's heat capacity and use this value to calculate its specific heat capacity if the block has a mass of 50 g.

- heat capacity = $Q / \Delta T$
- heat capacity = 3000 J / 60 K
- heat capacity = 50 J/K
- specific heat capacity = heat capacity of body / mass of body
- specific heat capacity = 50 J/K / 0.05 kg
- specific heat capacity = 1000 J/kg K

Figure 7.40 *Water's high specific heat capacity makes it useful for cooling systems.*

Why is the high specific heat capacity of water so important?

If you look back at the table of specific heat capacities of different substances, you will see that water has an especially high value. We should think about this further and see if it is important.

If a substance has a high specific heat capacity, it means that a large amount of heat energy is required to bring about a rise in temperature of 1 kg water by 1K. This is important when we remember that water is widely used in industry and in internal combustion engines for cooling.

If a liquid with a low specific heat capacity was used for cooling purposes, a given mass of this liquid would receive very little heat energy before its temperature increased to its boiling point, at which stage it would no longer act as a coolant.

So, the fact that a given mass of water will receive a large amount of heat energy compared to other liquids before it boils makes it very useful for cooling.

The reverse is also true. A large mass of hot water contains a very large amount of energy. This can then be pumped around the house and as the water cools is transfers this heat energy to rooms inside the house.

Summary

In this section you have learnt that:

- The specific heat capacity of a substance is the heat energy required to raise the temperature of 1 kg of a given substance by 1 K.
- The heat capacity of a body is the heat energy required to raise the temperature of the body by 1 K.
- The factors that affect the amount of heat absorbed or liberated by a body are the temperature change, the mass of the body and the material making up the body.
- The amount of heat energy absorbed or liberated by a body can be calculated using $Q = mc\Delta T$.
- The high specific heat capacity of water is significant as water is used for cooling.
- A calorimeter is used in specific heat capacity and heat capacity experiments.

Review questions

1. Define the term "specific heat capacity of water".
2. A metal bar of mass 100 g is warmed from 20 °C to 80 °C. How much heat is absorbed by the metal bar if the specific heat capacity of this metal is 450 J/kg K.
3. In an experiment to calculate the specific heat capacity of a metal, the following results were obtained:
   Mass of metal = 300 g.
   Start temperature = 20 °C.
   End temperature = 75 °C.
   Power rating of electrical heater = 100 W.
   Time of heating = 150 s.
   Use this data to calculate the specific heat capacity of the metal.
4. A hot metal block is placed into 50 g of water in an insulated container. The water increases in temperature from 20 °C to 32 °C. The specific heat capacity of water is 4200 J/kg K. Calculate the quantity of heat energy supplied to the water from the metal block.

2.Changes of state

By the end of this section you should be able to:

1.Define the terms latent heat, latent heat of fusion and latent heat of vaporisation.

2.Solve problems involving change of state.

Figure 7.41 *No matter how long you heat it the temperature of boiling water will not go over 100 °C.*

Heating and cooling curves

If we heat a solid, its particles gain energy and begin to vibrate faster and move further apart as its temperature increases. This continues until the solid melts. Even though the solid is continuously heated as it melts, its temperature will not increase until the entire solid has **melted**.

If the heat energy the substance gains is not used to increase the average kinetic energy of its particles (the temperature does not change) what is it being used for? We can see that the same effect occurs when a liquid boils. As the liquid is being heated at its boiling point, the temperature does not increase until all of the liquid has boiled.

The graph in Figure 7.42 shows how the temperature of a solid (A) varies with time as it is heated until melts (B) and finally forms a liquid (C). This is called a **heating curve**. The liquid is heated until it boils (D) until all of the liquid changes state into a gas (E), which continues to increase in temperature as it is heated.

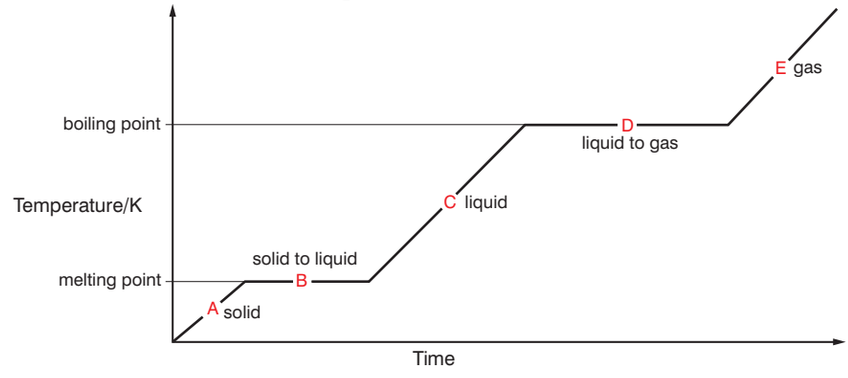


Figure 7.42 *This graph shows how the temperature of water changes as it is continuously heated.*

We cannot link the heat energy being absorbed during melting and boiling with an increase in temperature and so the heat energy appears to be hidden or latent.

If latent heat is not being used to increase the kinetic energy of the particles of a substance, what is it being used for? During a change in state, the forces of attraction holding the particles together have to be broken. This process requires energy and so, as a solid melts or a liquid boils, the heat supplied is used to separate the particles rather than to increase their kinetic energy. Consequently, the temperature of the substance does not change during a change in state. This is true for melting, boiling as well as condensing and freezing.

A similar shape is seen when the **cooling curve** of a substance is examined (Figure 7.43). When changes of state occur, the temperature remains constant as only potential energy is being lost as forces of attraction act between the particles again.

DID YOU KNOW?

The term **phase** is sometimes used instead of **state** – i.e. at room temperature and pressure water is in a liquid phase.

KEY WORDS

cooling curve *a graph showing the temperature of a substance against time as it loses heat energy and changes state*

heating curve *a graph showing the temperature of a substance against time as heat energy is applied and it changes state*

melted *when a substance has changed from a solid to a liquid state*

phase *the distinct form of a substance under different conditions e.g. solid, liquid, gas*

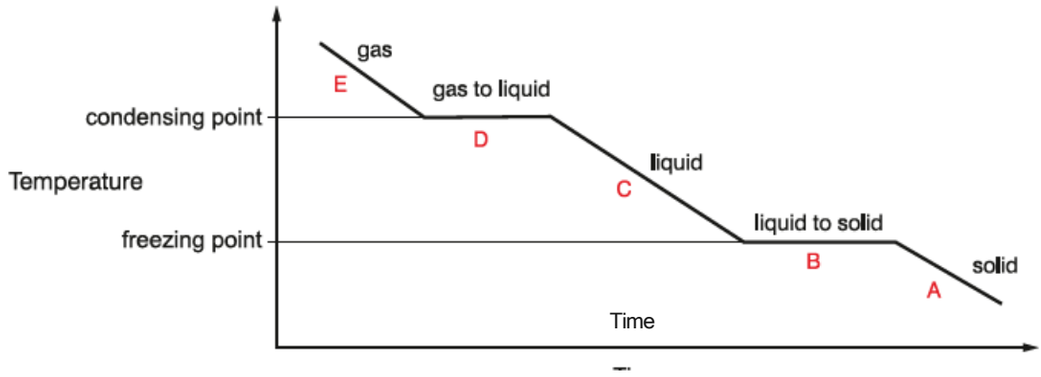


Figure 7.43 *A cooling curve for a gas cooling to eventually become a solid.*

Definition of specific latent heat

The specific latent heat *(L)* of a substance is defined as:

- **The quantity of heat energy required to change 1 kg of a substance from one state to another, at constant temperature.**

Specific latent heat has the units of J/kg. Notice again that we use the term 'specific' as this quantity is 'specific' for 1 kg of a substance.

In terms of an equation we have:

- $L = Q / m$

  L = specific latent heat in J/kg

  Q = energy required in J

  m = mass in kg

This is usually written as:

- $\boldsymbol{Q = mL}$

However, there are two changes of state to consider, liquid to gas and solid to liquid. We use two different versions of latent heat, the latent heat of fusion (melting) and latent heat of vaporisation (boiling).

Specific latent heat of fusion (L_f)

This is the quantity of heat energy required to change 1 kg of a substance from a solid to a liquid at constant temperature.

Specific latent heat of vaporisation (L_v)

This is the quantity of heat energy required to change 1 kg of a substance from a liquid to a gas at constant temperature.

Table 7.4 shows the values of the specific latent heat of fusion and specific latent heat of vaporisation for some elements and compounds.

Draw the shape of a cooling curve, starting at 90 °(, for a substance that has a boiling point of 83 °C and a melting point of 40 °C. Continue the cooling curve until the temperature reaches 25 °C.

KEY WORDS

state the distinct form of a substance under different conditions e.g. solid, liquid, gas

latent heat of fusion the amount of heat energy required to change the state of a substance from a solid to a liquid at constant temperature

latent heat of vaporisation the amount of heat energy required to change the state of a substance from a liquid to a gas at constant temperature

Worked example

1. Calculate the heat energy required to melt 1 kg of copper at its melting point.

   Heat change absorbed on melting:

   $Q = m \times L_f$ *State principle or equation to be used (from definition of specific latent heat)*

   Q = 1 kg × 209 000 J/kg *Substitute in known values and complete calculation*

   Q = 209 000 J = 209 kJ *Clearly state the answer with unit*

2. Calculate the mass of water that changes state if the water is at its boiling point and 500 kJ of energy is supplied.

   $Q = m \times L_v$ *State principle or equation to be used (from definition of specific latent heat)*

   $m = Q / L_v$ *Rearrange equation to make m the subject*

   m = 500 000 J / 2 500 000 J/kg *Substitute in known values and complete calculation*

   m = 0.2 kg *Clearly state the answer with unit*

3. Calculate the heat energy liberated when 0.025 kg of aluminium freezes at its freezing point.

   Heat change given out on freezing:

   $Q = m \times L_f$ *State principle or equation to be used (from definition of specific latent heat)*

   Q = 0.025 kg × 390 000 J/kg *Substitute in known values and complete calculation*

   Q = 9750 J = 9.75 kJ *Clearly state the answer with unit*

   In this case, this is the energy given out to the surroundings as the aluminium freezes.

Table 7.4 *Some specific latent heats of fusion and vaporisation*

Substance	Specific latent heat of fusion, L_f (J/kg)	Specific latent heat of vaporisation, L_v (J/kg)
aluminium	390 000	10 900 000
copper	209 000	4 730 000
gold	63 700	1 645 000
iron	245 000	6 080 000
water	334 000	2 500 000

Looking carefully at Table 7.4 we can see that the specific latent heat of vaporisation is always much higher than the specific latent heat of fusion. This means it takes a great deal more energy to turn a liquid to a gas than it does to turn a solid into a liquid. This is because when a substance changes from a liquid to a gas the bonds between all the molecules have to be broken apart, whereas the particles in a liquid remain bonded together.

The melting point of aluminium is 660 °C. So, at 660 °C, 1 kg of solid aluminium would require 390 000 J of heat energy to change its state into a liquid. This also means that 390 000 J of heat energy would be given out if 1 kg of liquid aluminium at 660 °C changed state into a solid.

Now we will work through some example calculations, using the specific latent heat data in Table 7.4.

Activity 7.10: Changes of state

Calculate the heat energy required to boil 100 g of aluminium at its boiling point. L_v (aluminium = 10 900 000 J/kg).

Calculate the heat energy liberated when 2 kg of water turns to a solid at its melting point. L_f (water) = 334 000 J/kg.

Now that we can use specific heat capacities and specific latent heats, we can calculate the heat energy required when substances are heated, taking into account changes in state.

- **Total energy required = energy required to increase temperature + energy required to change state**
- $Q_{total} = mc\Delta T + mL$

For example, we will now calculate the heat energy required to increase the temperature of 50 g of water from 25 °C to 125 °C.

Specific heat capacity of water = 4200 J/kg K.

Specific heat capacity of steam = 2080 J/kg K.
Specific latent heat of vaporisation of water = 2 501 000 J/kg.

We will need to calculate the heat energy required for this change in three stages:

1. Heat energy required to heat 50 g of water from 25 °C to 100 °C (ΔT = 75 °C = 75 K):
   - $Q = mc\Delta T$
   - Q = 0.05 kg × 4200 J/kg K × 75 K
   - Q_1 = 15 750 J
2. Heat energy required to boil 50 g of water at 100 °C:
   - $Q = mL_v$
   - Q = 0.05 kg × 2 501 000 J/kg
   - Q_2 = 125 050 J
3. Heat energy required to heat 50 g of steam from 100 °C to 125 °C (ΔT = 25 °C = 25 K):
   - $Q = mc\Delta T$
   - Q = 0.05 kg × 2080 J/kg K × 25 K
   - Q_3 = 2600 J.

   So, the total amount of heat energy required for this process:
   - $Q = Q_1 + Q_2 + Q_3$
   - Q = 15 750 J + 125 050 J + 2600 J
   - Q = 143 400 J = 143.4 kJ

Experiment to determine the specific latent heat of fusion of ice

The apparatus in Figure 7.44 can be used to determine the specific latent heat of fusion, L_f, of ice.

The copper calorimeter and stirrer is weighed before being half filled with water. The mass of the water present in the calorimeter is then determined before the water is heated to at least 10 °C above room temperature. Small quantities of ice are then added to the water, while stirring, until the temperature is below room temperature and all the ice has melted. The mass of the calorimeter, stirrer and water is then determined to find out the mass of ice added.

We will use the experimental data obtained using this method to determine the specific latent heat of fusion of ice.

Mass of calorimeter = 0.15 kg.
Mass of water = 2.00 kg.
Mass of ice added = 0.60 kg.
Start temperature of ice = –10 °C.
Start temperature of water = 49 °C.
Final temperature of water = 20 °C.
Specific heat capacity of water = 4200 J/kg K.

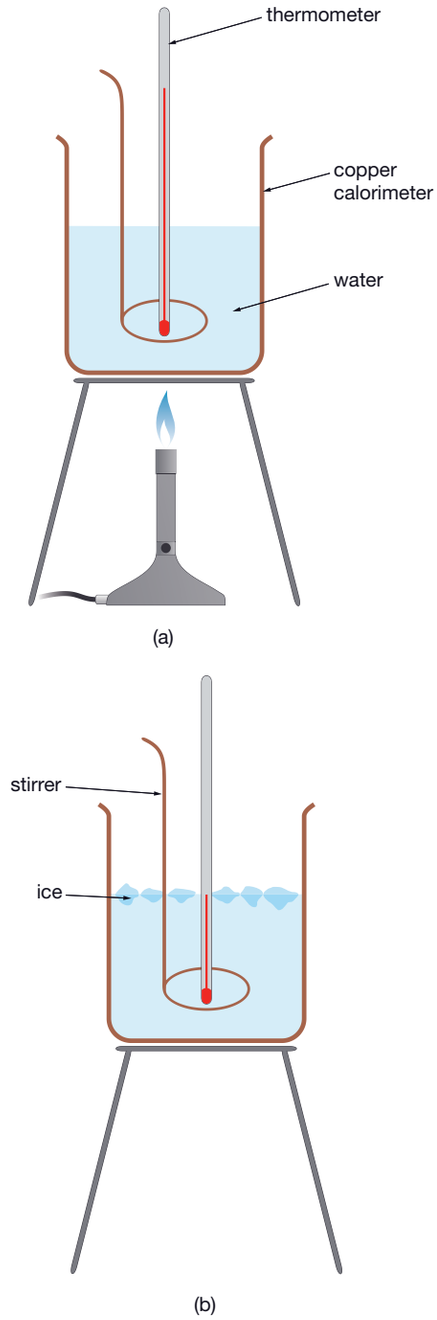


Figure 7.44 *A simple experiment to determine the specific latent heat of fusion of ice.*

Specific heat capacity of ice = 2100 J/kg K.
Specific heat capacity of copper = 420 J/kg K.

We will use the principle here that the heat energy lost from the water and the calorimeter will go to warming and melting the ice and then warming the cold water produced when the ice melts.

Heat energy absorbed by ice and cold water = Heat energy liberated from calorimeter/stirrer and water and water

1. **Heat energy lost by calorimeter/stirrer and water:**
   - ΔT = 49 °C – 20 °C = 29 °C = 29 K

   Heat energy lost by calorimeter:
   - $Q_{\text{lost calorimeter}} = m_{\text{calorimeter}} c_{\text{calorimeter}} \Delta T$
   - $Q_{\text{lost calorimeter}}$ = 0.15 kg × 420 J/kg K × 29 K = 1827 J

   Heat energy lost by water:
   - $Q_{\text{lost water}} = m_{\text{water}} c_{\text{water}} \Delta T$
   - $Q_{\text{lost water}}$ = 2.00 kg × 4200 J/kg K × 29 K = 243 600 J

   Total heat energy lost by calorimeter/stirrer and water = 243 600 + 1827 = 245 427 J.

2. **Heat energy absorbed by ice and water in warming from –10 °C to 20 °C:**
   Heat energy require to warm ice from –10 °C to 0 °C (ΔT = 0 °C – 10°C = 10 °C = 10 K).

   - $Q_{\text{ice (–10–0)}} = m_{\text{ice}} c_{\text{ice}} \Delta T$
   - $Q_{\text{ice (–10–0)}}$ = 0.60 kg × 2100 J/kg K × 10 K = 12 600 J Heat energy required to melt 0.60 kg of ice:
   - $\boldsymbol{Q_{\text{melt ice}} = mL_f}$
   - $\boldsymbol{Q_{\text{melt ice}}}$ **= 0.60 kg** × $\boldsymbol{L_f}$

   Heat energy require to warm cold water from 0 °C to 20 °C (ΔT = 20 °C – 0 °C = 20 °C = 20 K).

   - $Q_{\text{water (0–20)}} = m_{\text{ice}} c_{\text{water}} \Delta T$
   - $Q_{\text{water (0–20)}}$ = 0.60 kg × 4200 J/kg K × 20 K = 50 400 J Heat energy absorbed by ice and cold water= Heat energy liberated from calorimeter/stirrer and water

     12 600 J + 0.6L_f + 50 400 J = 245 427 J

   - 0.6L_f = 182 427 J
   - L_f = 182 427 J / 0.6
   - L_f = 304 045 J/kg = 304 kJ/kg

Summary

In this section you have learnt that:

- The terms specific latent heat of fusion and specific latent heat of vaporisation relate to the energy required to melt and vaporise 1 kg of a substance at constant temperature.
- It is possible to calculate the energy (Q) required or liberated on a change of state for a given mass, m, of a substance using the equation $Q = mL_f$ or $Q = mL_v$.

Review questions

1. Calculate the heat energy required to melt 10 g of copper at its melting point. (L_f for copper = 209 000 J/kg).
2. Calculate the heat energy required to melt 1.2 kg of gold at its melting point. (L_f for gold = 63 700 J/kg).
3. Calculate the heat energy liberated when 75 g of iron freezes at its freezing point. (L_f for iron = 245 000 J/kg).
4. Define the term specific latent heat of fusion of magnesium.
5. Calculate the heat energy required to increase the temperature of 0.1 kg of water from 10 °C to 150 °C.
   Specific heat capacity of water = 4200 J/kg K.
   Specific heat capacity of steam = 2080 J/kg K.
   Specific latent heat of vaporisation of water = 2 500 000 J/kg.

End of unit questions

1. Write a paragraph explaining the difference between the heat energy in a substance and the substances' temperature.
2. Explain, with reference to the appropriate laws of thermodynamics and particle movement, what happens when a cold object is in thermal contact with hot object.
3. Calculate the increase in length of an iron pipeline that is 30.00 m long at 20 °C when it is warmed to 45 °C. $\alpha_{iron} = 1.1 \times 10^{-5}\ K^{-1}$.
4. Calculate the increase in volume of ethanol that has a volume of $2.5 \times 10^{-4}\ m^3$ at 25 °C when it is warmed to 45 °C. Explain why the apparent expansion will be less than this calculated real expansion. $\gamma_{ethanol} = 75 \times 10^{-5}\ K^{-1}$.
5. In an experiment to calculate the specific heat capacity of a metal, the following data were obtained. Use the data to calculate the specific heat capacity of the metal.

Chapter 3: Electricity and Magnetism

Lesson 1: Electrostatics

Learning Objectives

- Describe and explain the charging processes: charging by rubbing, conduction and induction.
- Describe the distribution of charge on a conductor of variable shape.
- Explain how lightning is formed.
- Describe the use of a lightning rod.
- Describe how equipment works using electrostatics principles.
- Describe hazards and uses of electrostatics.

1. Electric charge

By the end of this section you should be able to:

- State the law of conservation of charge.
- Describe and explain the charging processes: charging by rubbing, conduction and induction.
- Perform an experiment to charge an electroscope by conduction and by induction.
- Describe the distribution of charge on a conductor of variable shape.
- Explain how lightning is formed.
- Describe the use of a lightning rod.
- Describe how equipment works using electrostatics principles.
- Describe hazards and uses of electrostatics.

KEY WORDS

electrostatics *the build up of electric charge on the surface of objects*

charge *an electric charge, positive, negative, or zero, found on the elementary particles of matter*

atoms *the smallest components of an element having the chemical properties of that element*

positive charge *having a deficiency of electrons*

negative charge *having a surplus of electrons*

Activity 2.1: Recalling other physical quantities that are conserved

In a small group, spend one minute writing a list of physical quantities that you have met that follow a conservation law. Hint: think back to Unit 1.

Electrostatics in everyday life

Have you ever felt a small electric shock when you have touched a metal door handle, or heard a crackling sound as you pull a woollen article of clothing over your head? Both these arise because of **electrostatics** – electric **charge** is transferred from one material to another. In this section you will learn more about this, its applications and its hazards.

What is electric charge?

All objects are made up from tiny building blocks called atoms. Individual atoms are made up of particles that possess both mass and electric charge. A material such as Perspex is made up from a

huge number of atoms, but although we are well aware of its mass, it does not seem to possess any charge at all. Why is this?

Every bit of mass in every atom adds up so that we feel its mass. In every single atom, however, the two sorts of electric charge cancel one another out: there is the same number of positive charges as negative ones, so we do not notice any overall charge. An 'uncharged' piece of Perspex contains a vast number of charges, but the numbers of the two kinds are equal. Charging it up involves upsetting the balance between positive and negative charges. Adding or removing a few more electrons would do this.

The elementary charge, $e = -1.6 \times 10^{-19}$ C, is the amount of charge on an electron (negative charge) or on a proton (positive charge). To find the total charge Q, you use $Q = N \times e$, where N is the number of electrons.

KEY WORDS

law of conservation of charge *the total electric charge of a system remains constant despite changes inside the system*

DID YOU KNOW?

The fact that some materials (such as amber) become charged after they are rubbed has been known for thousands of years. The Greek word for amber, *ήλεκτρον* (electron), is the source of the word 'electricity'.

2.Conservation of charge

Electric charge cannot be created or destroyed, only transferred. The number of **positive** charges on an atom of a substance is the same as the number of **negative** charges on the atom. The atom is neutral. The charges from some atoms can be removed from the atom and transferred to another material in the process known as charging but the overall number of positive and negative charges does not change. This is the **law of conservation of charge**.

Like charges will repel each other (for example, two positive charges will repel each other, two negative charges will repel each other). Unlike charges will attract each other (so a positive charge will attract a negative charge and a negative charge will attract a positive charge).

3.Charging materials by rubbing

It is possible to charge some materials by rubbing them. When you rub a piece of Perspex, for example, some of the charge is transferred from the surface of the Perspex to the material you are using to rub it, and so the overall charge on the Perspex becomes unbalanced.

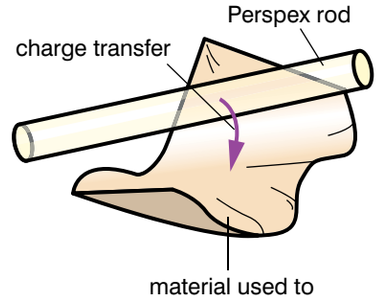


Figure 2.1 *Rubbing Perspex transfers charge from the surface of the Perspex to the material you are rubbing it with.*

Activity 2.2: Testing how charged bodies attract or repel one another

Take a bar of Perspex and pivot it. One way is to suspend it by a nylon thread (cotton, if the slightest bit damp, may allow the charges to leak off too readily). Charge the suspended rod by rubbing it. Bring a second charged Perspex rod up to the first one, and you will see the first one swing away.

Now charge the bar of Perspex again. Bring a charged rod of Polythene up to the suspended rod, and you will see the suspended rod moving towards it. What does this tell you about the charges on the Perspex and the Polythene?

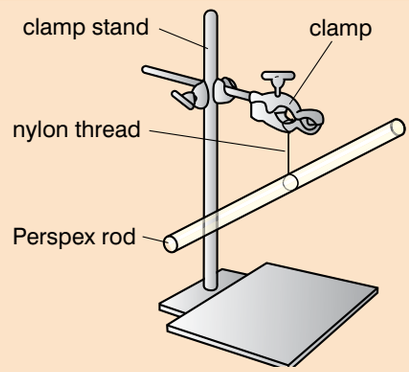


Figure 2.2 *Apparatus to test attraction or repulsion between charged bodies.*

Activity 2.3: The electrostatic attraction of water

Adjust a tap so a continuous but gentle stream of water is falling from it. Rub a plastic comb on your sleeve, and bring it up to the side of the water column. Describe and try to explain what happens.

Figure 2.3 *The electrostatic attraction of water.*

Activity 2.4: Hanging balloons using electrostatic attraction

Inflate a balloon and then briskly rub one side of it on your hair. Place the surface that you have rubbed towards a wall or door and release it when it appears to be sticking. What can you say about the nature of the surfaces to which the balloon sticks?

Figure 2.4 *Rubbing a balloon on hair.*

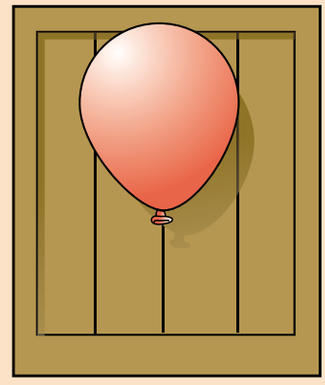


Figure 2.5 *A balloon sticking to a surface.*

Activity 2.5: Charging an electroscope by conduction

Charge a metal sphere. Bring it into contact with an electroscope. What happens? Draw a diagram to explain your results.

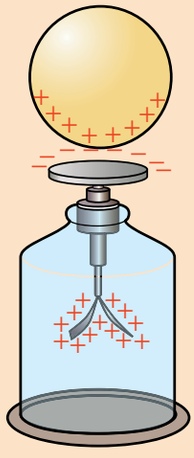


Figure 2.7 *Charging an electroscope by conduction.*

4.Charging materials by conduction

If a charged object is brought into contact with an uncharged (neutral) object, then the neutral object will become charged by **conduction**. Charge flows from the charged object to the neutral object. (This is what happens when you feel an electric shock when you touch a metal door handle – the door handle has become charged and the charge travels by conduction to you, a neutral object!)

Figure 2.6 *Charging a person by conduction.*

5.Charging by induction

Figure 2.8 shows a different way of charging a body known as **induction**.

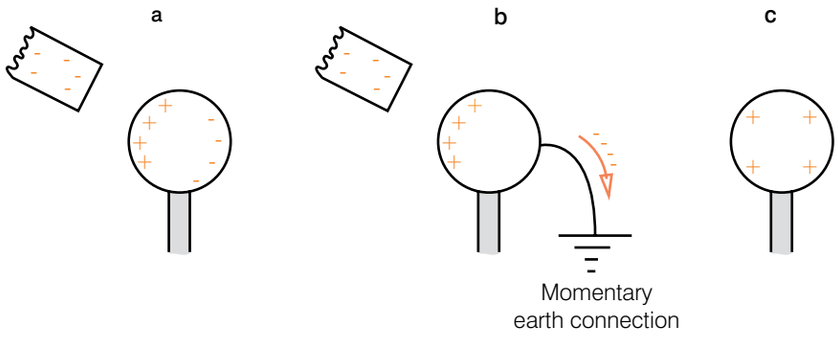


Figure 2.8 *Charging by induction.*

In Figure 2.8a, a negatively charged rod is brought near to a neutral metal sphere held on an insulating support. The negative charges in the metal sphere move to the side of the sphere furthest from the rod, leaving the side of the sphere nearest to the rod positively charged. In Figure 2.8b, while the rod is still nearby, you touch the sphere for a moment then let go. The negative charges now escape further, through you down to earth. In Figure 2.8c, the negative rod is taken away. The negative charges are unable to return from earth, so the sphere is left positively charged. These charges now distribute themselves evenly.

Activity 2.7: Charging by induction

Tear a sheet of newspaper into small pieces, approximately half a centimetre in diameter. Place the bits under a glass plate that is supported by two books (see Figure 2.10).

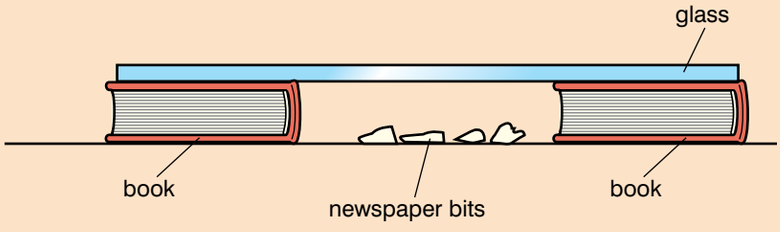


Figure 2.10 *Charging by induction.*

Rub the glass vigorously with a piece of silk and notice how the paper jumps up to the glass. In a small group, discuss what you see. Answer these questions:

- What attracts the paper to the glass?
- Why does it not stay attached to the glass?

Activity 2.6: Charging an electroscope by induction

From the information given about charging by induction, devise and carry out an experiment to charge an electroscope by induction. Draw diagrams to explain what happens.

DID YOU KNOW?

An electroscope is used to detect the presence and magnitude of electric charge. The British physician William Gilbert invented the first electroscope, which was called the *versorium*, around 1600. This was the first electrical measuring instrument.

Figure 2.9 *William Gilbert.*

KEY WORDS

conduction *the flow of charge from a charged object to an uncharged (neutral) object*

induction *a redistribution of electrical charge in an object, caused by the influence of nearby charges*

Figure 2.12 *Dr Robert Jemison Van de Graaf and Dr Karl Taylor Compton with a generator.*

KEY WORDS

conductor *a material that contains movable electric charges*

DID YOU KNOW?

The American physicist Robert J. Van de Graaf developed the **Van de Graaf generator** at Princeton University. The first model was demonstrated in October 1929 and used a silk ribbon bought at a local shop as the belt to carry the charge. Accelerating electrons are now used to sterilise food and process materials.

Activity 2.8: Using a Van de Graaf generator to make a fluorescent tube glow

Darken the room. Turn on the generator, and slowly move a fluorescent tube towards the generator. Why does the area between the tube, your hand and the generator glow?

6.Distribution of charges on the surfaces of conductors

Charges on the surface of a **conductor** are able to flow and spread themselves out evenly. The charges on the surface are all the same type and so they repel each other and move as far apart as possible. The result is that they move outwards until they reach the surface. They can get no further than that unless the insulation of the air breaks down to enable them to escape.

The charges on a sphere will distribute themselves evenly over its surface (see Figure 2.11a). If, however, the body has an irregular curvature (see 2.11b) the charge tends to cluster round the sharply curved parts, with the areas of gentle curvature storing far less of the charge. If the metal body has an actual point on it, the accumulation of charge there will be enormous, and the insulation of the air is more likely to break down there so that all the charge is able to escape from the surface. We refer to this is as the action of points.

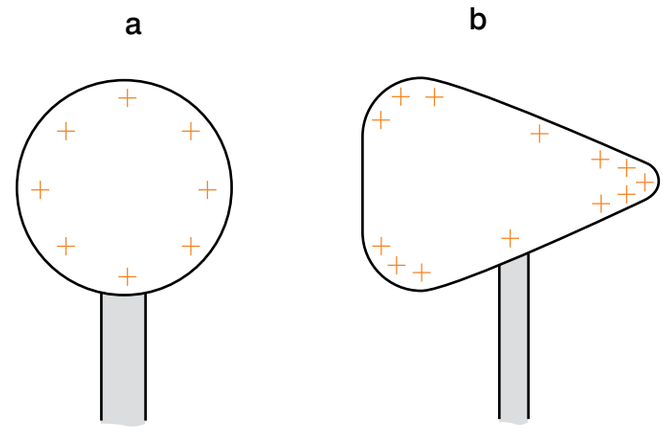


Figure 2.11 *Distribution of charges on conductors.*

As all the charge sits on the outside, it makes no difference whether the conductor, or metal object, is solid or hollow.

Thunderstorms

Scientists believe that thunderstorms in Ethiopia may sometimes trigger hurricanes as far away as across the Atlantic Ocean.

Thunderclouds form as a result of convection currents (warm air rises and cooler air goes down). Air contains water vapour which we often see as clouds.

As the air cools down, the water vapour in the air condenses. The tiny drops of water that condense experience very rough conditions and, in the process, tend to become charged. The mechanism by which this happens is more than just charging by friction, and we are still not entirely sure of all the details.

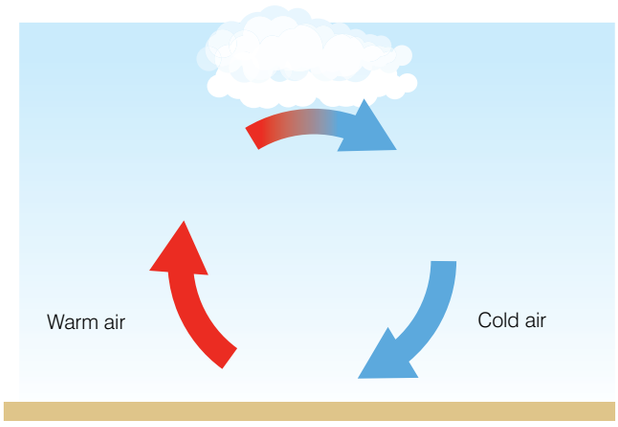


Figure 2.13 *A convection current.*

DID YOU KNOW?

Fluorescent lamps work because there is a flow of electrical current through a gas (usually argon) which has been charged.

KEY WORDS

Van de Graaf generator
a hollow sphere on which an electrostatic charge is accumulated using a moving belt

How lightning is formed

Charged water drops of one sign (positive or negative) may tend to collect in one part of the cloud. The result is lightning: sparks that jump often from one cloud region to another, but sometimes from the cloud to earth.

When a thundercloud passes overhead, if the base of the cloud is positively charged, then it will attract negative electrons in the earth underneath it (see Figure 2.15). Note that the charge carried by the base of the cloud may be negative or positive but the figure assumes that it is positive.

Should a spark jump to earth it will tend to jump the shortest air gap, such as to trees and tall buildings.

Figure 2.14 *Lightning.*

Lightning rods

To help to discharge the cloud safely, a lightning conductor is often fitted to the top of a building. This is a metal bar, pointed at the top, the other end of which is buried firmly in the ground. The idea is that it should get struck before the building and conduct the surge of charge harmlessly to the earth.

Figure 2.15 *How lightning occurs.*

Figure 2.16 *A lightning conductor will conduct electric charge from lightning safely away from a building to the earth.*

Activity 2.9: How might tall buildings be protected from damage by lightning?

In a small group, discuss how tall buildings might be protected from damage by lightning.

Activity 2.10: Why is the inside of a car safe in a storm?

Discuss with a partner why the inside of a car is a safe place to shelter during a storm.

Safety in storms

It is very rare that people are struck by lightning, and certainly you will not be struck while you are inside a car or an aeroplane – the metal shell around you would divert charge away. You are advised not to shelter under trees because they are particularly likely to get struck, and the spark could jump from the lower branches to you.

Activity 2.11: Design a safety poster

Use the information above to design a poster to show younger children how to stay safe during a thunderstorm.

7.Applications of electrostatics

Paint sprays

Electrostatic spray painting is often used to paint car body panels and bike frames. An electrostatic spray painting device sprays charged particles of paint through the air onto a surface. These devices are usually used for covering large surfaces with an even coating of liquid. They can be either automated or hand-held and have interchangeable heads to allow for different spray patterns.

There are three main technologies for charging the fluid (liquid or powders):

- *Direct charging* – an electrode (a charged metal plate) is placed in the paint supply reservoir or in the paint supply conduit.
- *Tribo charging* – this uses the friction of the fluid, which is forced through the barrel of the paint gun. It rubs against the side of the barrel and, in the process, builds up an electrostatic charge.
- *Post-atomisation charging* – the charged fluid comes into contact with an electrostatic field (see page 54) after leaving the spray head. The electrostatic field may be created by electrostatic induction, or by one or more electrodes (electrode ring, mesh or grid).

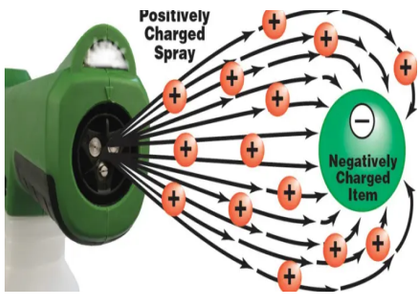


Figure 2.17 *An electrostatic spray gun in use.*

The charged paint particles repel each other, and so they spread themselves evenly as they leave the spray head. The object being painted is charged in the opposite way (for example, if the paint is positively charged, the object is negatively charged), or grounded. The paint is then attracted to the object, which gives a more even coat than the alternative technique, wet spray painting, and increases the percentage of paint that actually sticks to the object. This method also means that paint covers hard to reach areas. The object is then baked so that the paint sticks properly when the paint powder particles turn into a type of plastic.

Chimney filters

Smoke such as that from a large chimney contains many tiny solid particles, mainly soot. To help to keep the air clean, before reaching the chimney this smoke may be passed between two large upright charged metal plates. In a similar way to how a plastic comb attracts small pieces of paper, the solid specks in the smoke are trapped by sticking to one or other of the plates. From time to time the plates are shaken or scraped to remove the thick layer of dust that has formed on them.

Photocopiers

1. The surface of a cylindrical drum is charged using electrostatics. The drum has a coating of a material that conducts electricity when light shines on it.
2. A bright lamp shines light on the original document, and the white areas of the original document reflect the light onto the surface of the charged drum. The areas of the drum that are exposed to light (those areas that correspond to white areas of the original document) conduct electricity so they discharge. The areas of the drum that are not exposed to light (those areas that correspond to black portions of the original document) remain negatively charged. The result is a stored electrical image on the surface of the drum. (In digital machines, the original document is scanned and digitised and a laser is employed to discharge the drum in a similar fashion.)
3. The toner is positively charged. When it is applied to the drum to produce the image, it is attracted and sticks to the areas that are negatively charged (black areas), just as the balloon in Activity 2.4 stuck to the wall or door.
4. The toner image on the surface of the drum is transferred from the drum onto a piece of paper, which is charged more negatively than the drum.
5. The toner is melted and bonded to the paper by heat and pressure rollers.
6. The drum is wiped clean with a rubber blade and light is used to remove the charge.

Activity 2.12: Design a model chimney filter

In a small group, design a model of chimney filter.

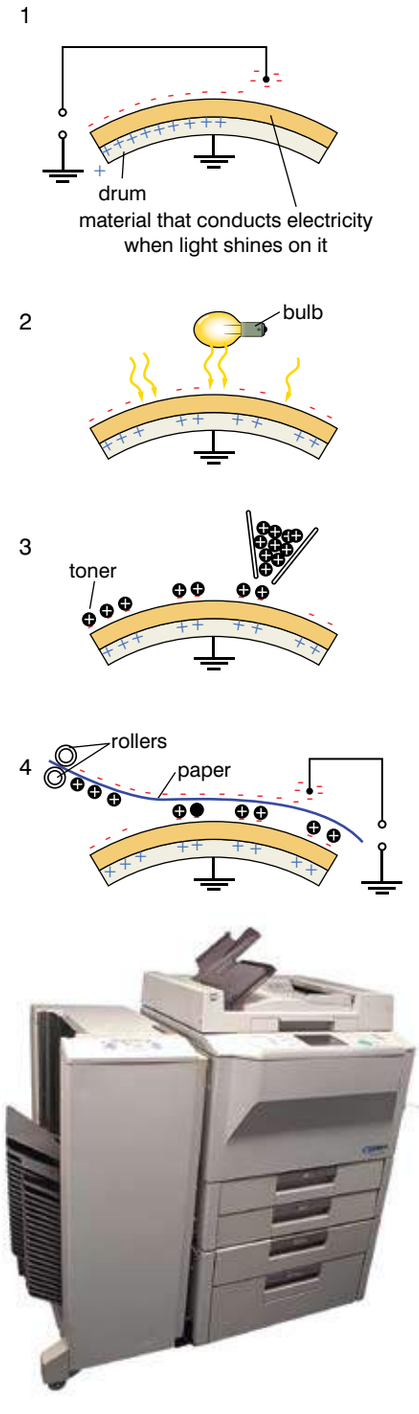


Figure 2.18 *How a photocopier works.*

When does electrostatic charge cause problems?

Electrostatic charge can be a problem or hazard when there is a sudden discharge. Electrostatic discharge gives a sudden current of electricity between two objects. On page 56, we found out that lightning (which is a form of electrostatic discharge) can be dangerous, especially if you are outside in a thunderstorm. If there is an electrostatic discharge in an area containing liquids or materials which catch fire easily (flammable materials), then the discharge may cause a fire. However, perhaps the main area where electrostatic discharge causes problems is in electronics, where circuits may be damaged by unwanted flows of charge. To protect circuits against this, antistatic devices have been developed. These include antistatic bags, clothing, mats and wrist straps.

Why is static electricity more apparent when the air is dry?

You notice static electricity much more when the air is dry because dry air is a relatively good electrical insulator, so if something is charged the charge tends to stay. When there is moisture in the air, the charged water molecules can remove charge quickly from a charged object.

Activity 2.13: Mind mapping uses and hazards of electrostatics

In a small group, discuss what you have learnt about the uses and hazards of electrostatics. You may like to do some research to find out how electrostatics is used in manufacturing, or more about antistatic devices. Make a summary of your discussion and research using mind maps such as the one in Figure 2.19.

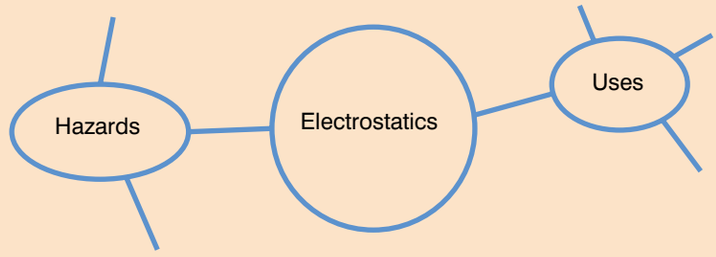


Figure 2.19 *Mind map to show uses and hazards of electrostatics.*

Prepare a presentation for the rest of your class.

Summary:In this section you have learnt that:

- Charge cannot be created or destroyed, it can only be transferred from one object to another. This is the law of conservation of charge.
- It is possible to charge objects by rubbing, conduction and induction.
- An electroscope can be charged by conduction and induction.
- The distribution of charge on a conductor with a constant surface such as a sphere is uniform (the charges spread out evenly). However, if there is a point on the surface, the charge will form clusters around the point.

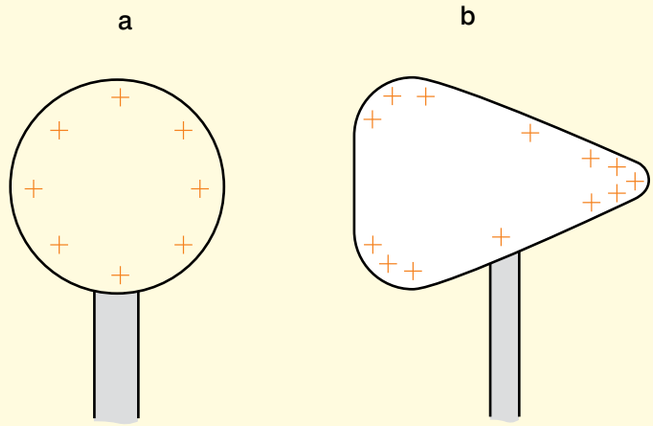


Figure 2.20 *Distribution of charge on conductors* *f* *arious shapes.*

- Lightning is formed when charged water drops of one sign (positive or negative) collect in one part of the cloud. There is electrostatic discharge (in the form of sparks) which jump often from one cloud region to another, but sometimes from the cloud to earth.
- A lightning rod may be used to protect tall buildings from lightning. This is a metal rod with a point at the top. It attracts the charge from the lightning and conducts it safely to earth rather than it going through the building and causing damage.
- Electrostatics is used in equipment such as paint sprayers and photocopiers. However, it can be a problem in some situations, such as where there are flammable materials, which may catch fire if there is an electrostatic spark, or in sensitive electronic devices. There are various antistatic devices available, which can be used in different situations.

Review questions

1. Why do charged objects soon lose their charge on a damp day?
2. All atoms that make up matter contain tiny charges, both positive and negative. How is it that most objects we meet in daily life appear to be uncharged?
3. Copy Figure 2.21 and show on it how the charges in the uncharged object will distribute themselves in the presence of the positively charged rod.

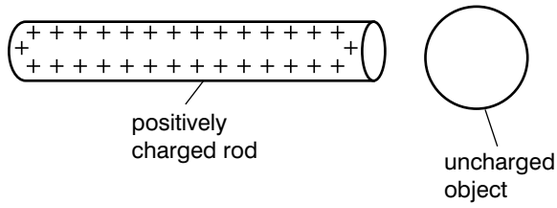


Figure 2.21

4. Why will a plastic comb that has been rubbed attract small pieces of paper?
5. a) Why do plastic objects frequently become covered with dust?
   b) Why is it not necessarily a good idea to rub such objects with a duster?
6. When car manufacturers apply paint to new cars they usually spray it in very fine droplets that are deliberately charged. Why do they do this?
7. a) An inflated toy balloon can be rubbed on your clothing then stuck on the ceiling. Explain why.
   b) Suggest why the balloon eventually drops off the ceiling.

Lesson 2 Current Electricity

Contents

Section	Learning Objectives
1. Electric current	• Define electric current and its SI unit. • Explain the flow of electric charges in a metallic conductor. • Calculate the number of electrons that pass a point in a given length of time when the current in the wire is known.
2. Combinations of resistors	• Identify combinations of resistors in series, parallel and series–parallel connection. • Derive an expression for the effective resistance of resistors connected in series. • Derive an expression for the effective resistance of resistors connected in parallel. • Calculate the effective resistance of resistors connected in series. • Calculate the effective resistance of resistors connected in parallel. • Calculate the current through each resistor in simple series, parallel and series–parallel combinations. • Calculate the voltage drop across each resistor in simple series, parallel and series–parallel connections.

1. Electric current

By the end of this section you should be able to:

- Define electric current and its SI unit.
- Explain the flow of electric charges in a metallic conductor.
- Calculate the number of electrons that pass a point in a given length of time when the current in the wire is known.

KEY WORDS

ampere *SI unit of electric current*

1.1 Electric current and its SI unit

An electric current is a flow of charge. Comparing it with water, a small current is like a trickle passing through a pipe; a really large current is like a river in flood.

The rate of flow of electric charge – that is, the electric current – is measured in amperes (A). The **ampere** is one of the fundamental units of the SI system. This means that the size of the ampere is not fixed in terms of other units: we simply compare currents to a 'standard ampere'.

Household appliances, such as toasters and kettles, run on a current of a few amperes. An ampere is quite a sizeable flow of charge; however, especially in electronic circuits, we also often deal in milliamperes (mA, thousandths of an ampere, 10^{-3}A) or even microamperes (μA, millionths of an ampere, 10^{-6} A).

If charge flows at a rate of one ampere, and continues to flow like that for a second, then the total amount of charge that has passed is one coulomb. This is how the size of the coulomb is fixed: in terms of the ampere and the second.

In practice it is probably best to picture coulombs of charge flowing at a rate of so many amperes. A current of 3 A, for example, is a flow rate of 3 coulombs of charge every second (3 C/s). With that

current, therefore, it should be obvious that in 10 seconds a total of 30 C of charge will pass, or that to supply 12 C of charge the current must flow for 4 seconds.

You are strongly advised to understand the previous paragraph, rather than just remember. When you need the formula linking amperes, coulombs and seconds, however, it is:

$\boldsymbol{Q = It}$ Units

Q coulombs (C)
I amperes (A)
t seconds (S)

Worked example 3.1

120 C of charge passes in 1 minute. What is the current?

$$I = \frac{Q}{t} = \frac{120\ \text{C}}{60\ \text{s}} = 2\ \text{C/s} = 2\ \text{A}$$

KEY WORDS

galvanoscope *an instrument for detecting the presence of an electric current*

electron *a negatively charged particle that orbits round the nucleus of an atom*

ion *an atom bearing unequal numbers of electrons and protons*

conduction electrons *electrons in the conduction band of a solid, free to move under the influence of an electric field*

insulator *a material that resists the flow of electric charge*

In the formula, Q stands for the quantity of charge that passes when a current I flows for a time t.

Worked example 3.2

How long will a current of 5 A take to pass 100 C of charge?

$Q = It$, so $t = \frac{Q}{I} = \frac{100\ \text{C}}{5\ \text{A}} = 20\ \text{s}$

Activity 3.1: Making a current tester

You can use a compass needle to make a simple **galvanoscope** (an instrument that detects the presence of an electric current). Simply wrap a few turns of thin insulated copper wire around a compass, connect the ends as shown in Figure 3.1 to a 1.5 V cell and a 1.5 V light bulb. When the wires
are connected as shown, the bulb will light and the compass needle will move around in one direction. If you want to investigate this further, disconnect the cell, turn it round and reconnect it the other way round. The needle should move in the opposite direction.

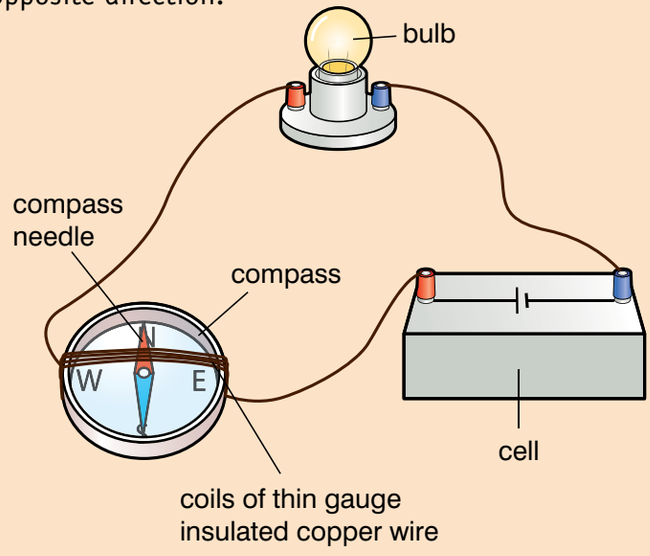


Figure 3.1 *Current tester.*

1.2 What are the charges that flow round a circuit?

To answer this question we must first look in rather more detail at the structure of an atom. All its positive charges are located in the central part, the nucleus. Each chemical element has a different number of positive charges in its nucleus, from one to nearly 100. Take copper as an example: it has 29 positive charges in the nucleus.

This means that an uncharged copper atom must have 29 negative electrons as well. They orbit round the nucleus, and each one has its own path. The first two electrons orbit in the innermost shell, the next eight fill the second shell and the following 18 just complete the third shell. That leaves a solitary electron as the first member of a new fourth shell.

It is this electron that accounts for the behaviour of copper as a conductor. It is comparatively easy to remove this electron, changing an uncharged copper atom into what we call a copper **ion** with an overall single positive charge (Figure 3.2).

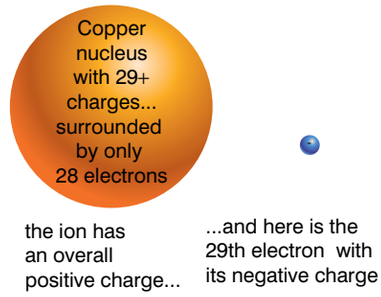


Figure 3.2 *The copper atom can be changed into a positive copper ion plus a negative electron.*

In a copper wire the atoms are packed close together just as in any other solid. More precisely, what are packed together are the positive ions; the complete atoms except for those single outer electrons. They are there as well, so the metal as a whole is uncharged.

In each copper atom 28 of the electrons are still firmly bound in orbit around their nucleus, fixed in its place in the solid. The 29th electrons we call the **conduction electrons**. They remain trapped within the metal as a whole, but otherwise are free to drift about inside it.

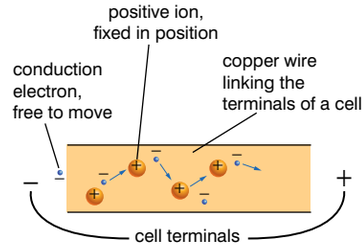


Figure 3.3 *The unattached electrons in the copper wire move when it is connected to the terminals of a cell.*

They are the charges that move when an electric current flows down the wire. Being negative they will be repelled from the cell's negative terminal and attracted to the positive one (Figure 3.3).

All metals have these extra one or two outer electrons, so they will all conduct electricity. With most other elements, every electron without exception is tightly bound to its own nucleus, so most other elements are **insulators**.

Conventional current

Electric currents have been investigated since the first cell was invented around 1800. It soon became obvious that charges were flowing round the circuit, but there was no way of telling whether they were positive charges going one way or negative charges going the other.

It was agreed to picture the flow as positive charges repelled from the positive plate, and this was referred to as conventional current.

When the electron was discovered in 1895, it was realised that the guess had been wrong. However, so firmly fixed was the idea that even today we still mark on our circuits the 'conventional current' flowing from the positive pole of the cell to the negative.

You must understand, however, that the electron flow is really negative charge going the opposite way.

Activity 3.2: Testing conductivity

Not all materials have conduction electrons in their structure, so not all materials will conduct electricity. Use your simple galvanoscope (from Activity 3.1) to test which materials conduct electricity and which do not.

Connect your galvanoscope to a copper rod as shown in Figure 3.4a. Observe that the compass needle moves, showing that the rod conducts electricity, allowing an electric current to flow. Connect your galvanoscope to a glass or plastic rod, as in Figure 3.4b. Observe that the compass needle does not move, showing that the rod does not allow an electric current to flow. Repeat this for a selection of different materials of the same size, and rank your materials in order of the size of the deflection of the compass needle.

You will see that some materials conduct electricity better than others, while some materials do not conduct electricity at all.

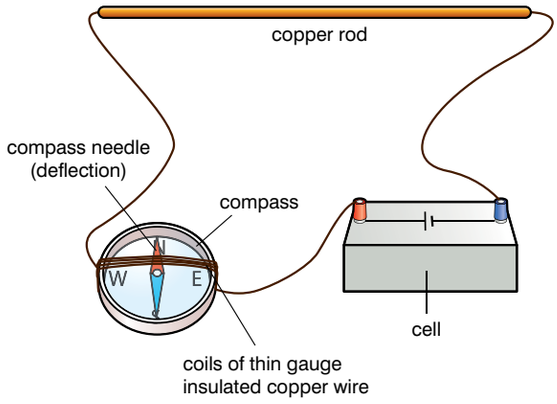


Figure 3.4a

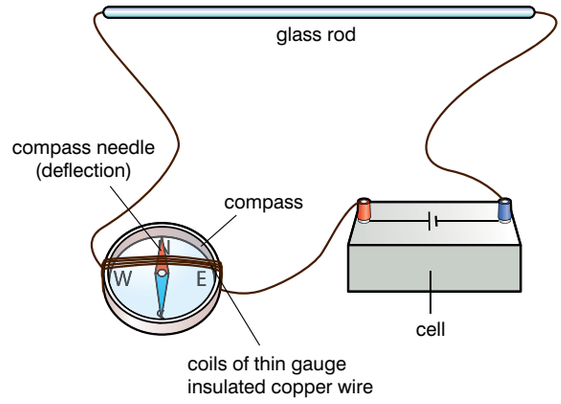


Figure 3.4b

KEY WORDS

drift speed *the average speed that an electron attains in an electric field*

1.3 Calculating the number of electrons that pass a point at a given length of time when the current in the wire is known

The wire shown in Figure 3.5 carries a current of I amperes. There are n free electrons per cubic metre of wire. Each electron carries a charge of e coulombs, the cross-sectional area of the wire is A square metres and the average **drift speed** of the electrons in this material is v metres per second.

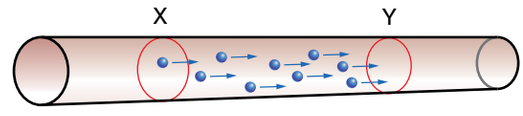


Figure 3.5

Assuming that it takes t seconds for an electron to pass from X to Y (and using the formula distance = speed × time), distance X–Y is vt metres. The volume of wire between X and Y is therefore Avt cubic metres.

The number of electrons between X and Y is therefore $nAvt$.

As each electron carries a charge of *e* coulombs, the total charge passing point Y in *t* seconds is *nAevt*.

The current, *I*, is the charge per second:

$$I = \frac{n.A.e.v.t}{t} = n.A.e.v$$

As the charge carried by an electron is known to be 1.6×10^{-19} C, the number of electrons passing through a wire can be calculated if the current in the wire is known.

Worked example 3.3

A current of 5 A is flowing through a 2 mm diameter wire. The drift speed of electrons in the wire is 10^{-5} m/s and the charge on the electron is 1.6×10^{-19} C. Calculate the number of electrons passing a point in the wire in a second.

The cross sectional area (*A*) of the wire is πr^2.

$$A = \pi \times \frac{d}{2} \times \frac{d}{2} = \pi \times 0.001 \times 0.001 = \pi \times 10^{-6}\ m^2.$$

Using $n = \frac{I}{Aev}$

$$\text{number of electrons} = = \frac{5}{\pi \quad 10^{-6} \times 1.6 \quad 10^{-5}} \times \times \times 10^{-19}$$

9.947×10^{29}

1.4.Study of electric charges in a metallic conductor

In Activities 3.1 and 3.2 you detected an electric current. Where does this electric current keep coming from and where does it keep going to? To start investigating the answer to this, look at Figure 3.6 which shows the top view of a corridor that ends up where it started. The whole place is filled with people who form a kind of endless queue. The door is shut, which means that nobody in the queue can move forward. Once the door is opened (once the circuit is switched on), all the people in that corridor can immediately start moving. They can keep circulating round and round the corridor, thus forming a continual current.

Of course when the door was opened the people filling the corridor found themselves free to move round the circuit, but they did not have to do so. Likewise charges will not circulate to give a current unless there is something that keeps 'pumping' them round. That is the job of the cell, as we shall see below.

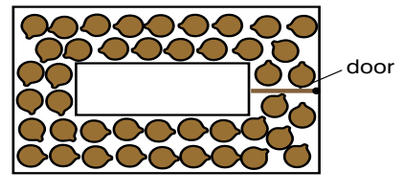


Figure 3.6

1.5.Electric energy from chemicals

KEY WORDS

electrochemical cell *a device capable of deriving electrical energy from chemical reactions*

electrodes *conductors used to make contact with part of a circuit*

electrolyte *a solution that conducts electricity*

electromotive force *a source of energy causing current to flow in an electrical circuit*

e.m.f. *electromotive force*

dry cell *electrochemical cell containing electrolyte in the form of a paste*

polarisation *the formation of a film of hydrogen gas on the positive plate of a dry cell*

In Activities 3.1 and 3.2 a flow of electric current is detected by a galvanoscope. What causes this current to flow around the circuit?

An electrochemical cell is one device that makes current flow round a circuit. Such cells have two **electrodes** made of two different **conductors**, an **electrolyte** solution (a solution that conducts electricity) which reacts with the electrodes, and a conductive wire through which electrons can flow.

Activity 3.3: Make your own electrochemical cell

Roll a lemon, orange, grapefruit, or other citrus on a firm surface to break the internal membranes.

Insert two rods – one copper, one zinc – into the fruit and connect as shown in Figure 3.7. Observe the size of the deflection of the needle on the instrument.

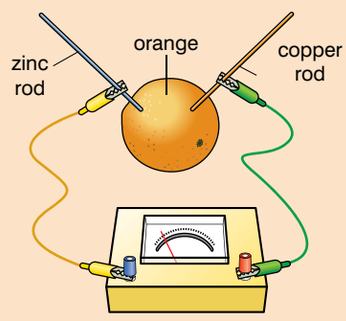


Figure 3.7

The electrodes in this cell are the copper rod and the zinc rod. The electrolyte here is the juice inside the orange.

Repeat the experiment using rods made from different materials and observe the deflection of the needle on the instrument. Do some pairs of materials produce a greater deflection than others?

The volt

A volt is defined as the potential difference (p.d.) required to produce a current of 1 A in a circuit with a resistance of 1 ohm (or Ω). It is named after the Italian scientist Alessandro Volta (1745–1827).

Some cells are more powerful than others. An ordinary torch cell (usually referred to as a battery) is rated at 1.5 volts. We call this the electromotive force of the cell, usually shortened to e.m.f. It is the cell that pumps the charge round the circuit, and for the time being it is sufficient to think of the e.m.f. as its 'electrical pumping strength'. All other things being equal, the greater the e.m.f. of the cell the greater the current that it will drive round a circuit. You will have got the right idea when you do not like to hear people talking about volts flowing round a circuit. Volts do not flow. They cause coulombs of charge to flow round a circuit at a rate of amperes.

Sources of electricity

There are many different types of cell. Figure 3.8 illustrates the principles of one common form, which we call a dry cell. Experiment shows that the carbon plate is the positive one, the zinc is the negative, and the e.m.f. is very close to 1.5 V.

Review questions

1. Charge flows along a wire at rate of 3 A. How many coulombs of charge will pass a given point in the circuit in:
   a) 1 s
   b) 12 s
   c) 2 minutes?
2. A very sensitive ammeter records a current of 30 μA. How long will it take before 6 μC of charge has flowed through it?
3. In an experiment to copperplate a coin, a current of 100 mA is passed through a solution of copper sulphate. If 30 C of charge must pass before the coin has enough copper deposited on it, for how long must you leave the current switched on?
4. a) A single electron has a charge of 1.6×10^{-19} C. How many electrons must pass round the circuit of question 5 to achieve the 30 C?
   b) For every two electrons that arrive at the coin which is being plated, one copper atom is deposited. If the mass of a copper atom is 1.1×10^{-25} kg, what is the total mass of the copper which now plates the coin?
5. What name do we give to the unit 'ampere second' (A s)? What physical quantity would be measured in such a unit?
6. You have two batteries, a large one and a tiny one, each consisting of a single dry cell. In terms of their performance, what would you expect to be the same for the two batteries and what would you expect to be different about them?
7. What is the difference between a primary cell and a secondary cell?
8. What temperature does the thermocouple in Figure 3.15 indicate for:
   a) air
   b) liquid B
   c) liquid C?

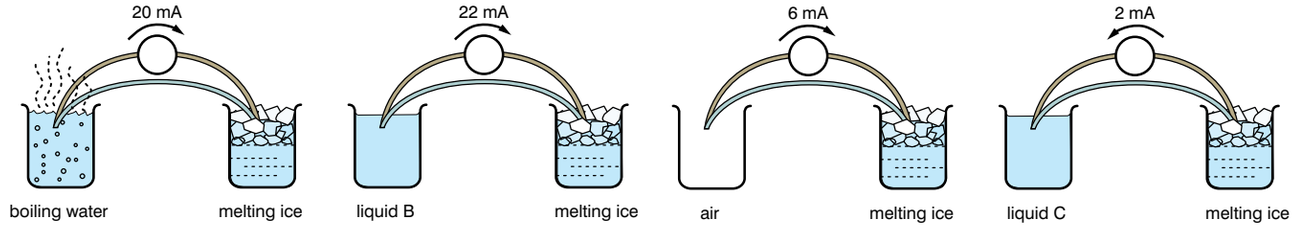


Figure 3.15

KEY WORDS

semiconductors materials that have an electrical conductivity between that of a conductor and an insulator

ammeter an instrument used to measure the electric current in a circuit

series circuit an electrical circuit in which the current passes through each point in turn

voltmeter an instrument used to measure potential difference in a circuit

potential difference the difference in electrical charge between two points in a circuit measured in volts

1.6.Constructing and drawing electric circuits

An electric current is a flow of charge. A circuit is made of conducting materials, with a device such as a cell, which keeps pumping the charges round the circuit.

Figure 3.22a shows the symbol for a cell. We often refer to it as a battery, though strictly speaking this is incorrect: 'battery' is the term reserved for a whole collection of cells acting together. The plus and minus signs are not usually marked on the drawing, so you must remember which is which. Remember too that **conventional current flows round the circuit from positive to negative**.

Figure 3.22b represents a battery of four cells. Notice that in order for them all to be pumping the charge the same way, the '+' terminal of one cell has to be joined to the '–' terminal of the next one.

Figure 3.24 *a) The symbol for a cell; b) a battery of four cells.*

The symbols for a cell, a light bulb, an ammeter and a switch are shown in the circuit in Figure 3.25. The **ammeter** is an instrument that measures the size of a current. It is fitted into the circuit so the current to be measured flows through it. In order that the ammeter does not reduce the current it has been put in to measure, it is essential that the instrument allows the current to flow through it freely (has a low resistance).

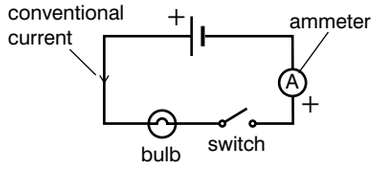


Figure 3.25 *A simple circuit.*

It is important that you connect the ammeter the correct way around, otherwise the current would try to twist it backwards. The terminal on it marked '+' needs to be joined to the positive side of the battery, as shown in Figure 3.25.

(Sometimes the terminals are colour-coded. The '+' terminal will be red, and the '–' terminal black.)

Figure 3.26 shows the symbols for a range of electrical components. If you have access to equipment, test what you have learnt so far by using some of these components to make circuits such as that shown in Figure 3.25.

Try constructing the circuit shown in Figure 3.16 (page 93). Use the ammeter and **voltmeter** to make measurements, varying the current using the variable resistor. Record your measurements and discuss your findings with the class.

If you have access to other equipment – such as resistors, variable resistors, ammeter, voltmeter, capacitors – use these and observe what happens when you construct a circuit. Using the symbols for each component, draw a circuit diagram.

The correct position for an ammeter in a circuit

Do you think the current would be the same both before and after the light bulb in Figure 3.25? We will investigate this for the slightly more complex circuit of Figure 3.27 where there are two bulbs.

We call a circuit like this a **series circuit**: the current has to pass first through one bulb and then through the other. Notice again the way the three ammeters have been connected, with their '+' terminals nearer the positive side of the battery.

The result may surprise you:

the current has the same value all the way round a series circuit.

The correct position for a voltmeter in a circuit

Consider the circuit of Figure 3.28, which shows two resistors in series in a circuit. What is the voltmeter doing? You may think of the voltmeter as being connected in parallel (see Section 3.3 for a description of parallel circuits) with the first resistor. This is true, but it is probably not the best way of looking at it.

Unlike an ammeter, the voltmeter is not part of the circuit. If you wish to measure a voltage drop, you place the voltmeter on the bench nearby. One lead comes from the voltmeter to see what conditions are like at one point in the circuit, while the other lead goes to a second point in the circuit to find out what things are like there: the voltmeter then indicates the difference between the two points. We call it the **potential difference** (p.d.) between the ends of the resistor, and measure it in volts. Sometimes we speak of it as the voltage drop (or just the voltage) across the resistor.

The voltmeter shows the difference in something – but the difference in what? It might help you to think of a comparison with heat: a temperature difference makes thermal energy flow, and a potential difference makes charge flow.

Ideally a voltmeter should draw no current from the circuit it is investigating. It must therefore have a very high resistance.

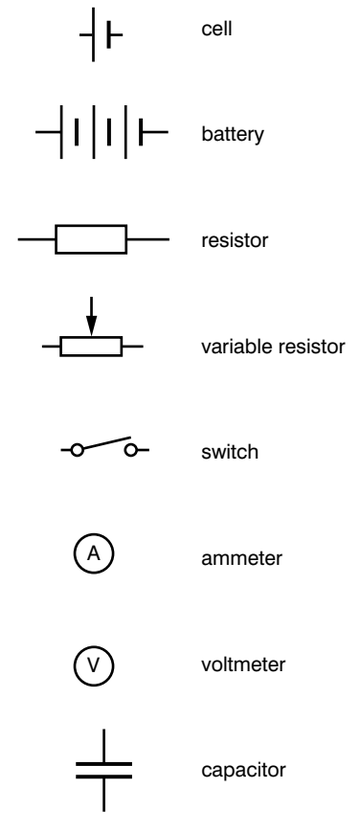


Figure 3.26 *Some circuit symbols.*

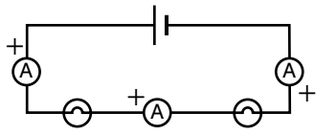


Figure 3.27 *Testing the current before and after it passes through the light bulbs.*

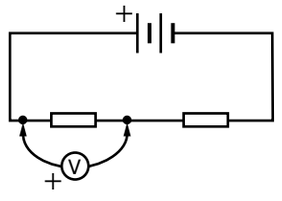


Figure 3.28 *A voltmeter connected in a circuit.*

2. Combinations of resistors

2.1 Series combination

Activity 3.6: Series and parallel circuits

Do these activities in groups.

Connect one bulb in series with a battery and note its brightness. Now connect a second and third bulb in series with the bulb.

Is there any change in the brightness of the first bulb when the second and third bulbs are added?

Once all bulbs are lit, remove one of the bulbs. What happens to the brightness of the others?

Connect one bulb to a battery and note its brightness. Now connect a second and third bulb in parallel with this bulb.

Is there any change in brightness of the first bulb when the second or third ones are added?

Once all bulbs are lit, remove one. Is there any change in the brightness of the remaining bulbs?

Look at the circuit in Figure 3.31. There are two resistors (1 Ω and 3 Ω) in series. Each resistor offers a different resistance to the current flow and there is therefore a different voltage drop across each resistor.

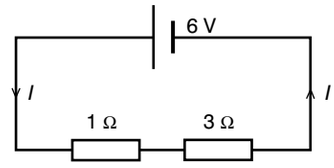


Figure 3.31 *Resistors in series in a circuit.*

With resistors in series, it is easy to work out the resistance of the circuit as a whole. You simply add up the separate values. In this case the circuit has a resistance of 4 Ω. (Be sure you could then work out that a current of 1.5 A will be drained from the battery, as

$I = \frac{V}{R} = \frac{6}{4}$ from Ohm's law.)

For a series circuit, the total resistance $R = R_1 + R_2$

where R_1 and R_2 are the resistances of the separate components.

Worked example 3.10

Consider the circuit shown in Figure 3.32.

a) Calculate the total resistance of the whole circuit.

b) Calculate the current in the circuit.

Since the circuit is a series one, the same current must flow through all three resistors. Consider the 3.5 Ω resistor.

c) Calculate the voltage drop necessary to send 2 A through 3.5 Ω.

d) Calculate the p.d. across the other two resistors.

You will need to use the relationship $R = \frac{V}{I}$ in the form of $V = IR$ and $I = \frac{V}{R}$.

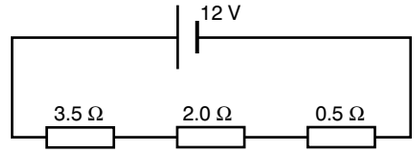


Figure 3.32

a) Total resistance in the circuit (series circuit)

= 3.5 + 2 + 0.5 = 6 Ω

b) Voltage supplied to the circuit is 12 V

Resistance in the circuit is 6 Ω

From Ohm's Law, $I = \frac{V}{R}$

$I = \frac{21}{6} = 2$ A

c) Resistance = 3.5 Ω

Current = 2 A

$V = IR$

$V = 2 \times 3.5 = 7$ V

The voltage drop across the 3.5 Ω resistor is 7 V.

d) For the 2 Ω resistor:

resistance = 2 Ω, current = 2 A

$V = IR = 2 \times 2 = 4$ V

For the 0.5 Ω resistor:

resistance = 0.5 Ω, current = 2 A

$V = IR = 0.5 \times 2 = 1$ V

Worked example 3.11

Calculate the size of resistor X in Figure 3.33.

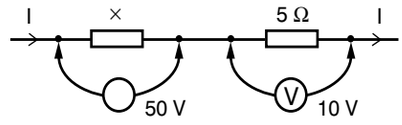


Figure 3.33

First look at the 5 Ω resistor. From $I = \frac{V}{R}$, a p.d. of 10 V across it must mean that a current I is flowing through it, given by:

$$I = \frac{V}{R} = \frac{10}{5} = 2 \text{ A}$$

Now look at resistor 'X'. Since they are in series the same 2 A is flowing through it. To achieve this there is a p.d. of 50 V across it.

$$R = \frac{V}{I} = \frac{50}{2} = 25\ \Omega$$

2.2 Parallel combination

It is less obvious how to work out the overall resistance of a circuit that consists of two resistors, not in series but in parallel (Figure 3.34).

If the two resistors are R_1 and R_2, then they behave like a single resistor R the value of which is given by:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

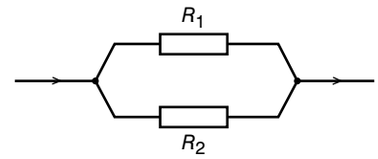


Figure 3.34 *Resistors in parallel.*

Worked example 3.12

A circuit consists of a 2 Ω resistor and a 3 Ω resistor in parallel. A 3 V battery is connected to the circuit. How big a current is drawn from it?

We must first calculate the effective resistance of the circuit using the formula for resistors in parallel.

Here,

$$\frac{1}{R} = \frac{1}{3} + \frac{1}{2} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$$

That gives us 'one over R', so to find R we must turn it over.

$$R = \frac{6}{5} = 1.2\ \Omega$$

The question now becomes 'What current will a 3 V battery send round a 1.2 Ω circuit?'

$$I = \frac{V}{R} = \frac{3}{1.2} = 2.5\ \text{A}$$

The voltage drop across resistors in parallel

Look at the circuit of Figure 3.35. Remembering that there is no voltage drop down a conducting lead, you should be able to see that the left-hand end of all three resistors is at +6 V and their right-hand ends are at 0 V.

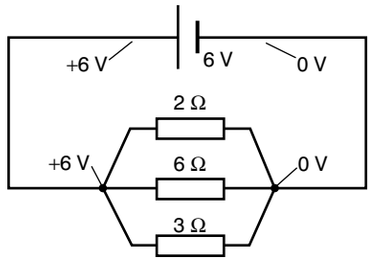


Figure 3.35

That illustrates what happens in this case:

if resistors are connected in parallel, they all have the same voltage drop across them.

This provides us with a means to work out the current in each branch of the circuit. Take the 2 Ω resistor as an example. The full 6 V of the battery is dropped across it, so the current I through it is given by:

$$I = \frac{V}{R} = \frac{6}{2} = 3\ \text{A}$$

Worked example 3.13

For the circuit shown in Figure 3.36:

a) calculate the voltage drop across each resistor

b) calculate the current through each resistor

c) calculate the effective resistance of the circuit.

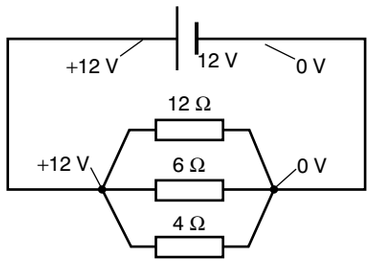


Figure 3.36

a) As the resistors are connected in parallel, the voltage drop will be the same across each, and will be equal to the full 12 V of the battery.

b) The voltage across the 12 Ω resistor is 12 V,

$I = \frac{V}{R} = \frac{12}{12} = 1$ A

The voltage across the 6 Ω resistor is 12 V,

$I = \frac{V}{R} = \frac{12}{6} = 2$ A

the voltage across the 4 Ω resistor is 12 V,

$I = \frac{V}{R} = \frac{12}{4} = 3$ A

c) Total resistance of the circuit is R.

$$\frac{1}{R} = \frac{1}{12} + \frac{1}{6} + \frac{1}{4}$$

$$= \frac{1 + 2 + 3}{12} = \frac{6}{12} = \frac{1}{2}$$

$R = 2\ \Omega$

Worked example 3.14

Two resistors of 10 Ω and 15 Ω are connected. What is their combined resistance if they are connected:

a) in series

b) in parallel?

a) In series

$R = R_1 + R_2$

$= 10 + 15 = 25\ \Omega$

b) In parallel

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$= \frac{1}{10} + \frac{1}{15}$$

$$= \frac{3 + 2}{30} = \frac{5}{30}$$

$$R = \frac{30}{50} = 6\ \Omega$$

- Resistors in series simply add.
- Resistors in parallel add by

  parallel will conduct better than either one on its own.
- The resistance of the whole circuit includes the resistance within the battery itself, and this may not always be negligible.

- Where the circuit branches in a parallel circuit, the sum of the currents approaching the junction equals the sum of the currents leaving it.
- In a series circuit, the separate voltage drops (potential differences) across all the resistors add up to the voltage of the battery.
- In a parallel circuit each branch will have the same voltage drop across it.

Review questions

1. What should ammeters (a), (b) and (c) read in Figure 3.37?

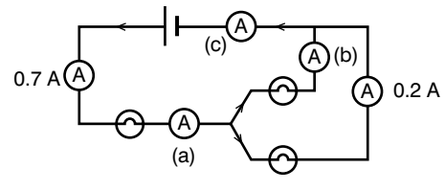


Figure 3.37

Which bulb should be the brightest? Why?

2. Give the missing ammeter readings a and b in Figure 3.38. Suggest why more current flows through some bulbs than through others.

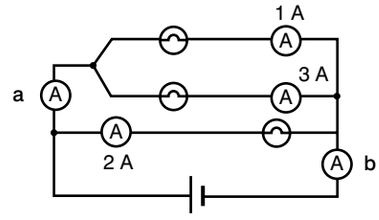


Figure 3.38

3. Give the missing ammeter readings c and d in Figure 3.39. Assume all the bulbs are identical ones. How would you expect the brightness of the different bulbs to compare?

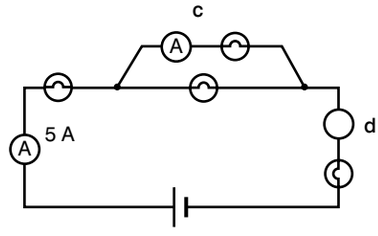


Figure 3.39

4. Draw a circuit diagram to show a battery of two cells with three light bulbs connected in parallel. Include two switches: one must turn one of the bulbs on and off, and the other must do the same for the other two bulbs together.
5. For the circuit diagram in Figure 3.40:
   a) What is the effective resistance of the two 4 Ω resistors in parallel?
   b) What is the total resistance of the circuit?
   c) Work out the current *I*.

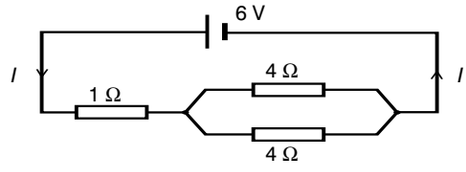


Figure 3.40

6. What single resistor could be placed in the box to replace the two shown in Figure 3.41?

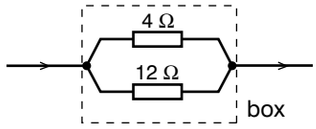


Figure 3.41

7. Calculate the effective resistance of a 25 Ω resistor and a 100 Ω resistor in parallel. What current will be drawn from a 50 V supply connected to the circuit?
8. a) Work out the total resistance of the circuit in Figure 3.42.
   b) What current will be drawn from the battery?

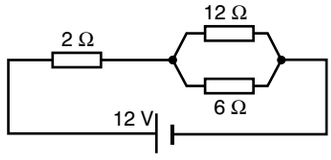


Figure 3.42

9.In the circuit of Figure 3.43 the battery has negligible internal resistance. Calculate:

a) the total resistance in the circuit
b) the current flowing in the circuit
c) the potential difference across the 50 Ω resistor.

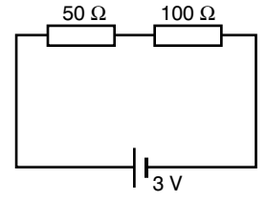


Figure 3.43

Practice

1. If 36 C of charge pass through a wire in 4 s, what current is it carrying?
2. An appliance has a resistance of 5 Ω and requires a current of 0.5 A for it to work. The only battery available has an e.m.f. of 12 V and negligible internal resistance. What extra resistance will you need to provide to limit the current to its correct value. Draw the circuit you would assemble.
3. A student incorrectly set up the circuit shown in Figure 3.53.

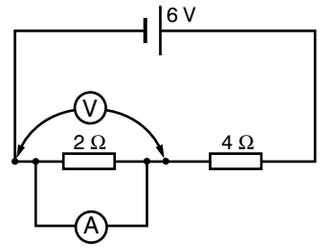


Figure 3.53

   a) What is wrong with it?
   b) Assume that both the ammeter and the battery have negligible resistance. Starting from one terminal of the battery, what route will all the current take as it flows round the circuit? How much resistance will this circuit have?What will the ammeter read?
   c) What will the voltmeter read?
   d) The student then reconnects the circuit so that the two resistors, the ammeter and the voltmeter are all in series with the battery. What would you expect the ammeter to read now? Give a reason for your answer.

4. A 12 V light bulb is put under test, with the following results:

Current through bulb (A)	0.4	0.6	0.8	1.0	1.2	1.4	1.6
p.d. across bulb (V)	1.4	2.6	3.9	5.5	7.4	9.7	12.6

   a) Draw the circuit you would use to obtain these readings.
   b) Copy the table out, and for each set of readings work out the resistance R of the bulb.
   c) Plot a graph of R against I. Why is it not valid to assume the graph goes through the origin?
   d) Use your graph to estimate the resistance of the bulb at room temperature. (Hint: in what circumstance will the bulb not be heated at all?)

5. Constantan is the name of an alloy that is sometimes used for making resistors in the laboratory. Its resistivity is 4.9×10^{-7} Ω m. Calculate the resistance of a 3 m long constantan wire with 1 mm^2 cross-sectional area.
6. Draw a labelled diagram of a dry cell. State the function of each of its parts.
7. The resistivity of copper is 2×10^{-8} Ω m. Work out the resistance of a copper wire of 1 mm^2 cross-sectional area and 3 m long.
8. A battery sends a current of 3 A through a 4 Ω resistor.
   a. What must the e.m.f. of the battery be?
   b. How many coulombs of charge will pass through the resistor in 1 minute?

9.Explain why it is wrong to write the units of resistivity as ohm/metre (Ω/m) with the stroke between the two quantities.

10.Draw a circuit diagram to show a battery of two cells and three light bulbs connected in series. Include a switch to turn the lights on and off. Does it matter where the switch is placed in the circuit?

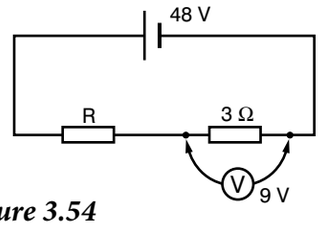


Figure 3.54

11. a) What current is flowing round the circuit in Figure 3.54?

b) How great must the p.d. be across resistor R?

c) What is the resistance of resistor R?

12.What should be the reading of ammeters A_1 and A_2 in Figure 3.55?

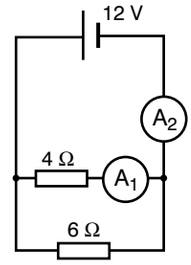


Figure 3.55

13.These questions refer to the circuit drawn in Figure 3.56.

a) What are the currents through the 6 Ω branch, through the 3 Ω branch and through the 2 Ω branch?

b) What is the total current drawn from the battery?

c) As far as the battery is concerned, what is the total resistance of its whole circuit?

d) Compare this with the value obtained from

$$\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}$$

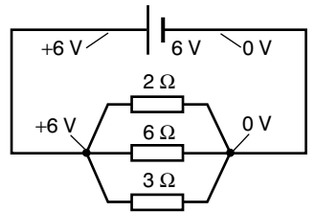


Figure 3.56

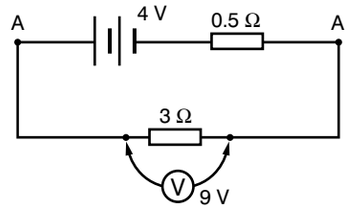


Figure 3.57

14. In Figure 3.57, A and B represent the terminals of a battery of e.m.f. 4 V and internal resistance 0.5 Ω. *R* is the total resistance of the circuit to which it is connected. Calculate the current that flows round the circuit and the p.d. between the terminals of the battery if the resistance *R* is

a) 7.5 Ω

b) 0 Ω.

Lesson 3 Magnetic Fields

Contents

Section	Learning Objectives
Concepts of magnetic field	• Describe a magnetic field. • Perform and describe an experiment to demonstrate the existence of a magnetic field around a current-carrying wire. • Sketch the resulting magnetic field lines pattern of a current-carrying wire. • Apply the right-hand rule to determine the direction of magnetic field lines around a straight current-carrying wire. • Calculate the magnetic field strength at a point due to a straight current-carrying wire. • Sketch the magnetic field lines pattern of a current loop. • Sketch the magnetic field lines pattern of a solenoid. • Specify the polarity of a solenoid using the right-hand rule. • Calculate the magnetic field strength at the centre of a solenoid.

1.The Magnetic Field due to a Current in a Straight Wire

- Magnetic field lines in a current-carrying wire are circular rings, centered on the wire
- The field lines are closer together near the wire, where the field is strongest
- The field lines become further apart with distance from the wire as the field becomes weaker
- Reversing the current, the direction of the field is also reversed.

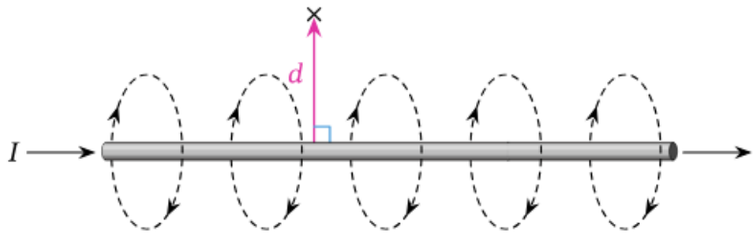
The Strength of the Magnetic Field due to a Current in a Straight Wire

The strength of a magnetic field, B, some distance d away from a straight wire carrying a current, I can be found using the equation

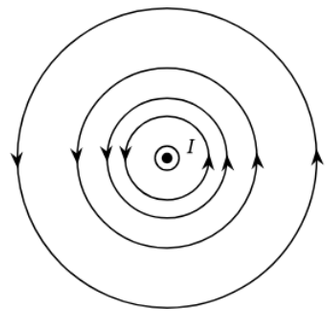
$$B = \mu_0 \frac{I}{4\pi d}$$

where the permeability of free space, $\mu_0 = 4\pi \times 10^{-7} T.m/A$.

It should be noted that the distance d must be measured perpendicular to the wire. A perpendicular distance measurement is shown in the diagram below.



The strength of the field, B decreases as the distance away from the wire, d increases. This is illustrated in the diagram below, which shows a view along the length of a straight current-carrying wire. It should be noted that the dot in the center of the wire indicates that the current points out of—and perpendicular to—the screen.

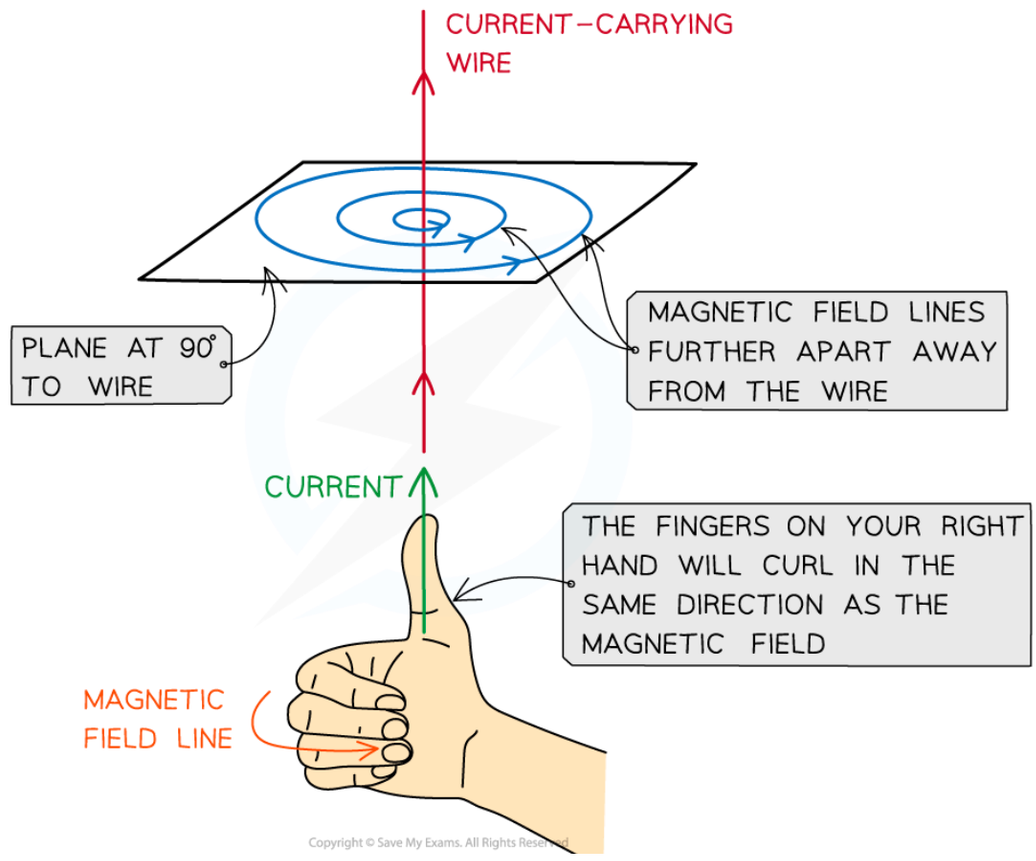


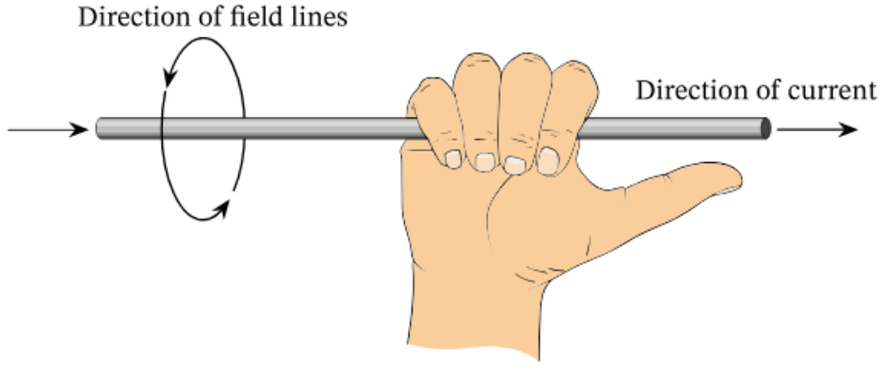
Thus far, we have only been concerned with the magnitude, or strength, of a magnetic field due to current in a wire. However, we must remember that magnetic field is a vector quantity, since it is defined using both a magnitude and a direction. We will use the right-hand rule to determine the direction of the magnetic field, as described below.

Rule: Right-Hand Rule for the Magnetic Field due to a Current in a Straight Wire

To determine the direction of the magnetic field due to a straight current-carrying wire, perform the following steps:

1. Using the right hand, point the thumb in the direction of the current.
2. "Grab" the wire, curling the fingers around its imaginary axis. The direction that the fingers curl in corresponds to the direction of the magnetic field.





Key Points

- A long, straight, current-carrying wire produces a magnetic field composed of concentric closed circles, and the field strength is given by $B = \frac{\mu_0 I}{2\pi d}$.
- The strength of the magnetic field, B, is inversely proportional to the distance away from the wire, d. Thus, the field strength falls off to zero as d gets very large.
- We can determine the direction of the magnetic field using the right-hand rule: point the thumb in the direction of the current and curl the fingers as if grabbing the wire. The direction that the fingers curl in corresponds to the direction of the magnetic field.

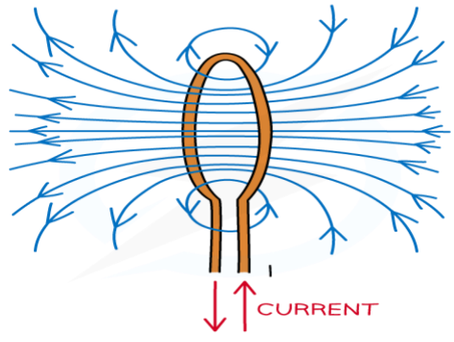
Practice: Calculating the Magnetic Field due to a Current in a Straight Wire

A long, straight cable in an industrial power plant carries a direct current of 100 A. Calculate the strength of the resulting magnetic field at a perpendicular distance of 0.06 m from this cable. the permeability of free space, $\mu_0 = 4\pi \times 10^{-7} T.m/A$..Give your answer in scientific notation to two decimal places.

2.The Magnetic Field due to a Current in a Circular Loop of Wire

In this explainer, we will learn how to calculate the magnetic field produced by a current in a circular loop of wire.

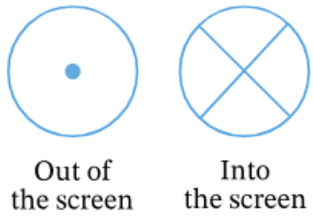
A wire curved to form a complete circle will produce a very strong magnetic field in one direction at its center.



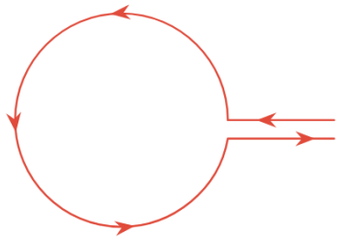
The direction of this produced magnetic field can be determined by the right-hand screw rule. A right-handed screw can only be turned in one direction to drive it into a surface.

The direction we must turn the screw in is the direction of the current in a loop. The point of the screw, the direction it is being driven in, is the direction of the magnetic field at the center of that loop.

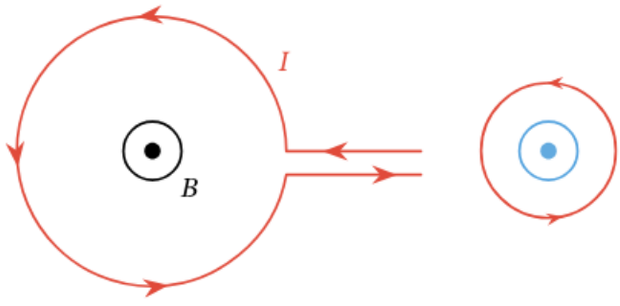
When a direction points into or out of the screen, we use the symbols below to show this.



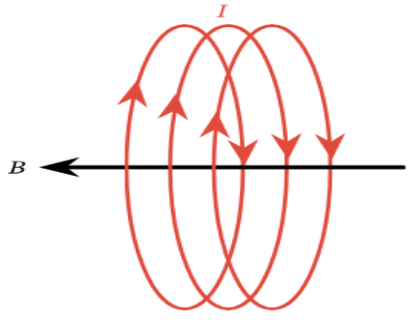
Let's look back at the direction of the magnetic field from the diagram above.



The direction of the current in the loop is counterclockwise. Thus, the direction the screw must turn in is counterclockwise. The direction it is being driven in must be out of the screen. This means the magnetic field direction is also out of the screen.



Magnetic Field Strength at the Center of N Current-Carrying Wire Loop



The magnetic field strength B at the center of N current-carrying wire loop is given by

$$B = \mu_0 \frac{\mathrm{NI}}{2\mathrm{r}}$$

where I is the current in the wire loop, r is the radius of the wire loop, N is the number of turns and the permeability of free space, $\mu_0 = 4\pi \times 10^{-7} T.m/A$.

Key Points

- The direction of a magnetic field at the center of a loop of wire is given by the right-hand screw rule.
- The magnetic field strength B at the center of a current-carrying loop of wire is given by the equation $B = \frac{\mu_0 I}{2r}$, where I is the current in the loop, r is the radius is the loop, and μ_0 is the permeability of free space, often expressed as $4\pi \times 10^{-7}$ T·m/A.
- When there are multiple current-carrying loops of the same radius in a coil, the magnetic field strength B is given by the equation $B = \frac{\mu_0 NI}{2r}$, where I is the current in the loop, r is the radius is the loop, N is the number of turns or loops in the wire, and μ_0 is the permeability of free space, often expressed as $4\pi \times 10^{-7}$ T·m/A.

Practice1: Strength of a Magnetic Field at the Center of a Loop

A 100turns circular loop of wire carries a constant current of 0.9 A. The radius of the loop is 13 mm. Calculate the strength of the magnetic field at the center of the loop. Give your answer in Tesla expressed in scientific notation to 1 decimal place.

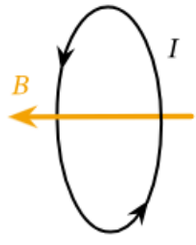
Practice2:

A circular loop of wire with a radius of 9.5 cm carries a constant current of I A.The strength of the magnetic field produced by the current is $5.2 \times 10^{-5} T$ at the center of the loop. Calculate I rounding your answer to 1 decimal place.

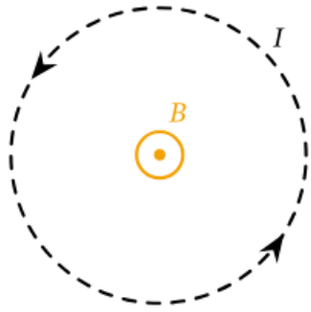
3.The Magnetic Field due to a Current in a Solenoid

We will learn how to calculate the magnetic field produced by a current in a solenoid.

Recall the direction of a magnetic field in a loop of a current-carrying wire. At the center of the loop, the magnetic field has one direction, as seen in the diagram below. The orange line is the magnetic field direction and black line is the wire loop.



The same loop seen from the front, with the magnetic field direction pointing out of the screen, would look like the diagram below.

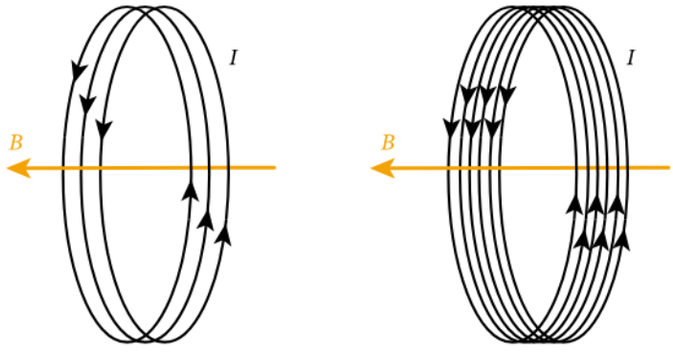


Recall that the symbols below are used to show that a direction is going out of or into the screen.

Out of the screen

Into the screen

The magnetic field strength at the center of a loop can be increased by placing more loops in line with it. The diagram below shows two sets of loops with the same current and the same radius lined up in this way.

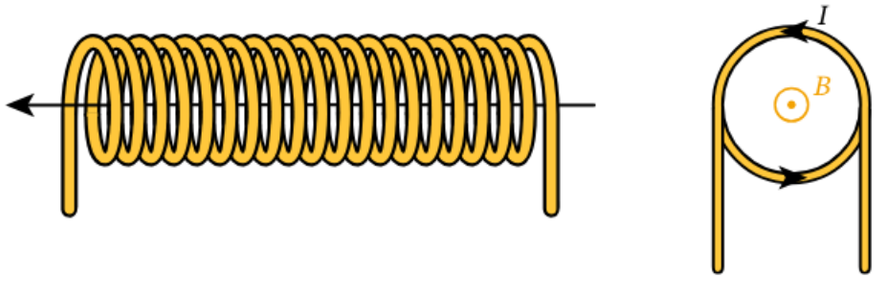


The set of loops on the right has a stronger magnetic field because it has more loops.

Instead of using a set of loops, strengthening the magnetic field at the center can be achieved using a single wire with multiple turns. The diagram below shows such a wire, with a side and front view.

Side View

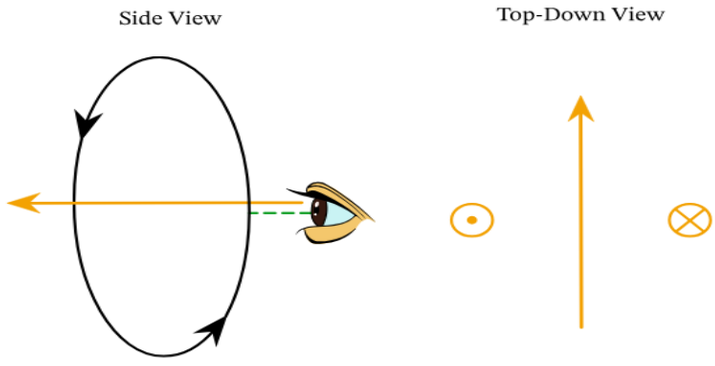
Front View



A wire with a series of turns like this is called a solenoid. Each turn of a solenoid contributes to the strength of the magnetic field at the center just like an additional loop would.

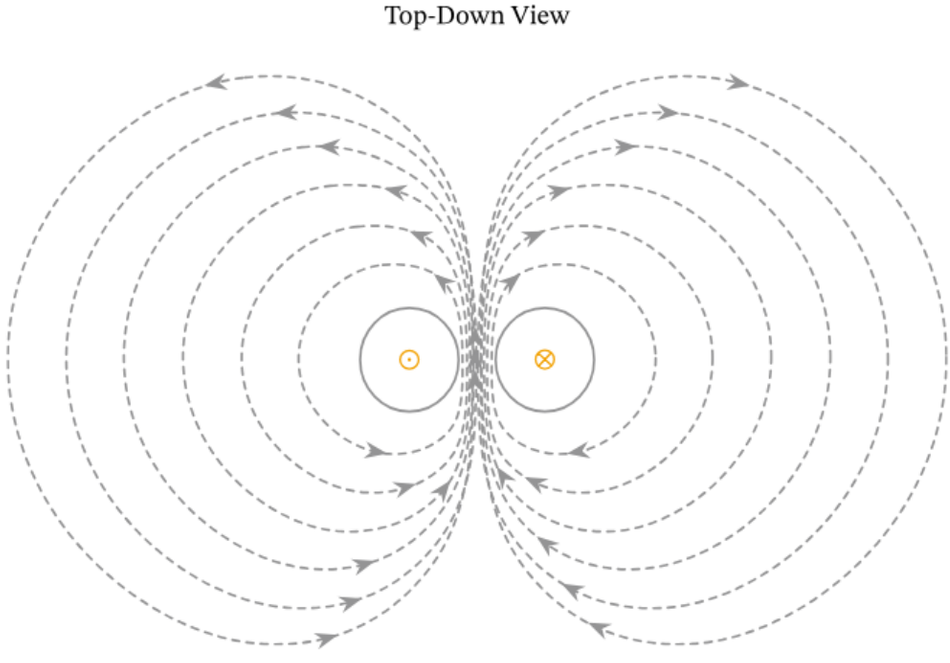
The magnetic field strength and direction at the very center of a solenoid is uniform. It has one direction and magnitude. Other points around the solenoid have different magnetic field directions and magnitudes.

Before looking at the magnetic field lines of a solenoid, let's consider the angle we will be viewing it from using a single loop. The diagram below shows a single loop of a current-carrying wire and its resultant magnetic field from two different angles.



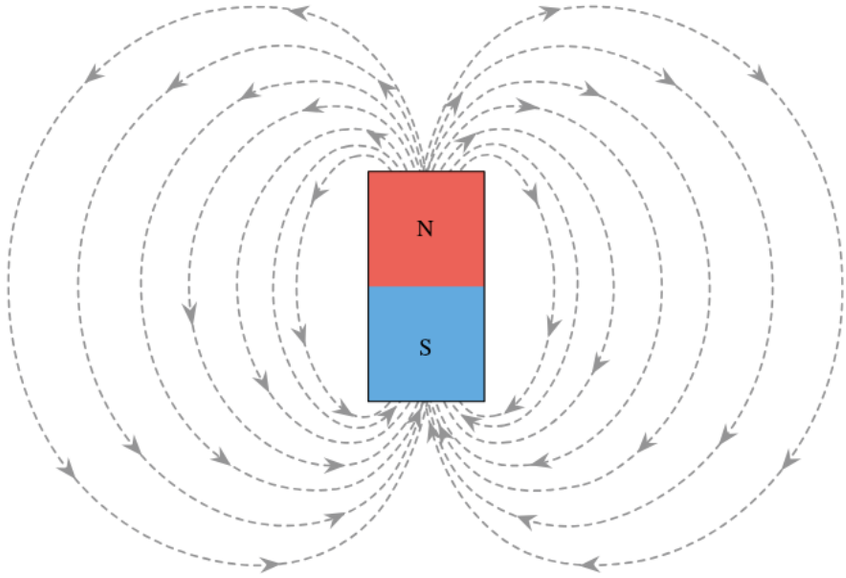
The side view of this loop shows the direction in which an observer looks, indicated by the eye, to obtain the top-down view. The top-down view still shows the direction of current, going into and out of the screen, but it does not show the bottom of the loop.

Now, let's look at the top-down view of a single loop with its magnetic field lines, in grey, in the diagram below.

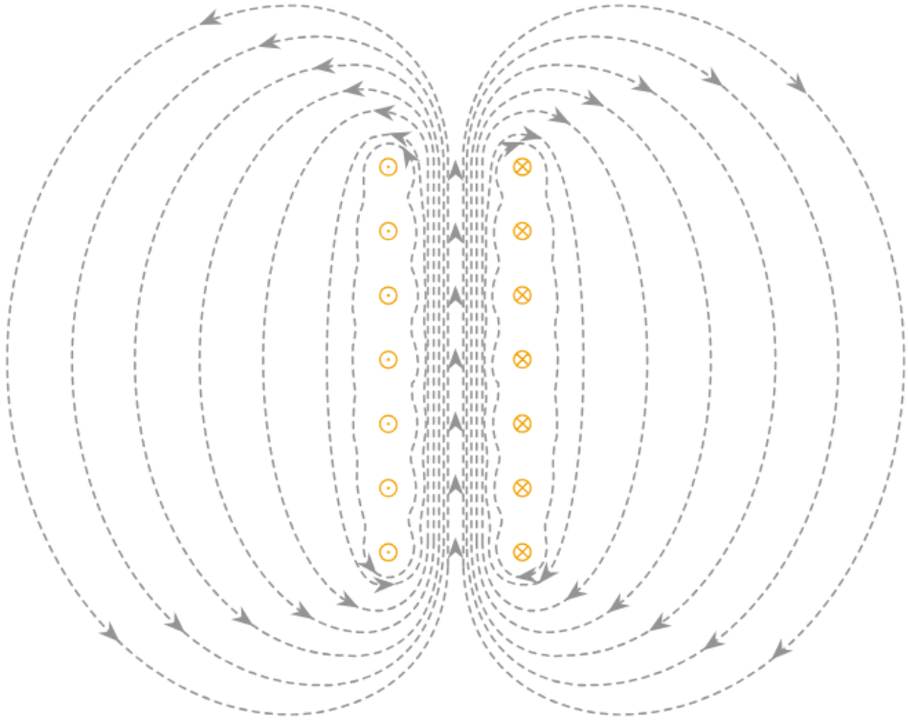


When there are more magnetic field lines close together, it means there is a stronger magnetic field. We can see that at the very center of the loop, the magnetic field lines are very close to each other with the same direction, meaning it has a strong magnetic field at that point.

Outside of the loop, the magnetic field lines resemble that of a bar magnet, as seen below.



Let's now look at a solenoid with seven turns from this angle. The diagram below shows this with its corresponding magnetic field.

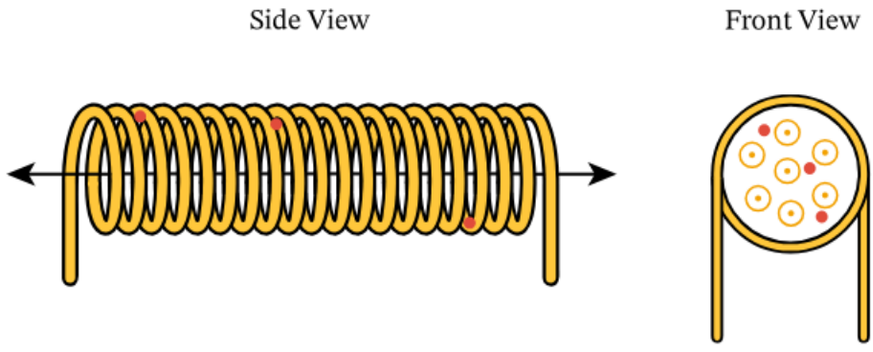


Note how the magnetic field lines are consistent and straight at the very center of these wire turns, but become less consistent toward the ends of the solenoid. The closer to these ends, the less uniform the field lines.

Now, let's consider a theoretical, very long solenoid. It is so long that we can treat it as if it had no ends. This would mean it has a perfectly uniform magnetic field within the turns at all points.

If we measured the magnetic field strength at different points within the turns of this theoretical solenoid, in each case the magnitude and direction of the magnetic field strength would be the same.

The diagram below shows a theoretical solenoid with three points, indicated by the red dots, that have equivalent magnetic field strengths and directions.



The magnetic field strength within the turns of this theoretical solenoid can be determined using an equation.

Equation: Magnetic Field Strength at the Center of a Solenoid

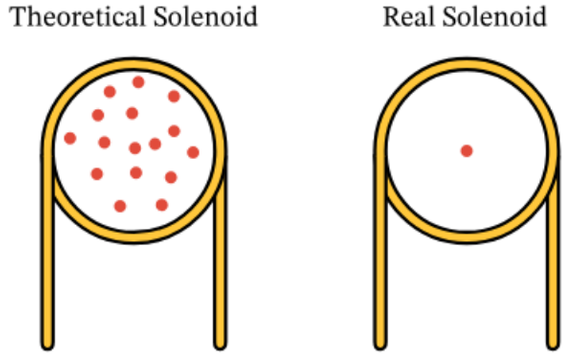
The magnetic field strength, B inside the center of a solenoid is found using the equation

$$B = \mu_0 \frac{NI}{L}$$

where I is the current of the solenoid, N is the number of turns the solenoid has, L is the length of the solenoid, and μ_0 is the permeability of free space, $\mu_0 = 4\pi \times 10^{-7} T.m/A$. $\frac{N}{L} = n$ is the number of turns per unit of length.

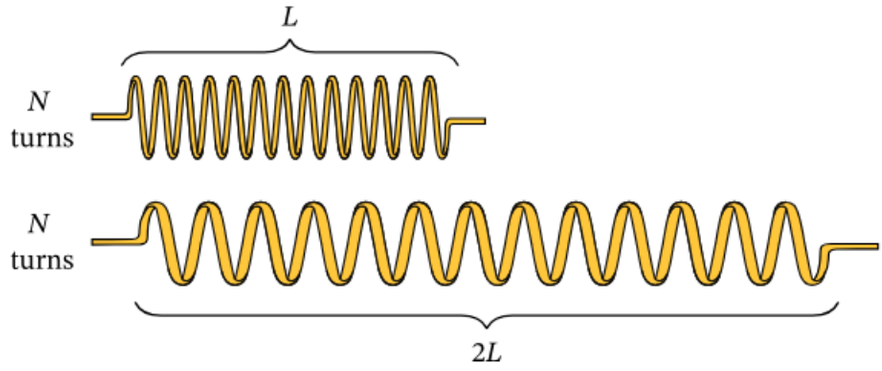
So, B can be written as $B = \mu_0 nI$

For a real solenoid with limited length, this equation is still useful to describe the magnetic field strength at the exact center of the turns, since this is where it is uniform. The diagram below shows points with the same magnetic field strength and direction, on both a theoretical and real solenoid.



Real solenoids have a fairly constant magnetic field direction inside the turns, but not magnetic field strength. Only the center has a consistent magnetic field strength.

Looking at the equation, we see that the length of a solenoid matters when finding the magnetic field strength at the center. Specifically, that magnetic field strength is inversely proportional to length. The diagram below shows two solenoids with the same current and number of turns but with different lengths.



Key Points

- A solenoid is a wire arranged in a series of turns or loops.
- When a solenoid carries a current, it produces a magnetic field that is strongest in the center of its loops.
- Within the solenoid loops, the magnetic field strength B is given by the equation $B = \mu_0 \frac{NI}{L}$ where N is the number of turns in the solenoid, I is the current in the solenoid, L is the length of the solenoid, and $\mu_0 = 4\pi \times 10^{-7} T.m/A$ is the permeability of free space.
- The equation for magnetic field strength B at the center of a solenoid using turns per unit of length is $B = \mu_0 nI$ where n is the number of turns per unit of length.

Practice1: Magnetic Field Strength at the Center of a Solenoid with Turns and Length

A solenoid has a length of 3.2 cm and consists of 90 turns of wire. The wire carries a constant current of 1.2 A. Calculate the strength of the magnetic field at the center of the solenoid. Give your answer in Tesla expressed in scientific notation to one decimal place. Use a value of $\mu_0 = 4\pi \times 10^{-7} T.m/A.$

Practice2: Magnetic Field Strength at the Center of a Solenoid

A wire that carries a constant current of 0.15 A is formed into a solenoid with 11 turns per centimeter. Calculate the strength of the magnetic field at the center of the solenoid. Give your answer in Tesla expressed in scientific notation to one decimal place. Use a value of $\mu_0 = 4\pi \times 10^{-7} T.m/A.$

Practice

1. The two wires shown in Figure P19.48 are separated by d =10.0 cm and carry currents of I=5.00A in opposite directions. Find the magnitude and direction of the net magnetic field
   (a) at a point midway between the wires;
   (b) at point P1, 10.0 cm to the right of the wire on the right; and;
   (c) at point P2, 2d = 20.0 cm to the left of the wire on the left.

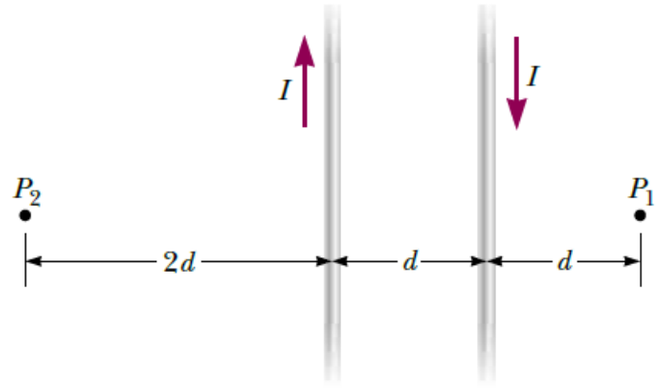


Figure P19.48

2. A thin, circular coil of wire with a radius 4.2 cm carries a constant current of 3.9 A. The coil has 35 turns of wire. What is the strength of the magnetic field at the center of the coil? Give your answer in Tesla expressed in scientific notation to 1 decimal place.

3.The magnetic field at the center of a 1.0cm diameter loop is 2.5mT.

A. hat is the current in the loop?

B. A long, s raight wire carries the same current you found in part A. At what distance from the wire is the magnetic 2.5mT?

4. A long solenoid that has 1000 turns uniformly distributed over a length of 0.400 m produces a magnetic field of magnitude $1.00 \times 10^{-4} T$ at its center. What current is required in the windings for that to occur?

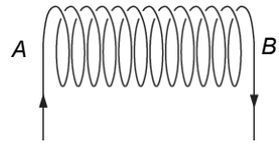
5. A certain superconducting magnet in the form of a solenoid of length 0.50 m can generate a magnetic field of 9.0 T in its core when its coils carry a current of 75 A. The windings, made of a niobium–titanium alloy. Find the number of turns in the solenoid.

6. A solenoid of 30 cm has 180 turns of coil. A current of 5 A passing through the solenoid.

(a). Identify the magnetic poles at the end *A* and *B*.

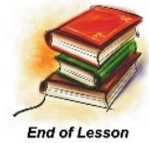
(b). Find the magnitude of the magnetic field at the centre of the solenoid.

(c). If the size of the current is halved, what happens to the answer in (b)?



7- A solenoid has 100 turns of coil. A current of 10 A passing through the solenoid. If the field strength is 0.000012T, Find the length of solenoid.

8- A solenoid has 150 turns of coil. The length of solenoid is 20cm. If the field strength is 0.000012T, Find the current.



End of Lesson

Lesson 4 Alternating Current

1.What is Alternating Current (AC)?

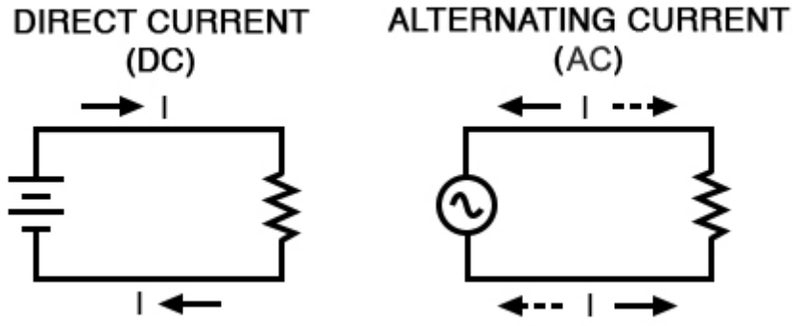
Most students of electricity begin their study with what is known as *direct current* (DC), which is electricity flowing in a constant direction, and/or possessing a voltage with constant polarity.

DC is the kind of electricity made by a battery (with definite positive and negative terminals), or the kind of charge generated by rubbing certain types of materials against each other.

2.Alternating Current vs Direct Current

As useful and as easy to understand as DC is, it is not the only "kind" of electricity in use. Certain sources of electricity naturally produce voltages alternating in polarity, reversing positive and negative over time.

Either as a voltage switching polarity or as a current switching direction back and forth, this "kind" of electricity is known as Alternating Current (AC):



Whereas the familiar battery symbol is used as a generic symbol for any DC voltage source, the circle with the wavy line inside is the generic symbol for any AC voltage source.

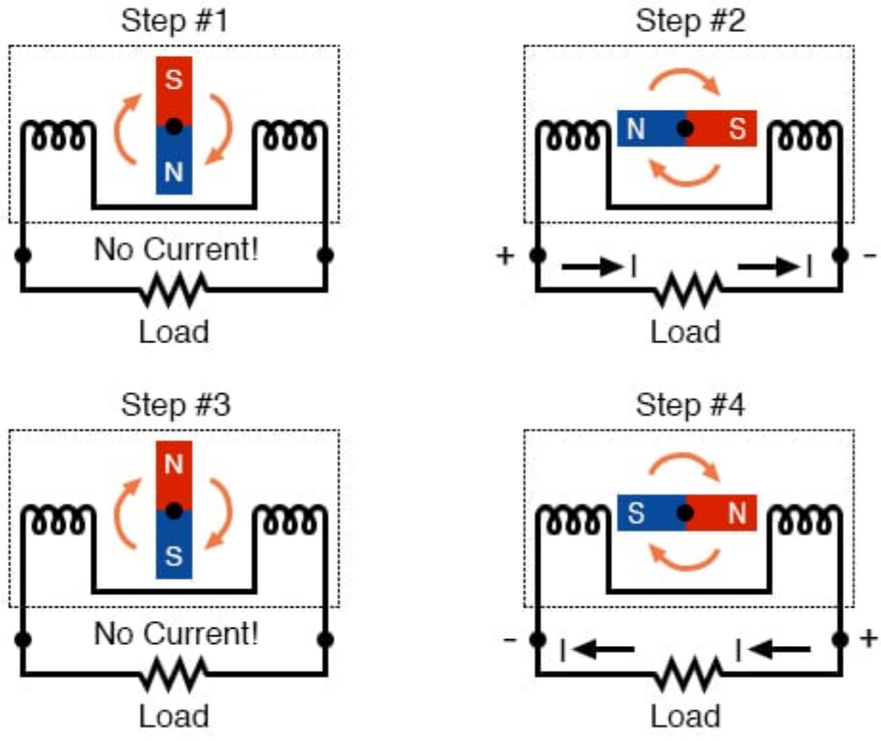
In applications where electricity is used to dissipate energy in the form of heat, the polarity or direction of current is irrelevant, so long as there is enough voltage and current to the load to produce the desired heat (power dissipation). However, with AC it is possible to build electric generators, motors and power distribution systems that are far more efficient than DC, and so we find AC being used predominantly across the world in high power applications.

To explain the details of why this is so, a bit of background knowledge about AC is necessary.

3.AC Alternators

If a machine is constructed to rotate a magnetic field around a set of stationary wire coils with the turning of a shaft, AC voltage will be produced across the wire coils as that shaft is rotated, in accordance with Faraday's Law of electromagnetic induction.

This is the basic operating principle of an AC generator, also known as an *alternator*: Figure below



Alternator operation

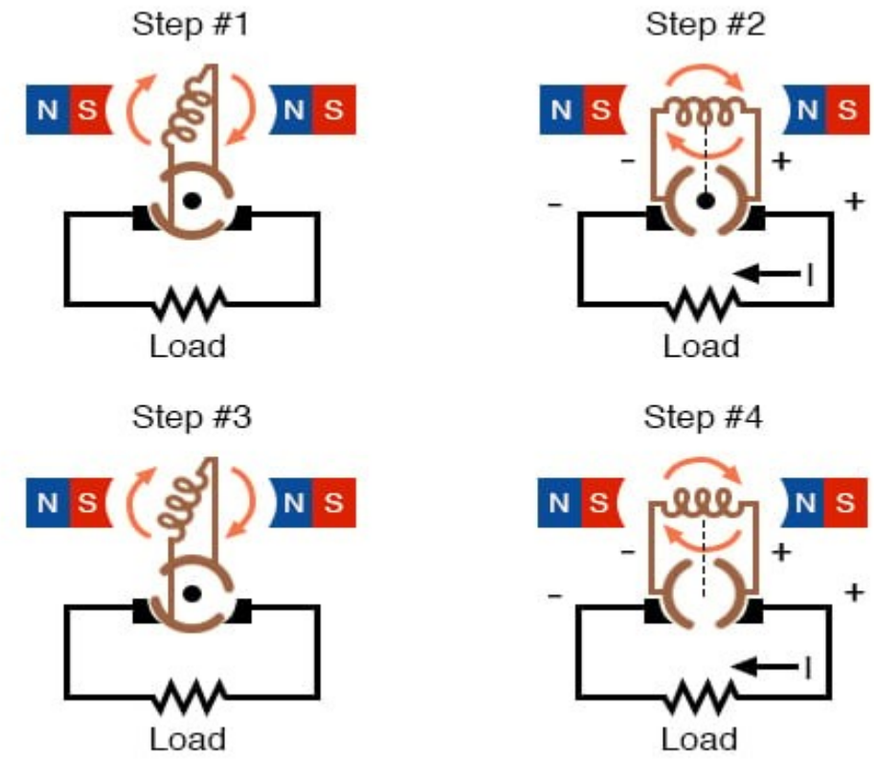
Notice how the polarity of the voltage across the wire coils reverses as the opposite poles of the rotating magnet pass by.

Connected to a load, this reversing voltage polarity will create a reversing current direction in the circuit. The faster the alternator's shaft is turned, the faster the magnet will spin, resulting in an alternating voltage and current that switches directions more often in a given amount of time.

While DC generators work on the same general principle of electromagnetic induction, their construction is not as simple as their AC counterparts.

With a DC generator, the coil of wire is mounted in the shaft where the magnet is on the AC alternator, and electrical connections are made to this spinning coil via stationary carbon "brushes" contacting copper strips on the rotating shaft.

All this is necessary to switch the coil's changing output polarity to the external circuit so the external circuit sees a constant polarity:

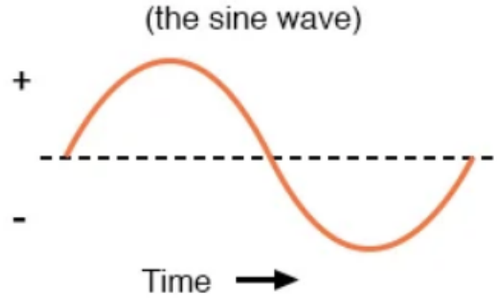


DC generator operation

The diagram shown above is a bit more simplified than what you would see in real life.

4.AC Waveforms

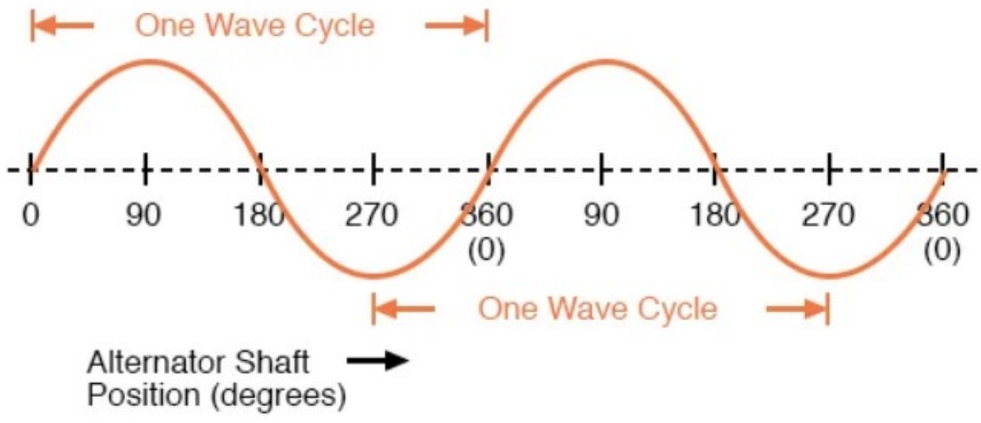
When an alternator produces AC voltage, the voltage switches polarity over time, but does so in a very particular manner. When graphed over time, the "wave" traced by this voltage of alternating polarity from an alternator takes on a distinct shape, known as a *sine wave*: Figure below



Graph of AC voltage over time (the sine wave).

5.Period vs Frequency

If we were to follow the changing voltage produced by a coil in an alternator from any point on the sine wave graph to that point when the wave shape begins to repeat itself, we would have marked exactly one *cycle* of that wave. This is most easily shown by spanning the distance between identical peaks, but may be measured between any corresponding points on the graph. The degree marks on the horizontal axis of the graph represent the domain of the trigonometric sine function, and also the angular position of our simple two-pole alternator shaft as it rotates: Figure below



Alternator voltage as function of shaft position (time).

Since the horizontal axis of this graph can mark the passage of time as well as shaft position in degrees, the dimension marked for one cycle is often measured in a unit of time, most often seconds or fractions of a second. When expressed as a measurement, this is often called the ***period*** of a wave.

The period of a wave in degrees is *always* 360, but the amount of time one period occupies depends on the rate voltage oscillates back and forth.

A more popular measure for describing the alternating rate of an AC voltage or current wave than *period* is the rate of that back-and-forth oscillation. This is called ***frequency***. The modern unit for frequency is the Hertz (abbreviated Hz), which represents the number of wave cycles completed during one second of time.

In the United States of America, the standard power-line frequency is 60 Hz, meaning that the AC voltage oscillates at a rate of 60 complete back-and-forth cycles every second. In Europe, and our country where the power system frequency is 50 Hz, the AC voltage only completes 50 cycles every second.

A radio station transmitter broadcasting at a frequency of 100 MHz generates an AC voltage oscillating at a rate of **100 *million*** cycles every second.

Prior to the canonization of the Hertz unit, frequency was simply expressed as "cycles per second." Older meters and electronic equipment often bore frequency units of "CPS" (Cycles Per Second) instead of Hz. Many people believe the change from self-explanatory units like CPS to Hertz constitutes a step backward in clarity.

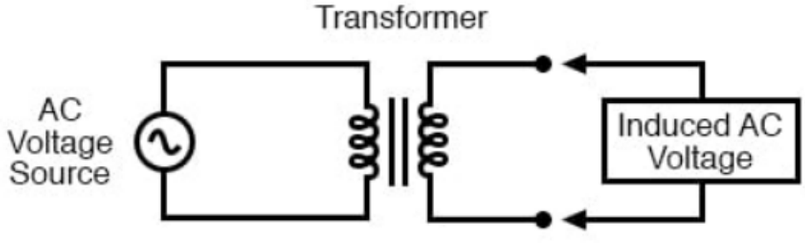
Period and frequency are mathematical reciprocals of one another. That is to say, if a wave has a period of 10 seconds, its frequency will be 0.1 Hz, or 1/10 of a cycle per second:

$$\text{Frequency in Hertz} = \frac{1}{\text{Period in secods}}$$

Or in symbol: $f = \frac{1}{T}$

6.Transformers

There is an effect of electromagnetism known as *mutual induction*, whereby two or more coils of wire placed so that the changing magnetic field created by one induces a voltage in the other. If we have two mutually inductive coils and we energize one coil with AC, we will create an AC voltage in the other coil. When used as such, this device is known as a ***transformer:***



Transformer "transforms" AC voltage and current.

The fundamental significance of a transformer is its ability to step voltage up or down from the powered coil to the unpowered coil. The AC voltage induced in the unpowered ("secondary") coil is equal to the AC voltage across the powered ("primary") coil multiplied by the ratio of secondary coil turns to primary coil turns.

If the secondary coil is powering a load, the current through the secondary coil is just the opposite: primary coil current multiplied by the ratio of primary to secondary turns.

The EMF induced in the secondary coil, called the secondary voltage, is proportional to the primary voltage. The secondary voltage also depends on the ratio of the number of turns on the secondary coil to the number of turns on the primary coil, as shown by the following expressions.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

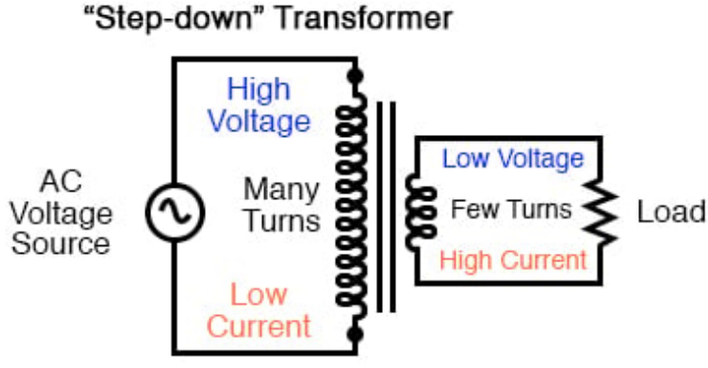
In an ideal transformer, the electric power delivered to the secondary circuit equals the power supplied to the primary circuit. An ideal trans-former dissipates no power itself, and can be represented by the following equations:

$$P_p = P_s$$

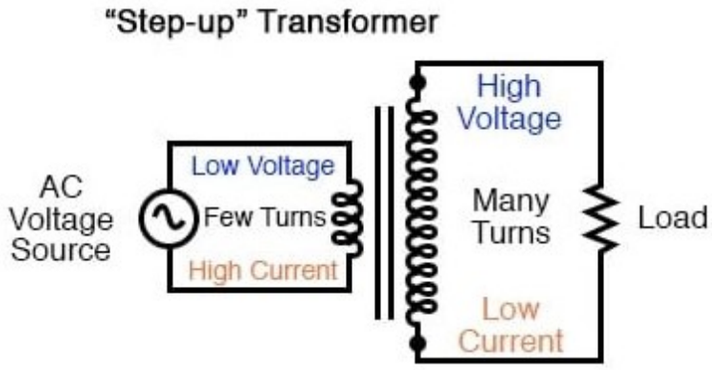
Or $$I_p \times V_p = I_s \times V_s$$

Rearranging the equation to find the ratio Vp /Vs shows that the current in the primary circuit depends on how much current is required by the secondary circuit. This relationship can be combined with the relationship shown earlier between voltage and the number of turns to result in the following.

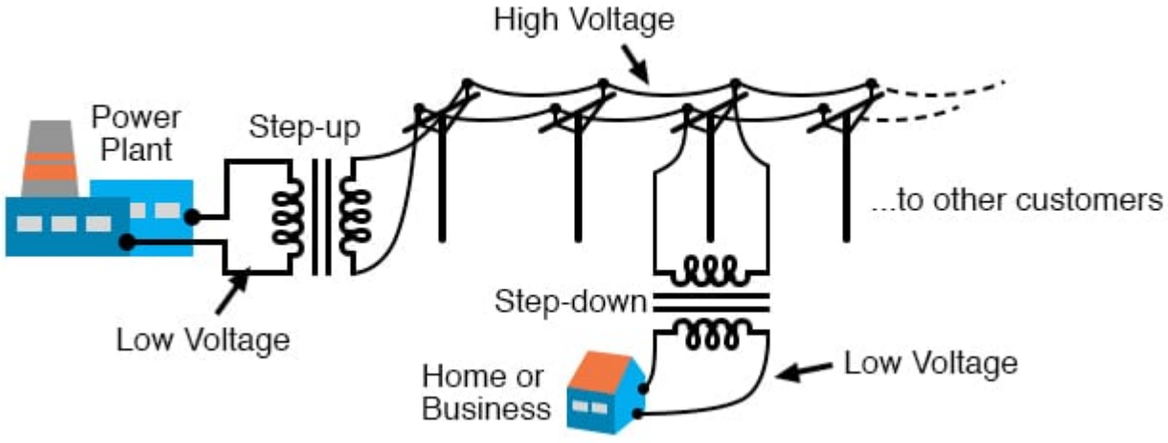
$$\frac{I_s}{I_p} = \frac{V_p}{V_s} = \frac{N_p}{N_s}$$



If the winding ratio is reversed so that the primary coil has fewer turns than the secondary coil, the transformer "steps up" the voltage from the source level to a higher level at the load:



When transmitting electrical power over long distances, it is far more efficient to do so with stepped-up voltages and stepped-down currents (smaller-diameter wire with less resistive power losses), then step the voltage back down and the current back up for industry, business, or consumer use.



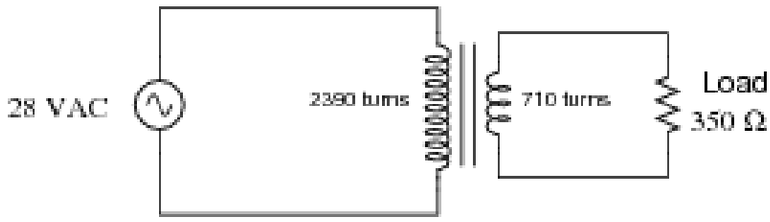
Transformer technology has made long-range electric power distribution practical. Without the ability to efficiently step voltage up and down, it would be cost-prohibitive to construct power systems for anything but close-range (within a few miles at most) use.

As useful as transformers are, they only work with AC, not DC. Because the phenomenon of mutual inductance relies on *changing* magnetic fields, and direct current (DC) can only produce steady magnetic fields, transformers simply will not work with direct current.

Practice

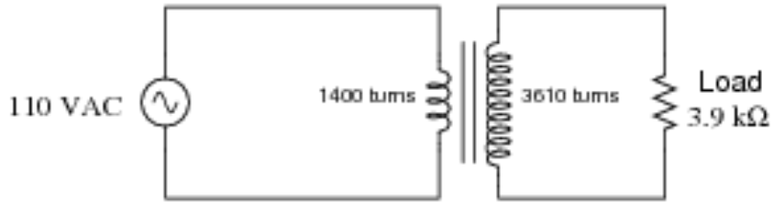
1. Calculate the voltage output by the secondary winding of a transformer if the primary voltage is 35 volts, the secondary winding has 4500 turns, and the primary winding has 355 turns.

2. Calculate the load current and load voltage in this transformer circuit:



3. Calculate the number of turns needed in the secondary winding of a transformer to transform a primary voltage of 300 volts down to a secondary voltage of 180 volts, if the primary winding has 1150 turns of wire.

4. Calculate the source current and load current in this transformer circuit:



9.Efficiency of Transformers

The ratio of the output power to the input power in a transformer is known as **efficiency of transformer**. The transformer efficiency is represented by (Rd).

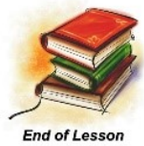
$$\boldsymbol{Rd = \frac{Output\ Power}{Input\ Power} \times 100\%}$$

Or

$$\boldsymbol{Rd = \frac{P_s}{P_p} \times 100\%}$$

Or

$$\boldsymbol{Rd = \frac{I_s \times V_s}{I_p \times V_p} \times 100\%}$$



End of Lesson

Lesson 5: Electronic Components

Objectives: You will be able to:

- Identify some basic electronic components
- Connect simple circuits.

1. Connecting wire

Crocodile Clips Male to Female connecting wires / jumper wires

2. Switches:

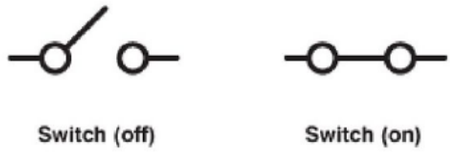
One of the most elementary and easy-to-overlook circuit component is the switch.

What is a Switch?

A switch is a component used to open close an electric circuit. They allow control over current flow in a circuit.

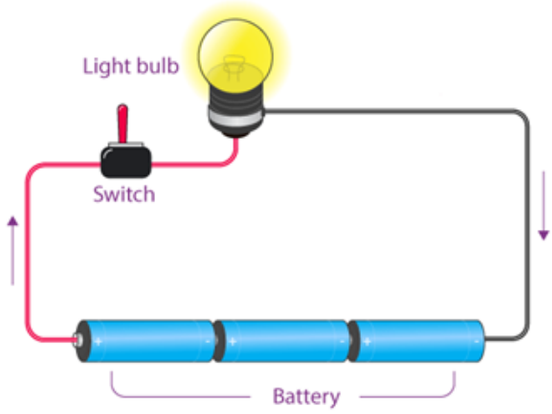
A switch can only exist in one of two states: open or closed. In the **off** state, a switch looks like an open gap in the circuit. This, in effect, looks like an **open circuit**, preventing current from flowing.

In the **on** state, a switch acts just like a piece of perfectly-conducting wire. A short. This **closes the circuit**, turning the system "on" and allowing current to flow unimpeded through the rest of the system.



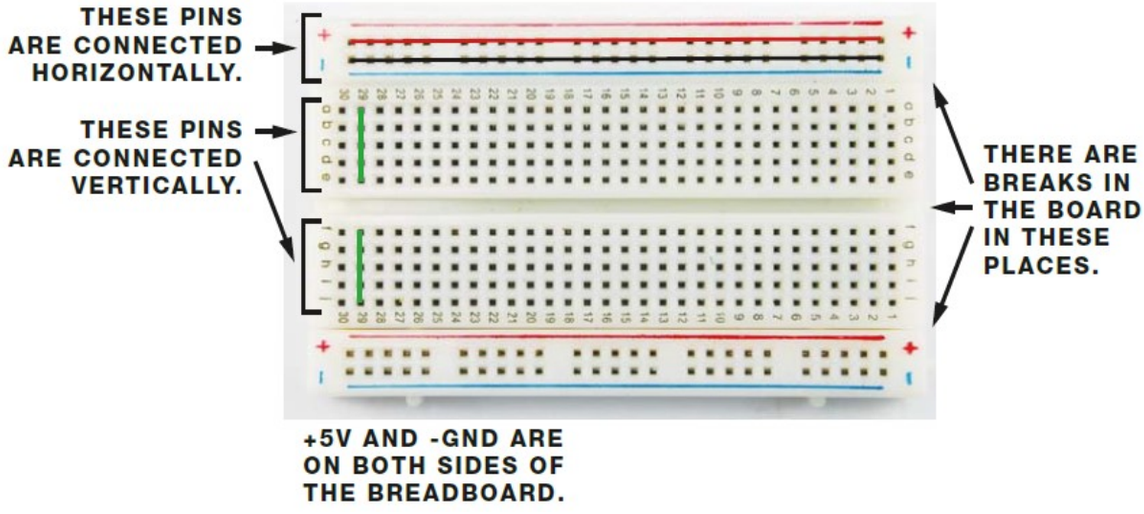
Switch (off) Switch (on)

Example of connecting switch in a simple circuit:



3. Breadboard

A board on which you can build electronic circuits. It's like a patch panel, with rows of holes that allow you to connect wires and components together.

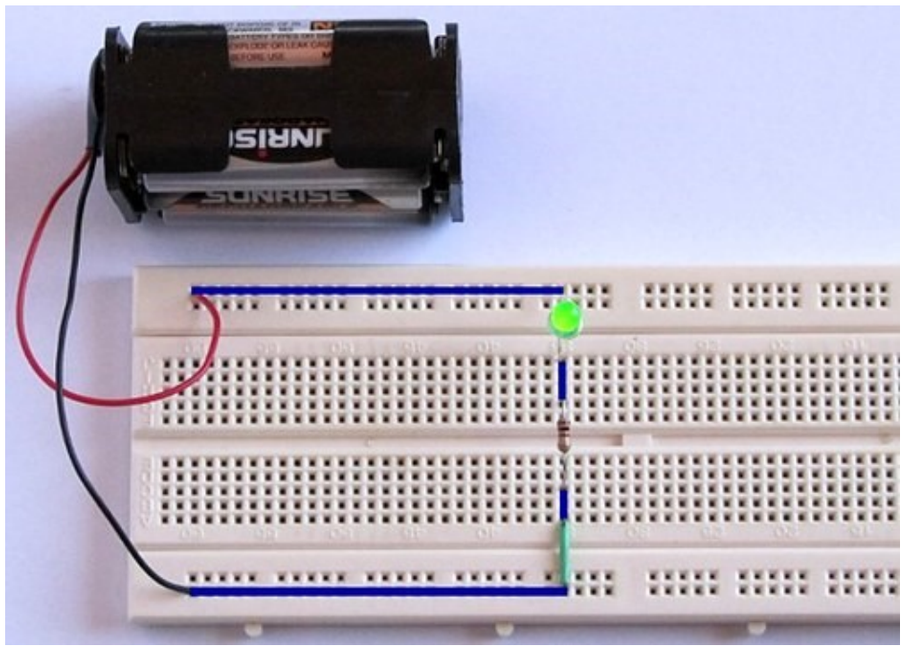


4. Light Emitting Diode (LED)

The LED (Light Emitting Diode) is exactly what its name suggests – a diode that emits light. LEDs are like small light bulbs and are available in different sizes and colors.

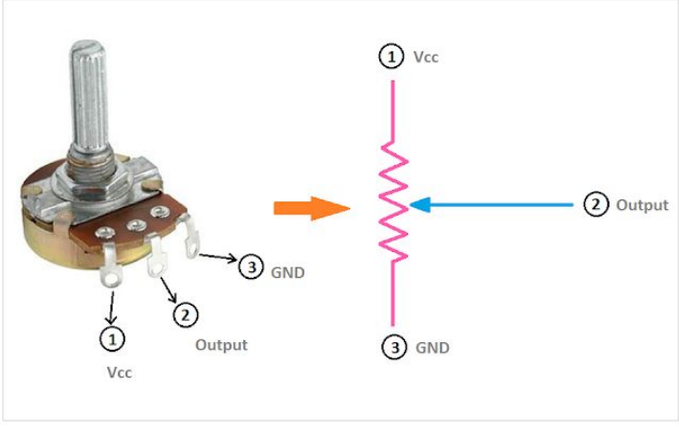
Example: LED Circuit on a Breadboard
Materials:

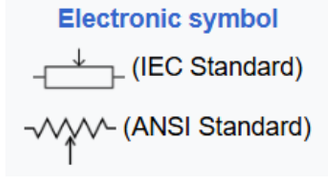
- Breadboard
- LED (color of your choice)
- Resistor (220-ohm or 330-ohm is common for LEDs)
- Jumper Wires
- 2 Batteries
- Battery Holder (for 2 batteries)



5. Potentiometer

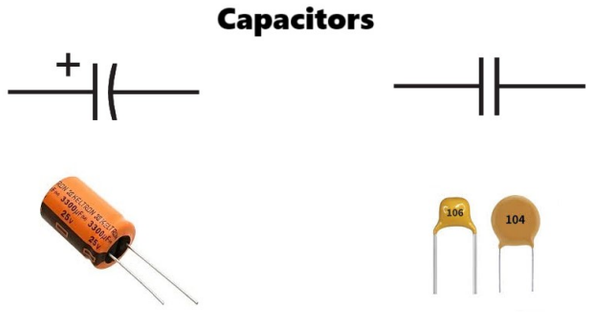
A variable resistor with three pins. Two of the pins are connected to the ends of a fixed resistor. The middle pin, or wiper, moves across the resistor, dividing it into two halves. When the external sides of the potentiometer are connected to voltage and ground, the middle leg will give the difference in voltage as you turn the knob. Often referred to as a pot.





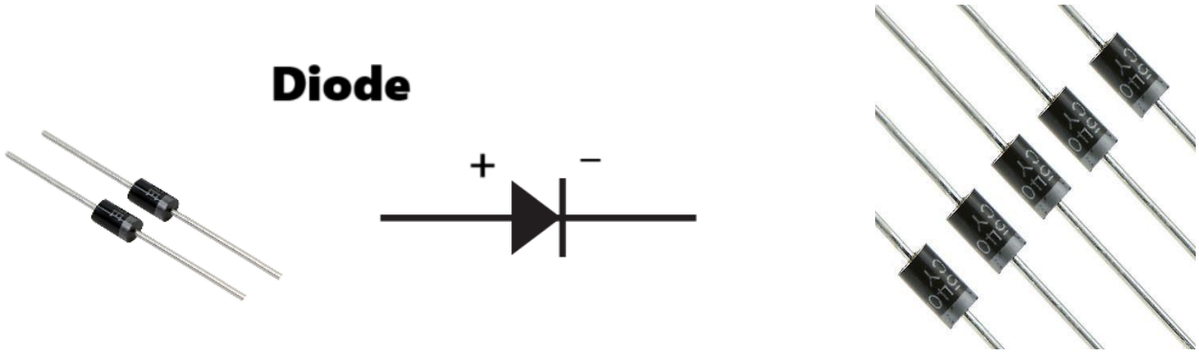
6.Capacitors

Capacitors store and release energy, sort of like tiny, rechargeable batteries. In some cases, they’ll smooth out fluctuations in power supply and work as filters. While in others, they’ll store up energy and then release it all at once. They’re the component that lets your camera flash go off without frying the whole system. Charging of capacitors can be done with the help of an external battery. A capacitor only allows AC components to pass through it and blocks the DC component.

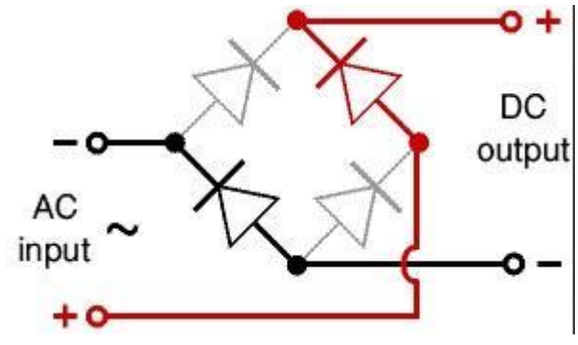


7.Diodes:

Diodes let current flow in one direction only, preventing those pesky reverse currents that mess with your circuits. The most famous type? The LED (Light Emitting Diode), which also happens to light up when electricity passes through. Polarity plays an important role in diodes, they are the very basic components having applications such as AC to DC conversion, clipping and clamping of signals and some most famous zener diodes used as voltage reference generators.



Function: the bridge converts the AC voltages to DC voltages. Rectifying Bridge consist of 4 diodes connected with each other in a specific way (as in the following figure). Each bridge has its own current that it can handle.



Chapter 4 : Optics

Lesson 1 Refraction of Light

OBJECTIVES

- **Recognize situations in which refraction will occur.**
- **Identify which direction light will bend when it passes from one medium to another.**
- **Solve problems using Snell's law.**

refraction

the bending of a wave front as the wave front passes between two substances in which the speed of the wave differs

1.REFRACTION OF LIGHT

Look at the tiny image of the flower that appears in the water droplet in **Figure 1.** The blurred flower can be seen in the background of the photo. Why does the flower look different when viewed through the droplet? This phenomenon occurs because light is bent at the boundary between the water and the air around it. The bending of light as it travels from one medium to another is called **refraction.**

If light travels from one transparent medium to another at any angle other than straight on (normal to the surface), the light ray changes direction when it meets the boundary. As in the case of reflection, the angles of the incoming and refracted rays are measured with respect to the normal. For studying refraction, the normal line is extended into the refracting medium, as shown in **Figure 2.** The angle between the refracted ray and the normal is called the *angle of refraction,* θ_r, and the angle of incidence is designated as θ_i.

Refraction occurs when light's velocity changes

Glass, water, ice, diamonds, and quartz are all examples of transparent media through which light can pass. The speed of light in each of these materials is different. The speed of light in water, for instance, is less than the speed of light in air. And the speed of light in glass is less than the speed of light in water.

Figure 1
The flower looks small when viewed through the water droplet. The light from the flower is bent because of the shape of the water droplet and the change in material as the light passes through the water.

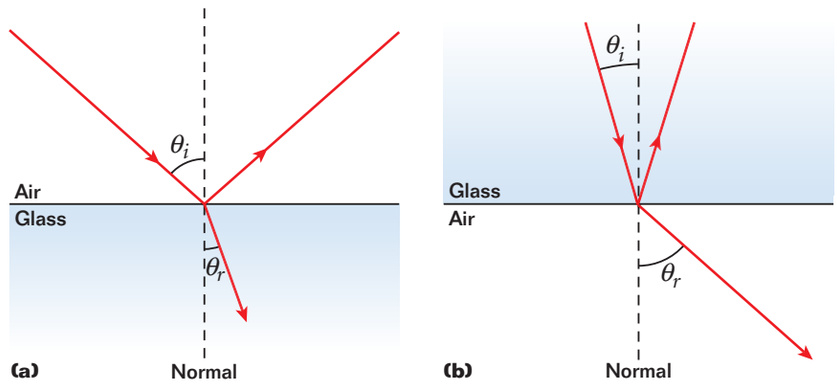


Figure 2
When light moves from one medium to another, part of it is reflected and part is refracted. **(a)** When the light ray moves from air into glass, the refracted portion is bent toward the normal, **(b)** whereas the path of the light ray moving from glass into air is bent away from the normal.

When light moves from a material in which its speed is higher to a material in which its speed is lower, such as from air to glass, the ray is bent toward the normal, as shown in Figure 2(a). If the ray moves from a material in which its speed is lower to one in which its speed is higher, as in Figure 2(b), the ray is bent away from the normal. If the incident ray of light is parallel to the nor-mal, rfcl lm pcdp_argml &`clbgle' maaspq gl cgrfcp a_qc, Lmrc rf_r rfc n_rf md _ jgefr p_w rf_r apmqqcq _ `ms lb_pw `crucc l rum bgd+dcpc l r k cbg_ gq pctcpqg`jc, Gd rfc p_w gl Dgespc O&_' mpgegl_rcb **inside the glass block, it would follow the same path as shown in the figure, but the reflected ray would be inside the block.**

2.THE LAW OF REFRACTION

index of refraction

the ratio of the speed of light in a vacuum to the speed of light in a given transparent medium

An important property of transparent substances is the [redacted]. The index of refraction for a substance is the ratio of the speed of light in a vacuum to the speed of light in that substance.

INDEX OF REFRACTION

$$n = \frac{c}{v}$$

$$\text{index of refraction} = \frac{\text{speed of light in vacuum}}{\text{speed of light in medium}}$$

Did you know?

The index of refraction of any medium can also be expressed as the ratio of the wavelength of light in a vacuum, λ_0, to the wavelength of light in that medium, λ_n, as shown in the following relation.

$$n = \frac{\lambda_0}{\lambda_n}$$

From this definition, we see that the index of refraction is a dimensionless number that is always greater than 1 because light always travels slower in a substance than in a vacuum. **Table 1** lists the indices of refraction for different substances. Note that the larger the index of refraction is, the slower light travels in that substance and the more a light ray will bend when it passes from a vacuum into that material.

Imagine, as an example, light passing between air and water. When light begins in the air (high speed of light and low index of refraction) and travels into the water (lower speed of light and higher index of refraction), the light rays are bent toward the normal. Conversely, when light passes from the water to the air, the light rays are bent away from the normal.

Note that the value for the index of refraction of air is nearly that of a vacuum. For simplicity, *use the value n = 1.00 for air when solving problems.*

Table 1 Indices of Refraction for Various Substances*

Solids at 20°C	*n*	Liquids at 20°C	*n*
Cubic zirconia	2.20	Benzene	1.501
Diamond	2.419	Carbon disulfide	1.628
Fluorite	1.434	Carbon tetrachloride	1.461
Fused quartz	1.458	Ethyl alcohol	1.361
Glass, crown	1.52	Glycerine	1.473
Glass, flint	1.66	Water	1.333
Ice (at 0°C)	1.309	**Gases at 0°C, 1 atm**	***n***
Polystyrene	1.49	Air	1.000 293
Sodium chloride	1.544	Carbon dioxide	1.000 450
Zircon	1.923		

*measured with light of vacuum wavelength = 589 nm

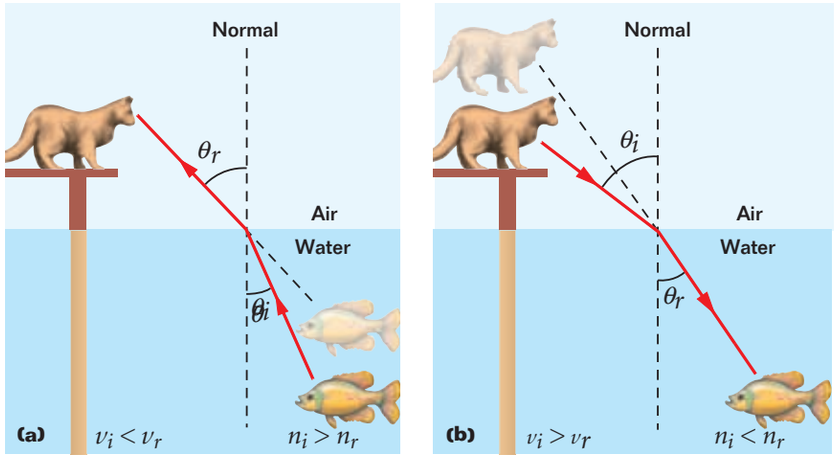


Figure 4
(a) To the cat on the pier, the fish looks closer to the surface than it really is. **(b)** To the fish, the cat seems to be farther from the surface than it actually is.

Objects appear to be in different positions due to refraction

When looking at a fish underwater, a cat sitting on a pier perceives the fish to be closer to the water's surface than it actually is, as shown in **Figure 4(a).** Conversely, the fish perceives the cat on the pier to be farther from the water's surface than it actually is, as shown in **Figure 4(b).**

Because of the reversibility of refraction, both the fish and the cat see along the same path, as shown by the solid lines in both figures. However, the light ray that reaches the fish forms a smaller angle with respect to the normal than does the light ray from the cat to the water's surface. The reason is that light is bent toward the normal when it travels from a medium with a lower index of refraction (the air) to one with a higher index of refraction (the water). Extending this ray along a straight line shows the cat's image to be above the cat's actual position.

On the other hand, the light ray that reaches the cat from the water's sur-face forms a larger angle with respect to the normal, because the light from the fish travels from a medium with a higher index of refraction to one with a lower index of refraction. Note that the fish's image is closer to the water's sur-face than the fish actually is. An underwater object seen from the air above appears larger than its actual size because the image, which is the same size as the object, is closer to the observer.

Wavelength affects the index of refraction

Note that the indices of refraction listed in **Table 1** are only valid for light that has a wavelength of 589 nm in a vacuum. The reason is that the amount that light bends when entering a different medium depends on the wavelength of the light as well as the speed. Thus, a spectrum is produced when white light passes through a prism. Each color of light has a different wavelength. There-fore, each color of the spectrum is refracted by a different amount.

Snell's law determines the angle of refraction

The index of refraction of a material can be used to figure out how much a ray of light will be refracted as it passes from one medium to another. As men-tioned, the greater the index of refraction, the more refraction occurs. But how can the angle of refraction be found?

In 1621, Willebrord Snell experimented with light passing through different media. He developed a relationship called Snell's law, which can be used to find the angle of refraction for light traveling between any two media.

SNELL'S LAW

$$n_i \sin \theta_i = n_r \sin \theta_r$$

index of refraction of first medium × sine of the angle of incidence = index of refraction of second medium × sine of the angle of refraction

SAMPLE PROBLEM A

Snell's Law

PROBLEM

A light ray of wavelength 589 nm (produced by a sodium lamp) traveling through air strikes a smooth, flat slab of crown glass at an angle of 30.0° to the normal. Find the angle of refraction, θ_r.

SOLUTION

Given: $\theta_i = 30.0°$ $n_i = 1.00$ $n_r = 1.52$

Unknown: $\theta_r = ?$

Use the equation for Snell's law.

$$n_i \sin \theta_i = n_r \sin \theta_r$$

$$\theta_r = \sin^{-1}\left[\frac{n_i}{n_r}(\sin \theta_i)\right] = \sin^{-1}\left[\frac{1.00}{1.52}(\sin 30.0°)\right]$$

$$\boxed{\theta_r = 19.2°}$$

PRACTICE A

Snell's Law

1. Find the angle of refraction for a ray of light that enters a bucket of water from air at an angle of 25.0° to the normal. (Hint: Use **Table 1.**)

2. For an incoming ray of light of vacuum wavelength 589 nm, fill in the unknown values in the following table. (Hint: Use **Table 1.**)

	from (medium)	to (medium)	θ_i	θ_r
a.	flint glass	crown glass	25.0°	?
b.	air	?	14.5°	9.80°
c.	air	diamond	31.6°	?

3. A ray of light of vacuum wavelength 550 nm traveling in air enters a slab of transparent material. The incoming ray makes an angle of 40.0° with the normal, and the refracted ray makes an angle of 26.0° with the normal. Find the index of refraction of the transparent material. (Assume that the index of refraction of air for light of wavelength 550 nm is 1.00.)

Lesson 2 Thin Lenses

1.TYPES OF LENSES

OBJECTIVES

- **Use ray diagrams to find the position of an image produced by a converging or diverging lens, and identify the image as real or virtual.**
- **Solve problems using the thin-lens equation.**
- **Calculate the magnification of lenses.**
- **Describe the positioning of lenses in compound microscopes and refracting telescopes.**

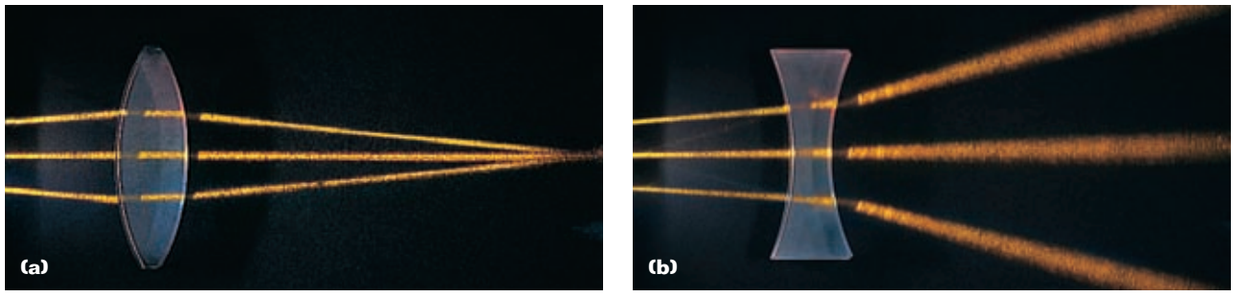
When light traveling in air enters a pane of glass, it is bent toward the normal. As the light exits the pane of glass, it is bent again. When the light exits, however, its speed increases as it enters the air, so the light bends away from the normal. Because the amount of refraction is the same regardless of whether light is entering or exiting a medium, the light rays are bent as much on exiting the pane of glass as they were on entering.

Curved surfaces change the direction of light

When the surfaces of a medium are curved, the direction of the normal line dif-fers for each spot on the surface of the medium. Thus, when light passes through a medium that has one or more curved surfaces, the change in the direction of the light rays varies from point to point. This principle is applied in media called **lenses.** Like mirrors, lenses form images, but lenses do so by refraction rather than by reflection. The images formed can be either real or virtual, depending on the type of lens and on the placement of the object. Recall that a real image is formed when rays of light actually intersect to form the image. Virtual images form at a point from which light rays appear to come but do not actually come. Real images can be projected onto a screen; virtual images cannot.

Lenses are commonly used to form images in optical instruments, such as cameras, telescopes, and microscopes. In fact, transparent tissue in the front of the human eye acts as a lens, converging light toward the light-sensitive retina, which lines the back of the eye.

A typical lens consists of a piece of glass or plastic ground so that each of its two refracting surfaces is a segment of either a sphere or a plane. **Figure 5** shows examples of lenses. Notice that the lenses are shaped differently. The lens that is thicker at the middle than it is at the rim, shown in **Figure 5(a),** is an example of a *converging* lens. The lens that is thinner at the middle than it



lens *a transparent object that refracts light rays such that they converge or diverge to create an image*

Figure 5

When rays of light pass through **(a)** a converging lens (thicker at the middle), they are bent inward. When they pass through **(b)** a diverging lens (thicker at the edge), they are bent outward.

is at the rim, shown in Figure 5(b), is an example of a diverging lens. The light rays show why the names converging and diverging are applied to these lenses.

Focal length is the image distance for an infinite object distance

As with mirrors, it is convenient to define a point called the *focal point* for a lens. Note that light rays from an object far away are nearly parallel. The focal point of a converging lens is the location where the image of an object at an infinite distance from the lens is focused. For example, in **Figure 6(a)** a group of rays parallel to the principal axis passes through a focal point, *F,* after being bent inward by the lens. Unlike mirrors, every lens has a focal point on each side of the lens because light can pass through the lens from either side, as illustrated in **Figure 6.** The distance from the focal point to the center of the lens is called the *focal length, f.* The focal length is the image distance that corresponds to an infi-nite object distance.

Rays parallel to the principal axis diverge after passing through a diverging lens, as shown in **Figure 6(b).** In this case, the focal point is defined as the point from which the diverged rays appear to originate. Again, the focal length is defined as the distance from the center of the lens to the focal point.

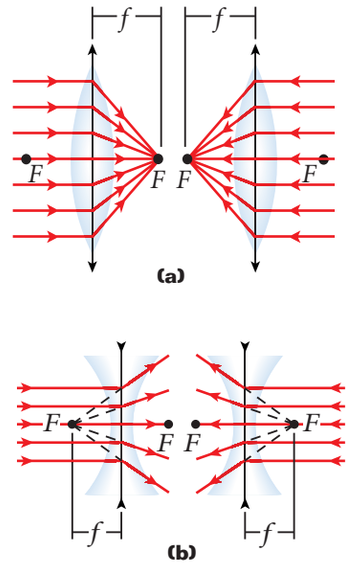


Figure 6
Both **(a)** converging lenses and **(b)** diverging lenses have two focal points but only one focal length.

Ray diagrams of thin-lens systems help identify image height and location

In the chapter on light and reflection, we used a set of standard rays and a ray diagram to predict the characteristics of images formed by spherical mirrors. A similar approach can be used for lenses.

We know, as shown in **Figure 5,** that refraction occurs at a boundary between two materials with different indexes of refraction. However, for *thin lenses* (lenses for which the thickness of the lens is small compared to the radius of curvature of the lens or the distance of the object from the lens), we can represent the front and back boundaries of the lens as a line segment pass-ing through the center of the lens. To draw ray diagrams in the thin-lens approximation, we will use a line segment with arrow ends to indicate a con-verging lens, as in **Figure 6(a).** To show a diverging lens, we will draw a line segment with "upside-down" arrow ends, as illustrated in **Figure 6(b).** We can then draw ray diagrams using the set of rules outlined in **Table 2.**

Table 2 Rules for Drawing Reference Rays

Ray	From object to lens	From *converging* lens to image	From *diverging* lens to image
Parallel ray	parallel to principal axis	passes through focal point, *F*	directed away from focal point, *F*
Central ray	to the center of the lens	from the center of the lens	from the center of the lens
Focal ray	passes through focal point, *F*	parallel to principal axis	parallel to principal axis

The reasons why these rules work relate to concepts already covered in this textbook. From the definition of a focal point, we know that light traveling parallel to the principal axis (parallel ray) will be focused at the focal point. For a converging lens, this means that light will come together at the focal point in back of the lens. (In this book, the *front* of the lens is defined as the side of the lens that the light rays first encounter. The *back* of the lens refers to the side of the lens opposite where the light rays first encounter the lens.) But a similar ray passing through a diverging lens will exit the lens as if it originat-ed from the focal point in front of the lens. Because refraction is reversible, a ray entering a converging lens from either focal point will be refracted so that it is parallel to the principal axis.

For both lenses, a ray passing through the center of the lens will continue in a straight line with no net refraction. This occurs because both sides of a lens are parallel to one another along any path through the center of the lens. As with a pane of glass, the exiting ray will be parallel to the ray that entered the lens. For ray diagrams, the usual assumption is that the lens is negligibly thin, so it is assumed that the ray is not displaced sideways but instead continues in a straight line.

2.CHARACTERISTICS OF LENSES

Table 3 summarizes the possible relationships between object and image posi-tions for converging lenses. The rules for drawing reference rays were used to create each of these diagrams. Note that applications are listed along with each ray diagram to show the varied uses of the different configurations.

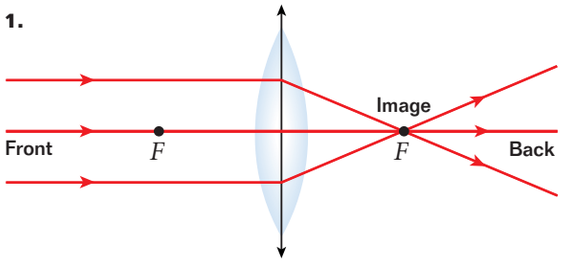
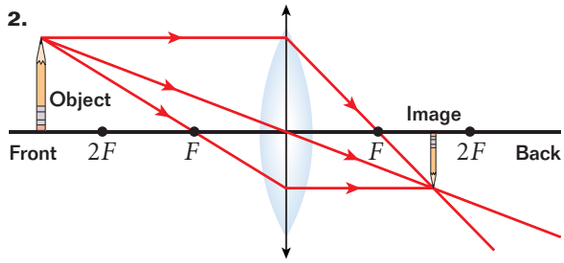
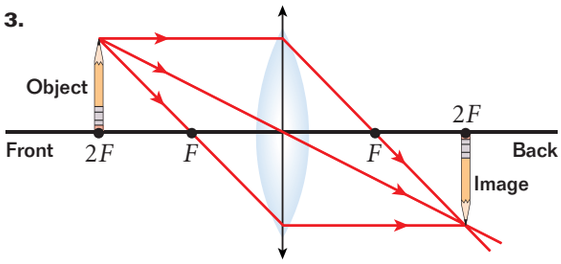
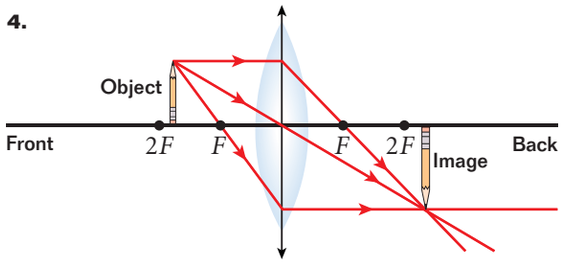
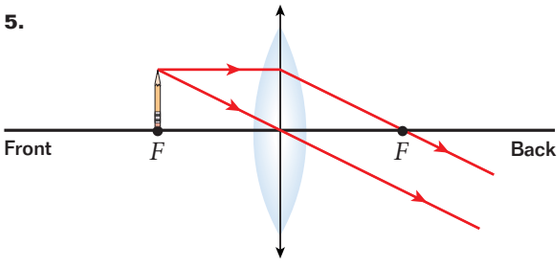
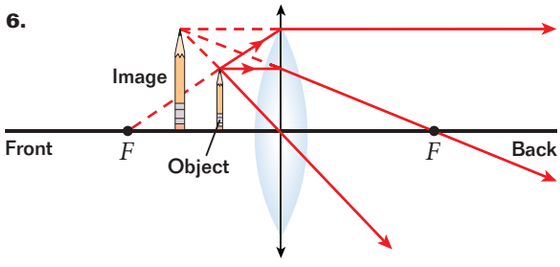
Converging lenses can produce real or virtual images of real objects

An object infinitely far away from a converging lens will create a point image at the focal point, as shown in the first diagram in **Table 3.** This image is real, which means that it can be projected on a screen.

As a distant object approaches the focal point, the image becomes larger and farther away, as shown in the second, third, and fourth diagrams in **Table 3.** When the object is at the focal point, as shown in the fifth diagram, the light rays from the object are refracted so that they exit the lens parallel to each other.(Because the object is at the focal point, it is impossible to draw a third ray that passes through that focal point, the lens, and the tip of the object.)

When the object is between a converging lens and its focal point, the light rays from the object diverge when they pass through the lens, as shown in the sixth diagram in **Table 3.** This image appears to an observer in back of the lens as being on the same side of the lens as the object. In other words, the brain interprets these diverging rays as coming from an object directly along the path of the rays that reach the eye. The ray diagram for this final case is less straight-forward than those drawn for the other cases in the table. The first two rays (parallel to the axis and through the center of the lens) are drawn.

Table 3 Images Created by Converging Lenses

Ray diagrams	
1. 	2. 
Configuration: object at infinity; point image at *F* **Applications:** burning a hole with a magnifying glass	**Configuration:** object outside 2*F*; real, smaller image between *F* and 2*F* **Applications:** lens of a camera, human eyeball lens, and objective lens of a refracting telescope
3. 	4. 
Configuration: object at 2*F*; real image at 2*F* same size as object **Applications:** inverting lens of a field telescope	**Configuration:** object between *F* and 2*F*; magnified real image outside 2*F* **Applications:** motion-picture or slide projector and objective lens in a compound microscope
5. 	6. 
Configuration: object at *F;* image at infinity **Applications:** lenses used in lighthouses and searchlights	**Configuration:** object inside *F;* magnified virtual image on the same side of the lens as the object **Applications:** magnifying with a magnifying glass; eye-piece lens of microscope, binoculars, and telescope

fashion. The third ray, however, is drawn so that if it were extended, it would connect the focal point in front of the lens, the tip of the object, and the lens in a straight line. To determine where the image is, draw lines extending from the rays exiting the lens back to the point where they would appear to have origi-nated to an observer on the back side of the lens (these lines are dashed in the sixth diagram in **Table 3**).

Diverging lenses produce virtual images from real objects

A diverging lens creates a virtual image of a real object placed anywhere with respect to the lens. The image is upright, and the magnification is always less than one; that is, the image size is reduced. Additionally, the image appears inside the focal point for any placement of the real object.

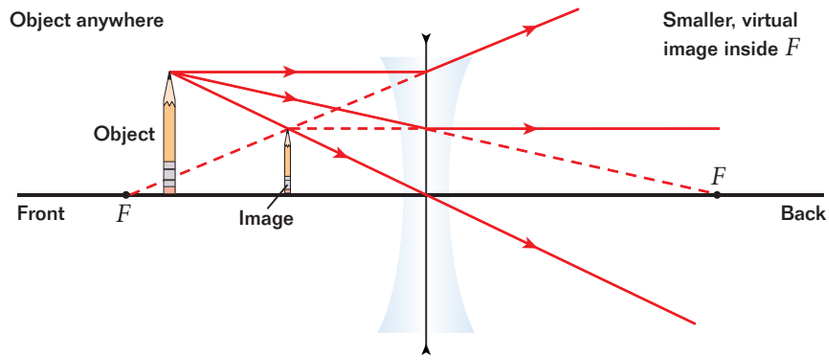


Figure 7 The image created by a diverging lens is always a virtual, smaller image.

The ray diagram shown in **Figure 7** for diverging lenses was created using the rules given in **Table 2.** The first ray, parallel to the axis, appears to come from the focal point on the same side of the lens as the object. This ray is indicated by the oblique dashed line. The second ray passes through the cen-ter of the lens and is not refracted. The third ray is drawn as if it were going to the focal point in back of the lens. As this ray passes through the lens, it is refracted parallel to the principal axis and must be extended backward, as shown by the dashed line. The location of the tip of the image is the point at which the three rays appear to have originated.

3.THE THIN-LENS EQUATION AND MAGNIFICATION

Ray diagrams for lenses give a good estimate of image size and distance, but it is also possible to calculate these values. The equation that relates object and image distances for a lens is called the *thin-lens equation* because it is derived using the assumption that the lens is very thin. In other words, this equation applies when the lens thickness is much smaller than its focal length.

THIN-LENS EQUATION

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f}$$

$$\frac{1}{\text{distance from object to lens}}+\frac{1}{\text{distance from image to lens}}=\frac{1}{\text{focal length}}$$

When using the thin-lens equation, we often illustrate it using the ray dia-gram model in which, for clarity, we magnify the vertical axis and show the lens position as a thin line. Always remember that the actual light rays bend at the lens surfaces and that our diagram showing bending at a single central line is an idealized model, which is quite good for thin lenses. But the model, and the equation, must be modified to deal properly with thick lenses, systems of lenses, and object and image points far from the principal axis.

The thin-lens equation can be applied to both converging and diverging lenses if we adhere to a set of sign conventions. **Table 4** gives the sign conven-tions for lenses. Under this convention, an image in back of the lens (that is, a real image) has a positive image distance, and an image in front of the lens, or a virtual image, has a negative image distance. A converging lens has a positive focal length and a diverging lens has a negative focal length. Therefore, con-verging lenses are sometimes called *positive lenses* and diverging lenses are sometimes called *negative lenses.*

Table 4
Sign Conventions for Lenses

	+	–
p	real object in front of the lens	real object in back of the lens
q	virtual image in back of the lens	virtual image in front of the lens
f	converging lens	diverging lens

Magnification by a lens depends on object and image distances

Recall that magnification (M) is defined as the ratio of image height to object height. The following equation can be used to calculate the magnification of both converging and diverging lenses.

MAGNIFICATION OF A LENS

$$M = \frac{h'}{h} = -\frac{q}{p}$$

$$\text{magnification} = \frac{\text{image height}}{\text{object height}} = -\frac{\text{distance from image to lens}}{\text{distance from object to lens}}$$

If close attention is given to the sign conventions defined in **Table 4,** then the magnification will describe the image's size and orientation. When the magnitude of the magnification of an object is less than one, the image is smaller than the object. Conversely, when the magnitude of the magnification is greater than one, the image is larger than the object.

Additionally, a negative sign for the magnification indicates that the image is real and inverted. A positive magnification signifies that the image is upright and virtual.

SAMPLE PROBLEM B

Lenses

PROBLEM

An object is placed 30.0 cm in front of a converging lens and then 12.5 cm in front of a diverging lens. Both lenses have a focal length of 10.0 cm. For both cases, find the image distance and the magnification. Describe the images.

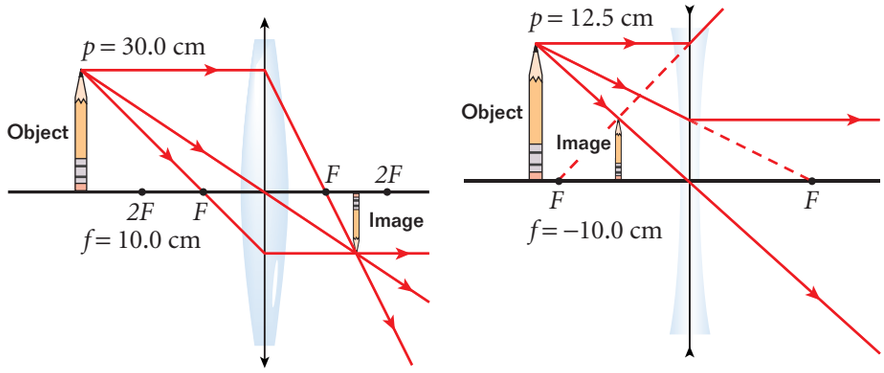
SOLUTION

1. DEFINE

Given: $f_{converging} = 10.0\text{ cm}$ $f_{diverging} = -10.0\text{ cm}$
$p_{converging} = 30.0\text{ cm}$ $p_{diverging} = 12.5\text{ cm}$

Unknown: $q_{converging} = ?$ $M = ?$ $q_{diverging} = ?$ $M = ?$

Diagrams:



2. PLAN

Choose an equation or situation:

The thin-lens equation can be used to find the image distance, and the equation for magnification will serve to describe the size and orientation of the image.

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \qquad M = -\frac{q}{p}$$

Rearrange the equation to isolate the unknown:

$$\frac{1}{q} = \frac{1}{f} - \frac{1}{p}$$

3. CALCULATE

For the converging lens:

$$\frac{1}{q} = \frac{1}{f} - \frac{1}{p} = \frac{1}{10.0\text{ cm}} - \frac{1}{30.0\text{ cm}} = \frac{2}{30.0\text{ cm}}$$

$$\boxed{q = 15.0\text{ cm}}$$

$$M = -\frac{q}{p} = -\frac{15.0\text{ cm}}{30.0\text{ cm}}$$

$$\boxed{M = -0.500}$$

REVIEW

1. What type of image is produced by the cornea and the lens on the retina?
2. What type of image, virtual or real, is produced in the following cases?
   a. an object inside the focal point of a camera lens
   b. an object outside the focal point of a refracting telescope's objective lens
   c. an object outside the focal point of a camera's viewfinder
3. Find the image position for an object placed 3.0 cm outside the focal point of a converging lens with a 4.0 cm focal length.
4. What is the magnification of the object from item 3?

5.An object is placed in front of a diverging lens with a focal length of 20.0 cm. For each object distance, find the image distance and the magnification. Describe each image.

a. 40.0 cm
b. 20.0 cm
c. 10.0 cm

6.A person looks at a gem using a converging lens with a focal length of 12.5 cm. The lens forms a vir-tual image 30.0 cm from the lens. Determine the magnification. Is the image upright or inverted?

SECTION OBJECTIVES

- **Predict whether light will be refracted or undergo total internal reflection.**
- **Recognize atmospheric conditions that cause refraction.**
- **Explain dispersion and phenomena such as rainbows in terms of the relationship between the index of refraction and the wavelength.**

5. Optical Phenomena

TOTAL INTERNAL REFLECTION

An interesting effect called **total internal reflection** can occur when light moves along a path from a medium with a *higher* index of refraction to one with a *lower* index of refraction. Consider light rays traveling from water into air, as shown in **Figure 10(a).** Four possible directions of the rays are shown in the figure.

At some particular angle of incidence, called the **critical angle,** the refracted ray moves parallel to the boundary, making the angle of refraction equal to 90°, as shown in **Figure 10(b).** For angles of incidence greater than the critical angle, the ray is entirely reflected at the boundary, as shown in **Figure 10.** This ray is reflected at the boundary as though it had struck a perfectly reflecting surface. Its path and the path of all rays like it can be predicted by the law of reflection; that is, the angle of incidence equals the angle of reflection.

total internal reflection

the complete reflection that takes place within a substance when the angle of incidence of light striking the surface boundary is greater than the critical angle

critical angle

the angle of incidence at which the refracted light makes an angle of 90° with the normal

In optical equipment, prisms are arranged so that light entering the prism is totally internally reflected off the back surface of the prism. Prisms are used in place of silvered or aluminized mirrors because they reflect light more effi-ciently and are more scratch resistant.

Snell's law can be used to find the critical angle. As mentioned above, when the

angle of incidence, θ_i, equals the critical angle, θ_C, then the angle of refraction, θ_r, equals 90°. Substituting these values into Snell's law gives the following relation.

$$n_i \sin \theta_C = n_r \sin 90°$$

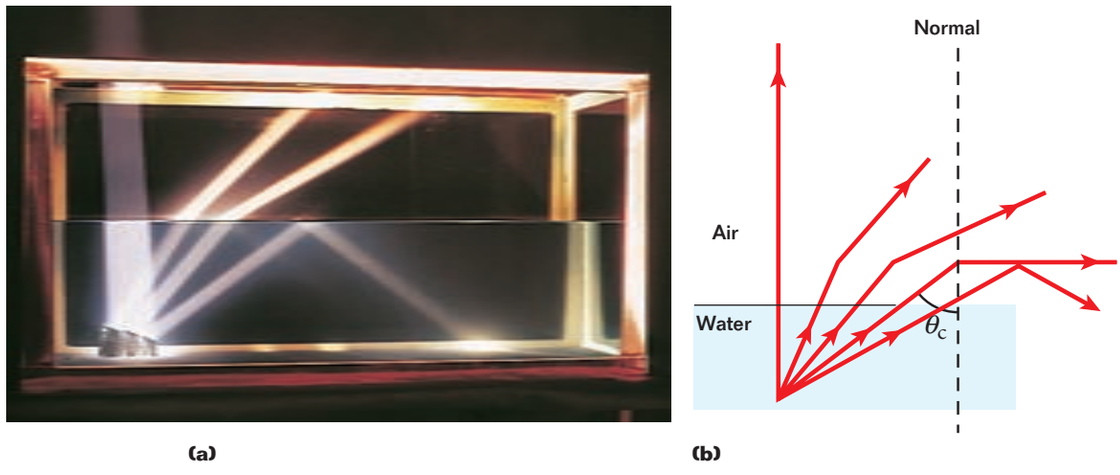


Figure 10

(a) This photo demonstrates several different paths of light radiated from the bottom of an aquarium. **(b)** At the critical angle, θ_C, a light ray will travel parallel to the boundary. Any rays with an angle of incidence greater than θ_C will be totally internally re-flected at the boundary.

Because the sine of 90° equals 1, the following relationship results.

CRITICAL ANGLE

$$\sin \theta_c = \frac{n_r}{n_i} \quad \text{for } n_i > n_r$$

$$\text{sine (critical angle)} = \frac{\text{index of refraction of second medium}}{\text{index of refraction of first medium}}$$

but only if index of refraction of first medium > index of refraction of second medium

Note that this equation can be used only when n_i is greater than n_r. In other words, *total internal reflection occurs only when light moves along a path from a medium of higher index of refraction to a medium of lower index of refraction.* If n_i were less than n_r, this equation would give sin $\theta_c > 1$, which is an impossible result because by definition the sine of an angle can never be greater than 1.

When the second substance is air, the critical angle is small for substances with large indices of refraction. Diamonds, which have an index of refraction of 2.419, have a critical angle of 24.4°. By comparison, the critical angle for crown glass, a very clear optical glass, where n = 1.52, is 41.0°. Because diamonds have such a small critical angle, most of the light that enters a cut diamond is totally internally reflected. The reflected light eventually exits the diamond from the most visible faces of the diamond. Jewelers cut diamonds so that the maximum light entering the upper surface is reflected back to these faces.

SAMPLE PROBLEM C

Critical Angle

PROBLEM

Find the critical angle for a water-air boundary if the index of refraction of water is 1.333.

SOLUTION

Given: $n_i = 1.333$ $n_r = 1.000$

Unknown: $\theta_c = ?$

Use the equation for critical angle on this page.

$$\sin \theta_c = \frac{n_r}{n_i}$$

$$\theta_c = \sin^{-1}\left(\frac{n_r}{n_i}\right) = \sin^{-1}\left(\frac{1.00}{1.333}\right)$$

$$\boxed{\theta_c = 48.6°}$$

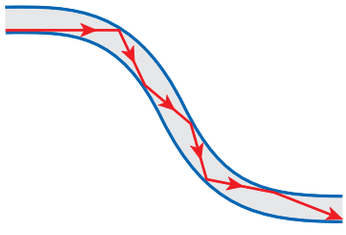
PRACTICE C

Critical Angle

1. Glycerine is used to make soap and other personal care products. Find the critical angle for light traveling from glycerine ($n = 1.473$) into air.
2. Calculate the critical angle for light traveling from glycerine ($n = 1.473$) into water ($n = 1.333$).
3. Ice has a lower index of refraction than water. Find the critical angle for light traveling from ice ($n = 1.309$) into air.
4. Which has a smaller critical angle in air, diamond ($n = 2.419$) or cubic zirconia ($n = 2.20$)? Show your work.

Fiber Optics

Another interesting application of total internal reflection is the use of glass or transparent plastic rods, like the ones shown in the photograph, to transfer light from one place to another. As indicated in the illustration below, light is confined to traveling within the rods, even around gentle curves, as a result of successive internal reflections. Such a *light pipe* can be flexible if thin fibers rather than thick rods are used. If a bundle of parallel fibers is used to construct an optical transmission line, images can be transferred from one point to another.



Light is guided along a fiber by multiple internal reflections.

This technique is used in a technology known as *fiber optics*. Very little light intensity is lost in these fibers as a result of reflections on the sides. Any loss of intensity is due essentially to reflections from the two ends and absorption by the fiber material. Fiber-optic devices are particularly useful for viewing images produced at inaccessible locations. For example, a fiber-optic cable can be threaded through the esophagus and into the stomach to look for ulcers.

Fiber optic cables are widely used in telecommunications because the fibers can carry much higher volumes of telephone calls and computer signals than can electrical wires.

ATMOSPHERIC REFRACTION

We see an example of refraction every day: the sun can be seen even after it has passed below the horizon. Rays of light from the sun strike Earth's atmos-phere and are bent because the atmosphere has an index of refraction differ-ent from that of the near-vacuum of space. The bending in this situation is gradual and continuous because the light moves through layers of air that have a continuously changing index of refraction. Our eyes follow them back along the direction from which they appear to have come.

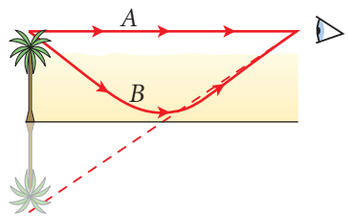


Figure 11
A mirage is produced by the bend-ing of light rays in the atmosphere when there are large temperature differences between the ground and the air.

Refracted light produces mirages

The *mirage* is another phenomenon of nature produced by refraction in the atmosphere. A mirage can be observed when the ground is so hot that the air directly above it is warmer than the air at higher elevations.

These layers of air at different heights above Earth have different densities and different refractive indices. The effect this can have is pictured in **Figure 11.** In this situation, the observer sees a tree in two different ways. One group of light rays reaches the observer by the straight-line path *A,* and the eye traces these rays back to see the tree in the normal fashion. A second group of rays travels along the curved path *B.* These rays are directed toward the ground and are then bent as a result of refraction. Consequently, the observer also sees an inverted image of the tree by tracing these rays back to the point at which they appear to have originated. Because both an upright image and an inverted image are seen when the image of a tree is observed in a reflecting pool of water, the observer subconsciously calls upon this past experience and concludes that a pool of water must be in front of the tree.

dispersion

the process of separating poly-chromatic light into its compo-nent wavelengths

DISPERSION

An important property of the index of refraction is that its value in anything but a vacuum depends on the wavelength of light. Because the index of refrac-tion is a function of wavelength, Snell's law indicates that incoming light of dif-ferent wavelengths is bent at different angles as it moves into a refracting material. This phenomenon is called **dispersion.** As mentioned in Section 1, the index of refraction decreases with increasing wavelength. For instance, blue light ($\lambda \approx 470$ nm) bends more than red light ($\lambda \approx 650$ nm) when passing into a refracting material.

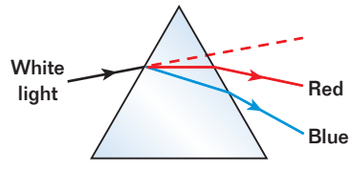


Figure 12
When white light enters a prism, the blue light is bent more than the red, and the prism disperses the white light into its various spectral components.

White light passed through a prism produces a visible spectrum

To understand how dispersion can affect light, consider what happens when light strikes a prism, as in **Figure 12.** Because of dispersion, the blue compo-nent of the incoming ray is bent more than the red component, and the rays that emerge from the second face of the prism fan out in a series of colors known as a *visible spectrum.* These colors, in order of decreasing wavelength, are red, orange, yellow, green, blue, and violet.

Rainbows are created by dispersion of light in water droplets

The dispersion of light into a spectrum is demonstrated most vividly in nature by a rainbow, often seen by an observer positioned between the sun and a rain shower. When a ray of sunlight strikes a drop of water in the atmosphere, it is first refracted at the front surface of the drop, with the violet light refracting the most and the red light the least. Then, at the back surface of the drop, the

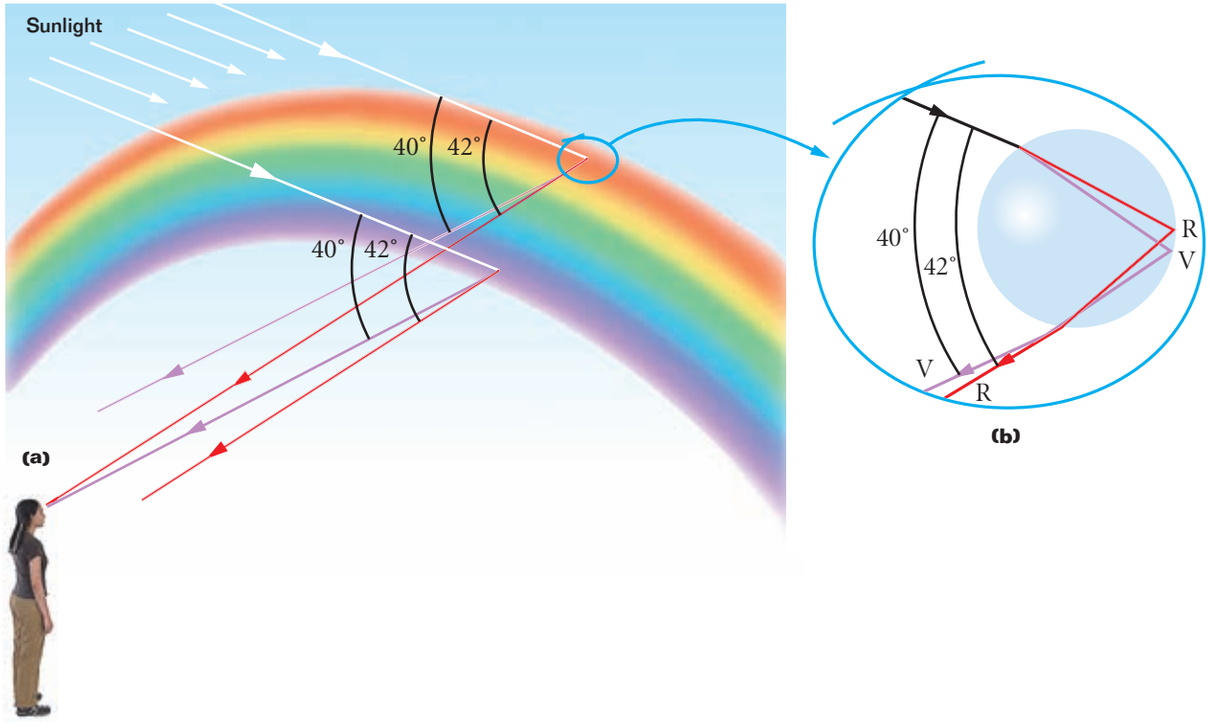


Figure 13
Rainbows **(a)** are produced because of dispersion of light in rain- drops. Sunlight is spread into a spectrum upon entering a spherical raindrop **(b),** then internally reflected on the back side of the raindrop. The perceived color of each water droplet then depends on the angle at which that drop is viewed.

light is reflected and returns to the front surface, where it again undergoes refraction as it moves from water into air. The rays leave the drop so that the angle between the incident white light and the returning violet ray is 40° and the angle between the white light and the returning red ray is 42°, as shown in **Figure 13(b).**

Now, consider **Figure 13(a).** When an observer views a raindrop high in the sky, the red light reaches the observer, but the violet light, like the other spectral colors, passes over the observer because it deviates from the path of the white light more than the red light does. Hence, the observer sees this drop as being red. Similarly, a drop lower in the sky would direct violet light toward the observer and appear to be violet. (The red light from this drop would strike the ground and not be seen.) The remaining colors of the spectrum would reach the observer from raindrops lying between these two extreme positions.

Note that rainbows are most commonly seen above the horizon, where the ends of the rainbow disappear into the ground. However, if an observer is at an elevated vantage point, such as on an airplane or at the rim of a canyon, a complete circular rainbow can be seen.

Summary

KEY IDEAS

1 . Refraction

- According to Snell's law, as a light ray travels from one medium into another medium where its speed is different, the light ray will change its direction unless it travels along the normal.
- When light passes from a medium with a smaller index of refraction to one with a larger index of refraction, the ray bends toward the normal. For the opposite situation, the ray bends away from the normal.

2. Thin Lenses

- The image produced by a converging lens is real and inverted when the object is outside the focal point and virtual and upright when the object is inside the focal point. Diverging lenses always produce upright, virtual images.
- The location of an image created by a lens can be found using either a ray diagram or the thin-lens equation.

3 . Optical Phenomena

- Total internal reflection can occur when light attempts to move from a material with a higher index of refraction to one with a lower index of refraction. If the angle of incidence of a ray is greater than the critical angle, the ray is totally reflected at the boundary.
- Mirages and the visibility of the sun after it has physically set are natural phe-nomena that can be attributed to refraction of light in Earth's atmosphere.

Variable Symbols

Quantities		Units	
θ_i	angle of incidence	°	degrees
θ_r	angle of refraction	°	degrees
n	index of refraction		
p	distance from object to lens	m	meters
q	distance from image to lens	m	meters
h'	image height	m	meters
h	object height	m	meters
θ_c	critical angle	°	degrees

Review questions

REFRACTION AND SNELL'S LAW

Review Questions

1. Does a light ray traveling from one medium into another always bend toward the normal?

2. As light travels from a vacuum ($n = 1$) to a medium such as glass ($n > 1$), does its wavelength change? Does its speed change? Does its frequency change?

3. What is the relationship between the speed of light and the index of refraction of a transparent substance?

4. Why does a clear stream always appear to be shallower than it actually is?

5. What are the three conditions that must be met for refraction to occur?

Conceptual Questions

6. Two colors of light (X and Y) are sent through a glass prism, and X is bent more than Y. Which color travels more slowly in the prism?

7. Why does an oar appear to be bent when part of it is in the water?

8. A friend throws a coin into a pool. You close your eyes and dive toward the spot where you saw it from the edge of the pool. When you reach the bottom, will the coin be in front of you or behind you?

9. The level of water ($n = 1.33$) in a clear glass container is easily observed with the naked eye. The level of liquid helium ($n = 1.03$) in a clear glass container is extremely difficult to see with the naked eye. Explain why.

Practice Problems

For problems 10–14, see Sample Problem A.

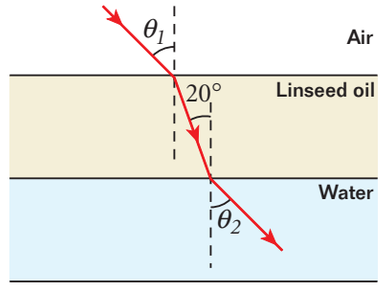
10. Light passes from air into water at an angle of incidence of 42.3°. Determine the angle of refraction in the water.

11. A ray of light enters the top of a glass of water at an angle of 36° with the vertical. What is the angle between the refracted ray and the vertical?

12. A narrow ray of yellow light from glowing sodium ($\lambda_0 = 589$ nm) traveling in air strikes a smooth surface of water at an angle of $\theta_i = 35.0°$. Determine the angle of refraction, θ_r.

13. A ray of light traveling in air strikes a flat 2.00 cm thick block of glass ($n = 1.50$) at an angle of 30.0° with the normal. Trace the light ray through the glass, and find the angles of incidence and refraction at each surface.

14. The light ray shown in the figure below makes an angle of 20.0° with the normal line at the boundary of linseed oil and water. Determine the angles θ_1 and θ_2. Note that $n = 1.48$ for linseed oil.



RAY DIAGRAMS AND THIN LENSES

Review Questions

15. Which type of lens can focus the sun's rays?

16. Why is no image formed when an object is at the focal point of a converging lens?

17. Consider the image formed by a thin converging lens. Under what conditions will the image be

- **a.** inverted?
- **b.** upright?
- **c.** real?
- **d.** virtual?
- **e.** larger than the object?
- **f.** smaller than the object?

18. Repeat a–f of item 17 for a thin diverging lens.

19. Explain this statement: The focal point of a converging lens is the location of an image of a point object at infinity. Based on this statement, can you think of a quick method for determining the focal length of a positive lens?

26. An object is placed in front of a converging lens with a focal length of 20.0 cm. For each object distance, find the image distance and the magnification. Describe each image.

- **a.** 40.0 cm
- **b.** 10.0 cm

TOTAL INTERNAL REFLECTION, ATMOSPHERIC REFRACTION, AND ABERRATIONS

Conceptual Questions

20. If a glass converging lens is submerged in water, will its focal length be longer or shorter than when the lens is in air?

21. In order to get an upright image, slides must be placed upside down in a slide projector. What type of lens must the slide projector have? Is the slide inside or outside the focal point of the lens?

22. If there are two converging lenses in a compound microscope, why is the image still inverted?

23. In a Jules Verne novel, a piece of ice is shaped into the form of a magnifying lens to focus sunlight and thereby start a fire. Is this possible?

Review Questions

27. Is it possible to have total internal reflection for light incident from air on water? Explain.

28. What are the conditions necessary for the occurrence of a mirage?

29. On a hot day, what is it that we are seeing when we observe a "water on the road" mirage?

30. Why does the arc of a rainbow appear with red colors on top and violet colors on the bottom?

31. What type of aberration is involved in each of the following situations?

- **a.** The edges of the image appear reddish.
- **b.** The central portion of the image cannot be clearly focused.
- **c.** The outer portion of the image cannot be clearly focused.
- **d.** The central portion of the image is enlarged relative to the outer portions.

Conceptual Questions

32. A laser beam passing through a nonhomogeneous sugar solution follows a curved path. Explain.

33. On a warm day, the image of a boat floating on cold water appears above the boat. Explain.

34. Explain why a mirror cannot give rise to chromatic aberration.

35. Why does a diamond show flashes of color when observed under ordinary white light?

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